Marine Water Quality in Hong Kong in 2024

Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Our Mission

To conduct a comprehensive and scientific monitoring programme that helps safeguard the health of Hong Kong's marine environment and achieve the Water Quality Objectives.

Acknowledgements

We acknowledge the following parties for their contribution to this report: Government Laboratory for undertaking chemical analyses of water and sediment samples; Marine Department for managing and operating the monitoring vessel *Dr. Catherine Lam* in conducting field sampling and measurement; Agriculture, Fisheries and Conservation Department for providing photos and information on marine parks, fish culture zones and red tides; Lands Department for providing information on geography and aerial photography of Hong Kong.

Abbreviations / 簡稱

Advance Disinfection Facilities	ADF	前期消毒設施
Ammonia Nitrogen	NH_4-N	氨氮
Conductivity-Temperature-Depth	CTD / 溫鹽深	溫度、鹽度、深度
Dissolved Oxygen	DO	溶解氧
Environmental Protection Department	EPD / 環保署	環境保護署
Escherichia coli	*E. coli*	大腸桿菌
Harbour Area Treatment Scheme	HATS	淨化海港計劃
Orthophosphate Phosphorus	PO_4-P	正磷酸鹽磷
Sewage Treatment Works	STW	污水處理廠
Stonecutters Island Sewage Treatment Works	SCISTW	昂船洲污水處理廠
Total Inorganic Nitrogen	TIN	總無機氮
Unionised Ammonia Nitrogen	NH_3-N	非離子化氨氮
Water Control Zone	WCZ	水質管制區
Water Quality Objective	WQO	水質指標
5-day Biochemical Oxygen Demand	BOD_5	五天生化需氧量

Content

1. **Introduction**
2. **The State of Hong Kong Marine Waters in 2024**
   - **2.1 Overall Compliance Rate of Water Quality Objectives (WQOs)**
   - **2.2 Highlight in 2024 – Water Quality of Victoria Harbour**
3. **Water Quality of Ten Water Control Zones (WCZs)**
   - **3.1 Eastern Waters**
   - **3.2 Central Waters**
   - **3.3 Western Waters**
   - **3.4 Southern Waters**
4. **Marine Sediment Quality**
5. **Typhoon Shelters**
6. **Phytoplankton**
   - **6.1 Phytoplankton Composition and Density**
   - **6.2 Phytoplankton Community Integrity Index (PCII)**
   - **6.3 Status of Red Tides**

List of Figures

Figure 1	Overall marine WQO compliance rates, 1986-2024
Figure 2	Compliance rates for four key marine WQOs, 1986-2024
Figure 3	Marine WQO compliance rates for the ten WCZs, 2022-2024
Figure 4	Water quality improvement in Victoria Harbour after the phased implementation of Harbour Area Treatment Scheme (HATS) since 2001
Figure 5	Overall WQO compliance rate for the Mirs Bay WCZ, 1986-2024
Figure 6	Long-term water quality trends in the Mirs Bay WCZ, 1986-2024
Figure 7	Overall WQO compliance rate for the Port Shelter WCZ, 1986-2024
Figure 8	Long-term water quality trends in the Port Shelter WCZ, 1986-2024
Figure 9	Overall WQO compliance rate for the Tolo Harbour and Channel WCZ, 1986-2024
Figure 10	Long-term water quality trends in the Tolo Harbour and Channel WCZ, 1986-2024

Figure 11	*E. coli* level at central (VM5) and eastern (VM1) parts of the Victoria Harbour WCZ, 1990-2024
Figure 12	Overall WQO compliance rate for the Victoria Harbour WCZ, 1986-2024
Figure 13	Long-term water quality trends in the Victoria Harbour WCZ, 1986-2024
Figure 14	Overall WQO compliance rate for the Eastern Buffer WCZ, 1986-2024
Figure 15	Long-term water quality trends in the Eastern Buffer WCZ, 1986-2024
Figure 16	Overall WQO compliance rate for the Junk Bay WCZ, 1986-2024
Figure 17	Long-term water quality trends in the Junk Bay WCZ, 1986-2024
Figure 18	Overall WQO compliance rate for the Western Buffer WCZ, 1986-2024
Figure 19	Long-term water quality trends in the Western Buffer WCZ, 1986-2024
Figure 20	Overall WQO compliance rate for the Deep Bay WCZ, 1986-2024
Figure 21	Long-term water quality trends in Inner Subzone of the Deep Bay WCZ, 1986-2024
Figure 22	Long-term water quality trends in Outer Subzone of the Deep Bay WCZ, 1986-2024
Figure 23	Overall WQO compliance rate for the North Western WCZ, 1986-2024
Figure 24	Long-term water quality trends in the North Western WCZ, 1986-2024
Figure 25	Overall WQO compliance rate for the Southern WCZ, 1986-2024
Figure 26	Long-term water quality trends in the Southern WCZ, 1986-2024
Figure 27	Long-term improvement in dissolved oxygen level in the Kwun Tong Typhoon Shelter, 1990-2024
Figure 28	PCII charts, index values and ranks for Mirs Bay WCZ
Figure 29	PCII charts, index values and ranks for Port Shelter WCZ
Figure 30	PCII charts, index values and ranks for Tolo Harbour and Channel WCZ
Figure 31	PCII charts, index values and ranks for Victoria Harbour WCZ
Figure 32	PCII charts, index values and ranks for Western Buffer WCZ
Figure 33	PCII charts, index values and ranks for Deep Bay WCZ
Figure 34	PCII charts, index values and ranks for Northwestern WCZ
Figure 35	PCII charts, index values and ranks for Southern WCZ
Figure 36	Concurrent reductions of nutrients levels and red tide occurrences in Tolo Harbour, 1986-2024

List of Appendices

Appendix A Background information of the marine water quality monitoring programme

The Water Control Zones in Hong Kong	A-1
The 76 water quality monitoring stations in the open waters of Hong Kong	A-2
The 45 sediment quality monitoring stations in the open waters of Hong Kong	A-3
The 18 water quality and 15 sediment quality monitoring stations in typhoon shelters, sheltered anchorages and Government Dockyard	A-4
Locations of marine water and sediment quality monitoring stations	A-5
Bathing beaches and secondary contact recreation subzones in Hong Kong	A-6
Fish and shellfish culture zones and marine conservation sites in Hong Kong	A-7
Water Quality Objective (WQO) for Total Inorganic Nitrogen (TIN) in the 10 Water Control Zones	A-8
Summary of Water Quality Objectives (WQOs) for marine waters of Hong Kong	A-9
Sediment quality criteria for classification of sediments	A-10
Summary of marine water quality parameters	A-11
Summary of marine sediment quality parameters	A12

Appendix B Summary of marine water quality monitoring data in 2024

Summary of water quality data for the Mirs Bay WCZ in 2024	B-1 – B-2
Summary of water quality data for the Port Shelter WCZ in 2024	B-3 – B-4
Summary of water quality data for the Tolo Harbour and Channel WCZ in 2024	B-5
Summary of water quality data for the Southern WCZ in 2024	B-6 – B-8
Summary of water quality data for the Victoria Harbour WCZ in 2024	B-9 – B-10
Summary of water quality data for the Eastern Buffer WCZ in 2024	B-11
Summary of water quality data for the Western Buffer WCZ in 2024	B-12
Summary of water quality data for the Junk Bay WCZ in 2024	B-13
Summary of water quality data for the Deep Bay WCZ in 2024	B-14

Summary of water quality data for the North Western WCZ in 2024 B-15

Appendix C WQO compliance rates for individual monitoring stations in each WCZ in 2024

WQO compliance rates for the Mirs Bay WCZ C-1 – C-5
WQO compliance rates for the Port Shelter WCZ C-6 – C-10
WQO compliance rates for the Tolo Harbour and Channel WCZ C-11 – C-13
WQO compliance rates of chlorophyll-*a* for the Tolo Harbour and Channel WCZ C-14
Total inorganic nitrogen and unionised ammonia levels in the Tolo Harbour and Channel WCZ C-15
WQO compliance rates for the Southern WCZ C-16 – C-22
WQO compliance rates for the Victoria Harbour WCZ C-23 – C-26
WQO compliance rates for the Eastern Buffer WCZ C-27 – C-28
WQO compliance rates for the Western Buffer WCZ C-29 – C-30
WQO compliance rates for the Junk Bay WCZ C-31
WQO compliance rates for the Deep Bay WCZ C-32 – C-33
WQO compliance rates for the North Western WCZ C-34 – C-35

Appendix D Results of water quality trend analysis for each WCZ

Long-term water quality trends in the Mirs Bay WCZ, 1991-2024 D-1
Long-term water quality trends in the Mirs Bay WCZ, 1986-2024 D-2
Long-term water quality trends in the Port Shelter WCZ, 1986-2024 D-3
Long-term water quality trends in the Tolo Harbour and Channel WCZ, 1986-2024 D-4
Long-term water quality trends in the Southern WCZ, 1986-2024 D-5 – D-6
Long-term water quality trends in the Victoria Harbour WCZ, 1986-2024 D-7 – D-8
Long-term water quality trends in the Eastern Buffer WCZ, 1986-2024 D-9
Long-term water quality trends sin the Western Buffer WCZ, 1986-2024 D-10
Long-term water quality trends in the Junk Bay WCZ, 1986-2024 D-11

Long-term water quality trends sin the Deep Bay WCZ, 1986-2024 D-12
Long-term water quality trends in the North Western WCZ, 1986-2024 D-13

Appendix E Summary of marine sediment quality monitoring data

Summary of marine sediment quality data for the Tolo Harbour and Channel WCZ and Southern WCZ, 2020-2024 E-1
Summary of marine sediment quality data for the Southern, Junk Bay WCZ and Deep Bay WCZ, 2020-2024 E-2
Summary of marine sediment quality data for the Port Shelter WCZ and Mirs Bay WCZ, 2020-2024 E-3
Summary of marine sediment quality data for the Mirs Bay WCZ, 2020-2024 E-4
Summary of marine sediment quality data for the North Western WCZ and Western Buffer WCZ, 2020-2024 E-5
Summary of marine sediment quality data for the Eastern Buffer WCZ and Victoria Harbour WCZ, 2020-2024 E-6

Appendix F Marine water and sediment quality in typhoon shelters, sheltered anchorages and Government Dockyard in 2024

Water quality in typhoon shelters, sheltered anchorages and Government Dockyard in 2024 F-1 – F-2
Long-term water quality trends in typhoon shelters, sheltered anchorages and Government Dockyard, 1986-2024 F-3 – F-4
Summary of water quality data for typhoon shelters, sheltered anchorages and Government Dockyard in 2024 F-5 – F-7
Summary of marine sediment quality data for typhoon shelters, sheltered anchorages and Government Dockyard, 2020-2024 F-8 – F-9

Appendix G Phytoplankton monitoring

The 26 phytoplankton monitoring stations in Hong Kong marine waters G-1
Composition (%) of phytoplankton groups in terms of total number of species in Hong Kong marine waters in 2024 G-2
Composition (%) of phytoplankton groups in terms of total density in Hong Kong marine waters in 2024 G-3

Annual mean total phytoplankton densities at 26 monitoring stations in Hong Kong marine waters in 2024 G-4
Annual mean diatoms densities at 26 monitoring stations in Hong Kong marine waters in 2024 G-5
Annual mean dinoflagellates densities at 26 monitoring stations in Hong Kong marine waters in 2024 G-6
Annual mean densities of other minor phytoplankton groups at 26 monitoring stations in Hong Kong marine waters in 2024 G-7
Sightings of red tide incidents in Hong Kong marine waters, 1975-2024 G-8
Occurrence of red tides in Hong Kong marine waters, 1975-2024 G-9
Seasonal distribution of red tides caused by different phytoplankton groups in Hong Kong marine waters, 1975-2024 G-10
Percentage abundance of the three most dominant phytoplankton species in Hong Kong marine waters in 2024 G-11

Appendix H Overall marine water quality in Hong Kong waters

Marine WQO compliance rates for the 10 WCZs, 2015-2024 H-1
Compliance rates for key marine WQOs, 2015-2024 H-2
Overall compliance rates for key marine WQOs, 1986-2024 H-2
Long-term changes in dissolved oxygen levels in Hong Kong marine waters,1986-2024 H-3
Long-term changes in 5-day biochemical oxygen demand levels in Hong Kong marine waters, 1986-2024 H-4
Long-term changes in *E. coli* levels in Hong Kong marine waters, 1986-2024 H-5
Long-term changes in ammonia nitrogen levels in Hong Kong marine waters, 1986-2024 H-6
Long-term changes in total inorganic nitrogen levels in Hong Kong marine waters, 1986-2024 H-7
Long-term changes in orthophosphate phosphorus levels in Hong Kong marine waters, 1986-2024 H-8
Long-term changes in chlorophyll-*a* levels in Hong Kong marine waters, 1986-2024 H-9
Long-term changes in temperature in Hong Kong marine waters, 1986-2024 H-10

1. Introduction

To protect the marine environment of Hong Kong for various beneficial and sustainable uses, the Environmental Protection Department (EPD) has been implementing a comprehensive marine water quality monitoring programme since 1986. The aims and objectives of the programme are to:

- evaluate the health state of marine waters;
- monitor long-term changes in water quality;
- provide a scientific basis for planning water pollution control strategies and evaluating their effectiveness; and
- assess the compliance with the Water Quality Objectives (WQOs).

On a monthly basis, the EPD monitors the marine water quality at 76 monitoring stations in open waters, and collects and examines phytoplankton samples taken from 26 of these stations. We also monitor, at bimonthly intervals, the water quality at 18 monitoring stations located in sheltered waters, including 14 typhoon shelters, three sheltered anchorages and the Government Dockyard. In addition, sediment samples are collected twice a year from 60 monitoring stations for analyses, including 45 stations in open waters and 15 stations in sheltered waters.

Most of the field work for water quality monitoring is conducted on board of *MV Dr. Catherine Lam* – EPD's marine monitoring vessel. An advanced conductivity-temperature-depth (CTD) profiler linked to a computer-controlled rosette water sampler is set up on the vessel to allow simultaneous depth profiling of *in situ* measurements and water sampling at specified depths. Marine sediments are collected by using a Van Veen sediment grab sampler. The water and sediment samples are analysed by the Government Laboratory and EPD's laboratory for over 80 physical, chemical and biological parameters. Details of the water quality and sediment parameters, their analytical methods as well as information of monitoring stations and WQOs are given in Appendix A.

2. The State of Hong Kong Marine Waters in 2024

2.1 Overall Compliance Rate of Marine Water Quality Objectives (WQOs)

The overall marine WQO compliance rate[1] for Hong Kong was 88% in 2024 (see Figure 1). The compliance rates for the four key WQO parameters and the overall WQO compliance rates for the ten water control zones (WCZs) are shown in Figures 2 and 3. Notably, full compliance with the WQOs on *Escherichia coli* (*E. coli*) and Unionised Ammonia Nitrogen (NH_3-N) in all applicable WCZs have been achieved for ten consecutive years since 2015. The compliance rates of the WQOs on Dissolved Oxygen (DO) and Total Inorganic Nitrogen (TIN) were 99% and 61% respectively in 2024.

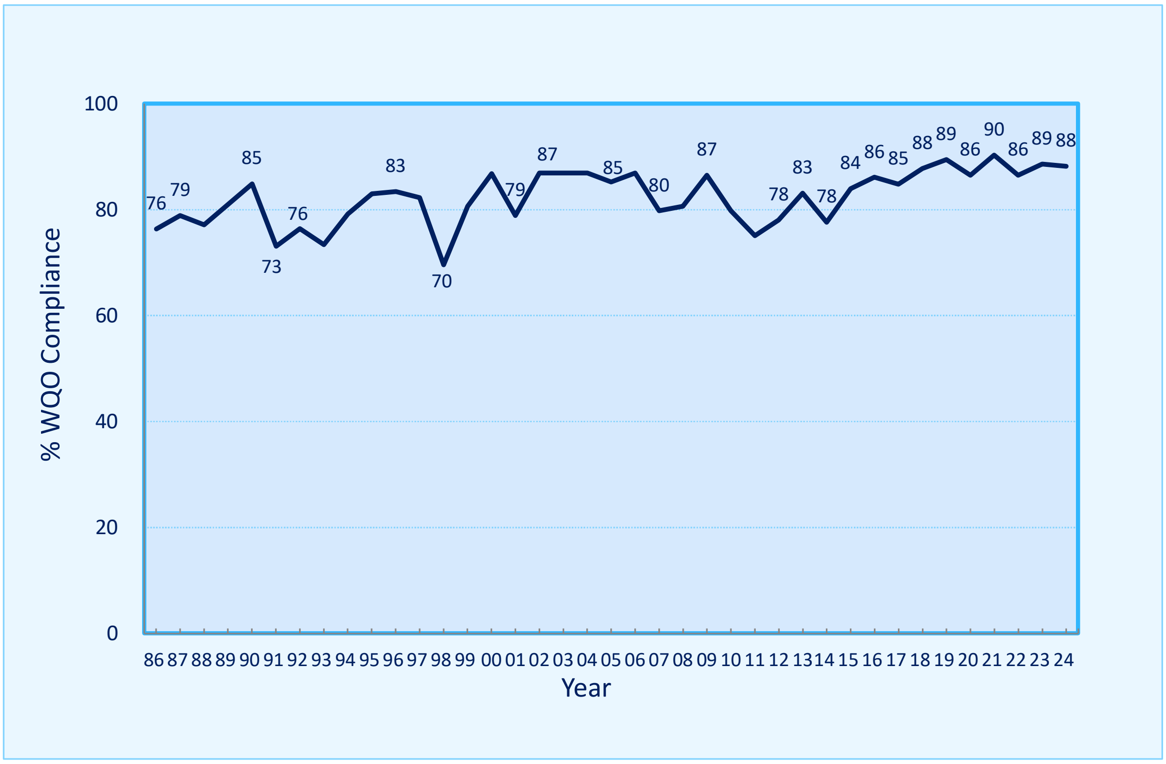


Figure 1. Overall marine WQO compliance rates, 1986-2024

For the DO WQO, it was not fully complied in certain areas in the Tolo Harbour and Channel WCZ. For the TIN WQO, non-compliance was observed in the waters with a relatively higher background level of TIN including those in the Deep Bay WCZ, the Southern WCZ and the North Western WCZ. The WQO compliance status of individual WCZs are presented in Chapter 3, with details of water quality monitoring data and WQO compliance status of individual monitoring stations given in Appendices B and C respectively.

[1] The overall marine WQO compliance rate for Hong Kong's marine waters is calculated based on the overall average of the compliance rates for all monitoring stations for the four key marine WQO parameters including DO, TIN, NH_3-N and *E. coli*.

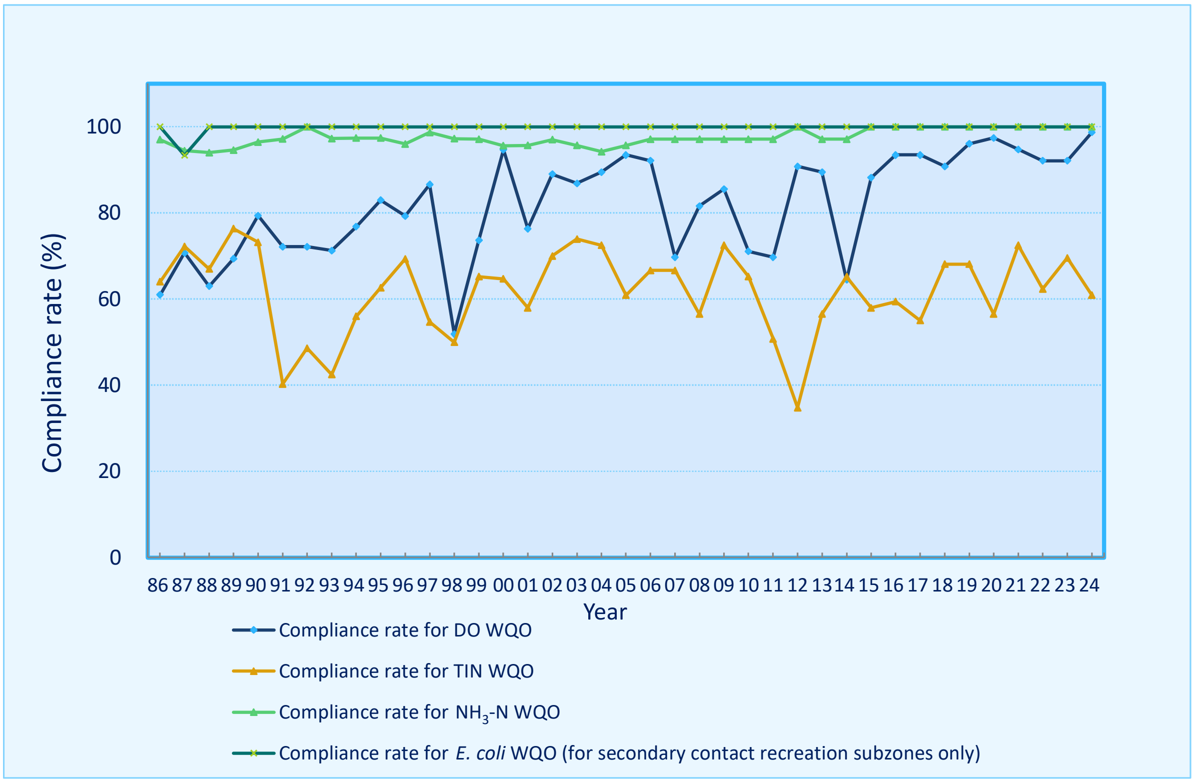


Figure 2. Compliance rates for four key marine WQOs, 1986-2024

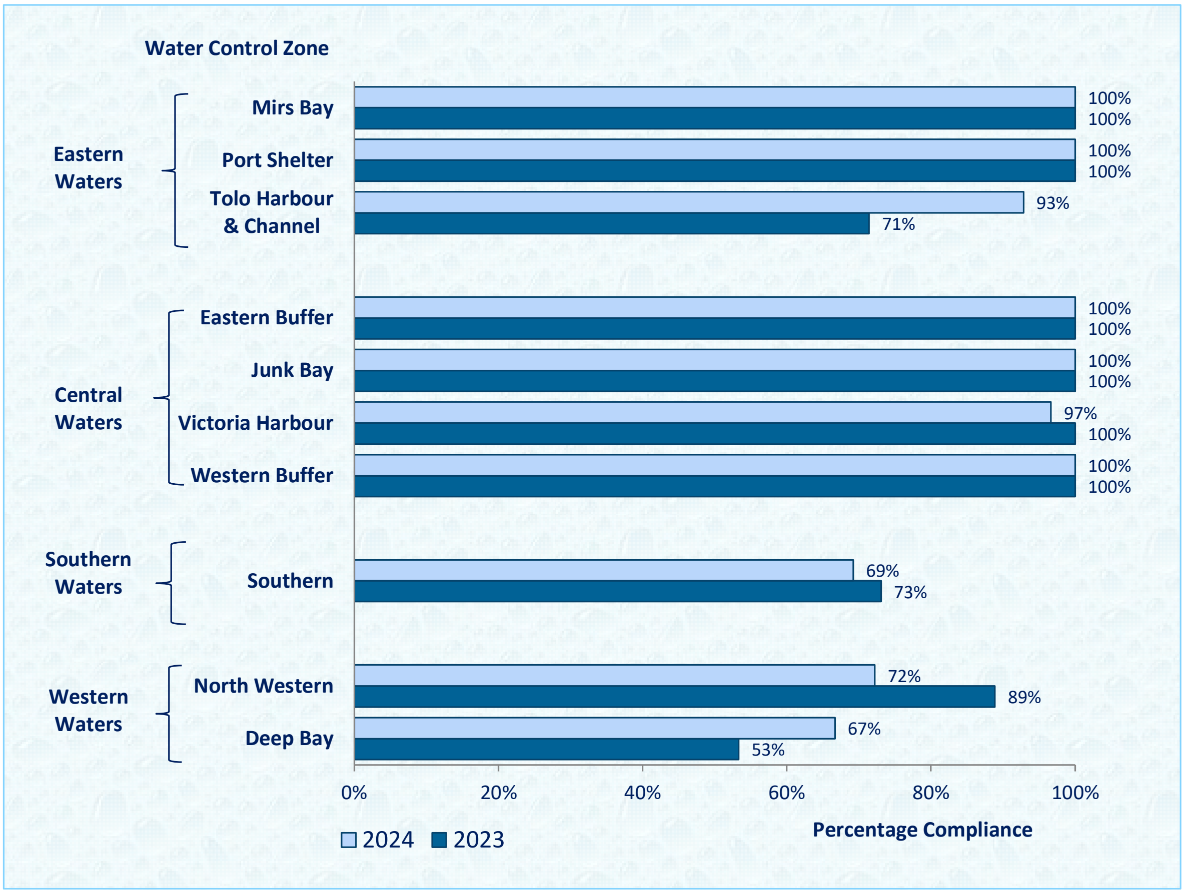


Figure 3. Marine WQO compliance rates for the ten WCZs, 2023-2024

2.2 Highlight in 2024 – Water Quality of Victoria Harbour

More than half of Hong Kong's population reside on both sides of the vibrant Victoria Harbour. The harbour has shown significant water quality improvement upon staged implementation of the Harbour Area Treatment Scheme (HATS) in the last 20 years. As illustrated in Figure 4, over the past nine years after the implementation of HATS Stage 2A, the average annual levels of *E. coli* and NH_3-N have significantly decreased to 424 counts per 100mL and 0.004 mg/L respectively, while that of DO has increased to 5.6 mg/L.

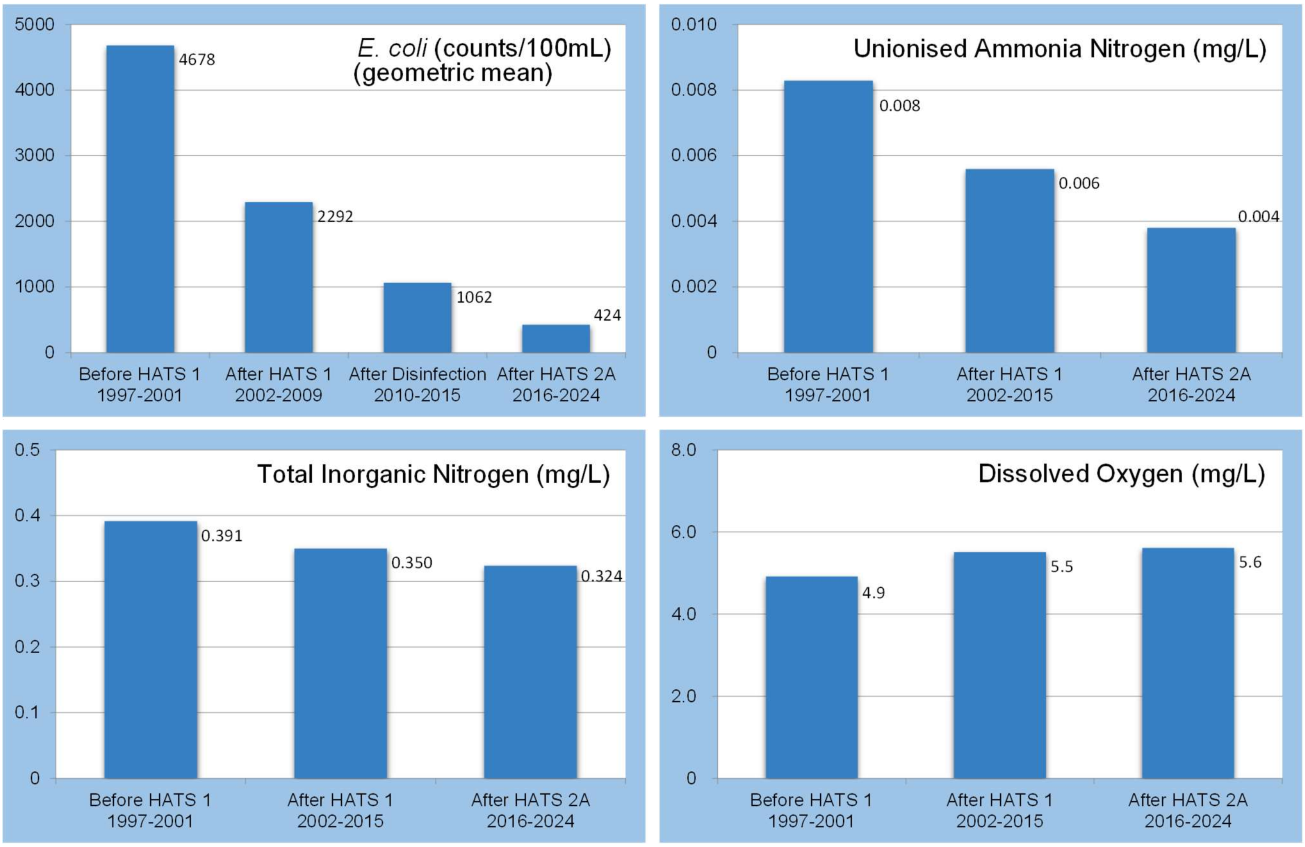
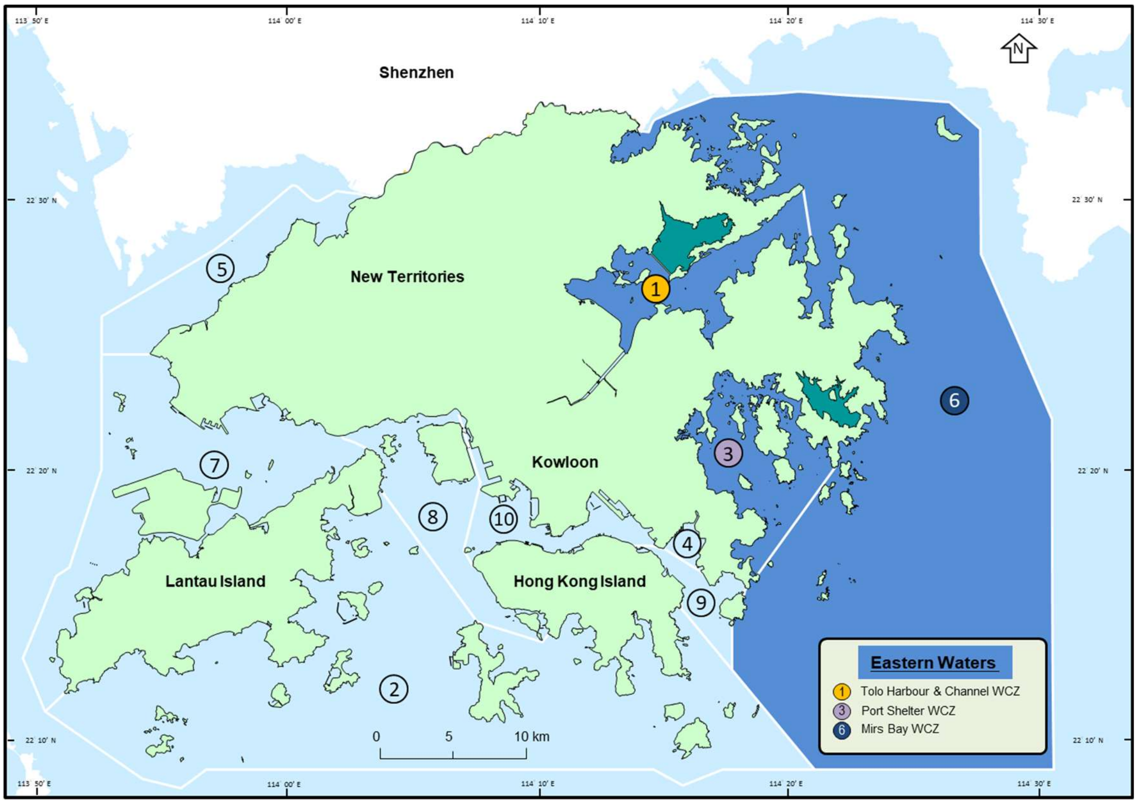


Figure 4. Water quality improvement in Victoria Harbour after the phased implementation of Harbour Area Treatment Scheme (HATS) since 2001

3. Water Quality of the Ten Water Control Zones (WCZs)

Based on the hydrographical conditions of Hong Kong waters and the proximity to the Pearl River Estuary, the ten WCZs are grouped into four main regions: namely Eastern, Central, Western and Southern Waters. Details of their respective water quality conditions are reported in the following sections.

3.1 Eastern Waters



The eastern waters comprise three WCZs: namely the Mirs Bay WCZ, the Port Shelter WCZ and the Tolo Harbour and Channel WCZ. These waterbodies contain seven gazetted bathing beaches, three marine parks (including Hoi Ha Wan Marine Park, Yan Chau Tong Marine Park and Tung Ping Chau Marine Park[1]), the Hong Kong Geopark, 22 fish culture zones and beautiful natural coastlines with pristine water quality supporting a diversified array of marine life, fisheries and recreation activities.

[1] Water quality in all marine parks and marine reserve are monitored by the Agriculture, Fisheries and Conservation Department (AFCD). For more details and water quality status of the marine parks and marine reserve, please visit the following webpages of AFCD.

https://www.afcd.gov.hk/english/country/cou_vis/cou_vis_mar/cou_vis_mar_des/cou_vis_mar_des.html

https://www.afcd.gov.hk/english/country/cou_vis/cou_vis_mar/cou_vis_mar_mon/cou_vis_mar_mon_wat.html

Mirs Bay Water Control Zone

In 2024, the Mirs Bay WCZ fully achieved the WQOs. The water quality was very good with high DO, and low nutrient and *E. coli* levels, fitting for various recreational and mariculture uses. Figures 5 and 6 illustrate the WQO compliance rates and some long-term water quality trends for the Mirs Bay WCZ in the past three decades.

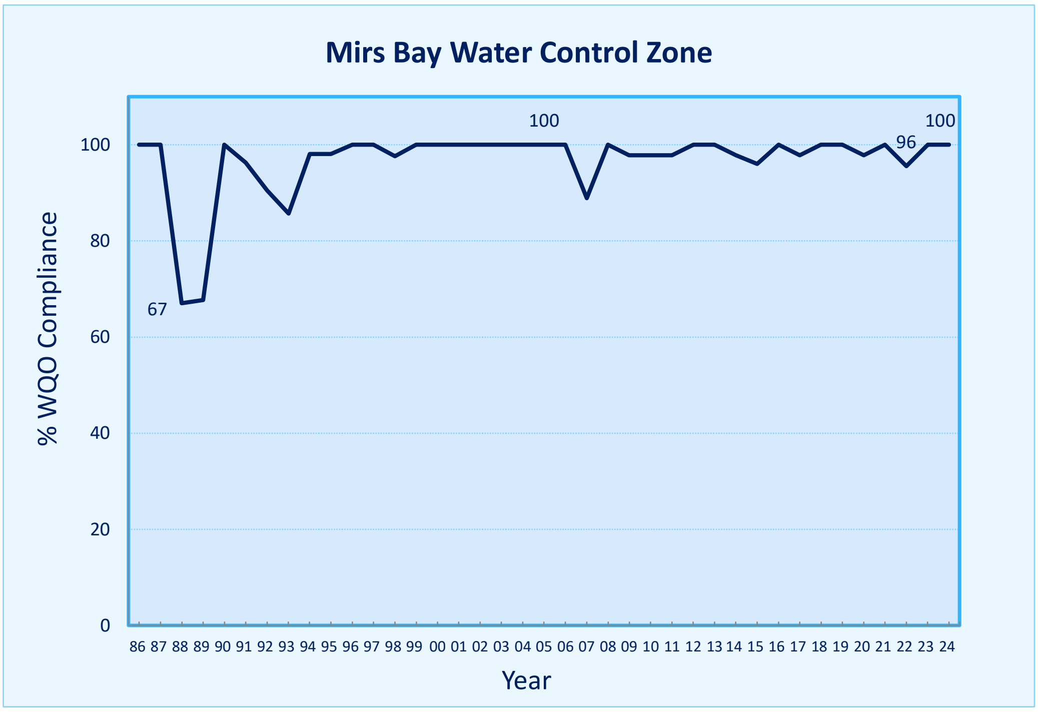


Figure 5. Overall WQO compliance rate for the Mirs Bay WCZ, 1986-2024

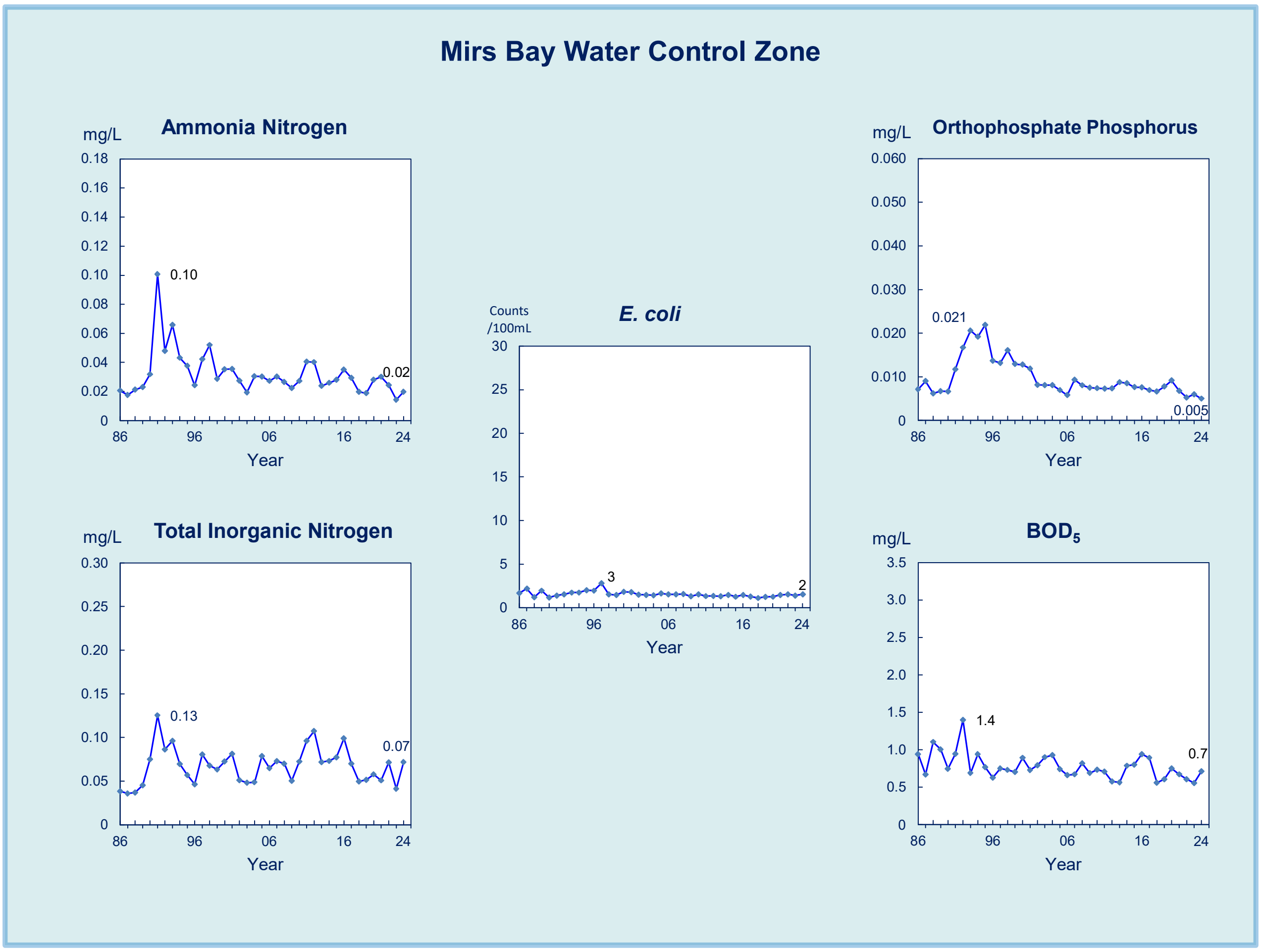


Figure 6. Long-term water quality trends in the Mirs Bay WCZ, 1986-2024

Port Shelter Water Control Zone

In 2024, the water quality of the Port Shelter WCZ remained excellent, fully achieving the marine WQOs.

The WQO compliance rates and long-term water quality trends for the Port Shelter WCZ since 1986 are illustrated in Figures 7 and 8. In addition to the generally low pollution levels, there was also a steady decrease in nutrients concentrations (including ammonia nitrogen (NH_4-N) and orthophosphate phosphorus (PO_4-P) in marine waters).

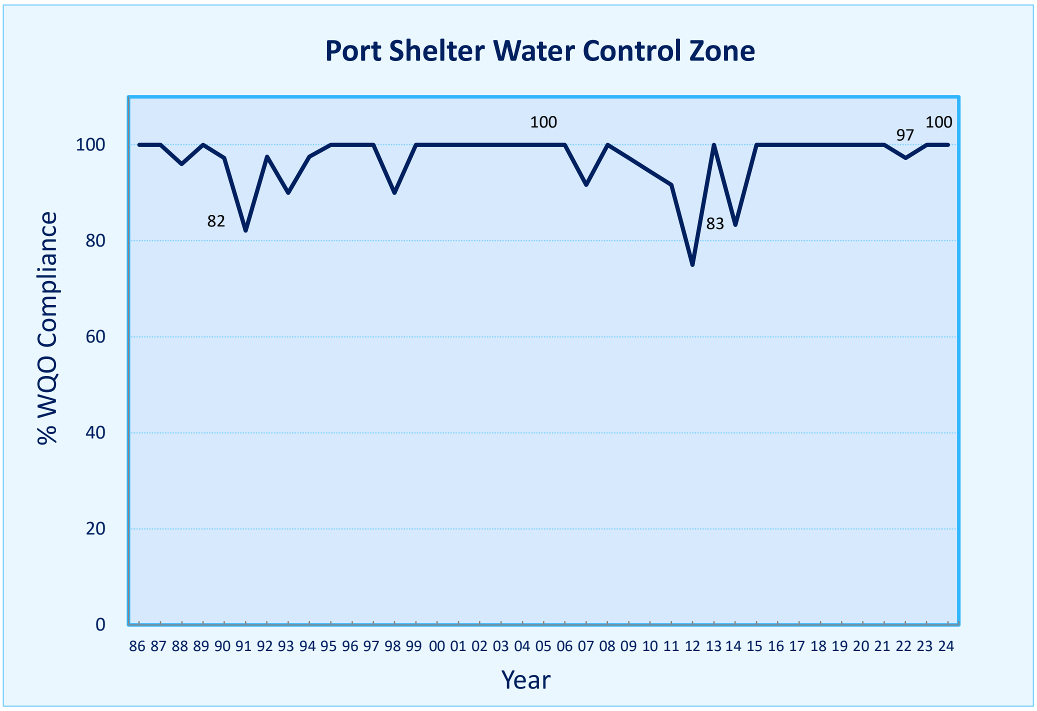


Figure 7. Overall WQO compliance rate for the Port Shelter WCZ, 1986-2024

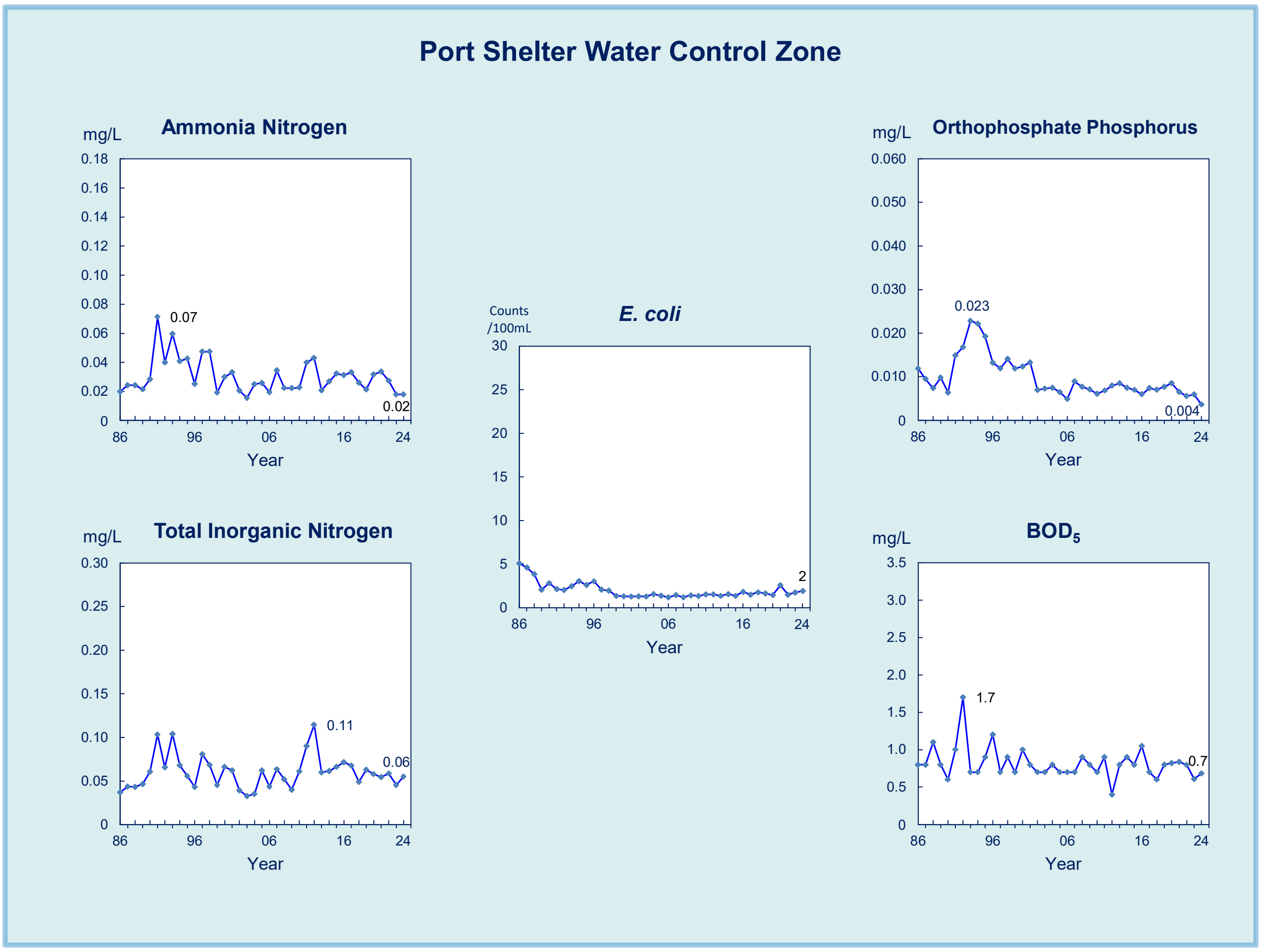


Figure 8. Long-term water quality trends in the Port Shelter WCZ, 1986-2024

Tolo Harbour and Channel Water Control Zone

The Tolo Harbour and Channel WCZ is highly land-locked, and hence its water body is generally subject to a natural hydrological phenomenon of water column stratification and associated formation of bottom layer water masses with relatively low DO level in summer period due to restricted water exchange with the open waters. In 2024, the overall marine WQO compliance rate for this WCZ was 93%, mainly ascribed to the influence on the DO WQO compliance rate in the Channel Subzone caused by the aforesaid natural hydrological phenomenon. On the other hand, the bacteriological WQO for secondary contact recreational uses in the WCZ has been consistently achieved, indicating a good water quaity suitable for the beneficial uses.

Upon the gradual implementation of the Tolo Harbour Action Plan (including the "Tolo Harbour Effluent Export Scheme"[1]) by the government since the mid-1980s, there has been substantial improvement in the water quality in Tolo Harbour in the past three decades, as shown in Figures 9 and10.

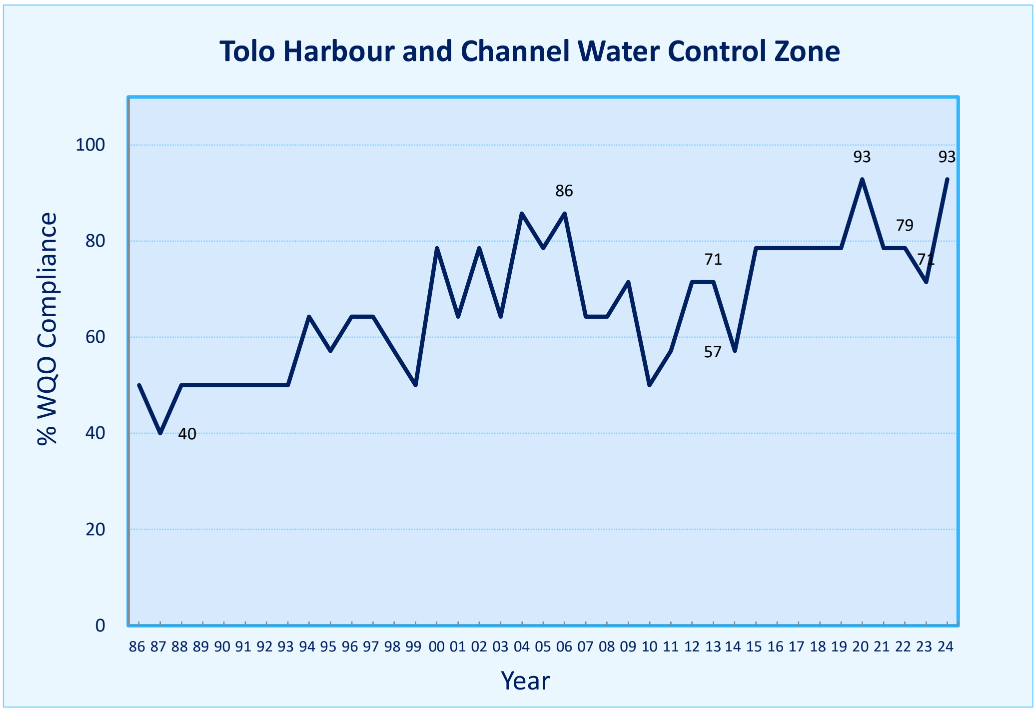


Figure 9. Overall WQO compliance rate for the Tolo Harbour and Channel WCZ, 1986-2024

[1] In 1998, the government fully implemented the "Tolo Harbour Effluent Export Scheme", transporting all treated effluent from Sha Tin and Tai Po Sewage Treatment Plants to Victoria Harbour for discharge.

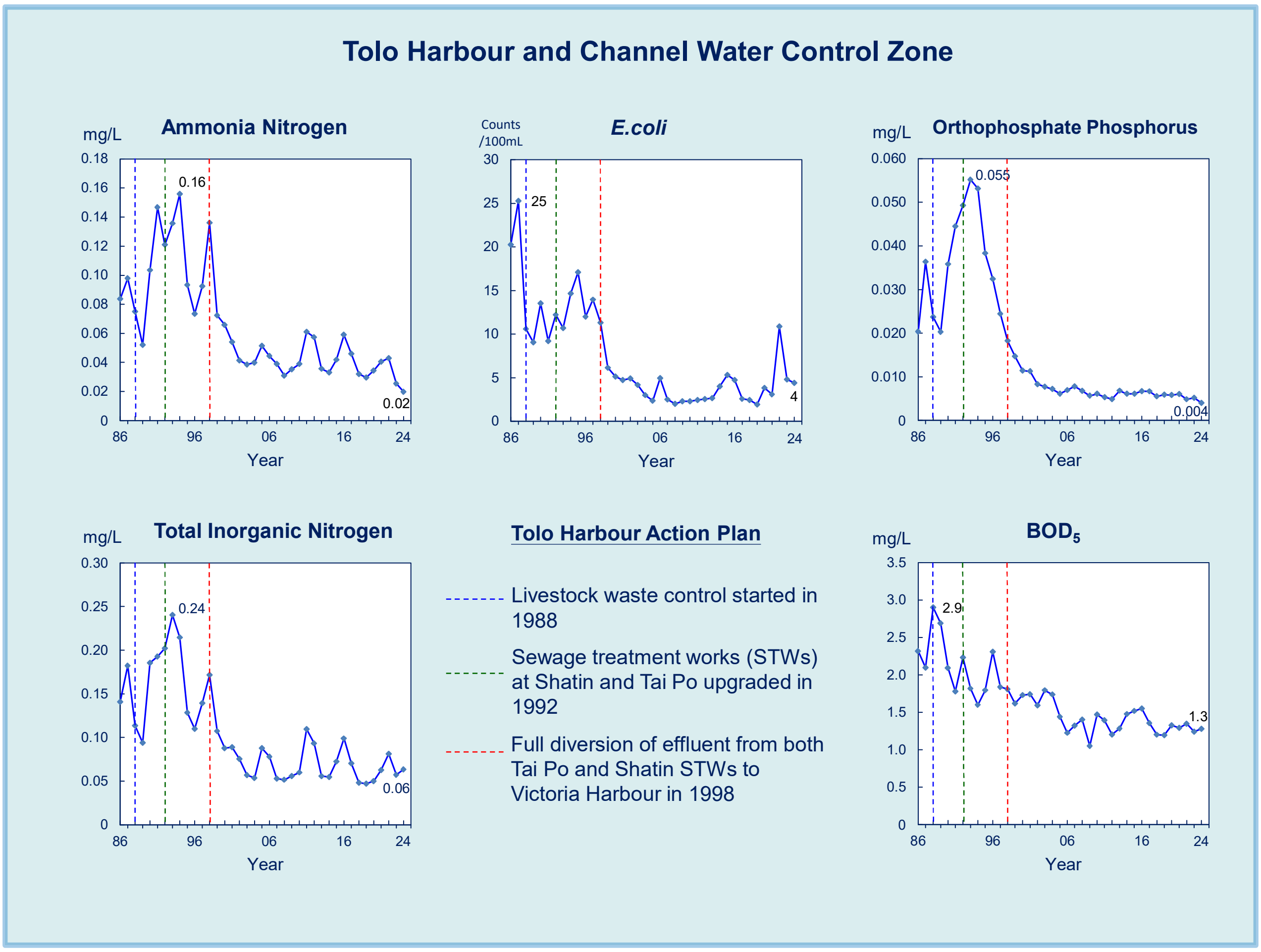
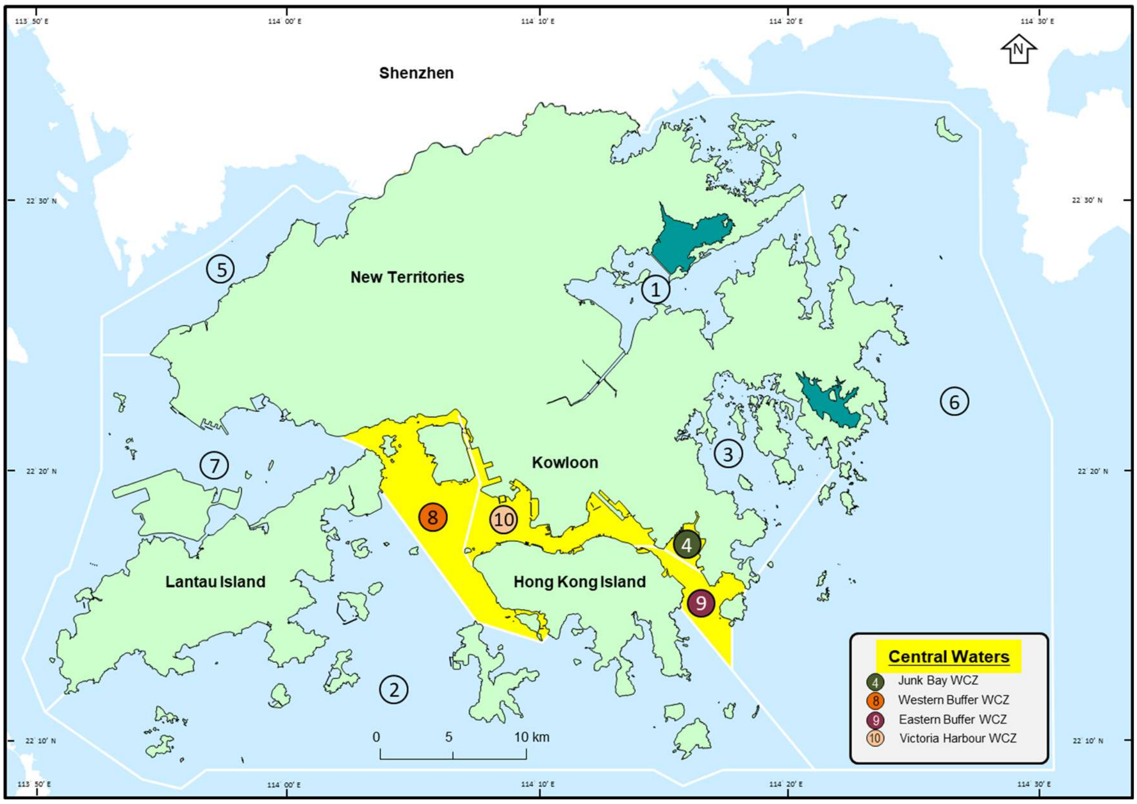


Figure 10. Long-term water quality trends in the Tolo Harbour and Channel WCZ, 1986-2024

3.2 Central Waters



The central waters of Hong Kong are important port areas and navigational channels covering four WCZs, i.e. the Victoria Harbour WCZ, the Eastern Buffer WCZ, the Western Buffer WCZ and the Junk Bay WCZ.

Victoria Harbour Water Control Zone

The Victoria Harbour WCZ achieved an overall WQO compliance rate of 97% in 2024.

As shown in Figure 11, the *E. coli* level in the eastern side of Victoria Harbour has decreased markedly since the implementation of HATS Stage 1 in 2001. The annual Cross Harbour Race, suspended since 1979 because of poor water quality, was resumed on the eastern side of the harbour in 2011 after implementation of the Advance Disinfection Facilities (ADF) of HATS. With full commissioning of the HATS Stage 2A, the *E. coli* level of the central harbour area has been further reduced. Since 2017, the race route of the event has returned to the traditional route in the central harbour.

Figures 12 and 13 show the WQO compliance rates and some long-term water quality trends for the Victoria Harbour WCZ in the period of 1986 to 2024.

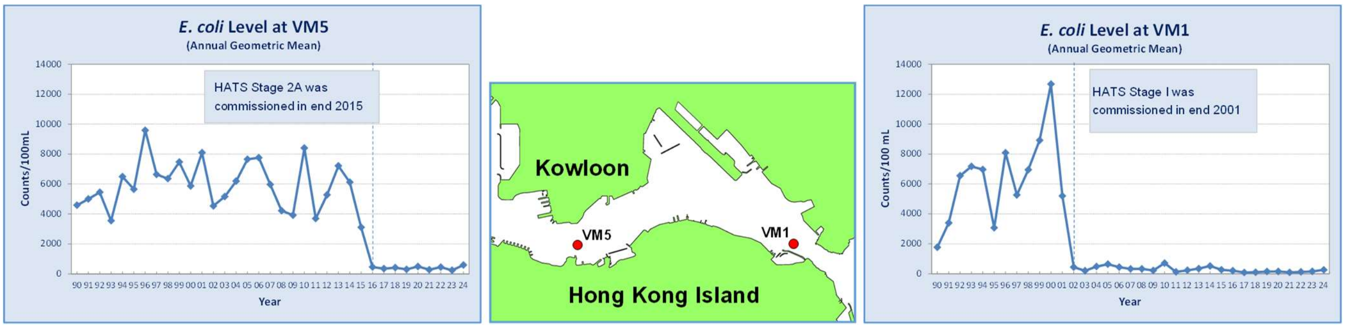


Figure 11. *E. coli* level at the central (VM5) and eastern (VM1) parts of the Victoria Harbour WCZ, 1990-2024

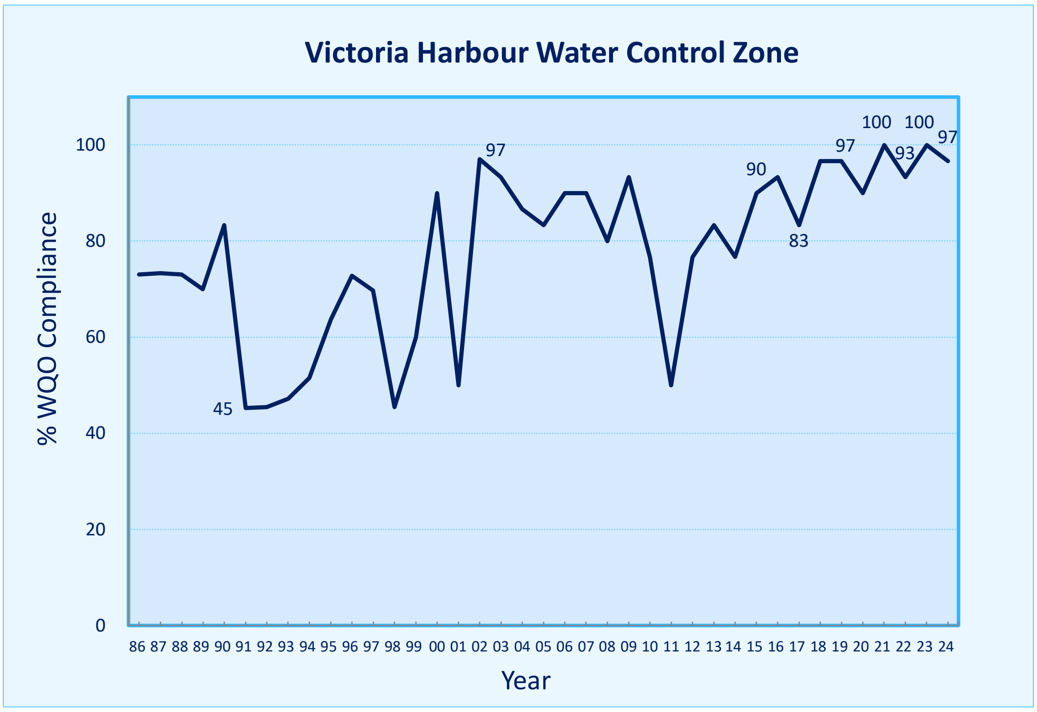


Figure 12. Overall WQO compliance rate for the Victoria Harbour WCZ, 1986-2024

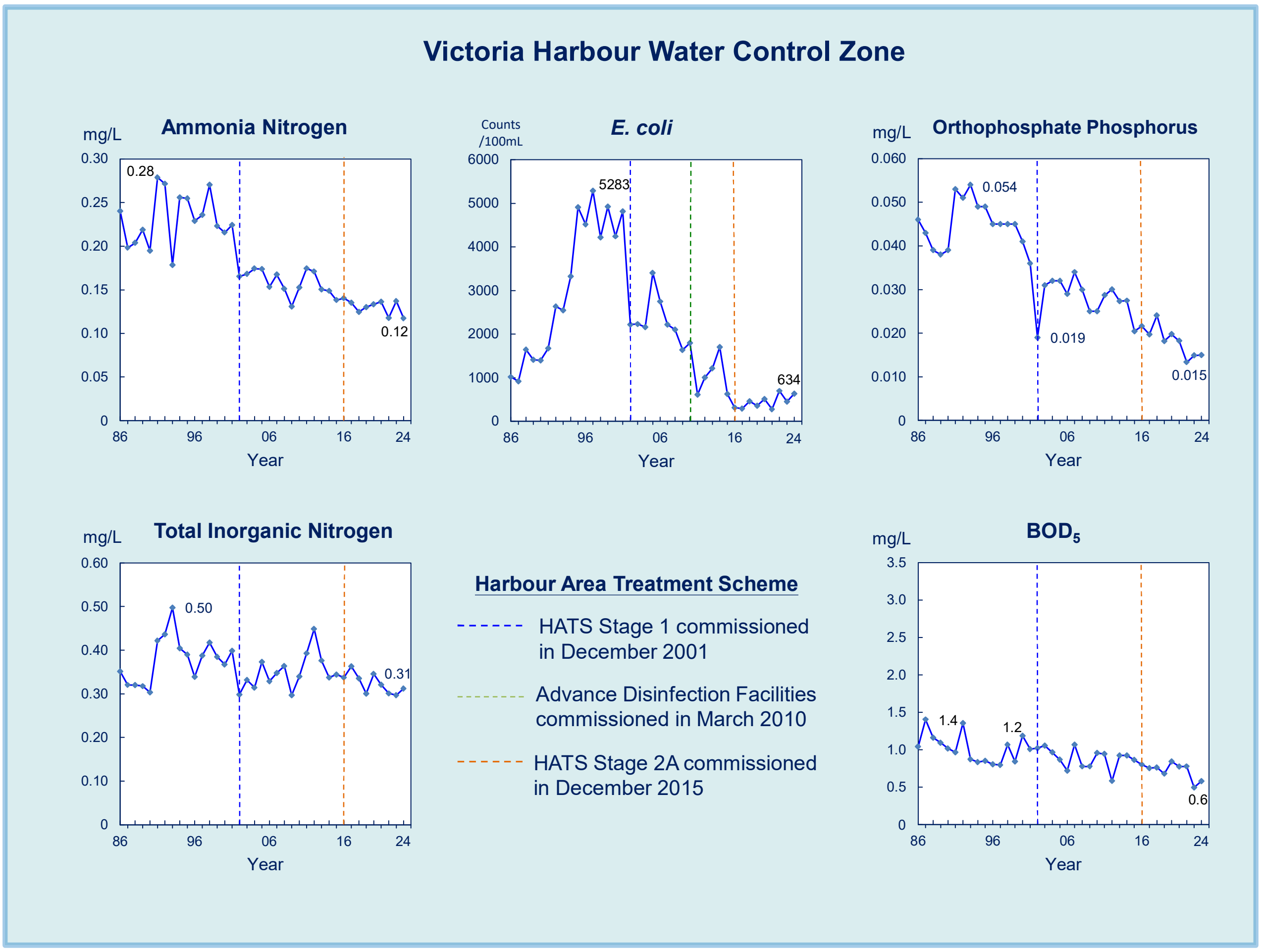


Figure 13. Long-term water quality trends in the Victoria Harbour WCZ, 1986-2024

Eastern Buffer Water Control Zone and Junk Bay Water Control Zone

Both the Eastern Buffer WCZ and the Junk Bay WCZ have fully achieved the marine WQOs for the past 25 years. Since the implementation of the HATS Stage 1 in 2001, the water quality of these two WCZs has improved noticeably with significant increase in DO level and decrease in nutrients and bacteria levels.

Figures 14 to 17 present the WQO compliance rates and long-term water quality improving trends for the Eastern Buffer WCZ and the Junk Bay WCZ over the past three decades.

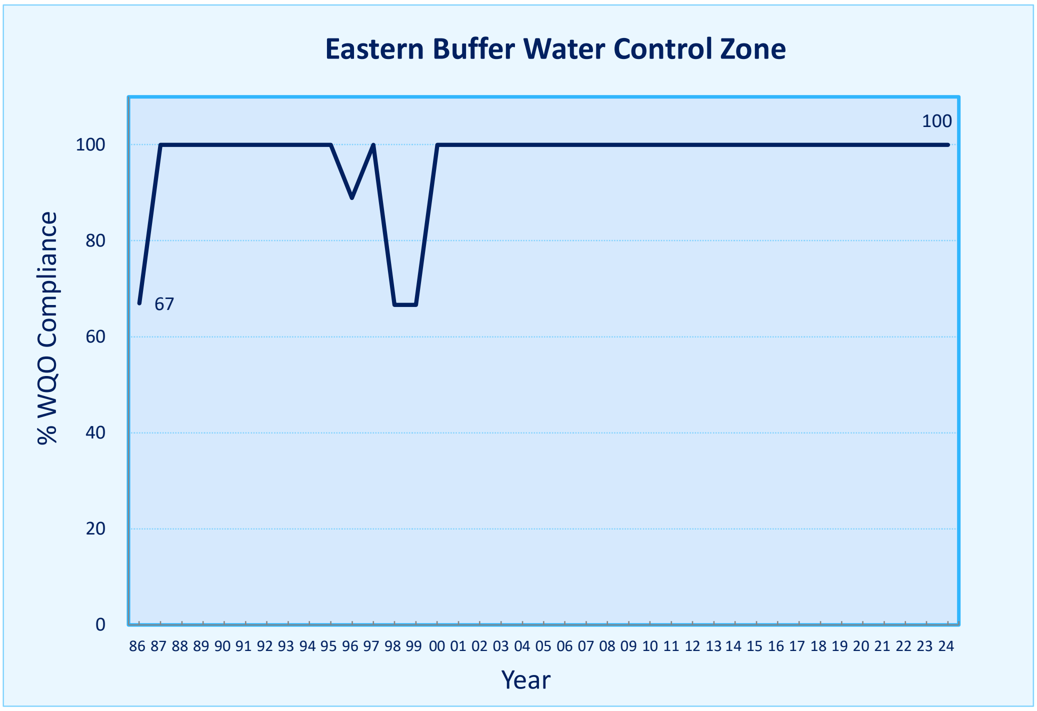


Figure 14. Overall WQO compliance rate for the Eastern Buffer WCZ, 1986-2024

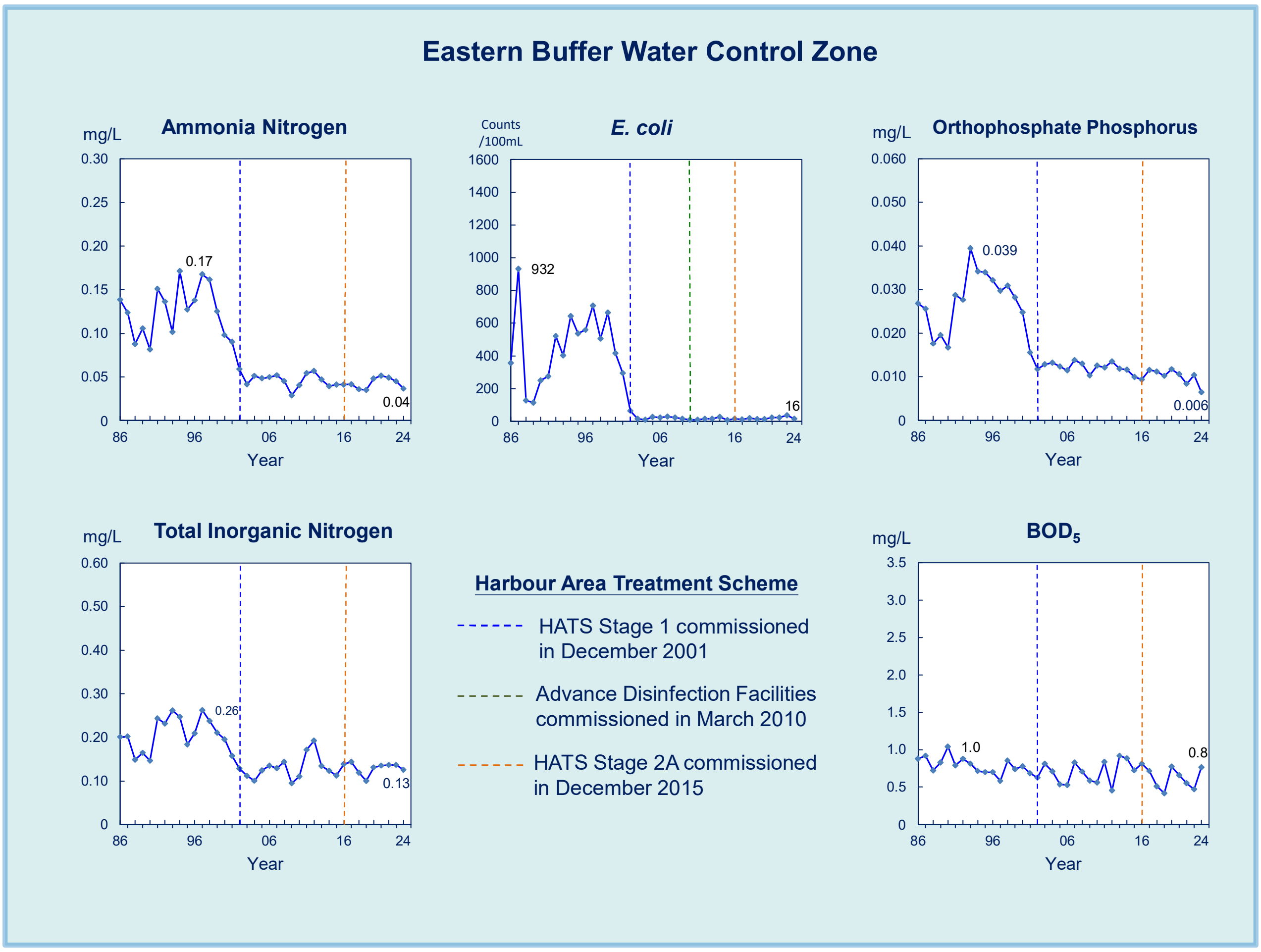


Figure 15. Long-term water quality trends in the Eastern Buffer WCZ, 1986-2024

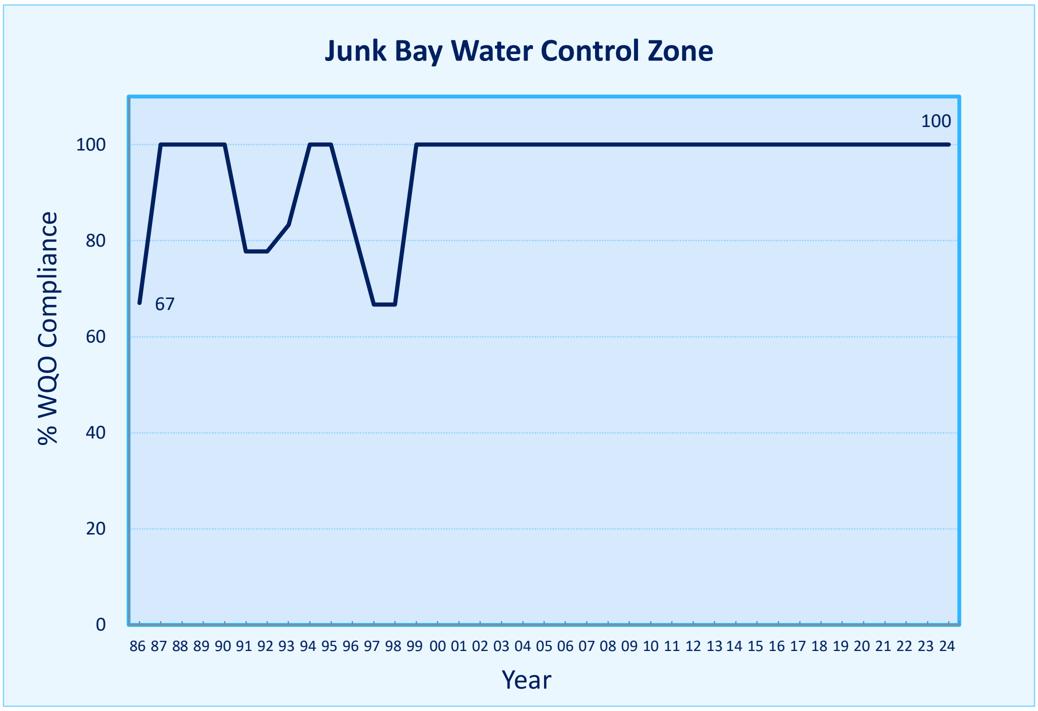


Figure 16. Overall WQO compliance rate for the Junk Bay WCZ, 1986-2024

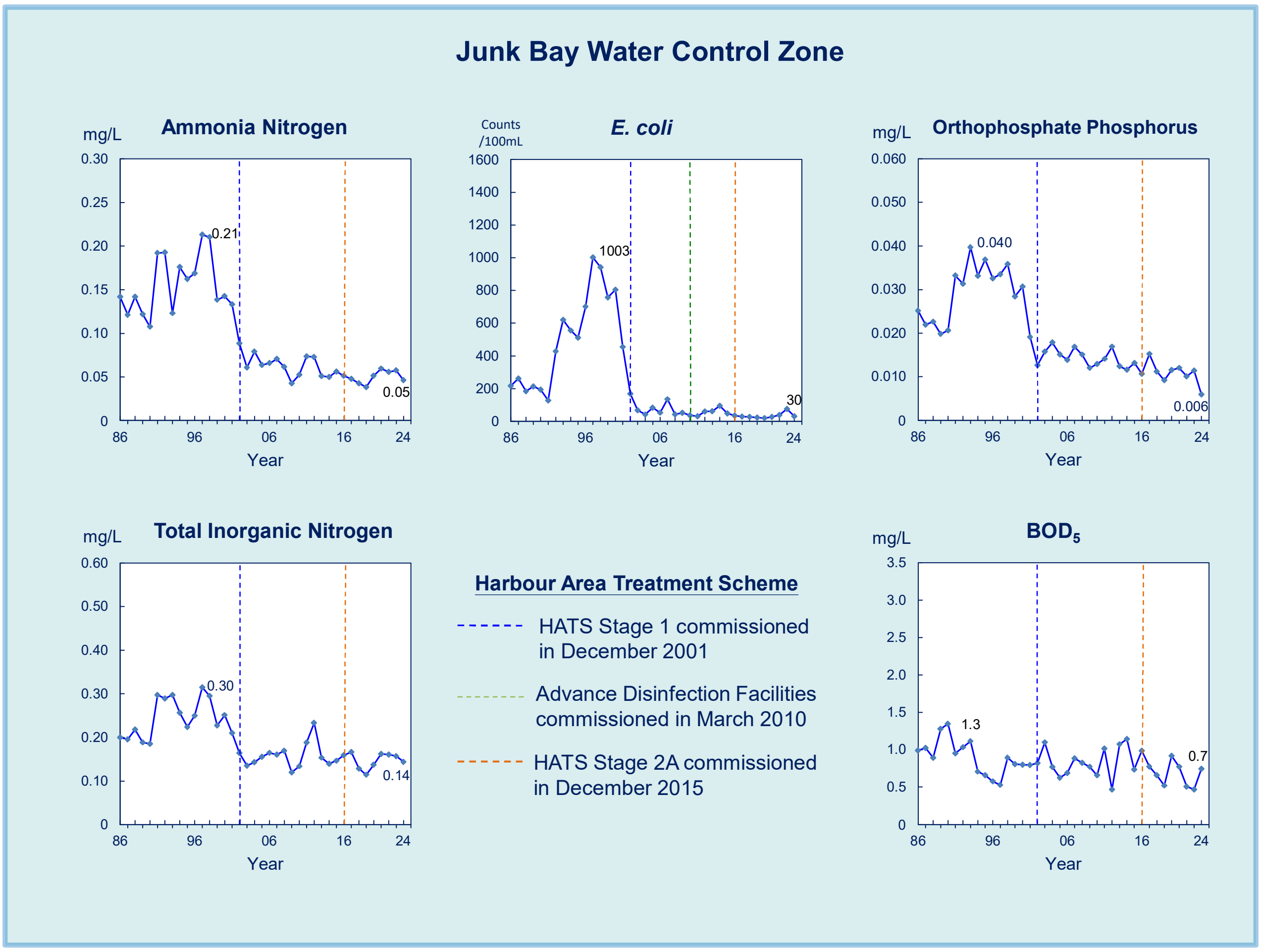


Figure 17. Long-term water quality trends in the Junk Bay WCZ, 1986-2024

Western Buffer Water Control Zone

The Western Buffer WCZ fully met the WQOs in 2024. Since the commissioning of the HATS ADF in 2010, the *E. coli* level in the WCZ decreased substantially.

Figures 18 and 19 illustrate the WQO compliance rate and some long-term water quality trends for the Western Buffer WCZ since 1986. Similar to other WCZs in the central waters, significant improvement of water quality as reflected in reduction in *E. coli* and PO_4-P levels has been observed.

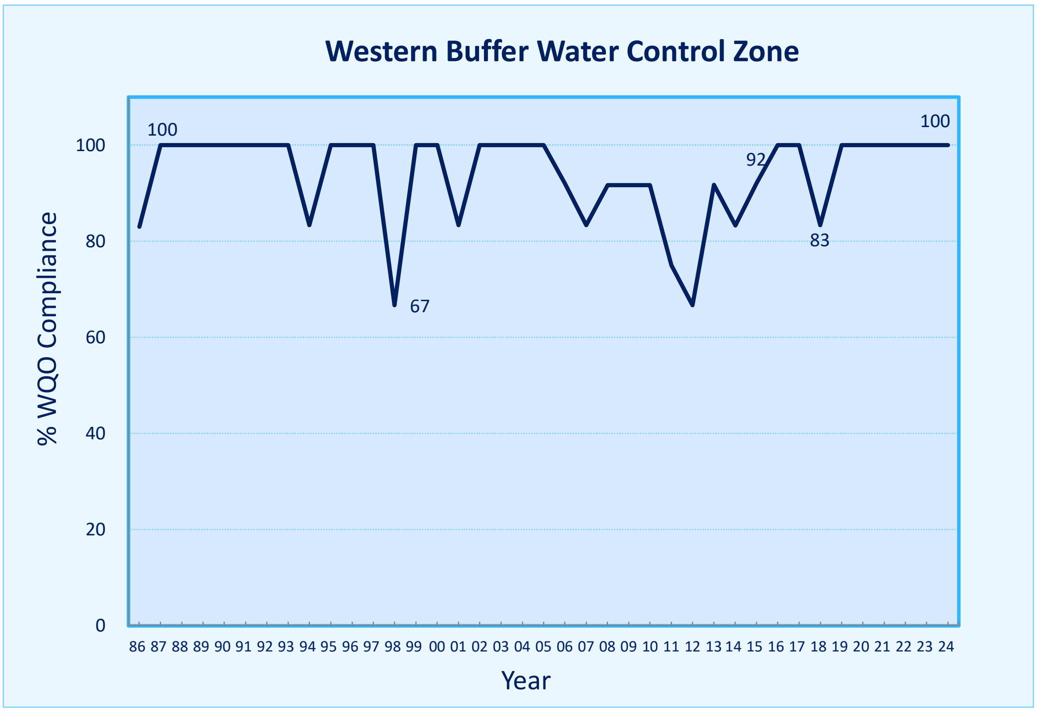


Figure 18. Overall WQO compliance rate for the Western Buffer WCZ, 1986-2024

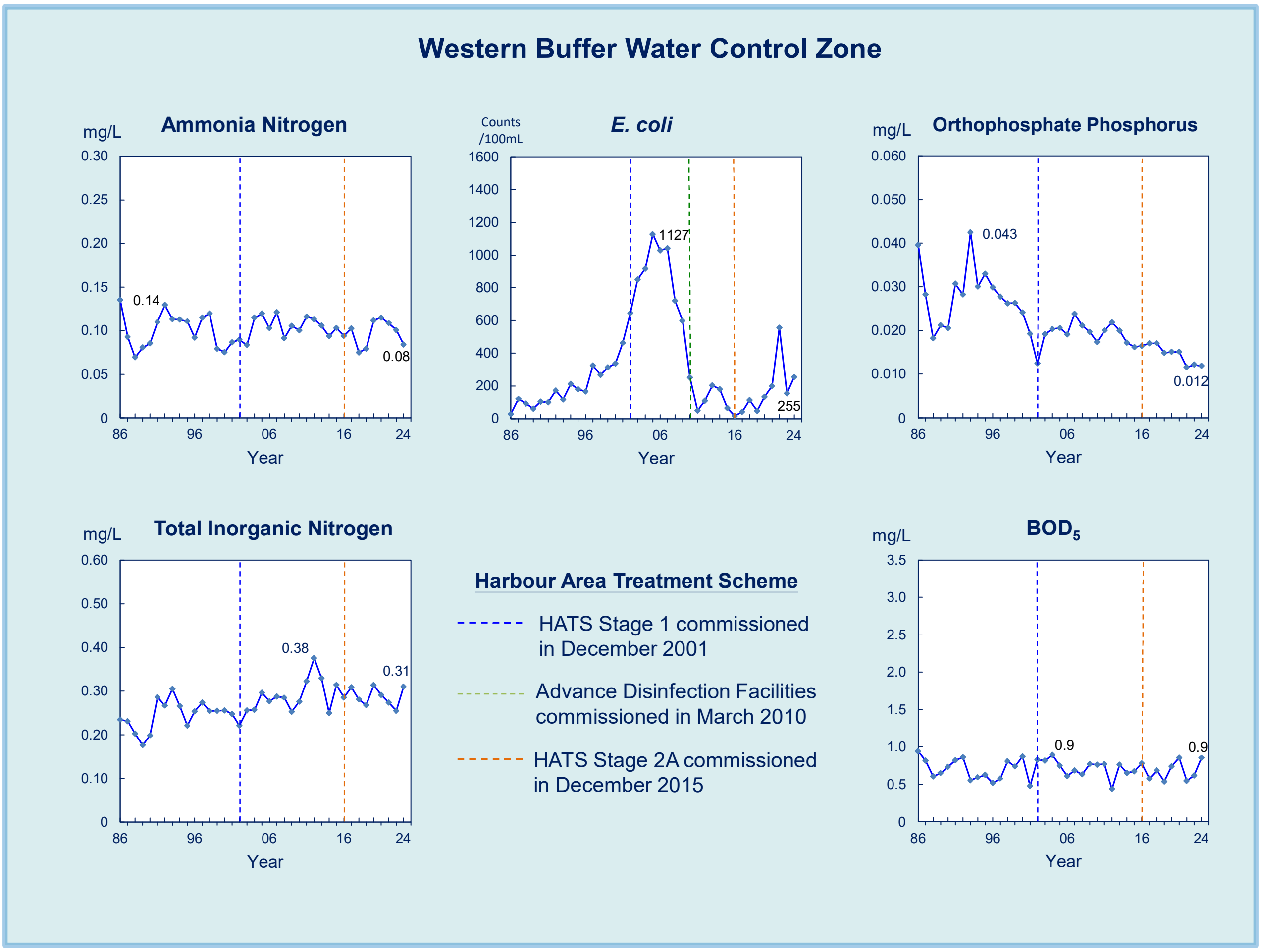
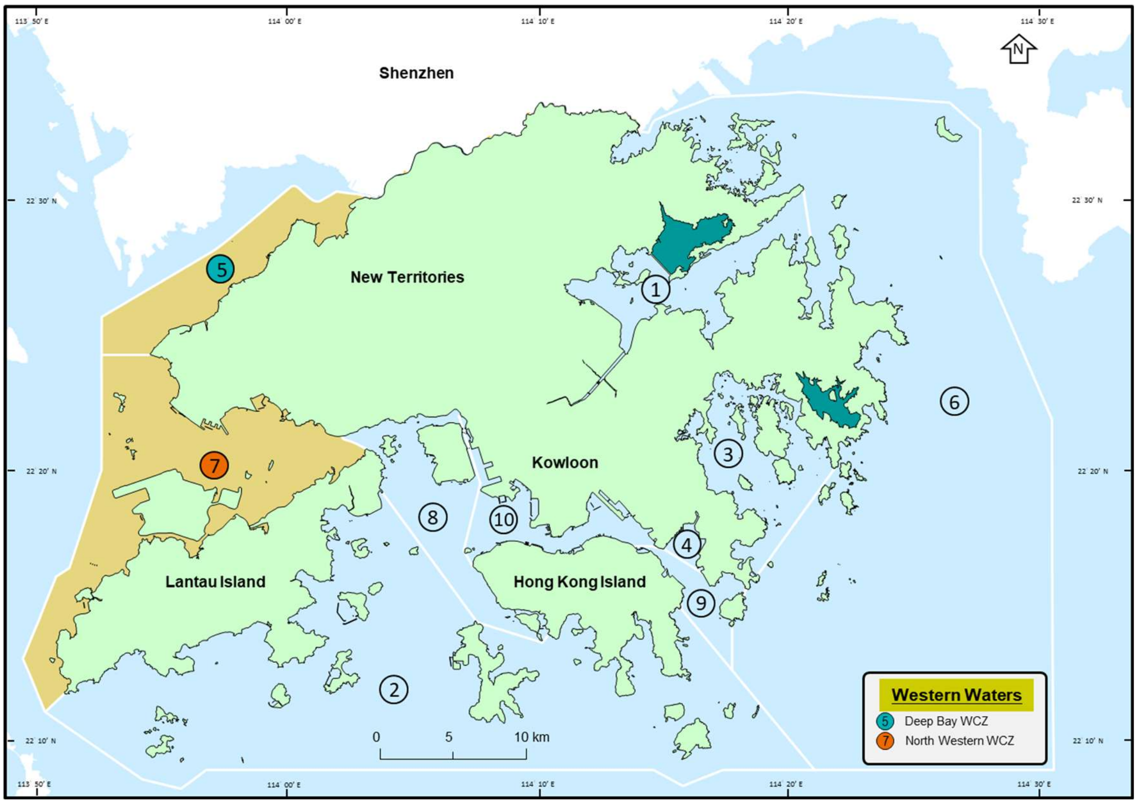


Figure 19. Long-term water quality trends in the Western Buffer WCZ, 1986-2024

3.3 Western Waters



The Deep Bay WCZ and the North Western WCZ are located in the western part of Hong Kong. The former includes the ecologically sensitive Mai Po Inner Deep Bay Ramsar Site and areas of oyster culture. The latter covers the waters around the North and Western side of Lantau Island, Tuen Mun, Sha Chau and Lung Kwu Chau with four marine parks inside, including Sha Chau and Lung Kwu Chau Marine Park, The Brothers Marine Park, part of the Southwest Lantau Marine Park and the new North Lantau Marine Park established in 2024.

Deep Bay Water Control Zone

In 2024, the overall WQO compliance rate for the Deep Bay WCZ was 67%, as compared with a ten-year average of 47% in 2009-2018. Overall, with the measures taken progressively by the governments of Hong Kong and Shenzhen under the Deep Bay Water Pollution Control Joint Implementation Programme, there have been significant water quality improvements in Deep Bay. In particular, the NH_3-N WQO has been fully met in the past ten years. Although Deep Bay, as compared with other WCZs, shows higher nutrient levels with annual depth-averaged TIN levels exceeding the respective TIN WQOs, a noticeable long-term decrease in TIN levels since mid-2000s has been seen. The marked improvements in the WQO compliance rates and long-term water quality trends for Deep Bay since mid-2000s are illustrated in Figures 20 to 22.

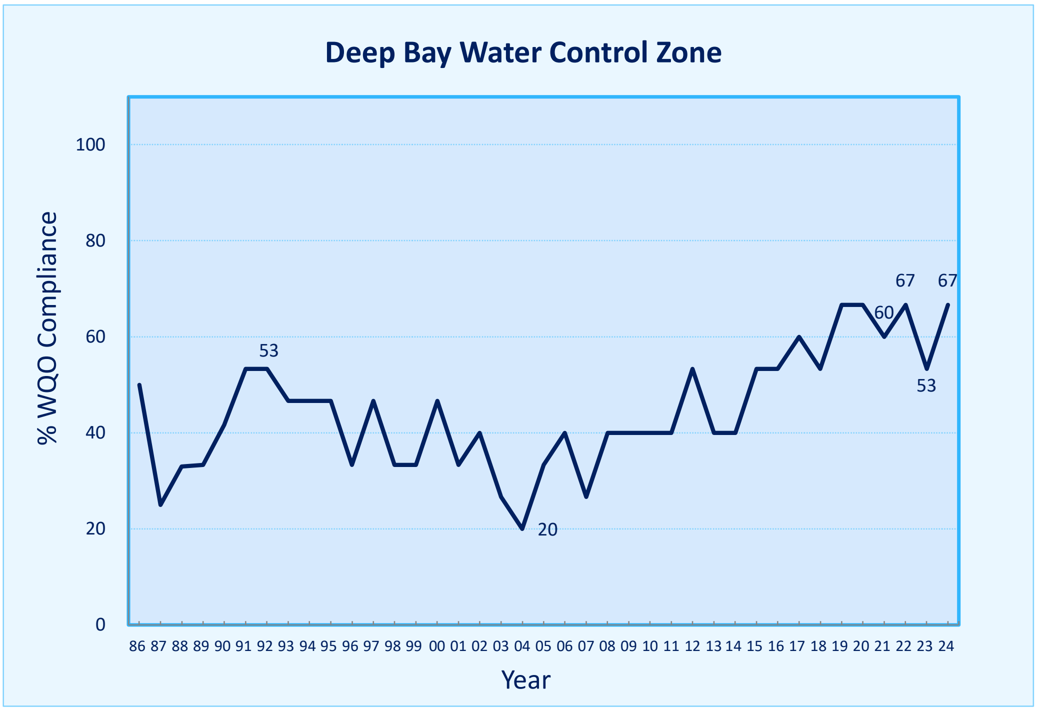


Figure 20. Overall WQO compliance for the Deep Bay WCZ, 1986-2024

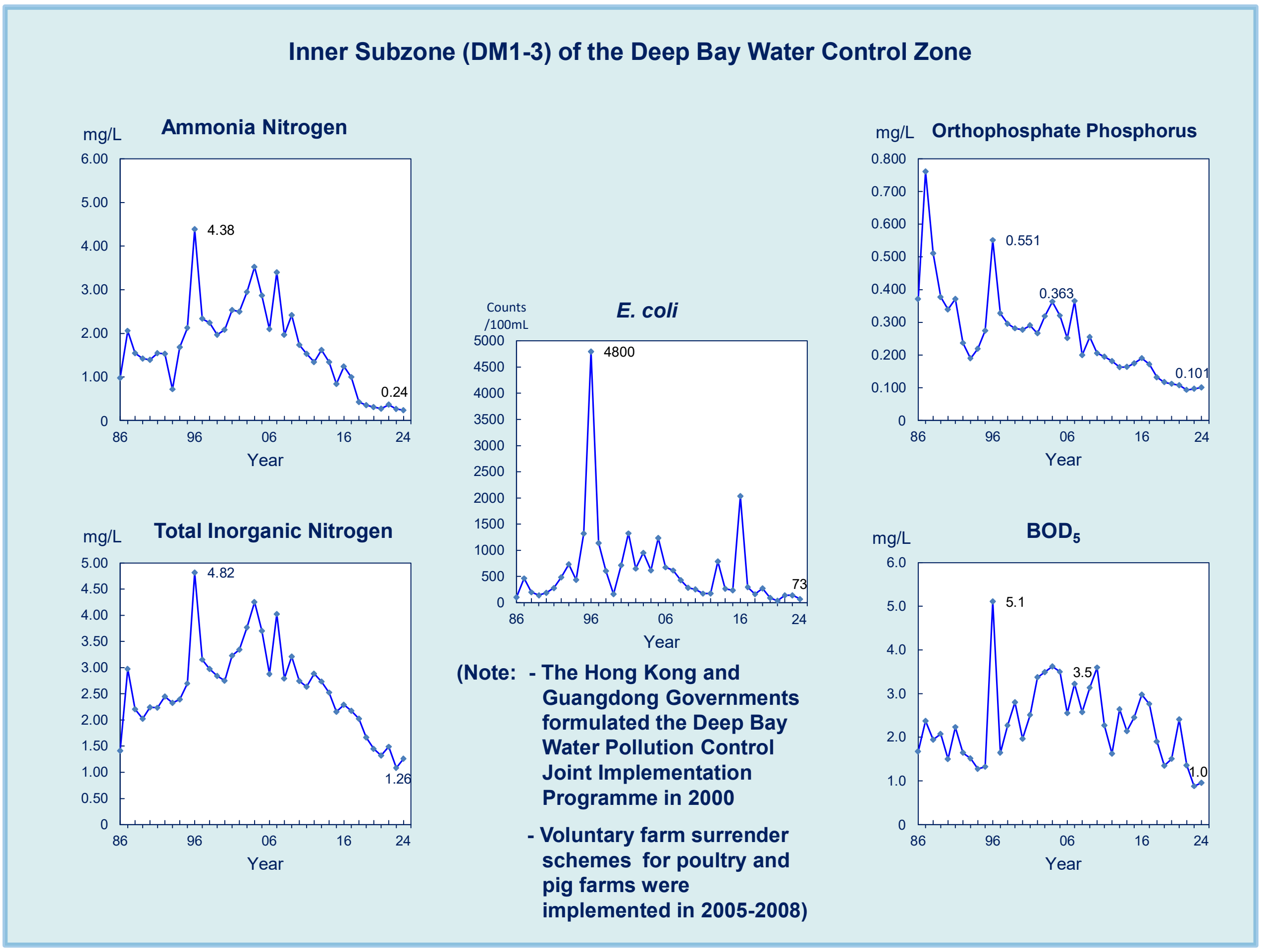


Figure 21. Long-term water quality trends in Inner Subzone of the Deep Bay WCZ, 1986-2024

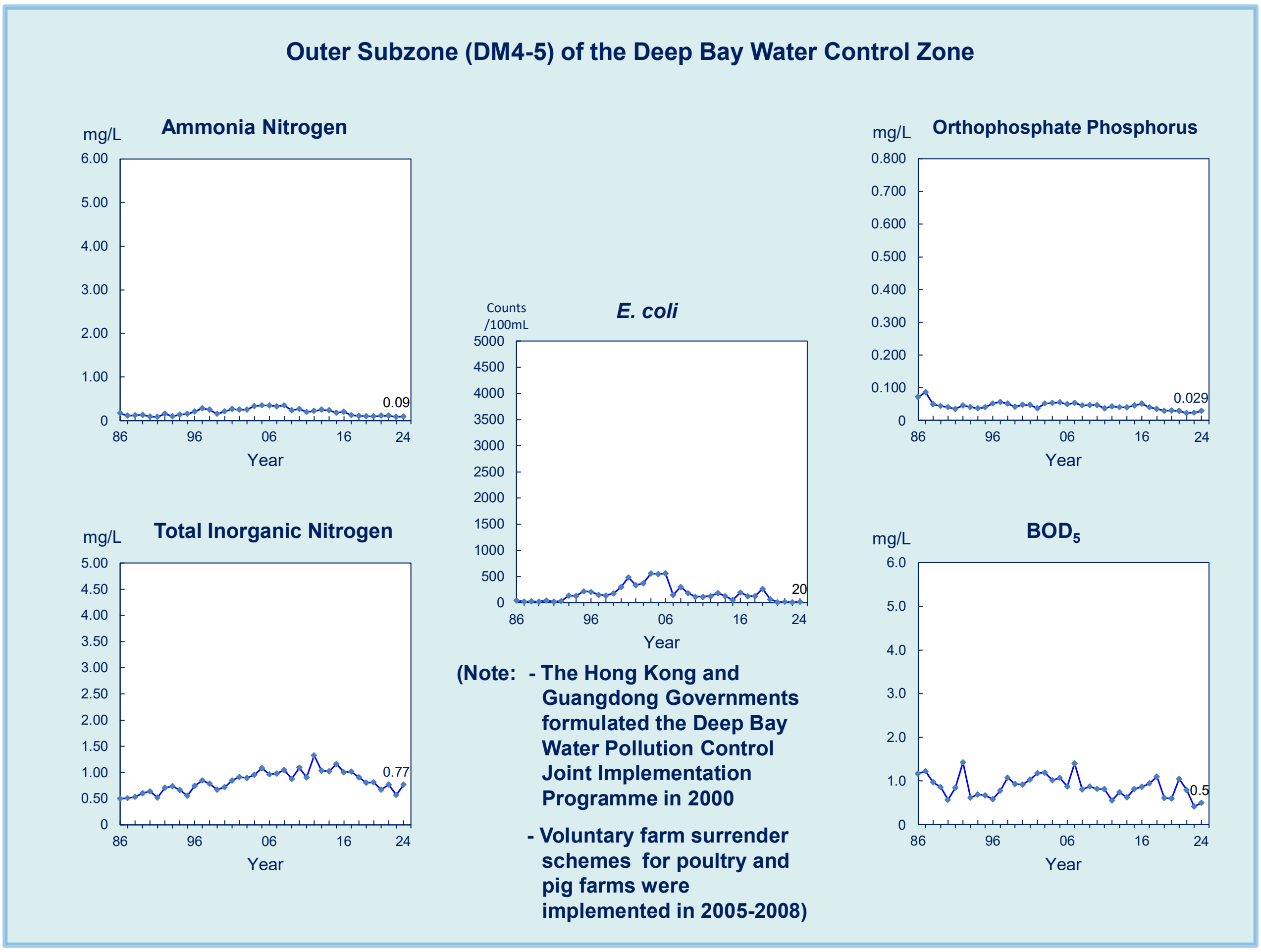


Figure 22. Long-term water quality trends Outer Subzone of the Deep Bay WCZ, 1986-2024

North Western Water Control Zone

In 2024, the overall WQO compliance rate of the North Western WCZ was 72%, with the DO and NH_3-N WQOs fully met. Under the influence of high seasonal background level in the Pearl River Estuary, the compliance rate for TIN WQO was 17%. The WQO compliance rate and long-term water quality trends for the North Western Water WCZ are illustrated in Figures 23 and 24.

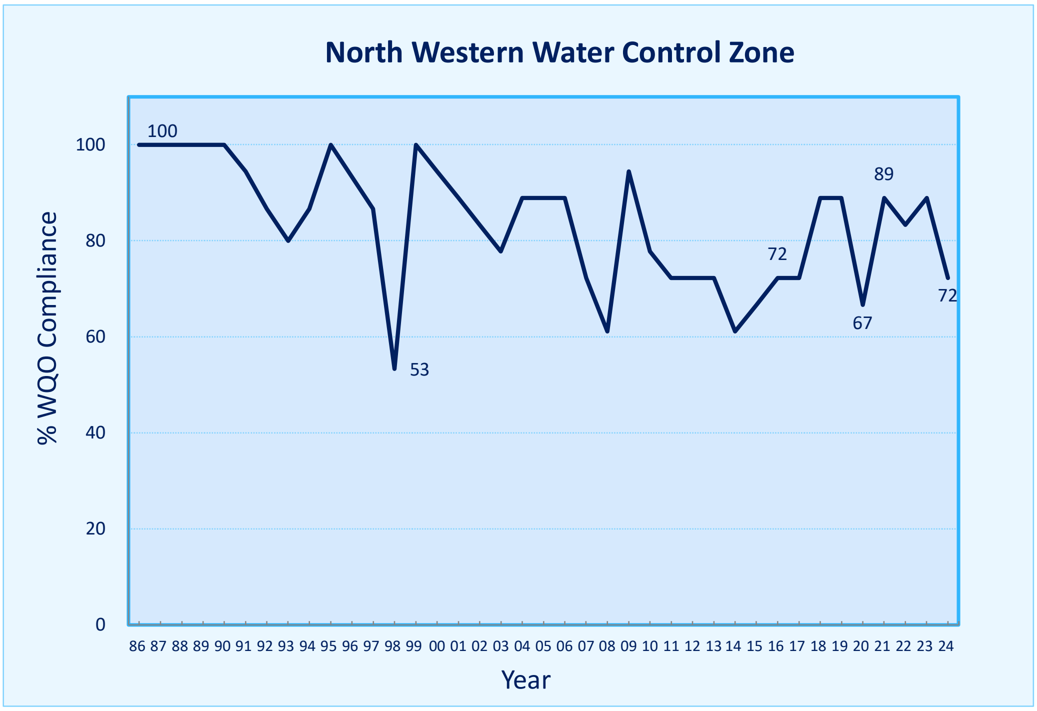


Figure 23. Overall WQO compliance for the North Western WCZ, 1986-2024

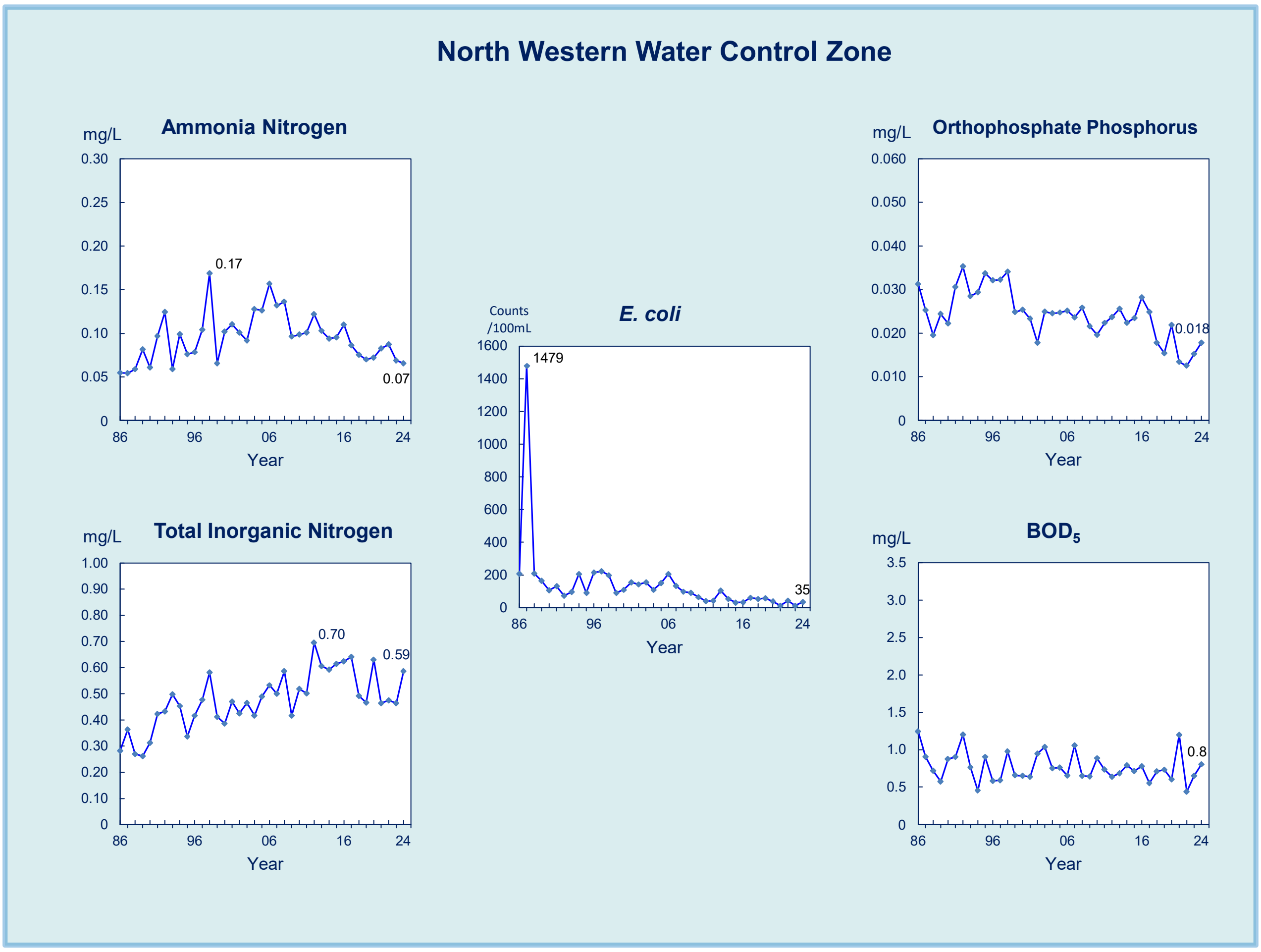
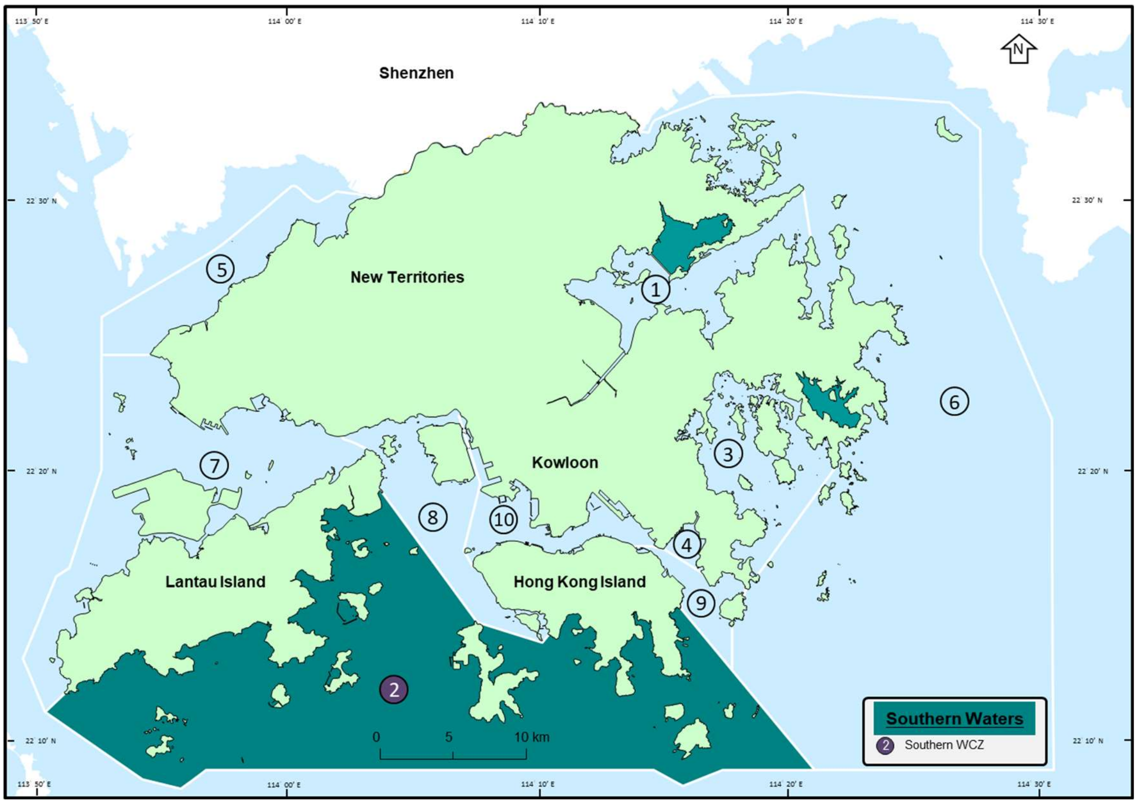


Figure 24. Long-term water quality trends in the North Western WCZ, 1986-2024

3.4 Southern Waters



The southern waters cover the Southern WCZ, stretching from south of Hong Kong Island to Lantau Island. These waters contain 21 gazetted beaches, two marine parks (including South Lantau Marine Park and part of the Southwest Lantau Marine Park), one marine reserve (Cape D'Aguilar Marine Reserve), and four fish culture zones.

Southern Water Control Zone

In 2024, the Southern WCZ achieved an overall WQO compliance rate of 69%, with full attainment of the DO and NH_3-N WQOs. Although the TIN level in the southern waters was generally lower than the upstream western and central waters, exceedance of the TIN WQO was recorded primarily due to the stringent WQO adopted for this WCZ.

The Southern WCZ also covers a number of secondary contact recreation subzones mainly located along the southern coast of Hong Kong Island and the outlying islands. Full compliance with the bacteriological WQO on *E. coli* for all these secondary contact recreation subzones has been consistently achieved. The WQO compliance rates and long-term water quality trends for the Southern WCZ are plotted in Figures 25 and 26.

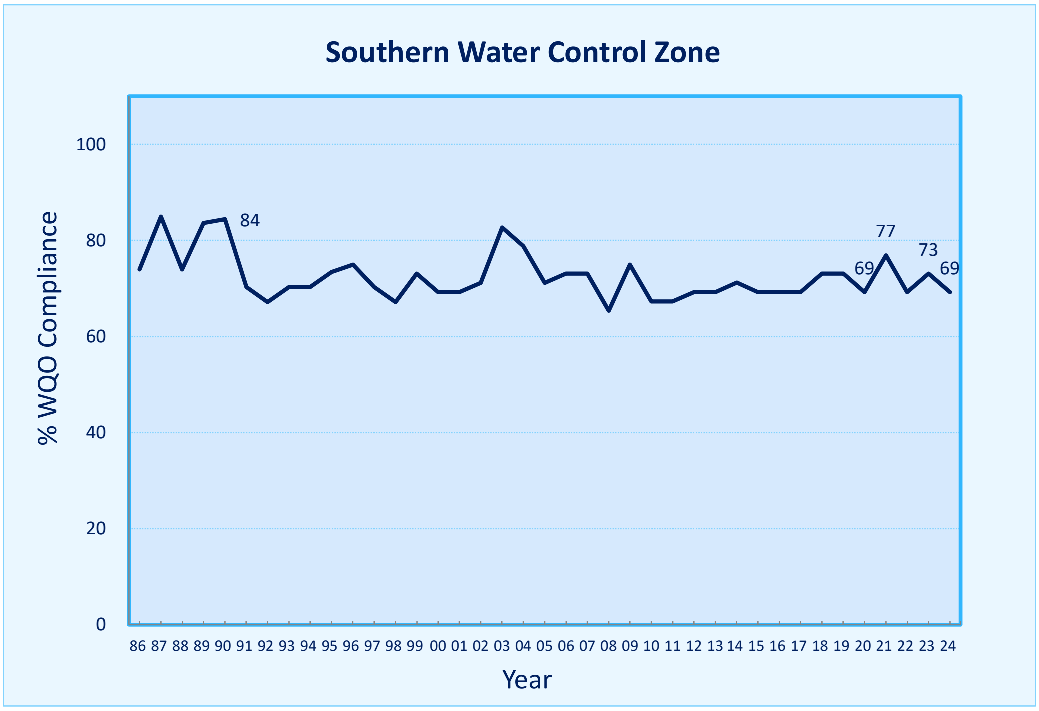


Figure 25. Overall WQO compliance for the Southern WCZ, 1986-2024

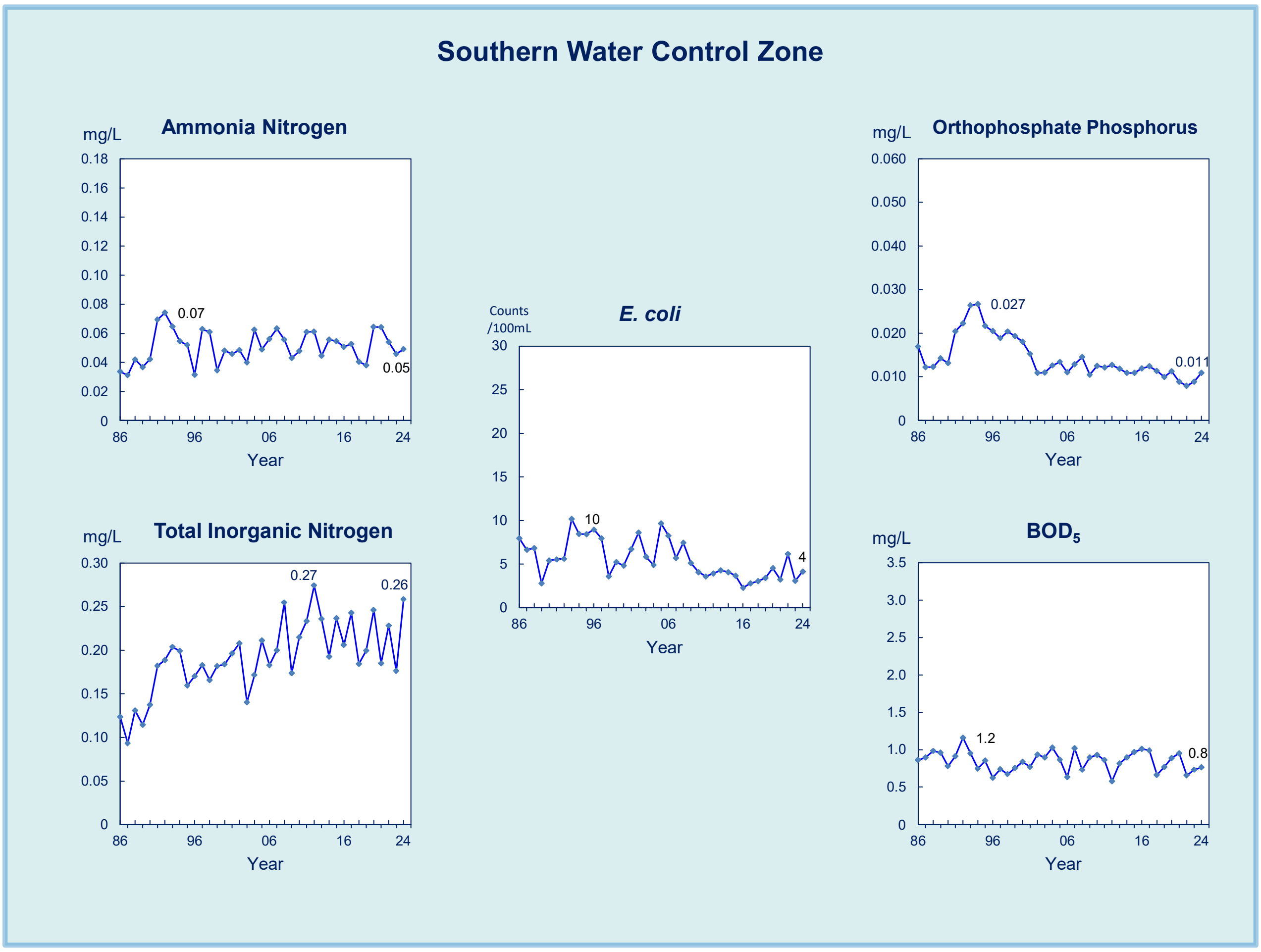


Figure 26. Long-term water quality trends in the Southern WCZ, 1986-2024

4. Marine Sediment Quality

Marine sediment samples taken in various locations of Hong Kong waters were analysed for over 60 physical, chemical and biological parameters. Details of marine sediment quality of individual WCZs for the last five years (2020-2024) are summarised in Appendix E.

5. Typhoon Shelters

The EPD monitored the water quality of 14 typhoon shelters, three sheltered anchorages and the Government Dockyard. Some of these typhoon shelters (e.g. Causeway Bay Typhoon Shelter and Kwun Tong Typhoon Shelter) are located adjacent to densely populated residential, commercial and/or industrial areas. Others (e.g. Cheung Chau Typhoon Shelter and Shuen Wan Typhoon Shelter) are located in outlying islands or relatively away from urban areas. There is no bacteriological WQO set for typhoon shelters which are used only for vessel mooring. In 2024, the NH_3-N levels recorded in the typhoon shelters were generally low and in compliance with the respective NH_3-N WQOs.

Overall, the water quality of the typhoon shelters in Hong Kong has been improving over the last decade. Figure 27 shows the long-term improving trend for the depth-averaged DO level in the Kwun Tong Typhoon Shelter.

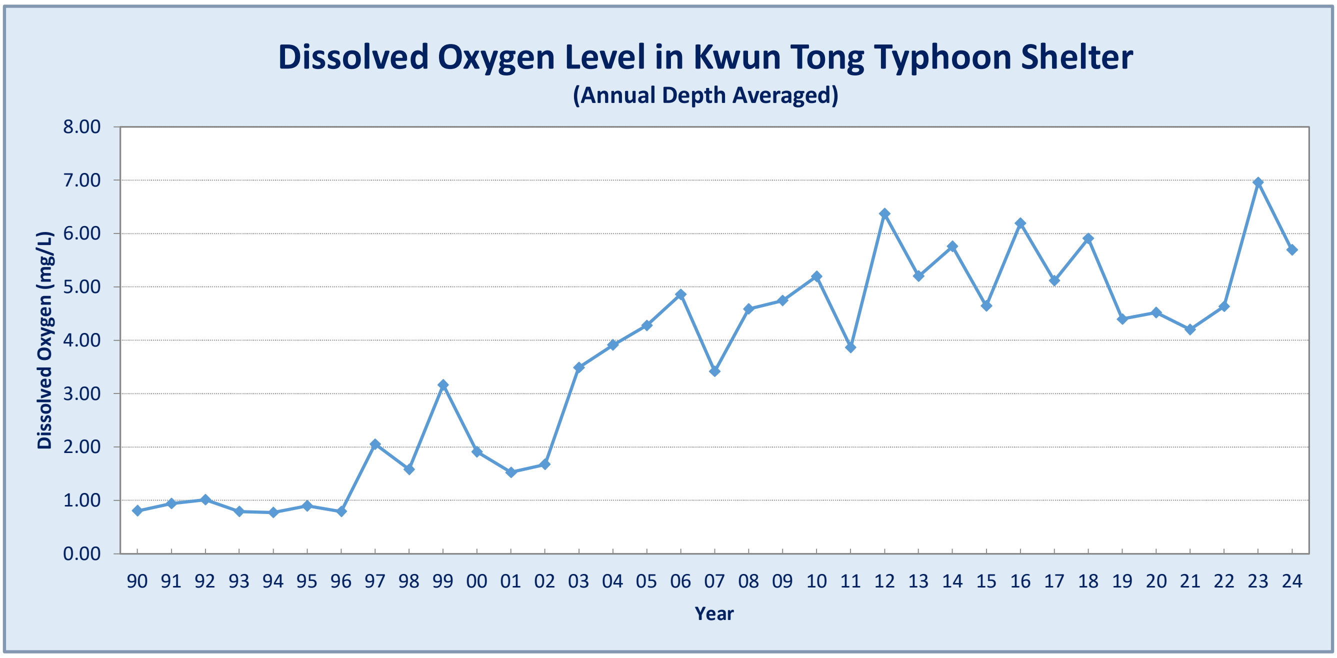


Figure 27. Long-term improvement in dissolved oxygen level in Kwun Tong Typhoon Shelter, 1990-2024

6. Phytoplankton

Phytoplankton is the most important primary producer that forms the foundation of aquatic food web. They serve as a vital food supply and are significant to maintain the balance of aquatic ecosystem. Populations of some phytoplankton are vulnerable to the direct impact of nutrient enrichment as well as hydrological conditions. When the environmental factors including weather, light intensity, wind speed, water temperature, currents, water clarity or turbidity, salinity, nutrient (such as nitrogen and phosphorus), trace elements and geographical location etc) reach the favorable conditions for the growth and accumulation of specific phytoplankton species, they may rapidly proliferate, aggregate and cause water discoloration, resulting in algal blooms (also known as red tides). This natural phenomenon may induce a change in the structure of phytoplankton community. In some cases, deoxygenation of bottom water may also result, which in turn impact the resilience and stability of the aquatic ecosystems. Therefore, changes in the composition and abundance of phytoplankton can, to a certain extent, reflect the health status of the marine ecosystems and can be used for the assessment of the eutrophication status of the marine waters and its associated biological impacts.

6.1 Phytoplankton Composition and Density

The EPD conducts monthly sampling of phytoplankton at 26 marine water quality monitoring stations to determine the long-term changes and trends in their composition and densities in Hong Kong marine waters. In 2024, a total of 92 phytoplankton species were recorded in Hong Kong waters, as compared with a five-year average of 95 phytoplankton species in 2019 – 2023. Of these, 55 species were diatoms (60%), 25 were dinoflagellates (27%) and 12 were from other minor algal groups[1] (13%). This phytoplankton composition profile was generally similar to those observed in the past five years. Of the samples examined in 2024, diatoms were the dominant group found in our coastal waters in terms of species richness (number of species identified). Diatoms were also the dominant group in terms of cell density. Details of the species richness and cell density of phytoplankton in local marine waters are given in Appendix G.

The composition of phytoplankton, including diatom, dinoflagellate and other minor algal group, was similar across all WCZs. Tolo Harbour and Channel WCZ is a highly enclosed waters and hence the total phytoplankton densities (as well as the densities of diatom, dinoflagellate and other minor algal group) were generally higher in this WCZ. The most abundant diatoms were *Chaetoceros* spp. which constituted 13% to 56% of the diatom population found in various WCZs. The most abundant dinoflagellates were *Gymnodinium* spp. which constituted 55% to 74% of the dinoflagellate population in various WCZs.

[1] Phytoplankton can be classified into different groups based on their morphological characteristics, photosynthetic pigments and nutrition modes. In this report, phytoplankton other than the two major groups, i.e., diatoms and dinoflagellates, are collectively regarded as other minor phytoplankton groups to facilitate the data presentation.

6.2 Phytoplankton Community Integrity Index (PCII)

Referencing the latest international practices and trends, the EPD has successfully developed and established a biological indicator namely Phytoplankton Community Integrity Index (PCII) in recent years. The index is used to supplement the prevailing Water Quality Objectives (WQOs) on nutrients (i.e. Total Inorganic Nitrogen (TIN) and Chlorophyll-*a*) to gauge the status of eutrophication in our local marine waters, and to assess the biological impacts of nutrient enrichment on the phytoplankton community therein in a more scientifically sound and robust manner. The development and applications of the PCII and the associated study findings were reviewed by academic scholars and published in an international scientific journal in February 2024[1], and are summarized as follows:

Diatoms and dinoflagellates are the two most dominant phytoplankton lifeform in global oceans, including the coastal waters of Hong Kong. PCII aims to monitor and gauge the change and trend of the abundance of these two major phytoplankton lifeforms. It is an objective and scientifically sound biological tool to effectively assess the ecological status of phytoplankton community and hence the associated biological impacts of nutrient enrichment on the local marine environment.

The computation and assessment method of PCII involve plotting the density data of diatom and dinoflagellate lifeform recorded in a three-year period when the TIN level was the lowest in a specific WCZ to result in a reference two-dimensional state space as described in Tett (2008)[2] for representing a “reference condition”. Such “reference condition” for each WCZ needs to be reviewed according to the latest trends of TIN levels to ensure that a more desirable status of phytoplankton communities with minimal impacts of nutrient enrichment is adopted for the PCII computation.

WCZ	Deep Bay	Northwestern	Southern	Victoria Harbour	Western Buffer	Port Shelter	Mirs Bay	Tolo Harbour and Channel
Period with lowest TIN level (“Reference condition”)	2021 – 2023	1999 – 2001	2002 – 2004	2022 – 2024	1997 – 1999	2002 – 2004	2002 – 2004	2007 – 2009
Number of monitoring stations	2	2	4	3	2	2	5	4

Period with the minimal TIN levels to represent “reference condition” and the number of monitoring stations at each WCZ

[1] Mak, Y.L., Tett, P., Yung, Y.K., Sun, W.C., Tsang, H.L., Chan, C.T., Liu, H., Chiu, W.L., Leung, K.F., Yang, R. and Chui, H.K. (2024) Phytoplankton community integrity index (PCII) – A potential supplementary tool for evaluating nutrient enrichment status of Hong Kong marine waters. *Marine Pollution Bulletin* 199: 115964.

[2] Tett, P., Carreira, C., Mills, D. K., Van Leeuwen, S., Foden, J., Bresnan, E. and Gowen, R. J. 2008. Use of a phytoplankton community index to assess the health of coastal waters. *ICES Journal of Marine Science* 65:1475-1482.

To evaluate changes in status of the phytoplankton community, the pairwise diatom and dinoflagellate abundance data collected during a specific three-year period are plotted into and compared with this two-dimensional state space of the "reference condition". The PCII is then calculated using the following equation:

$$\text{Phytoplankton Community Integrity Index (PCII)} = \frac{\text{No. of new points (n) between inner and outer envelopes}}{\text{Total number of new points (N)}}$$

The computed PCII values are between one and zero, with "1" representing no change from the reference condition and "0" representing a complete change. A five-tiered ranking system was established to describe the ecological status of phytoplankton community. A PCII value of 0.6 or above, which is equivalent to "Good" or "High" phytoplankton community status, is operationally set as the Biological Water Quality Criterion target of acceptable disturbance to the phytoplankton community in all WCZs of Hong Kong based on a local study conducted previously[1] as well as the latest international practices including European Union Water Framework Directive (EU WFD)[2] and the Convention for the Protection of the Marine Environment of the North-East Atlantic (OSPAR Convention)[3].

Rank	Range of PCII values	Status of Phytoplankton Community
1	≥0.8 to 1.0	High
2	≥0.6 to <0.8	Good
3	≥0.4 to <0.6	Fair
4	≥0.2 to <0.4	Poor
5	0 to <0.2	Bad

PCII ranking system

(Note: The color coding of PCII is based on the classification of the ecological status of various water bodies in the EU WFD[5])

[1] Lei, Y., Whyte, C., Davidson, K., Tett, P., and Yin, K. (2018). A change in phytoplankton community index with water quality improvement in Tolo Harbour, Hong Kong. *Marine Pollution Bulletin* 127: 823-830.

[2] European Union (2000). Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for common action in the field of water policy. *The Official Journal of the European Union* 43(L327): 1-72.

[3] OSPAR Commission (2013). *Common Procedure for the Identification of the Eutrophication Status of the OSPAR Maritime Area (Agreement 2013-08).*

In 2024, the PCII values at various WCZ ranged from 0.64 to 1.00, which met the Biological Water Quality Criterion target (≥ 0.6) and graded as “Good” or “High”. Six of them, namely Mirs Bay WCZ, Port Shelter WCZ, Tolo Harbour and Channel WCZ, Victoria Harbour WCZ, Western Buffer WCZ and Deep Bay WCZ were classified as “High”. When the Phytoplankton Monitoring Programme started to fully and systematically implement in 1997, the PCII scores in Hong Kong waters ranged from 0.51 to 0.72. None of the WCZs was graded as “High”, while four WCZs, namely Mirs Bay WCZ, Western Buffer WCZ, Northwestern WCZ and Southern WCZ achieved Rank 2 with “Good” status and met the Biological Water Quality Criterion target (≥ 0.6).

6.2.1 Eastern Waters

Mirs Bay Water Control Zone

In 2024, the PCII value for Mir Bay WCZ was 0.85 which achieved Rank 1 with "High" phytoplankton community status, indicating that there was no eutrophication in the WCZ. In contrast, the PCII values for Mir Bay ranged from 0.65 to 0.70 in the late 1990s, and were rated as Rank 2 with "Good" phytoplankton community status (Figure 28).

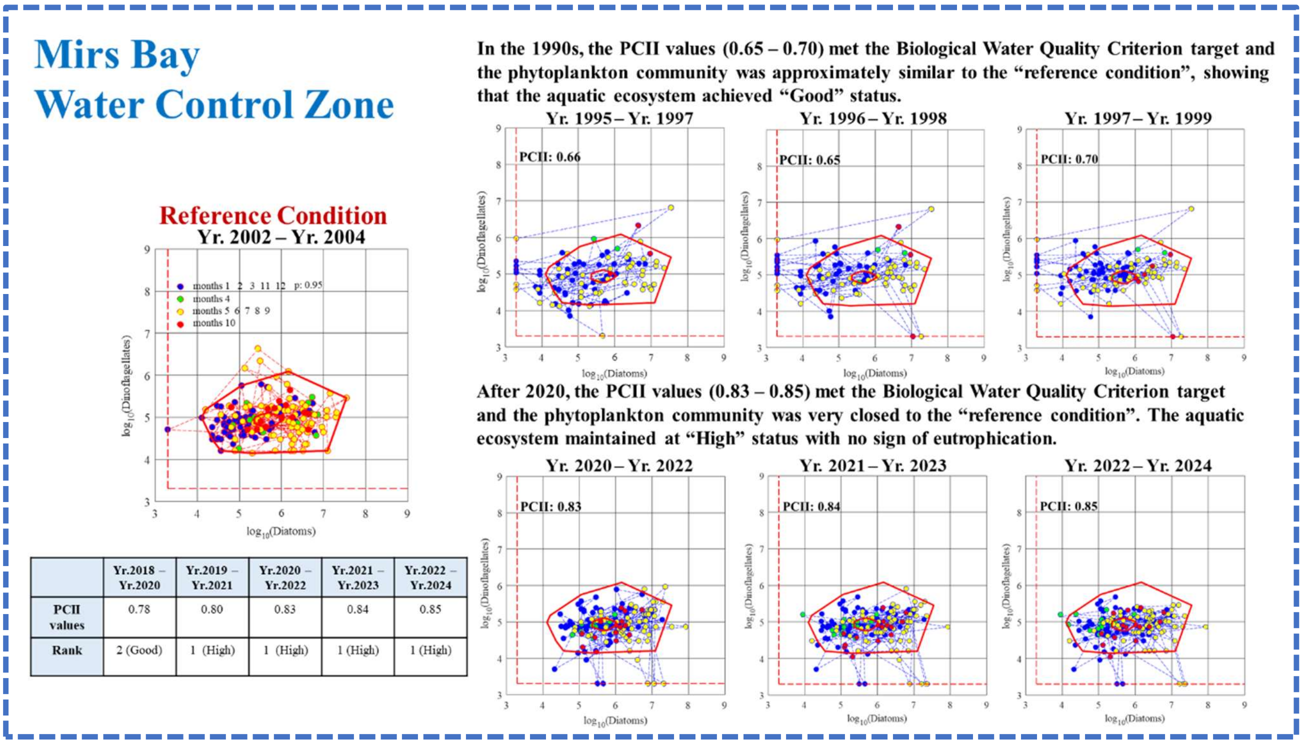
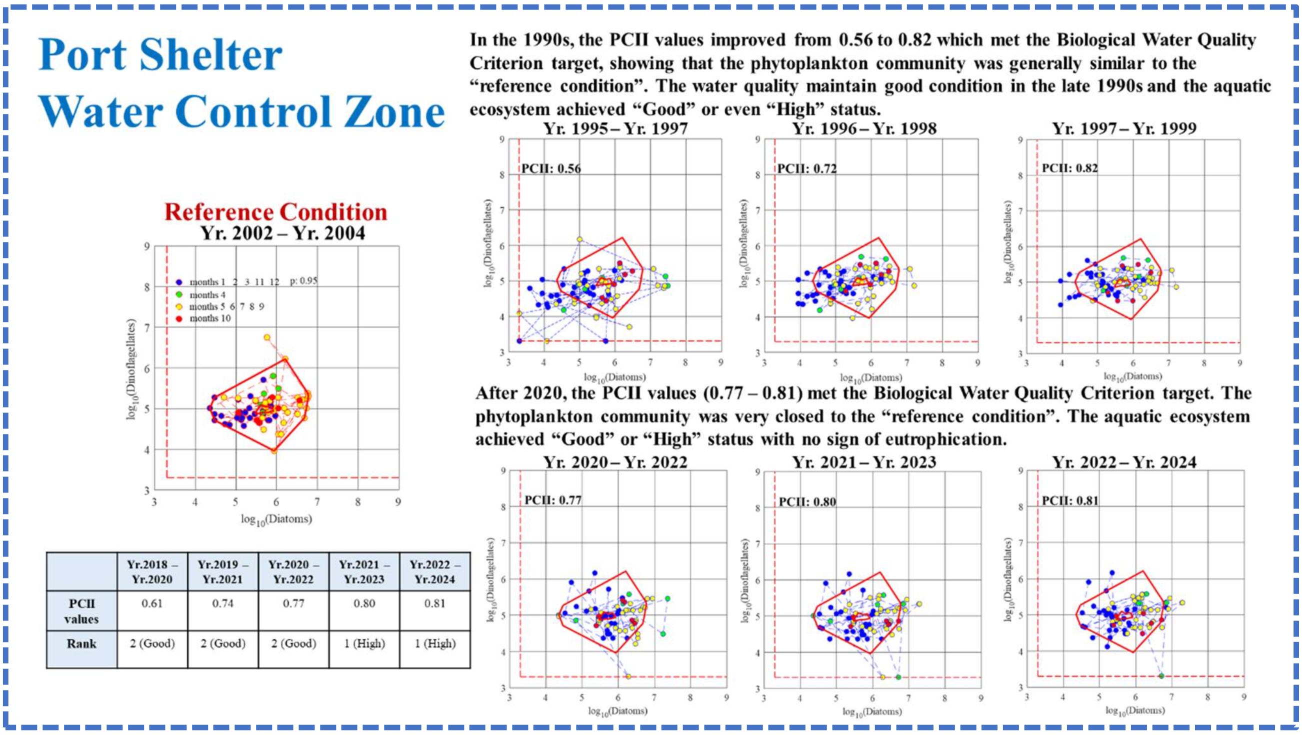


Figure 28. PCII charts, index values and ranks for Mirs Bay WCZ

Port Shelter Water Control Zone

In 2024, the PCII value for Port Shelter WCZ was 0.81 and classified as Rank 1 with "High" phytoplankton community status, reflecting the absence of eutrophication in this WCZ. During the late 1990s, the PCII values for Port Shelter WCZ improved from 0.56 (Rank 3) to 0.82 (Rank 1). As the water quality of the Port Shelter WCZ remained excellent, the PCII values continued to attain "Good" or "High" status (Figure 29).



	Yr.2018 – Yr.2020	Yr.2019 – Yr.2021	Yr.2020 – Yr.2022	Yr.2021 – Yr.2023	Yr.2022 – Yr.2024
PCII values	0.61	0.74	0.77	0.80	0.81
Rank	2 (Good)	2 (Good)	2 (Good)	1 (High)	1 (High)

Figure 29. PCII charts, index values and ranks for Port Shelter WCZ

Tolo Harbour and Channel Water Control Zone

In 2024, the PCII value for Tolo Harbour and Channel WCZ was 0.86 and attained Rank 1 with "High" phytoplankton community status, indicating no eutrophication in this WCZ. Based on the long-term phytoplankton data collected by the EPD, the PCII value in the 1990s occasionally fell below the Biological Water Quality Criterion target (0.60), and was corresponded to Rank 3 with "Fair" phytoplankton community status. After the full implementation of the "Tolo Harbour Action Plan" (including the "Tolo Harbour Effluent Export Scheme") in the late 1990s, the PCII values for Tolo Harbour and Channel WCZ showed significant improvement, rising to or above the Biological Water Quality Criterion target of 0.60 and maintaining at "Good" to "High" phytoplankton community status (Figure 30).

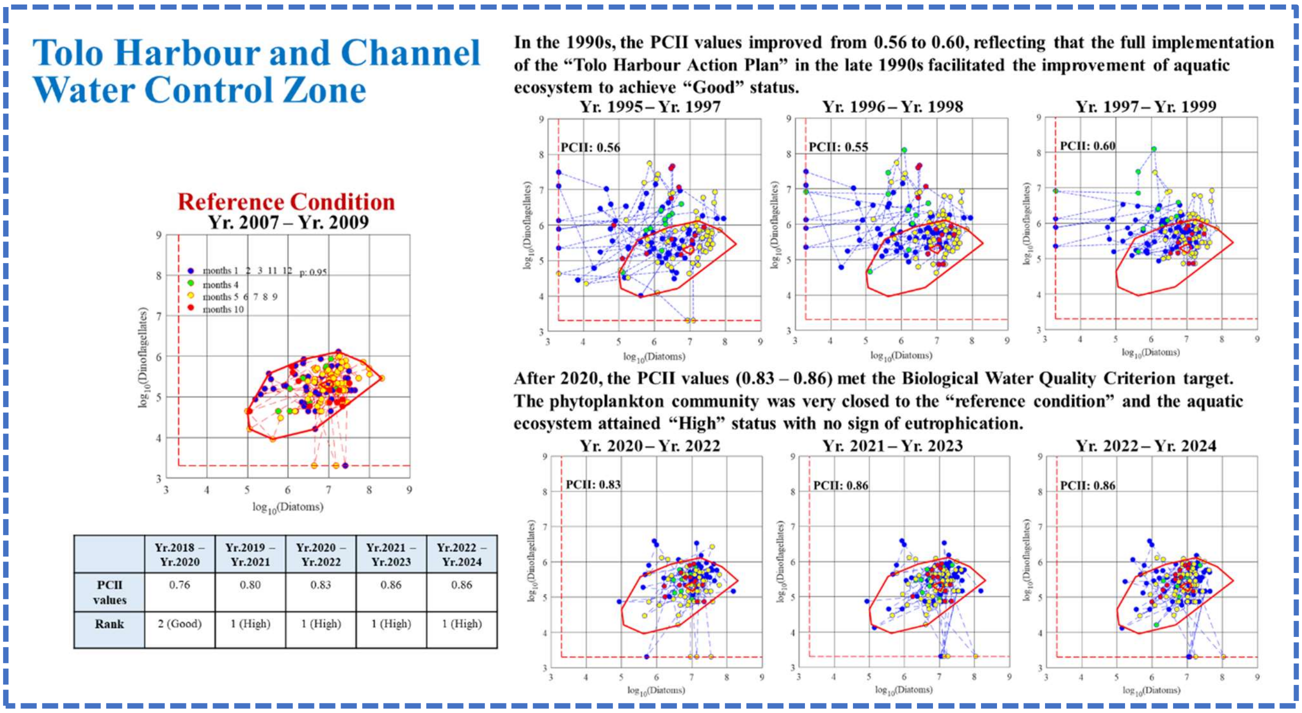


Figure 30. PCII charts, index values and ranks for Tolo Harbour and Channel WCZ

6.2.2 Central Waters

Victoria Harbour Water Control Zone

In 2024, the Victoria Harbour WCZ achieved a PCII value of 0.96 and was classified as Rank 1 with "High" phytoplankton community status, reflecting absence of any significant impact from nutrient enrichment in this WCZ. In contrast, the PCII values for Victoria Harbour WCZ ranged from 0.52 (Rank 3) to 0.70 (Rank 2) in the late 1990s. With the implementation of the "Harbour Area Treatment Scheme", the PCII values for the Victoria Harbour WCZ showed significant and steady improvement (Figure 31).

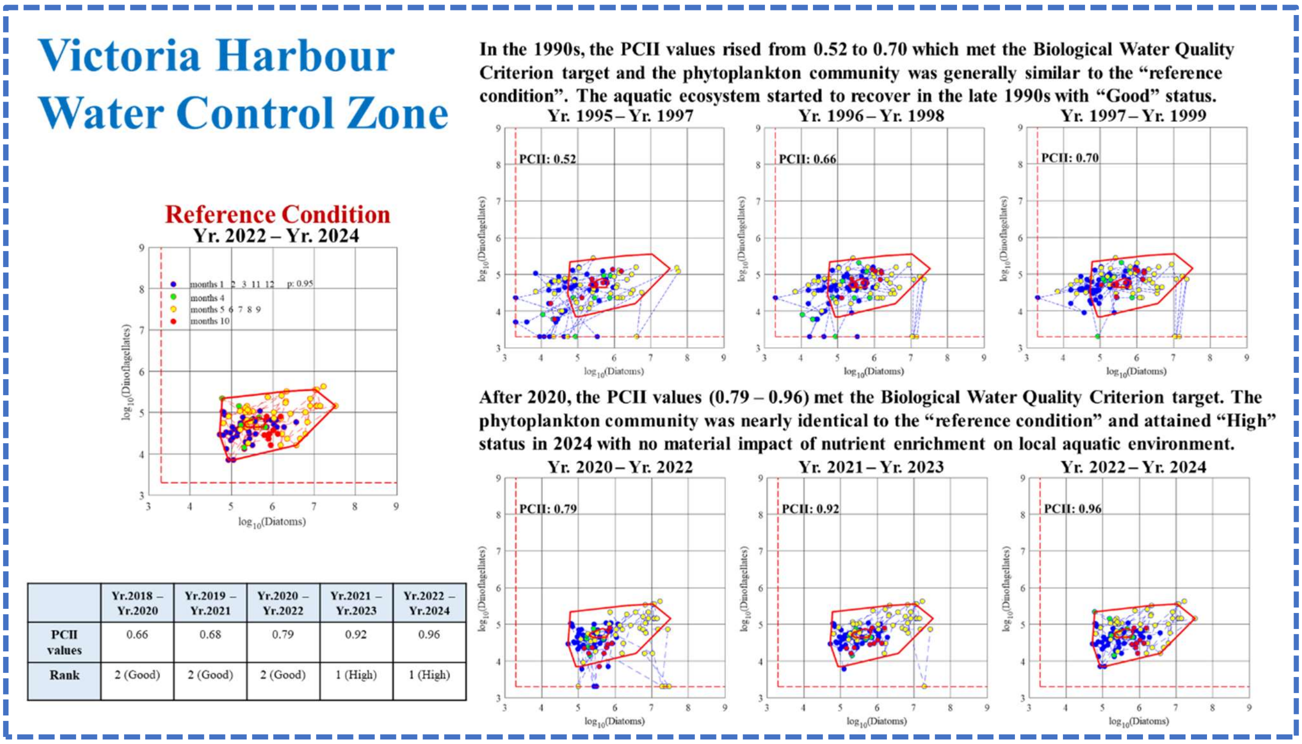
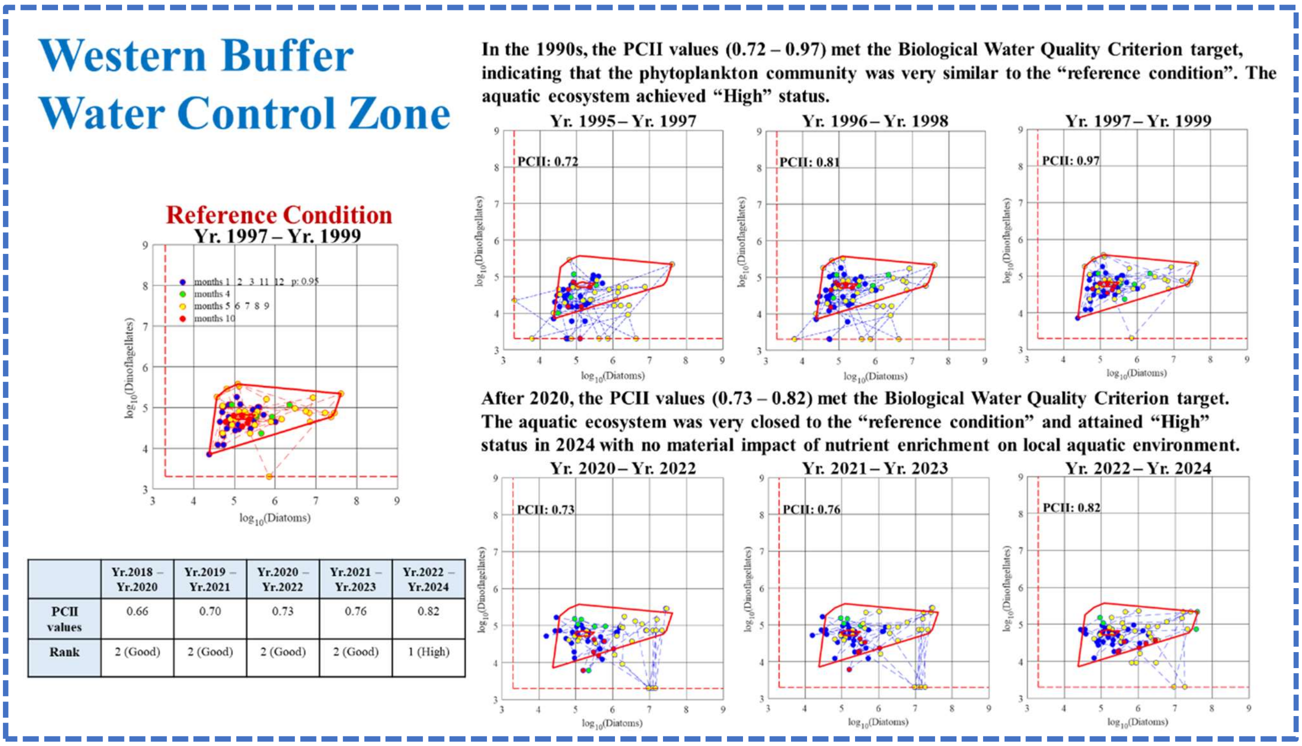


Figure 31. PCII charts, index values and ranks for Victoria Harbour WCZ

Western Buffer Water Control Zone

In 2024, the PCII value for Western Buffer WCZ was 0.82 (equivalent to Rank 1 with "High" phytoplankton community status), indicating that there was no material impact of nutrient enrichment on phytoplankton communities in this WCZ. During the late 1990s, the PCII values for Western Buffer WCZ ranged from 0.72 (Rank 2) to 0.97 (Rank 1). The phytoplankton community in this WCZ remained stable with achievement of "Good" to "High" status (Figure 32).



	Yr.2018 – Yr.2020	Yr.2019 – Yr.2021	Yr.2020 – Yr.2022	Yr.2021 – Yr.2023	Yr.2022 – Yr.2024
PCII values	0.66	0.70	0.73	0.76	0.82
Rank	2 (Good)	2 (Good)	2 (Good)	2 (Good)	1 (High)

Figure 32. PCII charts, index values and ranks for Western Buffer WCZ

6.2.3 Western Waters

Deep Bay Water Control Zone

The TIN levels in the Deep Bay WCZ continued to decline, and the“reference condition” period was set as year 2021 to 2023 when the TIN levels were the lowest compared to previous decades . In 2024, the PCII value for Deep Bay WCZ reached 1.00 and classified as Rank 1 with “High” phytoplankton community status. It indicated that the current nutrient levels did not have any significant impact on the phytoplankton community in this WCZ. In the late 1990s, the PCII values for Deep Bay WCZ ranged from 0.51 (Rank 3) to 0.68 (Rank 2). With the continuous joint efforts of both the Hong Kong and Shenzhen governments to implement of a series of water pollution control measures, the water quality of Deep Bay has improved significantly and the aquatic ecosystem has recovered correspondingly with achievement of “Good” to “High” phytoplankton community status (Figure 33).

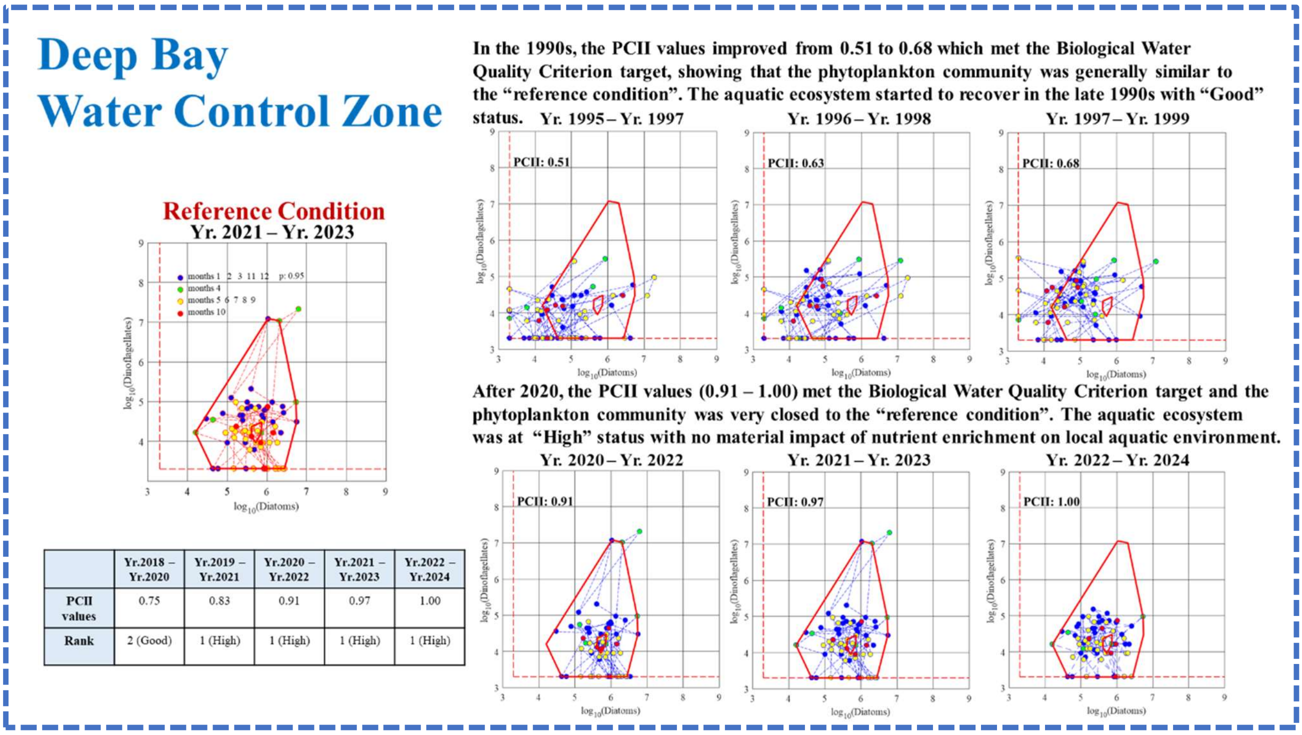
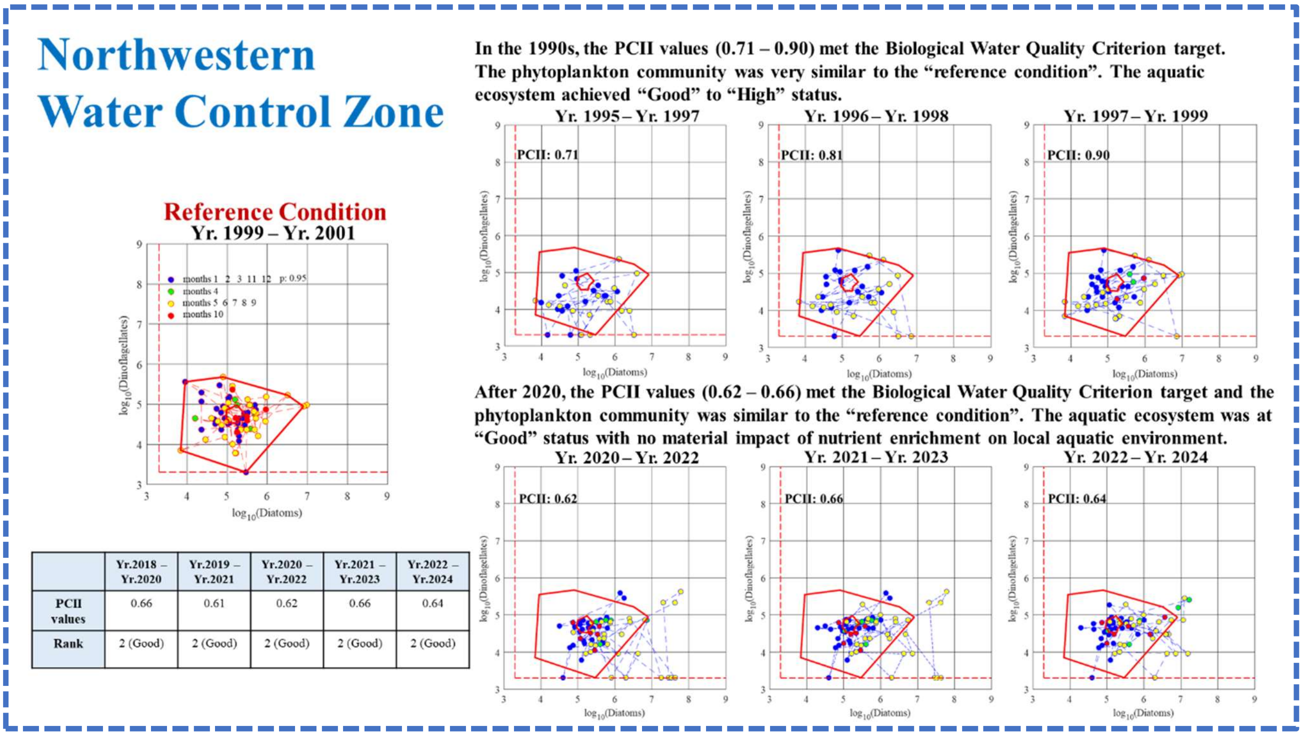


Figure 33. PCII charts, index values and ranks for Deep Bay WCZ

Northwestern Water Control Zone

In 2024, the PCII value for Northwestern WCZ was 0.64 (equivalent to Rank 2 with “Good” phytoplankton community status), indicating no marked impact from nutrient enrichment on the phytoplankton community in this WCZ. Despite the influence of seasonal high background nutrient levels from the Pearl River Estuary, the aquatic ecosystem remained stable and achieved

"Good" phytoplankton community status over the years with dominance of diatoms and reduced density of dinoflagellates (Figure 34).



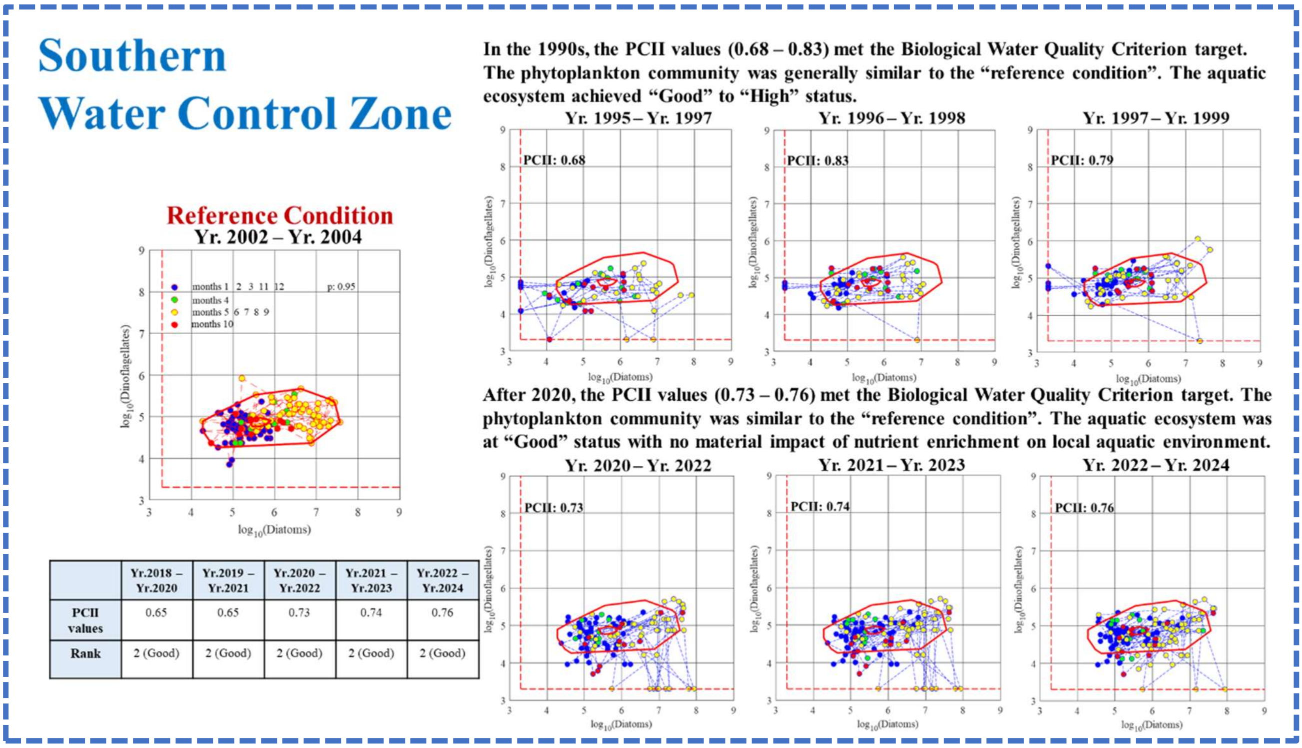
	Yr.2018 – Yr.2020	Yr.2019 – Yr.2021	Yr.2020 – Yr.2022	Yr.2021 – Yr.2023	Yr.2022 – Yr.2024
PCII values	0.66	0.61	0.62	0.66	0.64
Rank	2 (Good)	2 (Good)	2 (Good)	2 (Good)	2 (Good)

Figure 34. PCII charts, index values and ranks for Northwestern WCZ

6.2.4 Southern Waters

Southern Water Control Zone

In 2024, the PCII value for Southern WCZ was 0.76 which achieved Rank 2 with "Good" phytoplankton community status. It reflected that there was no significant impact from nutrient enrichment on the aquatic ecosystem. In contrast, the PCII values for Southern WCZ ranged from 0.68 to 0.83 in the late 1990s. It showed that the phytoplankton community in this WCZ maintained in the desirable range of "Good" to "High" status (Figure 35).



	Yr.2018 – Yr.2020	Yr.2019 – Yr.2021	Yr.2020 – Yr.2022	Yr.2021 – Yr.2023	Yr.2022 – Yr.2024
PCII values	0.65	0.65	0.73	0.74	0.76
Rank	2 (Good)	2 (Good)	2 (Good)	2 (Good)	2 (Good)

Figure 35. PCII charts, index values and ranks for Southern WCZ

6.3 Status of Red Tides

In 2024, there were a total of 11 reported red tide incidents in Hong Kong waters, as compared to an average of 10 incidents in the past five years. Among them, eight incidents involved only one WCZ; one incident involved four WCZs and two incidents involved five WCZs. The 11 red tide incidents were caused by five red tide species, all of which are non-toxic species except for *Heterosigma akashiwo and Pseudo-nitzschia pungens.* Nevertheless, no red tide related fish kill in Hong Kong waters was recorded in 2024.

The details of red tide incidents in 2024 are shown in the following table.

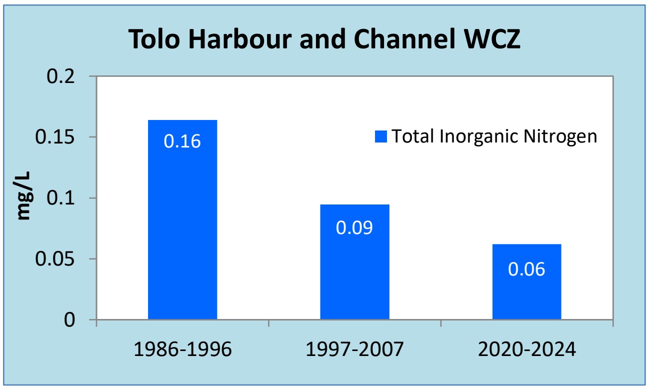
Incident No.	Sighting Period	Red Tide Species
1	19/1/2024	*Noctiluca scintillans*
2	25/1/2024 – 31/1/2024	*Phaeocystis globosa*
3	19/2/2024 – 21/2/2024	*Noctiluca scintillans*
4	6/3/2024 – 12/5/2024	*Noctiluca scintillans*
5	3/4/2024	*Heterosigma akashiwo**
6	5/4/2024 – 9/4/2024	*Heterosigma akashiwo** *Noctiluca scintillans*
7	6/6/2024 – 8/6/2024	*Noctiluca scintillans*
8	10/7/2024 – 21/7/2024	*Pseudo-nitzschia pungens**
9	22/8/2024 – 25/8/2024	*Levanderina fissa*
10	18/12/2024 – 4/2/2025	*Phaeocystis globosa*
11	26/12/2024 – 30/12/2024	*Noctiluca scintillans*

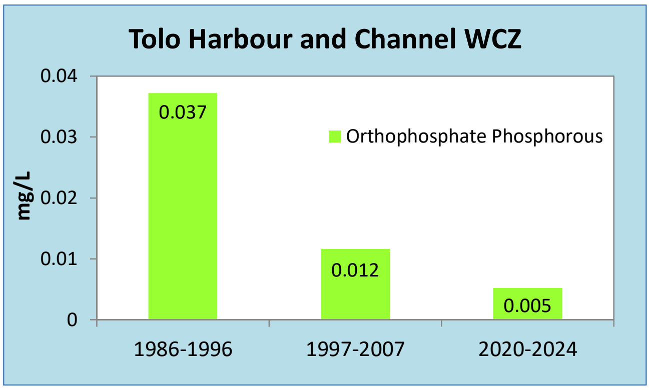
Red tide incidents in Hong Kong marine waters in 2024

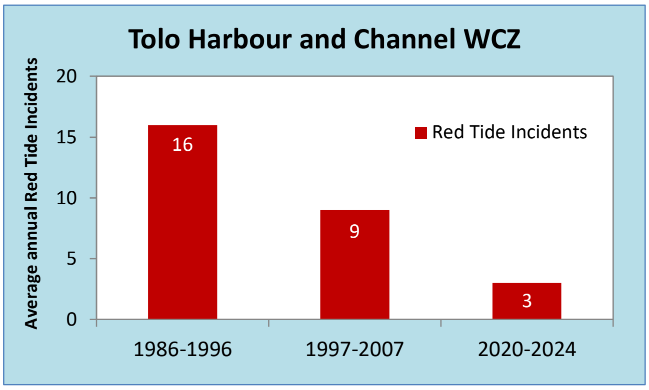
*: *Heterosigma akashiwo* and *Pseudo-nitzschia pungens* are potentially toxic species

(Source: Agriculture, Fisheries and Conservation Department)

The number of red tide incidents in Tolo Harbour has decreased significantly from the record high of 43 in 1988 to an annual average of about only three incidents in the recent five years. This is ascribed to the substantial improvement in water quality in the past decades following the successful implementation of the Tolo Harbour Action Plan (including the "Tolo Harbour Effluent Export Scheme"), as clearly shown in Figure 36.







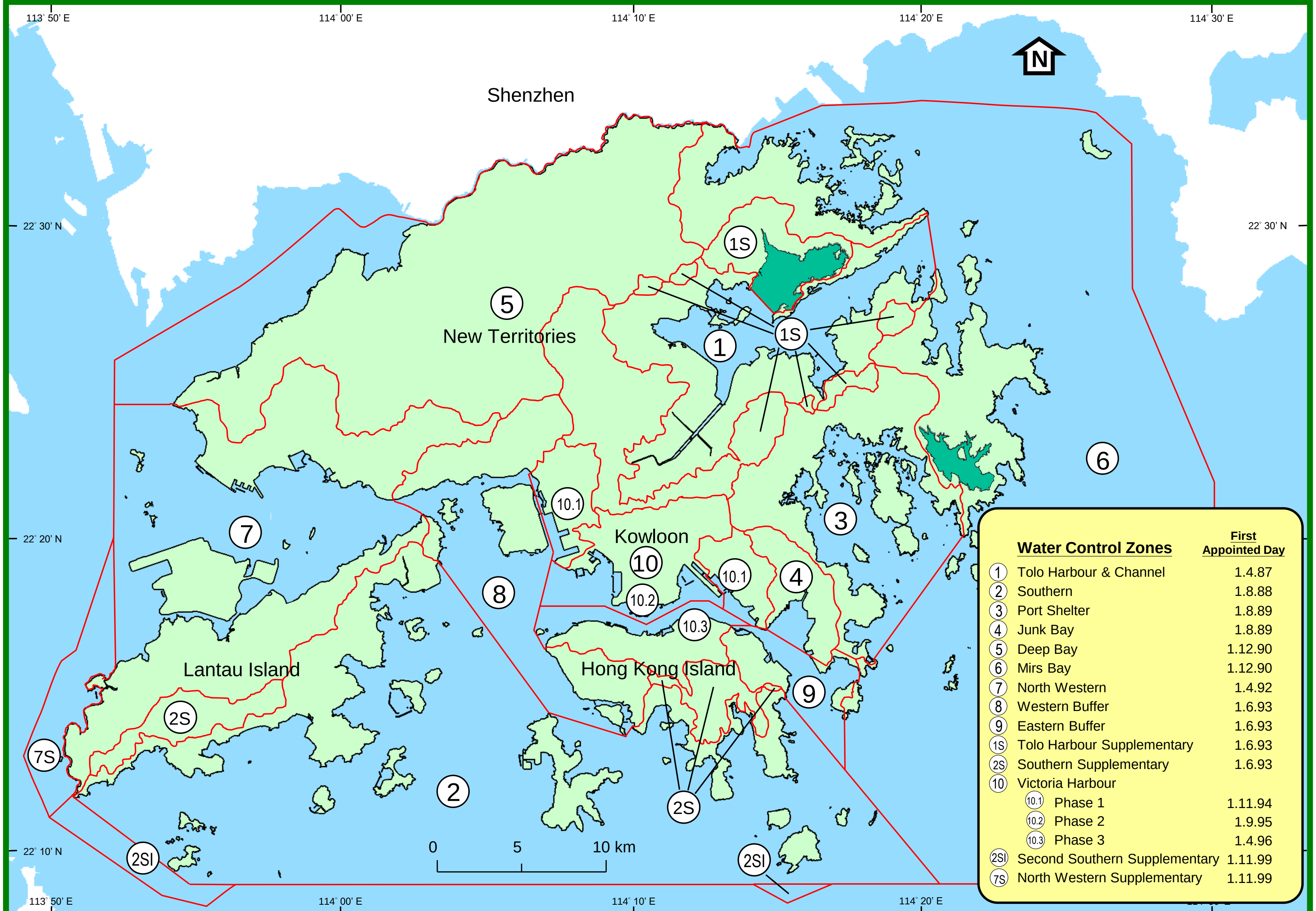
Tolo Harbour Action Plan:

1988 : Livestock Waste Control
Livestock Ban
1990 : Implementation of Water Pollution Control Ordinance
1992 : Sewage Treatment Works Process Modification
1993 : Sewerage First Aid Measures
1994 : Landfill Restoration
1995 : Effluent Export Scheme (Shatin) - Partial Diversion
1996 : Effluent Export Scheme (Tai Po) - Partial Diversion
1998 : Effluent Export Scheme - Full Diversion
Phased Implementation of Village Sewerage Plans

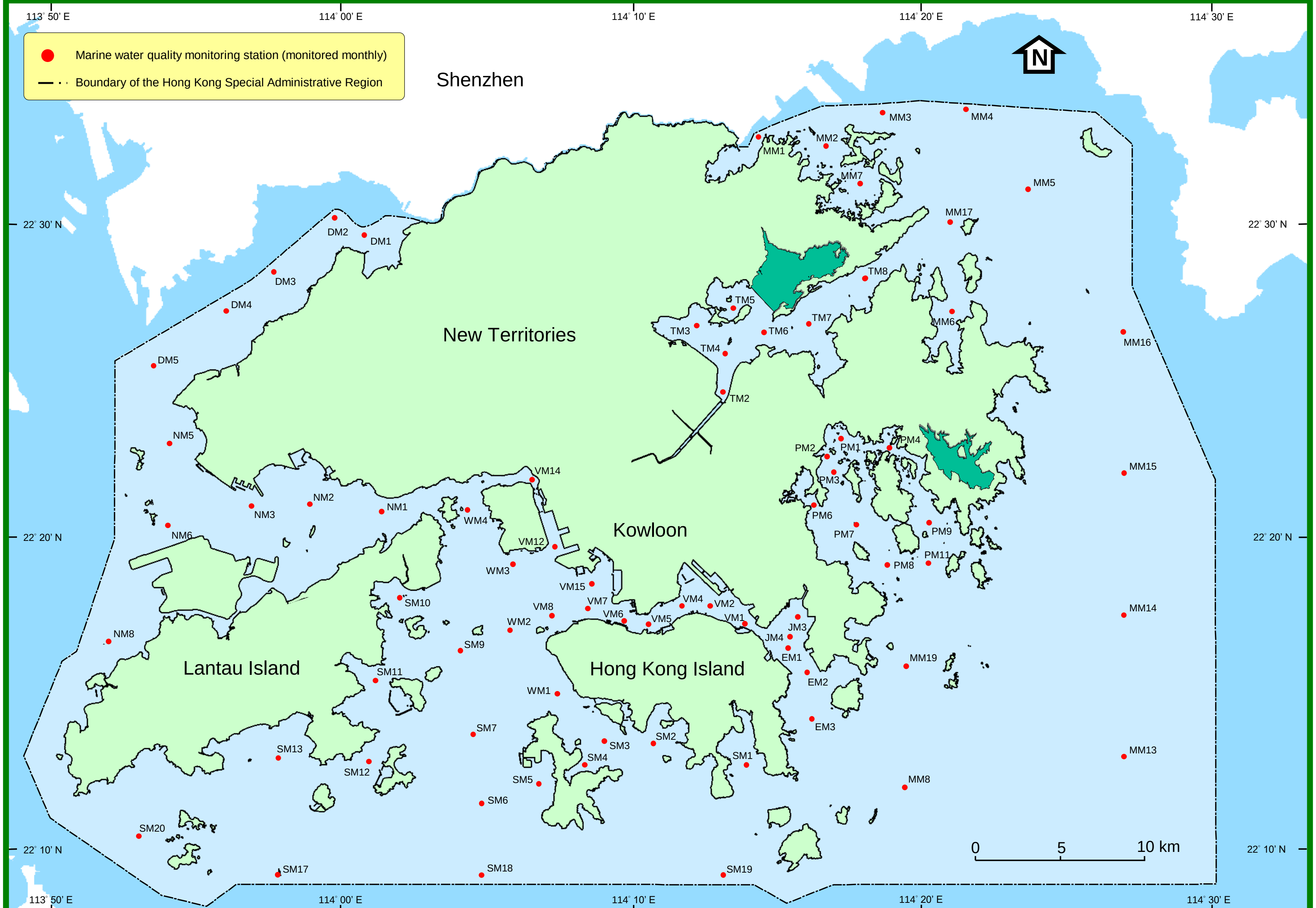
Figure 36. Concurrent reductions of nutrients levels and red tide occurrences in Tolo Harbour, 1986-2024

Appendices

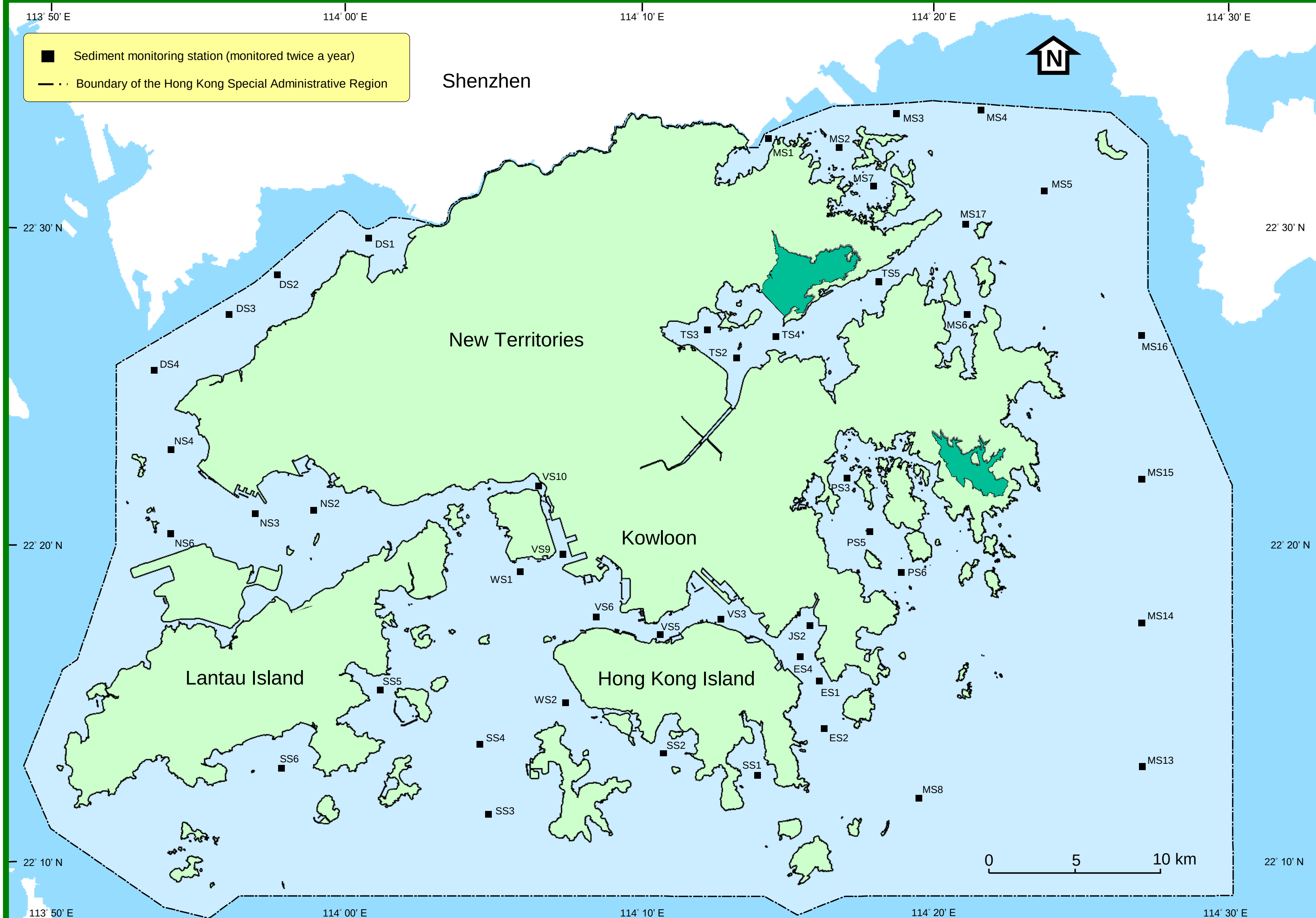
The Water Control Zones in Hong Kong
(Source: Environment Bureau– Plan No. WP/WP4/75, Oct 1999)



The 76 water quality monitoring stations in the open waters of Hong Kong

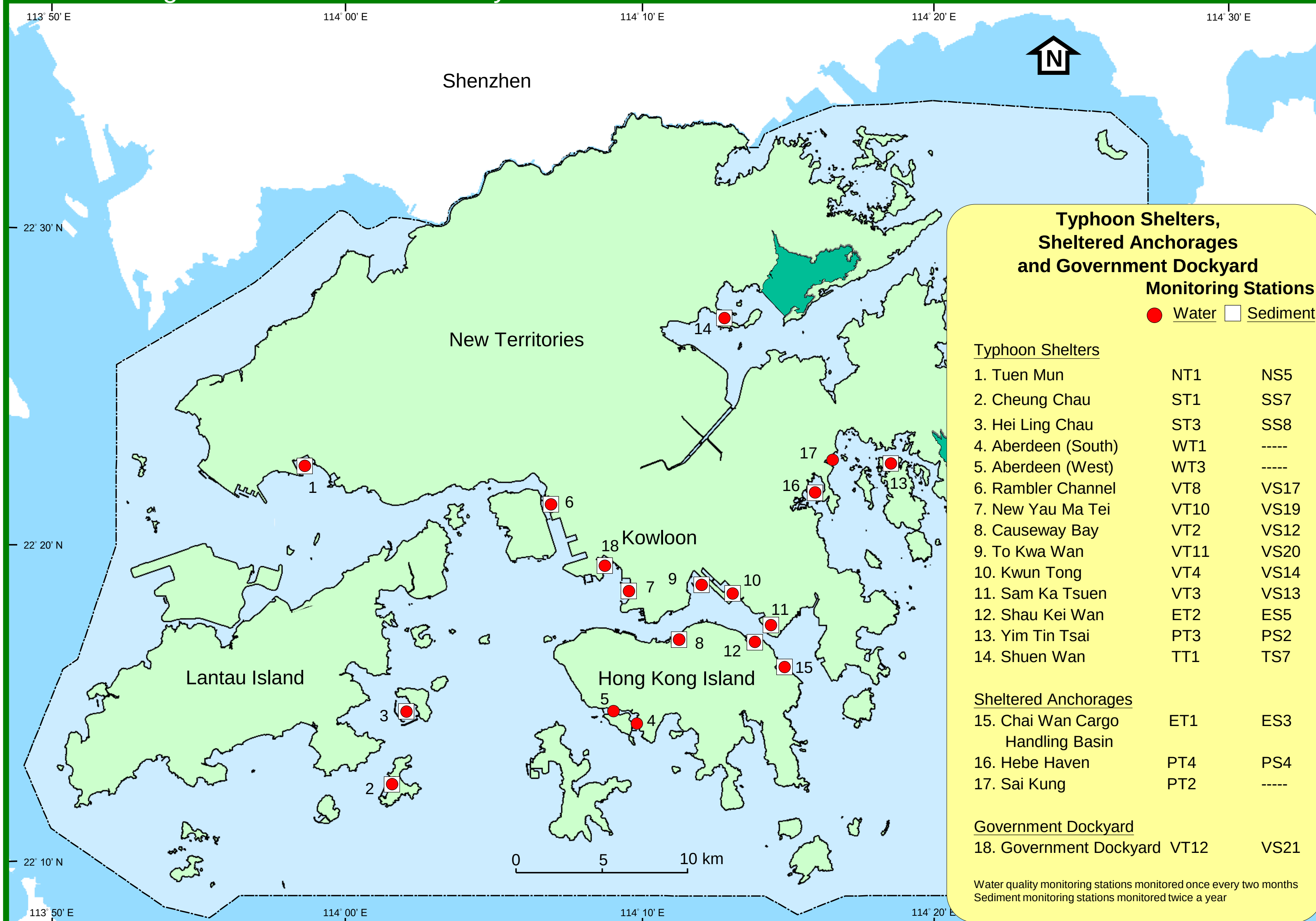


The 45 sediment monitoring stations in the open waters of Hong Kong



Appendix A

The 18 water quality and 15 sediment quality monitoring stations in typhoon shelters, sheltered anchorages and Government Dockyard

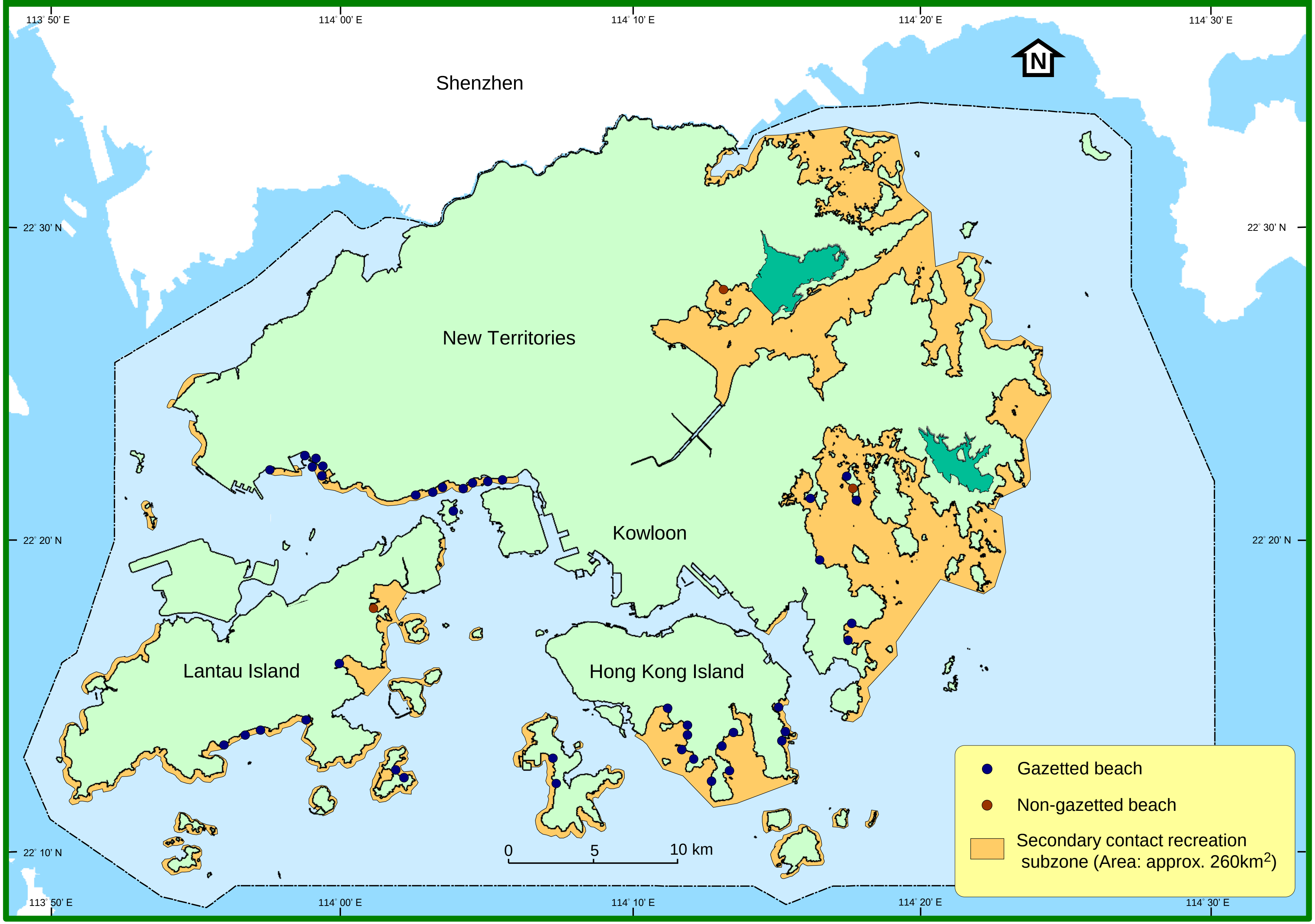


Locations of marine water and sediment quality monitoring stations

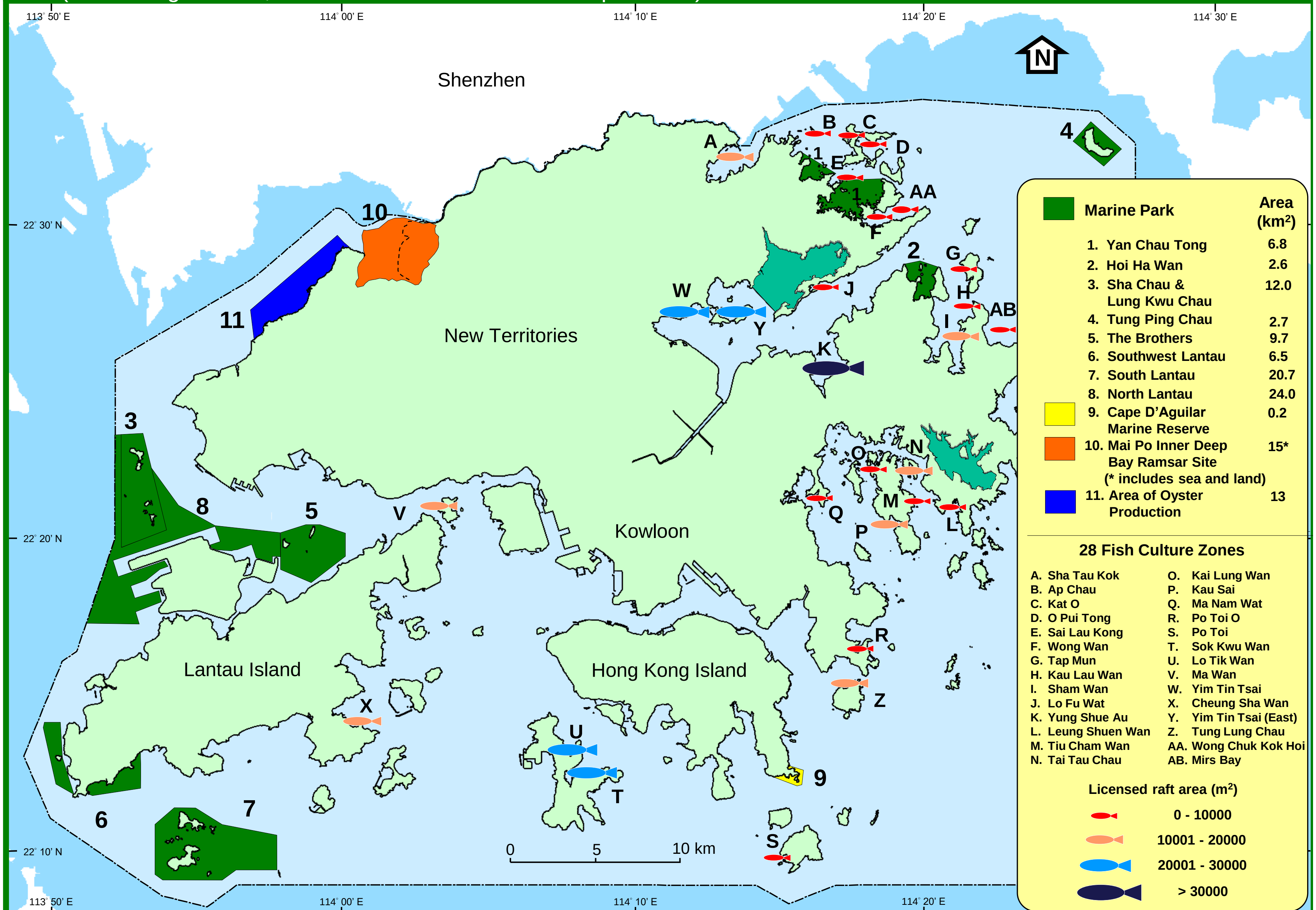
Water Control Zone	Station		Location				Depth (m)
	Water	Sediment	Latitude		Longitude		approx.
Tolo Harbour and Channel	TM2		22° 24.744'	N	114° 13.085'	E	4
	TM3	TS3	22° 26.857'	N	114° 12.181'	E	7
	TM4	TS2	22° 25.964'	N	114° 13.176'	E	8
	TM5		22° 27.426'	N	114° 13.456'	E	4
	TM6	TS4	22° 26.631'	N	114° 14.506'	E	12
	TM7		22° 26.907'	N	114° 16.057'	E	11
	TM8	TS5	22° 28.392'	N	114° 18.003'	E	22
	* TT1	* TS7	22° 27.270'	N	114° 12.717'	E	6
Southern	SM1	SS1	22° 12.738'	N	114° 13.885'	E	14
	SM2	SS2	22° 13.447'	N	114° 10.691'	E	14
	SM3		22° 13.527'	N	114° 8.980'	E	33
	SM4		22° 12.758'	N	114° 8.315'	E	11
	SM5		22° 12.141'	N	114° 6.728'	E	8
	SM6	SS3	22° 11.500'	N	114° 4.743'	E	14
	SM7	SS4	22° 13.740'	N	114° 4.473'	E	8
	SM9		22° 16.420'	N	114° 4.024'	E	8
	SM10		22° 18.125'	N	114° 1.919'	E	5
	SM11	SS5	22° 15.443'	N	114° 1.078'	E	8
	SM12		22° 12.861'	N	114° 0.869'	E	7
	SM13	SS6	22° 12.957'	N	113° 57.724'	E	6
	SM17		22° 9.211'	N	113° 57.727'	E	12
	SM18		22° 9.211'	N	114° 4.746'	E	21
	SM19		22° 9.211'	N	114° 13.077'	E	24
	SM20		22° 10.448'	N	113° 52.932'	E	7
	* ST1	* SS7	22° 12.517'	N	114° 1.493'	E	6
	* ST3	* SS8	22° 14.734'	N	114° 1.928'	E	6
Port Shelter	PM1		22° 23.242'	N	114° 17.145'	E	6
	PM2		22° 22.643'	N	114° 16.687'	E	8
	PM3	PS3	22° 22.156'	N	114° 16.910'	E	13
	PM4		22° 22.940'	N	114° 18.819'	E	6
	PM6		22° 21.102'	N	114° 16.213'	E	11
	PM7	PS5	22° 20.453'	N	114° 17.703'	E	17
	PM8	PS6	22° 19.168'	N	114° 18.745'	E	20
	PM9		22° 20.529'	N	114° 20.196'	E	15
	PM11		22° 19.240'	N	114° 20.163'	E	21
	* PT2		22° 22.798'	N	114° 16.540'	E	3
	* PT3	* PS2	22° 22.790'	N	114° 18.400'	E	6
	* PT4	* PS4	22° 21.728'	N	114° 15.879'	E	5
Junk Bay	JM3	JS2	22° 17.490'	N	114° 15.657'	E	10
	JM4		22° 16.873'	N	114° 15.378'	E	16
Deep Bay	DM1	DS1	22° 29.769'	N	114° 0.644'	E	2
	DM2		22° 30.454'	N	113° 59.549'	E	2
	DM3	DS2	22° 28.600'	N	113° 57.551'	E	3
	DM4	DS3	22° 27.335'	N	113° 55.937'	E	4
	DM5	DS4	22° 25.561'	N	113° 53.388'	E	8
North Western	NM1		22° 20.877'	N	114° 1.286'	E	34
	NM2	NS2	22° 21.130'	N	113° 58.815'	E	11
	NM3	NS3	22° 20.783'	N	113° 56.783'	E	14
	NM5	NS4	22° 23.051'	N	113° 53.972'	E	20
	NM6	NS6	22° 20.366'	N	113° 53.908'	E	8
	NM8		22° 16.695'	N	113° 51.886'	E	8
	* NT1	* NS5	22° 22.475'	N	113° 58.353'	E	4
Mirs Bay	MM1	MS1	22° 32.984'	N	114° 14.271'	E	6
	MM2	MS2	22° 32.626'	N	114° 16.648'	E	11
	MM3	MS3	22° 33.714'	N	114° 18.615'	E	16
	MM4	MS4	22° 33.817'	N	114° 21.483'	E	18
	MM5	MS5	22° 31.233'	N	114° 23.633'	E	20
	MM6	MS6	22° 27.334'	N	114° 20.997'	E	12
	MM7	MS7	22° 31.409'	N	114° 17.824'	E	13
	MM8	MS8	22° 12.021'	N	114° 19.345'	E	31
	MM13	MS13	22° 13.000'	N	114° 26.920'	E	28
	MM14	MS14	22° 17.560'	N	114° 26.920'	E	25
	MM15	MS15	22° 22.120'	N	114° 26.920'	E	24
	MM16	MS16	22° 26.670'	N	114° 26.920'	E	22
	MM17	MS17	22° 30.192'	N	114° 20.960'	E	17
	MM19		22° 15.921'	N	114° 19.411'	E	28
Western Buffer	WM1	WS2	22° 15.044'	N	114° 7.363'	E	35
	WM2		22° 17.074'	N	114° 5.730'	E	13
	WM3	WS1	22° 19.203'	N	114° 5.826'	E	20
	WM4		22° 20.940'	N	114° 4.256'	E	26
	* WT1		22° 14.494'	N	114° 9.737'	E	7
	* WT3		22° 14.811'	N	114° 8.918'	E	10
Eastern Buffer	EM1	ES4	22° 16.506'	N	114° 15.335'	E	16
	EM2	ES1	22° 15.732'	N	114° 15.971'	E	21
	EM3	ES2	22° 14.237'	N	114° 16.144'	E	21
	* ET1	* ES3	22° 16.203'	N	114° 14.624'	E	6
	* ET2	* ES5	22° 17.078'	N	114° 13.783'	E	12
Victoria Harbour	VM1		22° 17.280'	N	114° 13.839'	E	38
	VM2		22° 17.862'	N	114° 12.619'	E	12
		VS3	22° 17.631'	N	114° 12.526'	E	8
	VM4		22° 17.860'	N	114° 11.654'	E	12
	VM5		22° 17.266'	N	114° 10.510'	E	11
		VS5	22° 17.077'	N	114° 10.600'	E	8
	VM6		22° 17.371'	N	114° 9.665'	E	14
	VM7	VS6	22° 17.771'	N	114° 8.416'	E	10
	VM8		22° 17.564'	N	114° 7.175'	E	11
	VM12	VS9	22° 19.757'	N	114° 7.278'	E	14
	VM14	VS10	22° 21.935'	N	114° 6.527'	E	11
	VM15		22° 18.579'	N	114° 8.539'	E	11
	* VT2	* VS12	22° 17.194'	N	114° 11.304'	E	5
	* VT3	* VS13	22° 17.448'	N	114° 14.250'	E	5
	* VT4	* VS14	22° 18.734'	N	114° 12.814'	E	6
	* VT8	* VS17	22° 21.360'	N	114° 6.867'	E	5
	* VT10	* VS19	22° 18.590'	N	114° 9.430'	E	5
	* VT11	* VS20	22° 18.981'	N	114° 11.814'	E	6
	* VT12	* VS21	22° 19.429'	N	114° 8.587'	E	5

Note: 1. All locations are based on WGS84 datum
2. Water quality and sediment monitoring stations in typhoon shelters are marked with an asterisk *

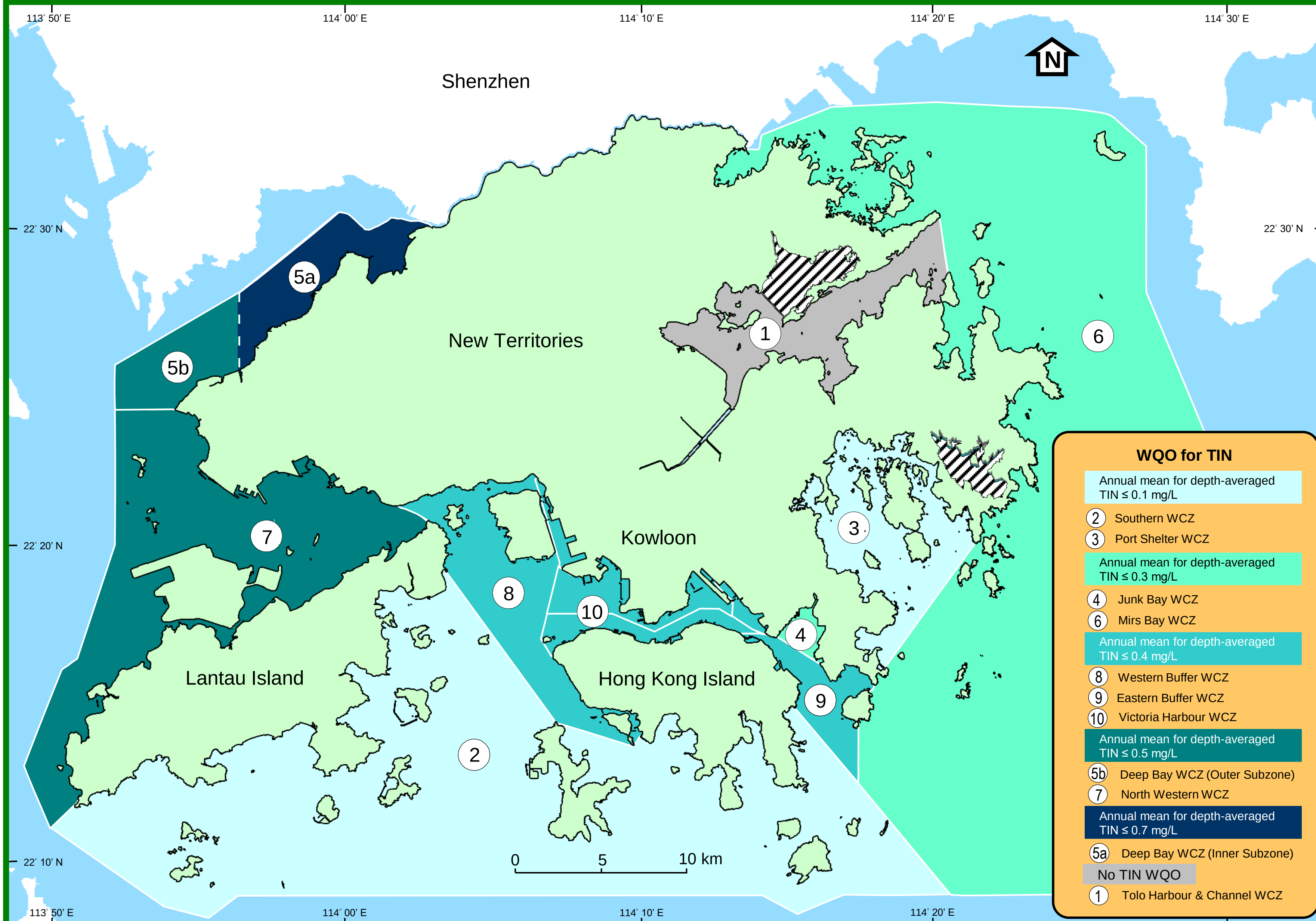
Bathing beaches and secondary contact recreation subzones in Hong Kong



Fish and shellfish culture zones and marine conservation sites in Hong Kong
(source: Agriculture, Fisheries and Conservation Department)



Water Quality Objective (WQO) for Total Inorganic Nitrogen (TIN) in the 10 Water Control Zones



Summary of Water Quality Objectives (WQOs) for marine waters of Hong Kong

Parameter	Water Quality Objective	Water Control Zone (WCZ) / Part(s) of zone / Subzone to which the WQO applies
Aesthetic Appearance	There should be no objectionable odours or discolouration of the water. Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent. Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam. There should be no recognisable sewage-derived debris. Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels or cause damage to vessels should be absent. The waters should not contain substances which settle to form objectionable deposits.	All WCZs (whole zone)
Dissolved Oxygen (bottom)	Not less than 2 mg/L for 90% of samples;	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
Dissolved Oxygen (Depth-averaged)	Not less than 4 mg/L for 90% of samples;	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
Dissolved Oxygen (bottom)	Not less than 2 mg/L	Harbour Subzone in Tolo Harbour & Channel WCZ
	Not less than 3 mg/L	Buffer Subzone in Tolo Harbour & Channel WCZ
	Not less than 4 mg/L	Channel Subzone in Tolo Harbour & Channel WCZ
Dissolved Oxygen (surface to 2m above bottom)	Not less than 4 mg/L	Harbour Subzone and Buffer Subzone in Tolo Harbour & Channel WCZ
Dissloved Oxygen (all depths)	Not less than 4 mg/L	Channel Subzone in Tolo Harbour & Channel WCZ
Nutrients	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.1 mg/L	Marine waters of Southern WCZ and Port Shelter WCZ
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.3 mg/L	Marine waters of Mirs Bay WCZ, Junk Bay WCZ, North Western WCZ (Castle Peak Subzone)
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.4 mg/L	Marine waters of Eastern Buffer WCZ, Western Buffer WCZ, Victoria Harbour WCZ
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.5 mg/L	Marine waters of Deep bay WCZ (Outer Subzone) and North Western WCZ (Whole zone except Castle Peak Subzone)
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.7 mg/L	Marine waters of Deep Bay WCZ (Inner Subzone)
Unionised Ammonia	Annual mean not to exceed 0.021 mg/L	All WCZs (whole zone) except Tolo Harbour & Channel WCZ
E.coli	Annual geometric mean not to exceed 610 cfu/100mL	Secondary contact recreation subzones in Tolo Harbour & Channel WCZ, Southern WCZ, Port Shelter WCZ, Mirs Bay WCZ, Deep Bay WCZ, North Western WCZ, Western Buffer WCZ
	Annual geometric mean not to exceed 610 cfu/100mL	Fish culture subzones in Tolo Harbour & Channel WCZ, Southern WCZ, Port Shelter WCZ, Junk Bay WCZ, Mirs Bay WCZ, Deep Bay WCZ, Eastern Buffer WCZ, Western Buffer WCZ
pH	To be in the range 6.5 - 8.5, change due to waste discharge not to exceed 0.2	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 0.5	Harbour Subzone in Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 0.3	Buffer Subzone in Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 0.1	Channel Subzone in Tolo Harbour & Channel WCZ
Salinity	Change due to waste discharge not to exceed 10% of natural ambient level	All WCZs (Whole zone) except Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 3ppt	Tolo Harbour & Channel WCZ
Temperature	Change due to waste discharge not to exceed 2^{o}C	All WCZs (Whole zone) except Tolo Harbour & Channel WCZ
	Change due to waste discharge not to exceed 1^{o}C	Tolo Harbour & Channel WCZ
Suspended Solids	Waste discharge not to raise the natural ambient level by 30% nor cause the accumulation of suspended solids which may adversely affect aquatic communities	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
Toxicants	Not to be present at levels producing significant toxic effect	All WCZs (Whole zone)
Chlorophyll-*a*	Not to exceed 20mg/m^3 (µg/L) calculated as running arithmetic mean of 5 daily measurements for any location and depth	Harbour Subzone in Tolo Harbour & Channel WCZ
	Not to exceed 10mg/m^3 (µg/L) calculated as running arithmetic mean of 5 daily measurements for any location and depth	Buffer Subzone in Tolo Harbour & Channel WCZ
	Not to exceed 6mg/m^3 (µg/L) calculated as running arithmetic mean of 5 daily measurements for any location and depth	Channel Subzone in Tolo Harbor & Channel WCZ

Sediment quality criteria for the classification of sediments [1]

Contaminants	Lower Chemical Exceedance Level (LCEL)	Upper Chemical Exceedance Level (UCEL)
Metals (mg/kg dry weight)		
Cadmium (Cd)	1.5	4
Chromium (Cr)	80	160
Copper (Cu)	65	110
Mercury (Hg)	0.5	1
Nickel (Ni) [2]	40	40
Lead (Pb)	75	110
Silver (Ag)	1	2
Zinc (Zn)	200	270
Metalloid (mg/kg dry weight)		
Arsenic (As)	12	42
Organic-PAHs (μg/kg dry weight)		
Low Molecular Weight PAHs [3]	550	3160
High Molecular Weight PAHs [4]	1700	9600
Organic-non-PAHs (μg/kg dry weight)		
Total PCBs	23	180
Organometallics (mg TBT/L in Interstitial water)		
Tributyltin [2]	0.15	0.15

Note: 1. The table is extracted from Appendix A of WBTC(W) No. 34/2002 Management of Dredged / Excavated Sediment (http://www.devb-wb.gov.hk)

2. When the LCEL and UCEL for a contaminant are the same, the contaminant level is considered to have exceeded UCEL if it is greater than the value shown.

3. Low molecular weight PAHs include acenaphthene, acenaphthylene, anthracene, fluorene, naphthalene, and phenanthrene.

4. High molecular weight PAHs include benzo[a]anthracene, benzo[a]pyrene, chrysene, dibenzo[a,h]anthracene, fluoranthene, pyrene, benzo[b]fluoranthene, benzo[k]fluoranthene, indeno[1,2,3-c,d]pyrene and benzo[g,h,i]perylene.

5. Total PCBs include 18 congeners: PCB 8, 18, 28, 44, 52, 66, 77, 101, 105, 118, 126, 128, 138, 153, 169, 170, 180, 187.

Summary of marine water quality parameters

Parameter		Unit	Reporting Limit	Sampling Depth	Standard Method / Techniques used [20]	Analysed by
Physical and Aggregate Properties	Temperature[1]	°C	0.1	Depth Profiling[10]	Instrumental (thermistor), SEACAT19+ CTD and Water Quality Profiler	MMT/EPD[15]
	Salinity [1,8]	---	0.1	Depth Profiling	Instrumental (electrical conductivity), SEACAT19+ CTD and Water Quality Profiler	MMT/EPD
	Dissolved Oxygen[1]	mg/L % saturation [9]	0.1 1	Depth Profiling	Instrumental (membrane elelectrode), SBE23Y dissolved oxygen sensor linked to SEACAT19+ CTD and Water Quality Profiler	MMT/EPD
	Turbidity[2]	NTU	0.1	Depth Profiling	Instrumental (nephelometric / infrared back scattering), OBS-3 turbidity sensor linked to SEACAT 19+ CTD and Water Quality Profiler	MMT/EPD
	pH[1]	---	0.1	Depth Profiling	Instrumental (electrodemetric), SBE18 pH sensor linked to SEACAT19 + CTD and Water Quality Profiler	MMT/EPD
	Secchi Disc Depth[2]	m	0.1	---	Manual	MMT/EPD
	Suspended Solids[2]	mg/L	0.5	S,M,B[11]	In-house method GL-PH-23 based on APHA 22ed 2540D (weighing)	GL[18]
	Volatile Suspended Solids[3]	mg/L	0.5	S,M,B	In-house method GL-PH-23 based on APHA 22ed 2540E (weighing)	GL
Aggregate Organic Constituents	5-day Biochemical Oxygen Demand (BOD_5)[4]	mg/L	0.1	S,M,B	In-house method based on APHA 20ed 5210B	EML/EPD[16]
Nutrients and Inorganic Constituents	Ammonia Nitrogen[5]	mg/L	0.005	S,M,B	In-house method GL-IN-15 based on ASTM D3590-11 Test method B	GL
	Unionised Ammonia[5]	mg/L	0.001	S,M,B	By calculation[12]	MMT/EPD
	Nitrite Nitrogen[5]	mg/L	0.002	S,M,B	In-house method GL-IN-18 based on APHA 22ed 4500-NO_2-B	GL
	Nitrate Nitrogen[5]	mg/L	0.002	S,M,B	In-house method GL-IN-18 based on APHA 22ed 4500-NO_3-I	GL
	Total Inorganic Nitrogen[5]	mg/L	0.01	S,M,B	By calculation[13]	MMT/EPD
	Total Kjeldahl Nitrogen[5] (soluble; soluble & particulate)	mg/L	0.05	S,M,B	In-house methods GL-IN-14 and GL-IN-15 based on ASTM D3590-11 Test method B	GL
	Total Nitrogen[5]	mg/L	0.05	S,M,B	By calculation[13]	MMT/EPD
	Orthophosphate Phosphorus[5]	mg/L	0.002	S,M,B	In-house method GL-IN-16 based on APHA 22ed 4500-P G	GL
	Total Phosphorus[5] (soluble; soluble & particulate)	mg/L	0.02	S,M,B	In-house methods GL-IN-14 and GL-IN-16 based on ASTM D515-88 Test method B and APHA 22ed 4500-P G	GL
	Silica (as SiO_2) (soluble) [5]	mg/L	0.05	S,M,B	In-house method GL-IN-17 based on APHA 22ed 4500-SiO_2 F	GL
Biological and Microbiological Examination	Chlorophyll-a [6]	µg/L	0.2	S,M,B	In-house method GL-OR-34 based on APHA 20ed 10200H 2 (spectrophotometric)	GL
	Phaeo-pigment [6]	µg/L	0.2	S,M,B	In-house method GL-OR-34 based on APHA 20ed 10200H 2 (spectrophotometric)	GL
	Escherichia coli (*E. coli*)[7]	cfu/100mL	1	S,M,B	In-house method, membrane filtration with CHROMagar Liquid *E. coli*-coliform culture[14]	EML/EPD
	Faecal Coliforms[7]	cfu/100mL	1	S,M,B	In-house method, membrane filtration with CHROMagar Liquid *E. coli*-coliform culture[14]	EML/EPD
	Phytoplankton	cell/mL	1	S	In-house method, 10 ml settled sub-sample using plankton chamber and inverted microscope[19]	WSL/EPD[17]

Note:
1. Indicate general oceanographic conditions of marine water.
2. Low transparency and light penetration would affect aesthetic value and photosynthesis in marine water.
3. Indicate the amount of particulate organic matters in marine water.
4. Indicate the amount of organic pollutants in marine water.
5. Major nutrients (nitrogen, phosphorus, silica) promoting algal growth in marine water.
6. Indicate the amount of algal biomass in marine water.
7. Sewage bacteria indicate the extent of faecal pollution in marine water.
8. Measuring and reporting of Salinity (S) are based on the Practical Salinity Scale and International Equation of State of Seawater. (UNESCO Technical Papers in Marine Science No. 30 (1981) ; No. 36 (1981) and No. 45 (1985))
9. Percent saturation of dissolved oxygen is calculated from dissolved oxygen in mg/L based on Weiss R.F. (1970); The solubility of nitrogen, oxygen and argon in water and seawater. Deep Sea Res. Vol. 17, pp.721-735.
10. Depth profiling - continuous measurements at downcast are processed and presented at 1m intervals from 1m below the surface to 1m above the seabed.
11. If water depth is 6m or above, sampling is taken at three depths: S - 1m below water surface; M - mid-depth of water column; B - 1m above seabed. If water depth is 4 to 5 m, "M" is skipped; If water depth is 3m or less, "M" and "B" are skipped.
12. i) Bower C.E. and Bidwell J.P. (1978), Ionization of ammonia in seawater: Effect of temperature, pH and salinity. J. Fish. Res. Board Can. Vol.35, pp.1012-1016;
    ii) K., Russo R.C. &et. al. (1975), Aqueous ammonia equilibrium calculations: effect of pH and temperature. J. Fish. Res. Board Can. Vol.32, pp.2379-2383.
13. Total Inorganic Nitrogen = Ammonia Nitrogen + Nitrite Nitrogen + Nitrate Nitrogen ; Total Nitrogen = Total Kjeldahl Nitrogen (soluble & particulate) + Nitrite Nitrogen + Nitrate Nitrogen
14. i) DoE, DHSS & PHLS (1983); The Bacteriological Examination of Drinking Water Supplies 1982, Sec.7.8 & 7.9;
    ii) B.S.W. Ho and T.Y. Tam (1997), Enumeration of E. coli in environmental waters and wastewater using a chromogenic medium. Wat. Sci. Tech.Vol.35, No.11-12, pp.409-413; method adopted in 1997.
15. MMT/EPD - Marine Monitoring Team, Water Policy and Science Group, Environmental Protection Department.
16. EML/EPD - Environmental Microbiology Laboratory, Water Policy and Science Group, Environmental Protection Department.
17. WSL/EPD - Water Sciences Laboratory, Water Policy and Science Group, Environmental Protection Department.
18. GL - Government Laboratory.
19. i) Lund, J.H., Kipling, C. and Le Cren, E.D. 1958. The inverted microscope method of estimating algal numbers, and the statistical basis of estimations by counting. Hydrobiologia Vol. 11, pp. 143-170.
    ii) Utermohl, H. 1958. Zur Vervollkommung der Quantitativen Phytoplankton-Methodik. Mitt. Inter. Verein. Lim. Vol. 9, pp. 1-38.
20. Mention of brand names and commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.

Summary of marine sediment[1] quality parameters

	Parameter	Unit [2]	Reporting Limit	Standard Method / Techniques used [8]	Analysed by
Physical and Aggregate Properties	Particle Size Fractionation	% w/w	1	In-house method, sieving and weighing ; 8 fractions : >4000 µm, <4000µm, <2000µm, <1000µm, <500µm, <250µm, <125µm and <63µm	MMT/EPD [6]
	Electrochemical Potential [4]	mV	1	Instrumental, Orion Model 250A pH/Redox Meter (electrodemetric)	MMT/EPD
	Total Solids (TS) [3]	% w/w	0.1	In-house method GL-PH-22 based on APHA 20ed 2540G (weighing)	GL [7]
	Total Volatile Solids (TVS) [3]	% TS	0.1	In-house method GL-PH-22 based on APHA 20ed 2540G (weighing)	GL
	Dry Wet Ratio	---	0.01	In-house method GL-PH-22 based on APHA 20ed 2540G (weighing)	GL
Aggregate Organic Constituents [3]	Chemical Oxygen Demand (COD)	mg/kg	2	In-house method GL-OR-47 based on ASTM D1252-00 Test method A (open reflux)	GL
	Total Carbon (TC)	% w/w	0.1	In-house method GL-OR-33 based on APHA 20ed 5310 B and BS EN 13137:2001	GL
Nutrients and Inorganic Constituents [3]	Ammonia-Nitrogen	mg/kg	0.05	In-house method GL-IN-19 based on ASTM D3590-89 Test method B	GL
	Total Kjeldahl Nitrogen	mg/kg	0.5	In-house methods GL-IN-14 and GL-IN-15 based on ASTM D3590-11 Test method B	GL
	Total Phosphorus	mg/kg	0.2	In-house methods GL-IN-14 and GL-IN-16 based on ASTM D515-88 Test method B and APHA 22ed 4500-P G	GL
	Total Sulphide	mg/kg	0.2	In-house method GL-IN-45 based on APHA 20ed 4500-S^{2-} A&D (spectrophotometric)	GL
	Total Cyanide	mg/kg	0.1	In-house method GL-IN-44 based on APHA 20ed 4500-CN^- A&E (distillation and amperometric)	GL
Metals & Metalloids [5]	Aluminium (Al)	mg/kg	1	In-house method GL-TE-103 (ICP-MS)	GL
	Arsenic (As)	mg/kg	0.1	In-house method GL-TE-103 (ICP-MS)	GL
	Barium (Ba)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
	Boron (B)	mg/kg	5	In-house method GL-TE-103 (ICP-MS)	GL
	Cadmium (Cd)	mg/kg	0.1	In-house method GL-TE-103 (ICP-MS)	GL
	Chromium (Cr)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
	Copper (Cu)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
	Iron (Fe)	mg/kg	5	In-house method GL-TE-103 (ICP-MS)	GL
	Lead (Pb)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
	Manganese (Mn)	mg/kg	1	In-house method GL-TE-103 (ICP-MS)	GL
	Mercury (Hg)	mg/kg	0.05	In-house method GL-TE-103 (ICP-MS)	GL
	Nickel (Ni)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
	Silver (Ag)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
	Vanadium (V)	mg/kg	0.1	In-house method GL-TE-103 (ICP-MS)	GL
	Zinc (Zn)	mg/kg	0.2	In-house method GL-TE-103 (ICP-MS)	GL
Trace Organic Compounds	Polychlorinated Biphenyls (PCBs)				
	18 PCB congeners : PCB 8, 18, 28, 44, 52, 66, 77, 101, 105, 118, 126, 128, 138, 153, 169, 170, 180, 187	µg/kg	2	In-house method GL-OR-25 based on Reference Method for the Analysis of Polychlorinated Biphenyls, Environmental Protection Series: Report EPS 1/RM/31, March 1997, Environment Canada (GC-MS)	GL
	Polyaromatic Hydrocarbons (PAHs)				
	- Acenaphthene	µg/kg	50	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Acenaphthylene	µg/kg	50	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Naphthalene	µg/kg	60	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Fluorene	µg/kg	10	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Phenanthrene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Anthracene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Fluoranthene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Pyrene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(*a*)anthracene	µg/kg	3	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Chrysene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(*b*)fluoranthene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(*k*)fluoranthene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(*a*)pyrene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Dibenzo(*a,h*)anthracene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(*g,h,i*)perylene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Indeno(*1,2,3-c,d*)pyrene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL

Note:
1. Birge-Ekman (0.023sq.m) grab / Van Veen (0.1sq.m) grab / Smith-McIntyre (0.1sq.m) grab is employed to collect sediment samples from the top 10cm of seabed.
2. All parameters are reported on a dry weight basis unless otherwise stated.
3. Determinants are reported on a wet weight basis.
4. Electrochemical potential (Eh) is measured "on-site" at 3cm below the surface of freshly collected sediment samples (Reference : Handbook of Techniques for Aquatic Sediment Sampling. By A. Mudrock & S.D. MacKnight, 1994, CRC Press).
5. Digestion procedure for metals and metalloids in sediment follows Government Laboratory's in-house method GL-TE-51.
6. MMT/EPD - Marine Monitoring Team, Water Policy and Science Group, Environmental Protection Department.
7. GL - Government Laboratory.
8. Mention of brand names and commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.

Summary of water quality data for the Mirs Bay WCZ in 2024

Parameter		Mirs Bay MM1	Crooked Island MM2	Crooked Island MM7	Port Island MM17	Mirs Bay North MM3	Mirs Bay North MM4	Mirs Bay North MM5
Number of samples		12	12	12	12	12	12	12
Temperature (°C)		25.1 (20.1 - 31.2)	25.0 (19.8 - 30.9)	24.8 (19.6 - 30.9)	24.5 (19.4 - 30.1)	24.7 (19.6 - 29.6)	24.4 (19.4 - 29.2)	24.6 (19.3 - 29.4)
Salinity		31.5 (25.5 - 33.9)	31.5 (25.6 - 34.1)	31.8 (26.0 - 34.0)	32.0 (26.6 - 34.1)	31.9 (26.4 - 34.2)	32.2 (28.1 - 34.2)	32.0 (27.5 - 34.2)
Dissolved Oxygen (mg/L)		6.3 (4.8 - 8.1)	6.1 (4.9 - 7.6)	6.0 (4.9 - 7.5)	6.1 (4.4 - 7.3)	6.0 (4.4 - 7.3)	6.0 (4.5 - 7.2)	5.9 (4.1 - 7.1)
	Bottom	5.9 (4.5 - 7.9)	5.8 (4.3 - 7.7)	5.4 (3.9 - 7.0)	5.8 (3.8 - 7.5)	5.8 (3.7 - 7.5)	5.5 (3.6 - 7.3)	5.5 (3.4 - 7.0)
Dissolved Oxygen (% Saturation)		90 (71 - 109)	88 (75 - 101)	86 (74 - 100)	88 (65 - 97)	86 (66 - 98)	86 (67 - 96)	84 (59 - 97)
	Bottom	85 (70 - 105)	84 (66 - 103)	77 (59 - 95)	83 (56 - 99)	82 (55 - 99)	78 (53 - 97)	78 (50 - 93)
pH		8.1 (7.8 - 8.4)	8.1 (7.8 - 8.3)	8.0 (7.8 - 8.2)	8.0 (7.8 - 8.2)	8.1 (7.8 - 8.3)	8.1 (7.9 - 8.3)	8.0 (7.8 - 8.2)
Secchi Disc Depth (m)		2.1 (1.5 - 2.7)	2.3 (1.7 - 3.0)	2.5 (1.6 - 3.2)	2.9 (1.8 - 6.5)	2.4 (1.5 - 3.7)	2.8 (1.7 - 3.9)	3.7 (1.6 - 6.6)
Turbidity (NTU)		2.4 (0.4 - 4.9)	2.1 (0.4 - 4.9)	2.0 (0.3 - 4.6)	1.8 (0.1 - 5.5)	2.2 (0.2 - 4.6)	2.2 (0.2 - 5.5)	2.5 (0.2 - 4.9)
Suspended Solids (mg/L)		4.6 (2.3 - 8.0)	4.5 (3.1 - 6.6)	3.5 (1.2 - 5.6)	3.3 (1.0 - 4.9)	3.7 (1.5 - 7.5)	3.8 (1.7 - 8.6)	3.9 (1.6 - 6.7)
5-day Biochemical Oxygen Demand (mg/L)		0.9 (0.2 - 2.0)	1.1 (0.1 - 2.7)	1.0 (<0.1 - 2.9)	1.0 (<0.1 - 3.0)	0.9 (<0.1 - 2.6)	0.9 (<0.1 - 2.5)	0.7 (<0.1 - 2.2)
Ammonia Nitrogen (mg/L)		0.035 (0.008 - 0.127)	0.023 (0.006 - 0.048)	0.021 (0.007 - 0.041)	0.018 (0.008 - 0.043)	0.020 (0.006 - 0.043)	0.019 (0.008 - 0.043)	0.017 (0.006 - 0.045)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.007)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.004)	<0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.003)
Nitrite Nitrogen (mg/L)		0.005 (<0.002 - 0.027)	0.004 (<0.002 - 0.020)	0.003 (<0.002 - 0.012)	0.004 (<0.002 - 0.020)	0.005 (<0.002 - 0.016)	0.005 (<0.002 - 0.029)	0.004 (<0.002 - 0.022)
Nitrate Nitrogen (mg/L)		0.059 (0.012 - 0.207)	0.049 (0.006 - 0.200)	0.038 (0.005 - 0.146)	0.038 (0.006 - 0.150)	0.045 (0.006 - 0.220)	0.040 (0.007 - 0.173)	0.039 (0.006 - 0.167)
Total Inorganic Nitrogen (mg/L)		0.10 (0.04 - 0.25)	0.08 (0.02 - 0.22)	0.06 (0.02 - 0.19)	0.06 (0.02 - 0.17)	0.07 (0.03 - 0.24)	0.06 (0.02 - 0.20)	0.06 (0.02 - 0.19)
Total Kjeldahl Nitrogen (mg/L)		0.21 (0.10 - 0.30)	0.18 (0.09 - 0.29)	0.18 (0.08 - 0.29)	0.16 (0.08 - 0.27)	0.16 (0.10 - 0.27)	0.16 (0.08 - 0.26)	0.16 (0.08 - 0.23)
Total Nitrogen (mg/L)		0.28 (0.12 - 0.51)	0.23 (0.10 - 0.49)	0.22 (0.08 - 0.44)	0.21 (0.09 - 0.42)	0.21 (0.11 - 0.50)	0.20 (0.09 - 0.45)	0.20 (0.09 - 0.40)
Orthophosphate Phosphorus (mg/L)		0.004 (<0.002 - 0.009)	0.004 (<0.002 - 0.012)	0.004 (<0.002 - 0.013)	0.005 (<0.002 - 0.012)	0.005 (<0.002 - 0.013)	0.005 (<0.002 - 0.012)	0.004 (<0.002 - 0.011)
Total Phosphorus (mg/L)		0.07 (0.02 - 0.18)	0.07 (0.03 - 0.14)	0.07 (0.03 - 0.14)	0.07 (0.03 - 0.15)	0.06 (0.03 - 0.13)	0.07 (0.03 - 0.13)	0.08 (0.03 - 0.15)
Silica (as SiO_2) (mg/L)		0.64 (0.15 - 1.31)	0.44 (0.16 - 0.86)	0.40 (0.15 - 0.71)	0.42 (0.06 - 0.73)	0.46 (0.06 - 0.90)	0.49 (<0.05 - 1.10)	0.44 (0.09 - 1.07)
Chlorophyll-*a* (μg/L)		4.3 (0.9 - 8.6)	3.0 (1.0 - 4.9)	2.6 (1.1 - 4.8)	2.1 (0.8 - 5.3)	2.2 (0.6 - 4.6)	1.7 (0.8 - 5.2)	2.0 (0.5 - 9.5)
E.coli (count/100mL)		16 (<1 - 9000)	2 (<1 - 44)	1 (<1 - 3)	1 (<1 - 2)	3 (<1 - 68)	1 (<1 - 16)	1 (<1 - 2)
Faecal Coliforms (count/100mL)		37 (1 - 21000)	5 (<1 - 130)	2 (<1 - 17)	2 (<1 - 18)	4 (<1 - 140)	2 (<1 - 80)	2 (<1 - 10)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Mirs Bay WCZ in 2024 (continued)

Parameter	Ninepin Group	Waglan Isalnd	Mirs Bay (South)	Mirs Bay (Central)			Long Harbour
	MM19	MM8	MM13	MM14	MM15	MM16	MM6
Number of samples	12	12	12	12	12	12	12
Temperature (°C)	24.1 (19.0 - 28.8)	24.1 (18.9 - 28.3)	24.1 (18.9 - 28.4)	24.2 (19.4 - 28.4)	24.2 (18.7 - 28.9)	24.2 (18.6 - 29.1)	24.6 (19.3 - 30.4)
Salinity	32.7 (27.0 - 34.4)	32.9 (30.3 - 34.4)	32.6 (27.3 - 34.4)	32.9 (29.8 - 34.4)	32.6 (29.2 - 34.4)	32.5 (28.1 - 34.3)	31.8 (26.9 - 34.1)
Dissolved Oxygen (mg/L)	5.9 (4.6 - 7.4)	6.0 (4.7 - 7.3)	6.0 (4.9 - 7.3)	5.9 (4.8 - 7.2)	5.9 (5.0 - 7.5)	6.0 (4.9 - 7.4)	6.2 (4.8 - 7.2)
Bottom	5.5 (3.7 - 7.1)	5.7 (3.6 - 7.0)	6.0 (4.6 - 7.2)	5.9 (4.5 - 7.0)	5.7 (3.8 - 7.0)	5.6 (4.0 - 7.0)	5.8 (3.9 - 7.6)
Dissolved Oxygen (% Saturation)	83 (69 - 100)	85 (71 - 99)	85 (70 - 106)	84 (72 - 98)	84 (73 - 101)	85 (75 - 101)	89 (71 - 102)
Bottom	77 (54 - 95)	81 (53 - 94)	85 (68 - 99)	83 (68 - 94)	81 (57 - 94)	79 (60 - 97)	83 (57 - 112)
pH	8.0 (7.8 - 8.2)	8.0 (7.8 - 8.2)	8.0 (7.8 - 8.2)	8.0 (7.8 - 8.2)	8.0 (7.7 - 8.2)	8.0 (7.7 - 8.1)	8.0 (7.7 - 8.2)
Secchi Disc Depth (m)	3.3 (2.3 - 4.6)	3.4 (1.9 - 5.9)	3.3 (1.9 - 6.0)	3.4 (2.1 - 5.3)	3.4 (1.9 - 5.9)	3.7 (2.5 - 6.3)	2.8 (1.7 - 3.8)
Turbidity (NTU)	2.9 (0.3 - 13.1)	2.6 (0.4 - 7.1)	2.8 (0.6 - 10.3)	2.3 (0.5 - 6.7)	2.9 (0.7 - 8.2)	2.1 (0.7 - 4.2)	2.1 (0.2 - 5.4)
Suspended Solids (mg/L)	2.8 (1.1 - 6.3)	3.7 (1.3 - 6.3)	4.2 (1.5 - 7.7)	3.4 (0.9 - 8.4)	3.5 (1.3 - 7.5)	3.8 (0.8 - 7.5)	3.8 (1.4 - 8.8)
5-day Biochemical Oxygen Demand (mg/L)	0.7 (<0.1 - 2.5)	0.4 (<0.1 - 1.0)	0.3 (<0.1 - 1.0)	0.3 (<0.1 - 0.9)	0.4 (<0.1 - 0.8)	0.6 (<0.1 - 2.2)	0.9 (0.1 - 3.2)
Ammonia Nitrogen (mg/L)	0.019 (<0.005 - 0.046)	0.018 (<0.005 - 0.039)	0.015 (<0.005 - 0.029)	0.015 (<0.005 - 0.026)	0.017 (<0.005 - 0.038)	0.020 (<0.005 - 0.037)	0.023 (<0.005 - 0.062)
Unionised Ammonia (mg/L)	<0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)	0.005 (<0.002 - 0.013)	0.006 (<0.002 - 0.019)	0.003 (<0.002 - 0.009)	0.004 (<0.002 - 0.010)	0.004 (<0.002 - 0.013)	0.005 (<0.002 - 0.027)	0.003 (<0.002 - 0.011)
Nitrate Nitrogen (mg/L)	0.060 (0.010 - 0.190)	0.066 (0.010 - 0.307)	0.056 (0.008 - 0.230)	0.055 (0.009 - 0.203)	0.047 (0.011 - 0.190)	0.039 (0.010 - 0.143)	0.037 (0.010 - 0.121)
Total Inorganic Nitrogen (mg/L)	0.08 (0.02 - 0.22)	0.09 (0.02 - 0.34)	0.07 (0.02 - 0.24)	0.07 (0.02 - 0.23)	0.07 (0.02 - 0.22)	0.06 (0.02 - 0.17)	0.06 (0.03 - 0.16)
Total Kjeldahl Nitrogen (mg/L)	0.15 (<0.05 - 0.27)	0.18 (0.07 - 0.31)	0.16 (0.08 - 0.27)	0.15 (0.07 - 0.21)	0.15 (0.07 - 0.24)	0.15 (0.07 - 0.24)	0.16 (0.08 - 0.26)
Total Nitrogen (mg/L)	0.21 (0.08 - 0.38)	0.25 (0.08 - 0.64)	0.22 (0.10 - 0.35)	0.21 (0.08 - 0.35)	0.20 (0.08 - 0.33)	0.20 (0.08 - 0.34)	0.20 (0.09 - 0.38)
Orthophosphate Phosphorus (mg/L)	0.006 (<0.002 - 0.018)	0.006 (<0.002 - 0.023)	0.006 (<0.002 - 0.023)	0.006 (<0.002 - 0.020)	0.006 (<0.002 - 0.019)	0.005 (<0.002 - 0.009)	0.004 (<0.002 - 0.013)
Total Phosphorus (mg/L)	0.07 (0.03 - 0.18)	0.07 (0.03 - 0.13)	0.08 (0.03 - 0.23)	0.08 (0.03 - 0.21)	0.10 (0.03 - 0.40)	0.07 (0.03 - 0.23)	0.07 (0.03 - 0.14)
Silica (as SiO_2) (mg/L)	0.65 (0.22 - 1.90)	0.69 (0.14 - 2.70)	0.60 (0.10 - 1.90)	0.59 (0.16 - 1.93)	0.61 (0.22 - 1.90)	0.51 (0.23 - 1.13)	0.45 (0.06 - 0.83)
Chlorophyll-*a* (µg/L)	1.4 (0.5 - 3.0)	2.1 (0.3 - 10.3)	1.6 (0.5 - 7.5)	1.5 (0.5 - 4.5)	1.5 (0.3 - 5.9)	1.6 (0.5 - 6.9)	2.4 (0.9 - 6.0)
E.coli (count/100mL)	1 (<1 - 2)	1 (<1 - 1)	1 (<1 - 1)	1 (<1 - 2)	1 (<1 - 8)	1 (<1 - 4)	1 (<1 - 2)
Faecal Coliforms (count/100mL)	1 (<1 - 4)	2 (<1 - 5)	1 (<1 - 3)	2 (<1 - 3)	2 (<1 - 8)	2 (<1 - 12)	3 (<1 - 20)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Port Shelter WCZ in 2024

Parameter		Inner Port Shelter PM1	PM2	PM3	PM4	Hebe Haven PM6
Number of samples		**12**	**12**	**12**	**12**	**12**
Temperature (°C)		25.4 (17.3 - 32.9)	25.3 (17.7 - 32.5)	24.7 (17.8 - 30.6)	25.1 (17.6 - 31.6)	24.7 (18.0 - 29.7)
Salinity		31.2 (25.9 - 33.9)	31.4 (26.4 - 34.1)	31.6 (27.4 - 34.1)	31.4 (26.4 - 34.0)	32.0 (29.7 - 33.9)
Dissolved Oxygen (mg/L)		6.4 (4.2 - 8.5)	6.1 (4.5 - 8.1)	6.0 (4.3 - 7.8)	6.2 (4.4 - 7.2)	6.1 (4.2 - 7.2)
	Bottom	6.5 (3.8 - 9.1)	6.1 (3.7 - 9.0)	5.7 (3.4 - 7.7)	6.1 (4.4 - 7.4)	5.7 (3.4 - 7.2)
Dissolved Oxygen (% Saturation)		92 (65 - 120)	88 (66 - 113)	86 (66 - 107)	89 (69 - 111)	88 (65 - 108)
	Bottom	93 (59 - 139)	87 (58 - 125)	80 (52 - 107)	87 (69 - 110)	81 (52 - 96)
pH		8.1 (7.8 - 8.3)	8.0 (7.7 - 8.3)	8.0 (7.7 - 8.3)	8.1 (7.8 - 8.2)	8.0 (7.7 - 8.1)
Secchi Disc Depth (m)		2.6 (1.7 - 3.6)	2.5 (1.4 - 4.0)	3.0 (2.0 - 4.3)	2.8 (1.3 - 4.5)	2.8 (1.5 - 5.3)
Turbidity (NTU)		1.6 (0.2 - 3.6)	1.3 (0.2 - 3.6)	1.3 (0.2 - 3.5)	1.7 (0.4 - 3.7)	1.4 (0.2 - 2.9)
Suspended Solids (mg/L)		4.0 (1.7 - 5.3)	3.9 (1.2 - 7.0)	3.4 (1.7 - 7.3)	3.9 (1.8 - 6.3)	3.3 (1.8 - 5.4)
5-day Biochemical Oxygen Demand (mg/L)		1.1 (0.3 - 2.6)	0.9 (0.3 - 2.2)	0.9 (0.2 - 2.4)	0.8 (0.2 - 2.4)	0.6 (0.1 - 1.1)
Ammonia Nitrogen (mg/L)		0.019 (0.005 - 0.059)	0.019 (0.007 - 0.053)	0.018 (0.007 - 0.042)	0.020 (0.006 - 0.054)	0.027 (0.006 - 0.075)
Unionised Ammonia (mg/L)		0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)		0.002 (<0.002 - 0.002)	0.004 (<0.002 - 0.014)	0.003 (<0.002 - 0.014)	0.002 (<0.002 - 0.002)	0.004 (<0.002 - 0.017)
Nitrate Nitrogen (mg/L)		0.025 (0.008 - 0.058)	0.031 (0.008 - 0.060)	0.026 (0.010 - 0.045)	0.032 (0.009 - 0.098)	0.038 (0.009 - 0.068)
Total Inorganic Nitrogen (mg/L)		0.05 (0.02 - 0.12)	0.05 (0.02 - 0.12)	0.05 (0.02 - 0.09)	0.05 (0.02 - 0.13)	0.07 (0.02 - 0.14)
Total Kjeldahl Nitrogen (mg/L)		0.21 (0.07 - 0.38)	0.24 (0.05 - 0.54)	0.21 (0.12 - 0.42)	0.20 (0.10 - 0.37)	0.21 (0.13 - 0.40)
Total Nitrogen (mg/L)		0.23 (0.08 - 0.42)	0.28 (0.06 - 0.60)	0.23 (0.14 - 0.46)	0.23 (0.12 - 0.43)	0.25 (0.15 - 0.48)
Orthophosphate Phosphorus (mg/L)		0.002 (<0.002 - 0.003)	0.003 (<0.002 - 0.005)	0.003 (<0.002 - 0.004)	0.003 (<0.002 - 0.009)	0.006 (<0.002 - 0.026)
Total Phosphorus (mg/L)		0.07 (0.02 - 0.18)	0.07 (0.03 - 0.16)	0.07 (0.03 - 0.14)	0.08 (0.03 - 0.18)	0.07 (0.03 - 0.15)
Silica (as SiO_2) (mg/L)		0.55 (0.10 - 1.12)	0.57 (0.13 - 1.12)	0.52 (0.17 - 0.95)	0.90 (0.10 - 3.67)	0.66 (0.28 - 1.31)
Chlorophyll-*a* (μg/L)		3.5 (0.8 - 8.0)	3.9 (0.8 - 8.5)	3.7 (0.6 - 10.7)	2.7 (0.5 - 5.1)	2.7 (0.7 - 8.1)
E.coli (count/100mL)		2 (<1 - 42)	8 (1 - 120)	2 (<1 - 25)	2 (<1 - 98)	3 (<1 - 36)
Faecal Coliforms (count/100mL)		5 (1 - 290)	39 (3 - 410)	5 (<1 - 110)	5 (<1 - 490)	7 (1 - 120)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Port Shelter WCZ in 2024 (continued)

Parameter		Outer Port Shelter PM7	Outer Port Shelter PM8	Rocky Harbour PM9	Bluff Island PM11
Number of samples		12	12	12	12
Temperature (°C)		24.5 (18.5 - 29.2)	24.3 (18.2 - 29.2)	24.4 (18.1 - 28.9)	24.3 (18.7 - 28.9)
Salinity		32.5 (30.4 - 34.3)	32.6 (30.3 - 34.1)	32.6 (31.0 - 34.3)	32.6 (30.5 - 34.3)
Dissolved Oxygen (mg/L)		5.9 (4.8 - 7.1)	5.8 (3.3 - 7.4)	5.9 (4.2 - 7.5)	5.8 (4.3 - 7.5)
	Bottom	5.6 (3.8 - 7.6)	5.5 (2.6 - 7.3)	5.4 (3.1 - 7.4)	5.5 (3.8 - 7.3)
Dissolved Oxygen (% Saturation)		84 (70 - 94)	83 (48 - 98)	85 (65 - 109)	83 (65 - 98)
	Bottom	79 (57 - 99)	78 (39 - 101)	76 (47 - 98)	78 (59 - 95)
pH		8.0 (7.7 - 8.2)	8.0 (7.6 - 8.2)	8.0 (7.6 - 8.2)	8.0 (7.7 - 8.2)
Secchi Disc Depth (m)		3.5 (1.6 - 6.8)	3.9 (1.9 - 7.6)	3.5 (2.1 - 5.8)	3.8 (1.8 - 6.5)
Turbidity (NTU)		1.3 (0.1 - 2.8)	2.7 (0.2 - 14.1)	1.4 (0.3 - 3.5)	2.2 (0.2 - 9.8)
Suspended Solids (mg/L)		2.8 (1.4 - 4.5)	3.4 (0.9 - 5.9)	3.7 (2.0 - 5.8)	3.2 (1.1 - 5.3)
5-day Biochemical Oxygen Demand (mg/L)		0.5 (<0.1 - 0.9)	0.4 (0.1 - 0.8)	0.5 (<0.1 - 1.1)	0.5 (<0.1 - 1.2)
Ammonia Nitrogen (mg/L)		0.017 (0.006 - 0.041)	0.014 (<0.005 - 0.042)	0.014 (0.007 - 0.035)	0.015 (0.006 - 0.045)
Unionised Ammonia (mg/L)		<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)
Nitrite Nitrogen (mg/L)		0.003 (<0.002 - 0.008)	0.007 (<0.002 - 0.029)	0.003 (<0.002 - 0.007)	0.004 (<0.002 - 0.014)
Nitrate Nitrogen (mg/L)		0.034 (0.010 - 0.124)	0.042 (0.007 - 0.170)	0.037 (0.008 - 0.193)	0.040 (0.007 - 0.200)
Total Inorganic Nitrogen (mg/L)		0.05 (0.02 - 0.16)	0.06 (0.01 - 0.19)	0.05 (0.02 - 0.21)	0.06 (0.02 - 0.22)
Total Kjeldahl Nitrogen (mg/L)		0.18 (0.12 - 0.28)	0.17 (0.10 - 0.34)	0.19 (0.10 - 0.45)	0.17 (0.07 - 0.51)
Total Nitrogen (mg/L)		0.22 (0.15 - 0.31)	0.21 (0.13 - 0.40)	0.23 (0.12 - 0.46)	0.21 (0.08 - 0.53)
Orthophosphate Phosphorus (mg/L)		0.003 (<0.002 - 0.009)	0.005 (<0.002 - 0.014)	0.004 (<0.002 - 0.018)	0.005 (<0.002 - 0.019)
Total Phosphorus (mg/L)		0.08 (0.03 - 0.16)	0.07 (0.03 - 0.15)	0.08 (0.03 - 0.17)	0.07 (0.03 - 0.14)
Silica (as SiO_2) (mg/L)		0.51 (0.24 - 1.17)	0.58 (0.20 - 1.47)	0.53 (0.23 - 1.87)	0.56 (0.20 - 1.83)
Chlorophyll-*a* (µg/L)		2.0 (0.5 - 4.7)	1.6 (0.6 - 3.4)	2.0 (0.8 - 4.6)	1.7 (0.6 - 5.1)
E.coli (count/100mL)		1 (<1 - 2)	1 (<1 - 7)	1 (<1 - 6)	1 (<1 - 6)
Faecal Coliforms (count/100mL)		2 (<1 - 10)	1 (<1 - 11)	1 (<1 - 9)	1 (<1 - 7)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Tolo Harbour and Channel WCZ in 2024

Parameter		Harbour Subzone			Buffer Subzone		Channel Subzone	
		TM2	TM3	TM4	TM5	TM6	TM7	TM8
Number of samples		12	12	12	12	12	12	12
Temperature (°C)		25.5 (20.0 - 31.6)	25.4 (20.1 - 31.2)	25.1 (19.9 - 30.9)	25.4 (20.2 - 31.2)	24.7 (19.3 - 30.7)	24.8 (19.2 - 30.4)	24.5 (19.0 - 29.9)
Salinity		30.1 (26.7 - 32.7)	30.8 (25.9 - 33.3)	31.1 (27.3 - 33.4)	30.5 (25.5 - 33.3)	31.1 (26.5 - 33.9)	31.6 (27.0 - 33.8)	32.0 (27.6 - 34.1)
Dissolved Oxygen (mg/L)		6.7 (4.6 - 8.6)	6.6 (5.6 - 8.9)	6.3 (5.4 - 7.7)	6.4 (5.0 - 8.1)	6.1 (5.2 - 7.3)	6.1 (4.6 - 7.6)	5.6 (4.7 - 6.7)
	Bottom	6.8 (4.8 - 8.8)	6.0 (4.6 - 8.6)	6.0 (5.1 - 8.0)	6.6 (5.3 - 8.5)	5.4 (4.4 - 6.7)	5.7 (4.3 - 7.2)	5.1 (3.5 - 6.8)
Dissolved Oxygen (% Saturation)		97 (67 - 115)	95 (82 - 119)	91 (81 - 103)	93 (75 - 108)	87 (77 - 97)	87 (69 - 101)	81 (69 - 89)
	Bottom	97 (71 - 117)	86 (67 - 115)	85 (74 - 105)	96 (79 - 117)	77 (64 - 89)	80 (64 - 95)	72 (51 - 90)
pH		8.0 (7.6 - 8.3)	8.1 (7.8 - 8.4)	8.0 (7.7 - 8.4)	8.1 (7.6 - 8.3)	8.0 (7.6 - 8.3)	8.0 (7.8 - 8.3)	8.0 (7.7 - 8.2)
Secchi Disc Depth (m)		2.1 (1.4 - 3.2)	2.4 (1.8 - 2.9)	2.3 (1.7 - 3.9)	2.8 (1.8 - 3.8)	2.8 (2.0 - 4.5)	2.9 (1.6 - 4.9)	3.0 (1.8 - 5.0)
Turbidity (NTU)		2.5 (0.3 - 10.9)	1.9 (0.4 - 8.5)	1.3 (0.3 - 2.8)	1.3 (0.2 - 2.9)	1.5 (0.5 - 2.6)	1.2 (0.2 - 2.3)	1.9 (0.3 - 3.2)
Suspended Solids (mg/L)		3.5 (2.4 - 5.2)	4.0 (2.7 - 6.3)	3.3 (1.3 - 4.5)	3.4 (1.7 - 7.5)	3.3 (1.1 - 6.5)	3.3 (2.0 - 4.3)	4.1 (2.4 - 6.2)
5-day Biochemical Oxygen Demand (mg/L)		1.6 (0.8 - 2.6)	1.6 (0.6 - 2.5)	1.5 (0.8 - 3.0)	1.1 (<0.1 - 2.3)	1.2 (0.6 - 2.9)	1.2 (0.4 - 2.0)	0.8 (0.3 - 1.5)
Ammonia Nitrogen (mg/L)		0.030 (0.010 - 0.068)	0.021 (0.006 - 0.040)	0.017 (<0.005 - 0.045)	0.017 (<0.005 - 0.044)	0.018 (<0.005 - 0.042)	0.015 (<0.005 - 0.027)	0.021 (0.005 - 0.079)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.009)	0.001 (<0.001 - 0.004)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.004)	<0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.002)
Nitrite Nitrogen (mg/L)		0.011 (<0.002 - 0.070)	0.004 (<0.002 - 0.013)	0.007 (<0.002 - 0.041)	0.003 (<0.002 - 0.009)	0.007 (<0.002 - 0.029)	0.005 (<0.002 - 0.023)	0.005 (<0.002 - 0.020)
Nitrate Nitrogen (mg/L)		0.073 (0.008 - 0.260)	0.036 (0.006 - 0.089)	0.032 (0.007 - 0.076)	0.027 (0.006 - 0.060)	0.031 (0.005 - 0.107)	0.033 (0.007 - 0.143)	0.032 (0.007 - 0.133)
Total Inorganic Nitrogen (mg/L)		0.11 (0.02 - 0.35)	0.06 (0.02 - 0.12)	0.06 (0.02 - 0.11)	0.05 (0.02 - 0.10)	0.06 (0.02 - 0.14)	0.05 (0.01 - 0.17)	0.06 (0.02 - 0.17)
Total Kjeldahl Nitrogen (mg/L)		0.30 (0.21 - 0.40)	0.27 (0.18 - 0.43)	0.24 (0.17 - 0.33)	0.20 (0.10 - 0.31)	0.24 (0.15 - 0.39)	0.21 (0.12 - 0.31)	0.19 (0.10 - 0.30)
Total Nitrogen (mg/L)		0.38 (0.22 - 0.67)	0.31 (0.19 - 0.44)	0.28 (0.18 - 0.37)	0.23 (0.11 - 0.32)	0.28 (0.16 - 0.40)	0.25 (0.13 - 0.38)	0.23 (0.11 - 0.36)
Orthophosphate Phosphorus (mg/L)		0.004 (<0.002 - 0.010)	0.003 (<0.002 - 0.011)	0.003 (<0.002 - 0.010)	0.005 (<0.002 - 0.012)	0.004 (<0.002 - 0.012)	0.004 (<0.002 - 0.010)	0.005 (<0.002 - 0.017)
Total Phosphorus (mg/L)		0.07 (0.03 - 0.16)	0.07 (0.03 - 0.15)	0.07 (0.03 - 0.15)	0.07 (0.03 - 0.19)	0.07 (0.03 - 0.15)	0.07 (0.03 - 0.15)	0.07 (0.03 - 0.16)
Silica (as SiO_2) (mg/L)		1.02 (0.08 - 2.10)	0.52 (0.08 - 1.30)	0.55 (0.09 - 1.50)	0.46 (0.18 - 1.30)	0.50 (0.11 - 1.23)	0.42 (0.09 - 1.06)	0.53 (0.18 - 1.19)
Chlorophyll-*a* (µg/L)		8.8 (3.9 - 17.0)	6.0 (3.3 - 11.9)	5.5 (2.7 - 11.1)	4.3 (1.7 - 10.4)	4.0 (2.4 - 9.8)	3.8 (1.3 - 7.8)	2.4 (1.1 - 5.4)
E.coli (count/100mL)		29 (<1 - 270)	11 (<1 - 370)	5 (<1 - 29)	2 (<1 - 35)	4 (<1 - 86)	2 (<1 - 14)	2 (<1 - 15)
Faecal Coliforms (count/100mL)		210 (2 - 4100)	56 (3 - 2200)	25 (1 - 120)	6 (1 - 110)	7 (<1 - 690)	4 (<1 - 72)	2 (<1 - 56)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Southern WCZ in 2024

Parameter		Hong Kong Island (South) SM1	Hong Kong Island (South) SM2	Hong Kong Island (South) SM19	East Lamma Channel SM3	East Lamma Channel SM4
Number of samples		**12**	**12**	**12**	**12**	**12**
Temperature (°C)		24.0 (19.2 - 28.9)	24.2 (19.4 - 28.8)	24.0 (19.2 - 29.0)	24.2 (19.8 - 28.8)	24.4 (19.8 - 28.8)
Salinity		31.8 (23.4 - 34.3)	31.3 (20.7 - 34.4)	31.5 (19.1 - 34.4)	31.8 (24.7 - 34.2)	30.9 (21.7 - 34.3)
Dissolved Oxygen (mg/L)		5.9 (4.6 - 7.1)	5.8 (4.6 - 7.3)	5.7 (4.2 - 7.3)	6.2 (5.0 - 7.7)	5.5 (3.0 - 7.8)
	Bottom	5.5 (3.1 - 7.3)	5.5 (3.0 - 7.3)	5.4 (3.2 - 7.3)	5.7 (2.5 - 8.0)	5.2 (2.4 - 7.4)
Dissolved Oxygen (% Saturation)		83 (67 - 100)	82 (66 - 98)	81 (60 - 98)	89 (72 - 103)	78 (43 - 114)
	Bottom	77 (44 - 97)	77 (42 - 102)	76 (46 - 97)	81 (36 - 107)	73 (34 - 108)
pH		8.1 (7.9 - 8.2)	8.0 (7.8 - 8.2)	8.1 (7.8 - 8.2)	7.8 (7.5 - 8.2)	7.9 (7.7 - 8.1)
Secchi Disc Depth (m)		3.0 (2.0 - 4.6)	2.9 (1.9 - 6.7)	3.3 (1.9 - 9.6)	2.9 (1.7 - 7.6)	2.8 (1.9 - 4.7)
Turbidity (NTU)		2.3 (0.2 - 5.5)	2.4 (0.3 - 6.5)	2.8 (0.2 - 9.3)	3.5 (0.1 - 7.7)	2.4 (0.2 - 5.9)
Suspended Solids (mg/L)		3.5 (1.1 - 6.0)	3.9 (1.6 - 7.2)	3.5 (1.1 - 7.4)	4.2 (1.3 - 8.0)	3.7 (1.7 - 6.4)
5-day Biochemical Oxygen Demand (mg/L)		0.7 (<0.1 - 1.8)	0.7 (<0.1 - 2.3)	0.5 (0.2 - 0.9)	0.9 (0.2 - 2.6)	0.8 (<0.1 - 2.4)
Ammonia Nitrogen (mg/L)		0.026 (0.006 - 0.066)	0.038 (0.011 - 0.082)	0.023 (0.007 - 0.067)	0.045 (0.009 - 0.095)	0.071 (0.011 - 0.140)
Unionised Ammonia (mg/L)		0.001 (<0.001 - 0.004)	0.002 (<0.001 - 0.005)	0.001 (<0.001 - 0.004)	0.001 (<0.001 - 0.004)	0.003 (<0.001 - 0.005)
Nitrite Nitrogen (mg/L)		0.015 (<0.002 - 0.060)	0.016 (<0.002 - 0.069)	0.014 (<0.002 - 0.055)	0.015 (<0.002 - 0.048)	0.020 (<0.002 - 0.085)
Nitrate Nitrogen (mg/L)		0.104 (0.008 - 0.370)	0.126 (0.011 - 0.427)	0.102 (0.007 - 0.330)	0.123 (0.021 - 0.361)	0.156 (0.021 - 0.407)
Total Inorganic Nitrogen (mg/L)		0.15 (0.03 - 0.40)	0.18 (0.05 - 0.47)	0.14 (0.03 - 0.35)	0.18 (0.04 - 0.45)	0.25 (0.06 - 0.46)
Total Kjeldahl Nitrogen (mg/L)		0.15 (0.05 - 0.32)	0.16 (0.07 - 0.28)	0.13 (0.05 - 0.20)	0.19 (0.10 - 0.27)	0.21 (0.09 - 0.36)
Total Nitrogen (mg/L)		0.26 (0.08 - 0.71)	0.30 (0.11 - 0.72)	0.25 (0.07 - 0.52)	0.33 (0.13 - 0.61)	0.39 (0.19 - 0.79)
Orthophosphate Phosphorus (mg/L)		0.008 (<0.002 - 0.025)	0.009 (0.002 - 0.027)	0.009 (<0.002 - 0.027)	0.008 (<0.002 - 0.013)	0.011 (<0.002 - 0.026)
Total Phosphorus (mg/L)		0.07 (0.03 - 0.16)	0.07 (0.03 - 0.16)	0.08 (0.03 - 0.16)	0.09 (0.03 - 0.19)	0.08 (0.04 - 0.17)
Silica (as SiO_2) (mg/L)		0.93 (0.26 - 1.90)	0.94 (0.26 - 1.97)	0.95 (0.24 - 1.87)	0.83 (0.19 - 2.21)	1.02 (0.26 - 2.23)
Chlorophyll-*a* (µg/L)		2.5 (0.6 - 9.5)	2.8 (0.4 - 15.0)	1.8 (0.3 - 4.7)	4.5 (0.6 - 11.5)	3.8 (0.4 - 26.3)
E.coli (count/100mL)		2 (<1 - 15)	3 (<1 - 25)	1 (<1 - 3)	10 (<1 - 1300)	8 (<1 - 92)
Faecal Coliforms (count/100mL)		5 (<1 - 54)	7 (1 - 140)	2 (<1 - 7)	24 (1 - 1600)	17 (2 - 160)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Southern WCZ in 2024 (continued)

Parameter		West Lamma Channel SM5	SM6	SM7	SM9	SM18
Number of samples		12	12	12	12	12
Temperature (°C)		24.6 (19.6 - 29.6)	24.4 (19.5 - 29.2)	24.6 (19.8 - 29.2)	24.6 (19.7 - 28.4)	24.2 (19.5 - 29.0)
Salinity		30.6 (15.4 - 34.4)	30.5 (16.7 - 34.4)	29.6 (17.2 - 34.1)	29.3 (17.3 - 33.5)	31.3 (19.5 - 34.4)
Dissolved Oxygen (mg/L)		6.4 (4.6 - 8.5)	6.0 (4.6 - 7.9)	6.4 (4.4 - 9.3)	6.0 (4.8 - 8.2)	6.1 (4.6 - 7.7)
	Bottom	6.2 (4.1 - 10.3)	5.6 (3.2 - 7.5)	6.3 (4.0 - 11.3)	5.4 (2.9 - 8.1)	5.6 (3.3 - 7.3)
Dissolved Oxygen (% Saturation)		91 (68 - 123)	85 (67 - 109)	91 (64 - 134)	85 (71 - 114)	86 (66 - 111)
	Bottom	88 (63 - 148)	78 (46 - 103)	90 (61 - 163)	75 (41 - 108)	80 (47 - 96)
pH		8.1 (7.9 - 8.2)	8.0 (7.9 - 8.2)	8.0 (7.8 - 8.2)	7.8 (7.5 - 8.1)	8.1 (7.8 - 8.2)
Secchi Disc Depth (m)		2.4 (1.7 - 5.7)	2.6 (1.7 - 7.3)	2.4 (1.5 - 4.8)	2.3 (1.5 - 3.8)	3.4 (2.0 - 8.2)
Turbidity (NTU)		3.2 (0.5 - 7.3)	4.1 (0.2 - 8.2)	4.3 (0.2 - 11.6)	4.4 (0.4 - 6.8)	3.4 (0.2 - 8.8)
Suspended Solids (mg/L)		4.6 (1.6 - 9.3)	4.3 (1.1 - 8.8)	4.8 (1.1 - 10.0)	4.2 (1.7 - 7.2)	3.9 (0.8 - 12.9)
5-day Biochemical Oxygen Demand (mg/L)		0.9 (<0.1 - 2.2)	0.7 (<0.1 - 1.7)	0.8 (<0.1 - 2.8)	1.1 (<0.1 - 2.4)	0.6 (<0.1 - 1.0)
Ammonia Nitrogen (mg/L)		0.038 (0.009 - 0.127)	0.051 (<0.005 - 0.132)	0.078 (0.009 - 0.143)	0.087 (0.027 - 0.170)	0.029 (0.006 - 0.081)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.008)	0.003 (<0.001 - 0.008)	0.004 (<0.001 - 0.008)	0.003 (<0.001 - 0.006)	0.002 (<0.001 - 0.005)
Nitrite Nitrogen (mg/L)		0.019 (<0.002 - 0.069)	0.020 (<0.002 - 0.073)	0.028 (<0.002 - 0.099)	0.028 (0.003 - 0.073)	0.016 (<0.002 - 0.076)
Nitrate Nitrogen (mg/L)		0.149 (0.007 - 0.553)	0.153 (0.009 - 0.453)	0.222 (0.045 - 0.547)	0.255 (0.043 - 0.743)	0.118 (0.009 - 0.390)
Total Inorganic Nitrogen (mg/L)		0.21 (0.03 - 0.61)	0.23 (0.04 - 0.51)	0.33 (0.09 - 0.66)	0.37 (0.13 - 0.87)	0.16 (0.03 - 0.42)
Total Kjeldahl Nitrogen (mg/L)		0.16 (0.05 - 0.35)	0.19 (0.07 - 0.31)	0.23 (0.09 - 0.38)	0.27 (0.19 - 0.35)	0.13 (0.06 - 0.23)
Total Nitrogen (mg/L)		0.33 (0.13 - 0.92)	0.36 (0.13 - 0.77)	0.48 (0.16 - 0.95)	0.55 (0.25 - 1.03)	0.27 (0.07 - 0.62)
Orthophosphate Phosphorus (mg/L)		0.010 (0.003 - 0.027)	0.011 (<0.002 - 0.026)	0.015 (0.004 - 0.029)	0.012 (0.002 - 0.020)	0.009 (0.002 - 0.027)
Total Phosphorus (mg/L)		0.07 (0.03 - 0.15)	0.08 (0.03 - 0.14)	0.08 (0.04 - 0.20)	0.09 (0.04 - 0.18)	0.08 (0.03 - 0.15)
Silica (as SiO_2) (mg/L)		1.14 (0.24 - 2.40)	1.19 (0.22 - 2.63)	1.41 (0.21 - 3.00)	1.34 (0.18 - 4.70)	1.17 (0.20 - 2.53)
Chlorophyll-*a* (µg/L)		2.7 (0.7 - 12.5)	2.7 (0.7 - 10.6)	5.2 (0.8 - 32.3)	8.0 (0.7 - 45.3)	1.8 (0.4 - 7.5)
E.coli (count/100mL)		2 (<1 - 45)	2 (<1 - 25)	5 (1 - 110)	100 (14 - 950)	1 (<1 - 4)
Faecal Coliforms (count/100mL)		2 (1 - 89)	4 (<1 - 54)	11 (2 - 170)	190 (31 - 1400)	2 (1 - 9)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Southern WCZ in 2024 (continued)

Parameter		Lantau Island (East) SM10	Lantau Island (East) SM11	Lantau Island (South) SM12	Lantau Island (South) SM13	Lantau Island (South) SM17	Soko Islands SM20
Number of samples		12	12	12	12	12	12
Temperature (°C)		24.7 (18.8 - 29.9)	24.8 (18.0 - 29.8)	24.7 (18.0 - 29.6)	24.9 (19.6 - 29.7)	24.6 (19.5 - 29.3)	24.8 (19.8 - 29.3)
Salinity		29.1 (19.0 - 33.4)	29.1 (19.6 - 33.6)	28.4 (14.8 - 33.6)	28.7 (15.2 - 34.4)	29.4 (18.3 - 34.2)	28.8 (17.6 - 34.2)
Dissolved Oxygen (mg/L)		6.6 (5.1 - 8.9)	6.8 (4.2 - 10.0)	6.7 (4.5 - 9.5)	7.0 (4.8 - 9.1)	6.1 (4.4 - 8.0)	6.7 (4.9 - 8.2)
	Bottom	6.6 (4.4 - 10.0)	6.8 (3.0 - 12.2)	6.3 (3.5 - 9.2)	6.5 (3.5 - 9.2)	5.5 (3.5 - 7.9)	6.5 (3.6 - 9.9)
Dissolved Oxygen (% Saturation)		94 (78 - 128)	96 (61 - 144)	94 (68 - 127)	99 (70 - 126)	86 (63 - 107)	95 (71 - 117)
	Bottom	93 (66 - 144)	95 (43 - 175)	89 (51 - 123)	92 (51 - 123)	78 (52 - 105)	91 (51 - 140)
pH		7.9 (7.6 - 8.3)	8.0 (7.7 - 8.3)	8.0 (7.7 - 8.3)	8.1 (7.7 - 8.4)	8.0 (7.8 - 8.2)	8.0 (7.8 - 8.3)
Secchi Disc Depth (m)		2.0 (1.4 - 3.0)	2.2 (1.5 - 3.7)	2.1 (1.6 - 3.1)	2.1 (1.6 - 3.0)	2.4 (1.6 - 4.0)	2.0 (1.4 - 3.4)
Turbidity (NTU)		4.2 (0.4 - 11.4)	3.5 (0.4 - 9.3)	3.5 (0.6 - 14.0)	3.3 (0.6 - 9.5)	3.9 (0.4 - 11.2)	3.7 (0.5 - 12.9)
Suspended Solids (mg/L)		4.7 (1.6 - 9.2)	4.7 (2.3 - 10.6)	4.5 (2.0 - 9.6)	4.8 (2.7 - 10.0)	4.4 (1.0 - 7.1)	5.0 (1.9 - 14.2)
5-day Biochemical Oxygen Demand (mg/L)		0.8 (<0.1 - 1.9)	1.0 (<0.1 - 2.5)	0.8 (<0.1 - 1.9)	0.8 (<0.1 - 2.0)	0.5 (<0.1 - 1.6)	0.6 (<0.1 - 1.9)
Ammonia Nitrogen (mg/L)		0.081 (0.017 - 0.170)	0.070 (0.022 - 0.157)	0.050 (<0.005 - 0.153)	0.038 (0.005 - 0.137)	0.029 (0.005 - 0.086)	0.031 (0.010 - 0.110)
Unionised Ammonia (mg/L)		0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.007)	0.002 (<0.001 - 0.005)	0.002 (<0.001 - 0.005)	0.001 (<0.001 - 0.005)	0.002 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)		0.035 (0.004 - 0.105)	0.037 (0.004 - 0.120)	0.039 (0.003 - 0.120)	0.032 (<0.002 - 0.095)	0.026 (<0.002 - 0.086)	0.028 (<0.002 - 0.096)
Nitrate Nitrogen (mg/L)		0.226 (0.085 - 0.500)	0.235 (0.036 - 0.487)	0.261 (0.065 - 0.713)	0.239 (0.018 - 0.763)	0.238 (0.020 - 0.737)	0.254 (0.017 - 0.743)
Total Inorganic Nitrogen (mg/L)		0.34 (0.18 - 0.61)	0.34 (0.07 - 0.63)	0.35 (0.10 - 0.79)	0.31 (0.04 - 0.86)	0.29 (0.04 - 0.78)	0.31 (0.04 - 0.85)
Total Kjeldahl Nitrogen (mg/L)		0.26 (0.16 - 0.37)	0.25 (0.15 - 0.35)	0.22 (0.11 - 0.32)	0.22 (0.11 - 0.37)	0.17 (0.09 - 0.29)	0.19 (0.11 - 0.35)
Total Nitrogen (mg/L)		0.52 (0.33 - 0.85)	0.53 (0.20 - 0.84)	0.52 (0.20 - 1.03)	0.49 (0.13 - 1.20)	0.44 (0.13 - 1.05)	0.47 (0.13 - 1.07)
Orthophosphate Phosphorus (mg/L)		0.016 (<0.002 - 0.029)	0.013 (<0.002 - 0.028)	0.014 (<0.002 - 0.026)	0.010 (0.002 - 0.024)	0.009 (<0.002 - 0.023)	0.010 (<0.002 - 0.024)
Total Phosphorus (mg/L)		0.10 (0.04 - 0.22)	0.09 (0.03 - 0.21)	0.08 (0.03 - 0.20)	0.07 (0.03 - 0.17)	0.07 (0.03 - 0.17)	0.08 (0.03 - 0.18)
Silica (as SiO_2) (mg/L)		1.24 (0.14 - 3.05)	1.34 (0.13 - 3.70)	1.57 (0.05 - 4.27)	1.43 (0.09 - 4.43)	1.66 (0.31 - 4.13)	1.53 (0.19 - 4.63)
Chlorophyll-*a* (µg/L)		8.3 (0.9 - 25.5)	8.9 (0.5 - 27.3)	8.1 (0.9 - 21.0)	8.7 (0.8 - 28.0)	3.9 (1.2 - 14.7)	5.7 (1.1 - 24.3)
E.coli (count/100mL)		12 (<1 - 350)	4 (1 - 33)	10 (1 - 440)	4 (<1 - 22)	2 (<1 - 12)	2 (<1 - 21)
Faecal Coliforms (count/100mL)		23 (1 - 750)	9 (2 - 210)	20 (1 - 680)	8 (<1 - 56)	4 (1 - 47)	4 (<1 - 71)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Victoria Harbour WCZ in 2024

Parameter		Victoria Harbour (East) VM1	Victoria Harbour (East) VM2	Victoria Harbour (Central) VM4	Victoria Harbour (Central) VM5	Victoria Harbour (Central) VM6
Number of samples		**12**	**12**	**12**	**12**	**12**
Temperature (°C)		24.0 (18.9 - 29.2)	24.1 (19.0 - 29.2)	23.9 (18.9 - 29.3)	24.1 (18.9 - 29.4)	24.2 (19.0 - 29.4)
Salinity		32.4 (27.9 - 33.9)	32.2 (29.6 - 34.0)	32.0 (28.7 - 33.9)	31.4 (26.4 - 33.9)	31.5 (27.9 - 33.9)
Dissolved Oxygen (mg/L)		5.8 (3.8 - 7.1)	5.6 (4.3 - 8.0)	6.0 (4.5 - 8.9)	5.8 (4.7 - 7.8)	5.6 (4.2 - 6.6)
	Bottom	5.2 (2.5 - 7.2)	5.1 (3.3 - 6.9)	5.2 (3.4 - 6.4)	5.1 (3.3 - 6.7)	4.8 (2.6 - 6.6)
Dissolved Oxygen (% Saturation)		82 (55 - 94)	80 (64 - 116)	83 (55 - 119)	83 (69 - 109)	79 (65 - 90)
	Bottom	74 (36 - 96)	73 (52 - 94)	73 (53 - 87)	72 (52 - 90)	68 (38 - 88)
pH		7.7 (7.2 - 8.1)	7.8 (7.3 - 8.1)	7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)
Secchi Disc Depth (m)		2.9 (2.1 - 3.9)	2.7 (1.7 - 4.0)	2.7 (1.5 - 4.0)	2.7 (1.9 - 3.3)	2.5 (1.5 - 3.9)
Turbidity (NTU)		3.8 (0.7 - 6.6)	3.9 (0.7 - 9.6)	4.3 (1.2 - 9.9)	5.5 (1.0 - 22.4)	5.2 (1.0 - 23.8)
Suspended Solids (mg/L)		5.1 (1.2 - 8.7)	5.1 (1.9 - 9.3)	4.7 (1.2 - 10.7)	4.1 (1.6 - 6.4)	4.2 (2.2 - 8.1)
5-day Biochemical Oxygen Demand (mg/L)		0.5 (<0.1 - 1.0)	0.5 (<0.1 - 1.3)	0.5 (<0.1 - 1.0)	0.5 (<0.1 - 1.1)	0.7 (<0.1 - 2.6)
Ammonia Nitrogen (mg/L)		0.061 (0.012 - 0.210)	0.086 (0.016 - 0.233)	0.098 (0.018 - 0.207)	0.121 (0.021 - 0.293)	0.131 (0.016 - 0.273)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.011)	0.003 (<0.001 - 0.015)	0.003 (<0.001 - 0.013)	0.004 (<0.001 - 0.016)	0.004 (<0.001 - 0.014)
Nitrite Nitrogen (mg/L)		0.017 (0.005 - 0.052)	0.019 (0.004 - 0.053)	0.018 (0.005 - 0.051)	0.018 (0.006 - 0.048)	0.019 (0.006 - 0.049)
Nitrate Nitrogen (mg/L)		0.101 (0.027 - 0.233)	0.127 (0.025 - 0.263)	0.124 (0.024 - 0.300)	0.132 (0.034 - 0.343)	0.143 (0.036 - 0.333)
Total Inorganic Nitrogen (mg/L)		0.18 (0.05 - 0.31)	0.23 (0.05 - 0.40)	0.24 (0.05 - 0.38)	0.27 (0.06 - 0.44)	0.29 (0.07 - 0.44)
Total Kjeldahl Nitrogen (mg/L)		0.20 (0.14 - 0.30)	0.23 (0.13 - 0.37)	0.26 (0.17 - 0.44)	0.29 (0.20 - 0.43)	0.30 (0.21 - 0.48)
Total Nitrogen (mg/L)		0.32 (0.20 - 0.42)	0.38 (0.21 - 0.52)	0.40 (0.20 - 0.55)	0.44 (0.26 - 0.63)	0.46 (0.25 - 0.64)
Orthophosphate Phosphorus (mg/L)		0.011 (0.004 - 0.027)	0.013 (0.005 - 0.027)	0.014 (0.004 - 0.031)	0.015 (0.005 - 0.025)	0.014 (<0.002 - 0.023)
Total Phosphorus (mg/L)		0.10 (0.04 - 0.16)	0.11 (0.05 - 0.25)	0.10 (0.05 - 0.25)	0.11 (0.05 - 0.24)	0.11 (0.05 - 0.21)
Silica (as SiO_2) (mg/L)		0.78 (0.25 - 1.50)	0.82 (0.19 - 1.57)	0.73 (0.19 - 1.20)	0.78 (0.19 - 1.33)	0.81 (0.15 - 1.40)
Chlorophyll-*a* (μg/L)		2.8 (0.7 - 10.5)	3.3 (0.6 - 15.4)	3.1 (0.5 - 18.0)	3.4 (0.5 - 19.9)	4.3 (0.4 - 22.7)
E.coli (count/100mL)		260 (73 - 1500)	230 (74 - 2900)	240 (30 - 880)	580 (96 - 13000)	570 (16 - 6800)
Faecal Coliforms (count/100mL)		520 (130 - 3200)	460 (120 - 4900)	600 (160 - 1900)	1400 (180 - 22000)	1300 (87 - 15000)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Victoria Harbour WCZ in 2024 (continued)

Parameter		Victoria Harbour (West) VM7	Victoria Harbour (West) VM8	Stonecutters Island VM15	Rambler Channel VM12	Rambler Channel VM14
Number of samples		12	12	12	12	12
Temperature (°C)		24.6 (19.5 - 29.3)	24.5 (19.3 - 29.3)	24.4 (19.4 - 28.5)	24.2 (19.4 - 27.2)	24.4 (19.3 - 28.1)
Salinity		31.2 (24.9 - 33.7)	31.2 (27.0 - 34.0)	29.9 (19.1 - 33.9)	29.9 (21.3 - 33.8)	28.3 (16.1 - 33.8)
Dissolved Oxygen (mg/L)		5.6 (4.1 - 7.5)	5.7 (4.2 - 8.5)	5.6 (4.2 - 8.2)	5.5 (2.8 - 8.9)	5.6 (4.0 - 7.7)
	Bottom	5.3 (3.5 - 7.1)	4.7 (2.9 - 6.5)	5.1 (2.0 - 8.4)	4.6 (1.8 - 6.3)	4.8 (3.8 - 6.2)
Dissolved Oxygen (% Saturation)		79 (61 - 110)	80 (58 - 113)	80 (63 - 118)	78 (41 - 118)	78 (58 - 107)
	Bottom	76 (54 - 103)	67 (43 - 90)	72 (29 - 121)	65 (26 - 89)	67 (56 - 88)
pH		7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)	7.7 (7.4 - 8.1)	7.7 (7.3 - 8.2)	7.8 (7.4 - 8.2)
Secchi Disc Depth (m)		2.7 (2.0 - 4.2)	2.6 (2.0 - 3.4)	2.7 (1.7 - 3.4)	2.4 (1.7 - 2.9)	2.3 (1.3 - 3.2)
Turbidity (NTU)		3.6 (0.8 - 5.2)	4.0 (0.6 - 6.7)	3.6 (0.9 - 6.2)	3.6 (0.8 - 8.4)	3.7 (1.0 - 5.6)
Suspended Solids (mg/L)		5.0 (2.3 - 9.6)	5.3 (2.2 - 11.5)	5.9 (2.4 - 10.3)	5.1 (2.7 - 13.5)	4.8 (1.9 - 11.7)
5-day Biochemical Oxygen Demand (mg/L)		0.7 (0.1 - 2.8)	0.7 (<0.1 - 1.9)	0.7 (<0.1 - 2.6)	0.5 (<0.1 - 0.9)	0.5 (<0.1 - 1.1)
Ammonia Nitrogen (mg/L)		0.153 (0.039 - 0.233)	0.135 (0.020 - 0.257)	0.156 (0.082 - 0.263)	0.131 (0.078 - 0.210)	0.101 (0.043 - 0.170)
Unionised Ammonia (mg/L)		0.005 (0.001 - 0.012)	0.004 (<0.001 - 0.013)	0.004 (0.001 - 0.011)	0.003 (0.001 - 0.010)	0.002 (0.001 - 0.005)
Nitrite Nitrogen (mg/L)		0.021 (0.005 - 0.048)	0.024 (0.003 - 0.058)	0.028 (0.005 - 0.060)	0.034 (0.005 - 0.093)	0.041 (0.004 - 0.100)
Nitrate Nitrogen (mg/L)		0.163 (0.040 - 0.380)	0.165 (0.030 - 0.370)	0.202 (0.058 - 0.507)	0.225 (0.065 - 0.480)	0.329 (0.065 - 0.920)
Total Inorganic Nitrogen (mg/L)		0.34 (0.09 - 0.50)	0.32 (0.09 - 0.47)	0.39 (0.17 - 0.71)	0.39 (0.17 - 0.62)	0.47 (0.19 - 1.04)
Total Kjeldahl Nitrogen (mg/L)		0.34 (0.25 - 0.54)	0.30 (0.27 - 0.35)	0.37 (0.22 - 0.71)	0.31 (0.18 - 0.40)	0.28 (0.19 - 0.43)
Total Nitrogen (mg/L)		0.52 (0.33 - 0.79)	0.49 (0.33 - 0.69)	0.60 (0.30 - 0.87)	0.56 (0.29 - 0.79)	0.65 (0.31 - 1.19)
Orthophosphate Phosphorus (mg/L)		0.017 (<0.002 - 0.033)	0.015 (<0.002 - 0.031)	0.016 (0.009 - 0.027)	0.017 (0.010 - 0.025)	0.017 (0.005 - 0.035)
Total Phosphorus (mg/L)		0.12 (0.05 - 0.23)	0.12 (0.04 - 0.24)	0.09 (0.05 - 0.17)	0.09 (0.05 - 0.21)	0.09 (0.05 - 0.22)
Silica (as SiO_2) (mg/L)		0.94 (0.14 - 1.70)	1.08 (0.11 - 2.24)	1.08 (0.19 - 3.33)	1.33 (0.22 - 3.50)	1.94 (0.26 - 6.13)
Chlorophyll-*a* (µg/L)		4.8 (0.4 - 22.8)	4.9 (0.6 - 23.0)	5.9 (0.4 - 30.5)	3.1 (0.6 - 10.9)	3.1 (0.5 - 12.9)
E.coli (count/100mL)		1200 (120 - 11000)	1300 (23 - 12000)	2000 (200 - 12000)	1700 (220 - 7000)	430 (97 - 3800)
Faecal Coliforms (count/100mL)		2800 (430 - 23000)	2700 (71 - 22000)	3900 (390 - 24000)	3700 (490 - 17000)	1100 (150 - 6800)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Eastern Buffer WCZ in 2024

Parameter		Chai Wan EM1	Tathong Channel EM2	EM3
Number of samples		12	12	12
Temperature (°C)		24.0 (19.1 - 29.0)	24.1 (19.1 - 28.7)	24.0 (19.1 - 28.6)
Salinity		32.4 (30.2 - 34.2)	32.5 (30.2 - 34.3)	32.6 (30.2 - 34.4)
Dissolved Oxygen (mg/L)		5.6 (4.1 - 6.9)	5.7 (4.2 - 7.4)	6.0 (4.8 - 7.2)
	Bottom	5.3 (2.9 - 7.0)	5.3 (3.2 - 7.8)	5.9 (3.7 - 7.6)
Dissolved Oxygen (% Saturation)		80 (55 - 93)	80 (60 - 100)	86 (73 - 98)
	Bottom	75 (42 - 94)	75 (46 - 104)	84 (53 - 102)
pH		7.9 (7.4 - 8.2)	7.9 (7.4 - 8.2)	7.9 (7.4 - 8.2)
Secchi Disc Depth (m)		2.9 (1.6 - 5.0)	2.8 (1.8 - 4.9)	2.9 (1.6 - 4.5)
Turbidity (NTU)		2.9 (0.3 - 4.8)	3.1 (0.3 - 5.1)	2.6 (0.3 - 4.9)
Suspended Solids (mg/L)		5.8 (1.5 - 14.7)	5.6 (1.5 - 15.7)	4.3 (0.7 - 12.8)
5-day Biochemical Oxygen Demand (mg/L)		0.8 (0.1 - 1.9)	0.8 (<0.1 - 1.9)	0.8 (<0.1 - 2.0)
Ammonia Nitrogen (mg/L)		0.045 (0.006 - 0.163)	0.037 (0.007 - 0.133)	0.028 (0.008 - 0.103)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.009)	0.001 (<0.001 - 0.007)	0.001 (<0.001 - 0.006)
Nitrite Nitrogen (mg/L)		0.015 (<0.002 - 0.051)	0.015 (<0.002 - 0.050)	0.011 (<0.002 - 0.036)
Nitrate Nitrogen (mg/L)		0.089 (0.007 - 0.230)	0.081 (0.010 - 0.211)	0.056 (0.006 - 0.151)
Total Inorganic Nitrogen (mg/L)		0.15 (0.01 - 0.29)	0.13 (0.02 - 0.27)	0.10 (0.02 - 0.19)
Total Kjeldahl Nitrogen (mg/L)		0.20 (0.11 - 0.36)	0.19 (0.14 - 0.32)	0.17 (0.11 - 0.26)
Total Nitrogen (mg/L)		0.31 (0.18 - 0.48)	0.29 (0.17 - 0.45)	0.24 (0.12 - 0.37)
Orthophosphate Phosphorus (mg/L)		0.007 (0.002 - 0.022)	0.007 (<0.002 - 0.020)	0.005 (<0.002 - 0.010)
Total Phosphorus (mg/L)		0.09 (0.03 - 0.26)	0.09 (0.03 - 0.24)	0.09 (0.03 - 0.22)
Silica (as SiO_2) (mg/L)		0.57 (0.09 - 1.05)	0.54 (0.10 - 1.10)	0.47 (0.10 - 0.78)
Chlorophyll-*a* (μg/L)		4.7 (0.7 - 15.5)	4.4 (0.6 - 16.6)	4.3 (0.5 - 13.9)
E.coli (count/100mL)		32 (1 - 220)	23 (<1 - 280)	6 (<1 - 97)
Faecal Coliforms (count/100mL)		69 (1 - 620)	52 (1 - 760)	9 (1 - 130)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Western Buffer WCZ in 2024

Parameter		Hong Kong Island (West) WM1	Hong Kong Island (West) WM2	Tsing Yi (South) WM3	Tsing Yi (West) WM4
Number of samples		12	12	12	12
Temperature (°C)		24.2 (19.8 - 28.9)	24.4 (19.6 - 28.5)	24.2 (19.7 - 28.7)	24.3 (19.8 - 28.6)
Salinity		31.6 (25.6 - 34.1)	29.6 (20.7 - 32.8)	31.2 (26.0 - 33.5)	30.3 (22.2 - 33.4)
Dissolved Oxygen (mg/L)		6.1 (4.1 - 8.1)	6.2 (3.0 - 9.2)	5.6 (4.0 - 8.1)	5.3 (4.1 - 7.9)
	Bottom	5.1 (2.1 - 8.0)	5.8 (2.6 - 8.3)	5.0 (1.8 - 7.8)	4.7 (2.0 - 7.7)
Dissolved Oxygen (% Saturation)		87 (57 - 108)	87 (43 - 121)	79 (54 - 108)	75 (59 - 105)
	Bottom	72 (24 - 107)	82 (37 - 120)	70 (18 - 104)	65 (29 - 103)
pH		7.9 (7.5 - 8.2)	7.8 (7.5 - 8.1)	7.8 (7.3 - 8.1)	7.8 (7.3 - 8.1)
Secchi Disc Depth (m)		2.9 (1.6 - 7.7)	2.5 (1.2 - 5.5)	2.3 (1.3 - 3.9)	2.4 (1.2 - 5.4)
Turbidity (NTU)		3.9 (0.2 - 6.3)	3.4 (0.2 - 7.4)	3.8 (0.5 - 6.6)	4.8 (0.3 - 7.4)
Suspended Solids (mg/L)		4.2 (1.5 - 6.8)	4.9 (2.2 - 7.9)	4.9 (1.9 - 7.6)	5.1 (2.6 - 8.0)
5-day Biochemical Oxygen Demand (mg/L)		1.0 (<0.1 - 2.7)	0.8 (<0.1 - 2.4)	0.7 (<0.1 - 2.2)	0.9 (<0.1 - 2.6)
Ammonia Nitrogen (mg/L)		0.050 (0.007 - 0.131)	0.079 (0.013 - 0.170)	0.109 (0.027 - 0.220)	0.098 (0.021 - 0.220)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.006)	0.003 (<0.001 - 0.007)	0.003 (<0.001 - 0.008)	0.003 (<0.001 - 0.007)
Nitrite Nitrogen (mg/L)		0.017 (<0.002 - 0.051)	0.026 (0.005 - 0.081)	0.024 (0.004 - 0.065)	0.029 (0.005 - 0.082)
Nitrate Nitrogen (mg/L)		0.141 (0.018 - 0.362)	0.233 (0.023 - 0.700)	0.193 (0.038 - 0.453)	0.241 (0.035 - 0.620)
Total Inorganic Nitrogen (mg/L)		0.21 (0.03 - 0.46)	0.34 (0.04 - 0.82)	0.33 (0.14 - 0.62)	0.37 (0.12 - 0.69)
Total Kjeldahl Nitrogen (mg/L)		0.21 (0.10 - 0.30)	0.24 (0.13 - 0.32)	0.27 (0.15 - 0.34)	0.26 (0.16 - 0.38)
Total Nitrogen (mg/L)		0.37 (0.17 - 0.62)	0.50 (0.17 - 0.96)	0.48 (0.31 - 0.78)	0.53 (0.29 - 0.85)
Orthophosphate Phosphorus (mg/L)		0.008 (0.002 - 0.018)	0.012 (0.003 - 0.021)	0.013 (0.003 - 0.020)	0.014 (0.003 - 0.024)
Total Phosphorus (mg/L)		0.09 (0.03 - 0.19)	0.09 (0.04 - 0.21)	0.09 (0.04 - 0.22)	0.09 (0.04 - 0.20)
Silica (as SiO_2) (mg/L)		0.87 (0.16 - 2.10)	1.22 (0.18 - 4.13)	1.14 (0.19 - 2.73)	1.36 (0.18 - 3.80)
Chlorophyll-*a* (µg/L)		5.8 (0.6 - 17.6)	4.9 (0.5 - 17.0)	3.8 (0.6 - 11.4)	4.0 (0.5 - 16.8)
E.coli (count/100mL)		27 (<1 - 1400)	180 (2 - 2700)	1400 (150 - 6000)	610 (130 - 13000)
Faecal Coliforms (count/100mL)		50 (1 - 2100)	310 (3 - 4700)	2500 (360 - 11000)	1200 (280 - 23000)

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2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Junk Bay WCZ in 2024

Parameter		Junk Bay JM3	Junk Bay JM4
Number of samples		12	12
Temperature (°C)		24.5 (19.2 - 29.4)	24.0 (19.2 - 29.0)
Salinity		31.9 (25.2 - 34.1)	32.2 (27.4 - 34.2)
Dissolved Oxygen (mg/L)		6.5 (4.8 - 9.2)	5.8 (4.1 - 7.8)
	Bottom	6.0 (3.8 - 8.2)	5.4 (3.3 - 7.7)
Dissolved Oxygen (% Saturation)		93 (74 - 123)	82 (56 - 111)
	Bottom	85 (55 - 114)	76 (46 - 103)
pH		7.9 (7.5 - 8.2)	7.8 (7.4 - 8.2)
Secchi Disc Depth (m)		2.6 (1.6 - 3.8)	2.9 (1.8 - 4.7)
Turbidity (NTU)		2.5 (0.3 - 4.4)	3.2 (0.4 - 5.3)
Suspended Solids (mg/L)		3.6 (1.0 - 5.6)	4.2 (0.9 - 6.8)
5-day Biochemical Oxygen Demand (mg/L)		0.8 (<0.1 - 2.0)	0.7 (<0.1 - 1.9)
Ammonia Nitrogen (mg/L)		0.049 (0.013 - 0.113)	0.044 (0.011 - 0.111)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.007)	0.001 (<0.001 - 0.006)
Nitrite Nitrogen (mg/L)		0.014 (<0.002 - 0.055)	0.016 (<0.002 - 0.053)
Nitrate Nitrogen (mg/L)		0.082 (0.013 - 0.217)	0.085 (0.012 - 0.180)
Total Inorganic Nitrogen (mg/L)		0.15 (0.03 - 0.26)	0.14 (0.02 - 0.26)
Total Kjeldahl Nitrogen (mg/L)		0.23 (0.17 - 0.33)	0.20 (0.13 - 0.32)
Total Nitrogen (mg/L)		0.33 (0.19 - 0.50)	0.30 (0.15 - 0.50)
Orthophosphate Phosphorus (mg/L)		0.006 (<0.002 - 0.011)	0.006 (<0.002 - 0.012)
Total Phosphorus (mg/L)		0.08 (0.03 - 0.19)	0.10 (0.03 - 0.30)
Silica (as SiO_2) (mg/L)		0.44 (0.11 - 0.84)	0.55 (0.10 - 0.88)
Chlorophyll-*a* (µg/L)		6.8 (0.9 - 24.7)	5.3 (0.9 - 20.3)
E.coli (count/100mL)		25 (7 - 90)	37 (1 - 230)
Faecal Coliforms (count/100mL)		56 (19 - 160)	83 (5 - 640)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for the Deep Bay WCZ in 2024

Parameter		Inner Deep Bay DM1	Inner Deep Bay DM2	Inner Deep Bay DM3	Outer Deep Bay DM4	Outer Deep Bay DM5
Number of samples		**12**	**12**	**12**	**12**	**12**
Temperature (°C)		24.9 (16.8 - 31.3)	24.8 (16.5 - 31.0)	24.8 (17.8 - 30.8)	24.9 (19.0 - 30.2)	24.7 (18.7 - 29.8)
Salinity		16.1 (6.4 - 24.9)	18.8 (8.6 - 24.9)	22.1 (12.7 - 28.8)	23.4 (13.4 - 30.5)	25.2 (16.3 - 31.7)
Dissolved Oxygen (mg/L)		7.7 (5.0 - 9.9)	7.0 (5.0 - 8.4)	6.5 (4.2 - 8.2)	6.2 (4.1 - 8.6)	5.9 (4.0 - 8.3)
	Bottom	#N/A #N/A	#N/A #N/A	#N/A #N/A	6.0 (4.2 - 8.6)	5.6 (4.0 - 8.7)
Dissolved Oxygen (% Saturation)		101 (65 - 123)	93 (67 - 113)	89 (61 - 110)	85 (61 - 113)	82 (59 - 111)
	Bottom	#N/A #N/A	#N/A #N/A	#N/A #N/A	83 (63 - 114)	77 (58 - 115)
pH		7.6 (7.1 - 8.0)	7.7 (7.3 - 8.2)	7.8 (7.6 - 8.3)	7.8 (7.7 - 8.2)	7.9 (7.7 - 8.2)
Secchi Disc Depth (m)		1.1 (0.7 - 1.4)	1.2 (0.4 - 1.6)	1.3 (0.5 - 1.7)	1.6 (1.1 - 2.5)	1.8 (1.3 - 2.3)
Turbidity (NTU)		24.8 (2.9 - 38.2)	17.4 (2.6 - 43.8)	11.0 (1.4 - 20.8)	8.0 (0.9 - 21.7)	5.3 (0.5 - 9.8)
Suspended Solids (mg/L)		22.2 (5.6 - 61.0)	16.3 (6.8 - 48.0)	10.0 (3.2 - 20.0)	7.5 (3.1 - 16.0)	5.7 (1.8 - 10.0)
5-day Biochemical Oxygen Demand (mg/L)		1.2 (0.3 - 2.6)	1.0 (0.2 - 3.2)	0.6 (<0.1 - 1.9)	0.5 (<0.1 - 1.8)	0.5 (<0.1 - 1.8)
Ammonia Nitrogen (mg/L)		0.374 (0.082 - 0.620)	0.224 (0.049 - 0.390)	0.117 (0.030 - 0.190)	0.098 (0.007 - 0.175)	0.089 (0.007 - 0.163)
Unionised Ammonia (mg/L)		0.007 (0.004 - 0.014)	0.005 (0.001 - 0.013)	0.004 (<0.001 - 0.007)	0.003 (<0.001 - 0.007)	0.003 (<0.001 - 0.007)
Nitrite Nitrogen (mg/L)		0.118 (0.068 - 0.280)	0.084 (0.032 - 0.280)	0.064 (0.016 - 0.190)	0.080 (0.014 - 0.285)	0.068 (0.009 - 0.170)
Nitrate Nitrogen (mg/L)		1.210 (0.560 - 1.700)	0.906 (0.360 - 1.300)	0.684 (0.160 - 1.200)	0.650 (0.205 - 1.200)	0.556 (0.210 - 1.020)
Total Inorganic Nitrogen (mg/L)		1.70 (0.75 - 2.34)	1.21 (0.55 - 1.77)	0.87 (0.27 - 1.42)	0.83 (0.29 - 1.41)	0.71 (0.30 - 1.20)
Total Kjeldahl Nitrogen (mg/L)		0.80 (0.54 - 1.20)	0.55 (0.34 - 1.10)	0.32 (0.24 - 0.46)	0.26 (0.17 - 0.42)	0.25 (0.11 - 0.41)
Total Nitrogen (mg/L)		2.13 (1.20 - 2.92)	1.54 (0.78 - 2.48)	1.07 (0.45 - 1.66)	0.99 (0.40 - 1.61)	0.87 (0.40 - 1.34)
Orthophosphate Phosphorus (mg/L)		0.138 (0.094 - 0.180)	0.106 (0.026 - 0.170)	0.058 (0.008 - 0.110)	0.034 (0.008 - 0.056)	0.025 (0.005 - 0.040)
Total Phosphorus (mg/L)		0.26 (0.14 - 0.38)	0.21 (0.12 - 0.33)	0.13 (0.05 - 0.25)	0.09 (0.04 - 0.18)	0.08 (0.03 - 0.15)
Silica (as SiO_2) (mg/L)		5.70 (0.17 - 11.00)	4.33 (0.11 - 11.00)	3.36 (0.26 - 8.20)	3.23 (0.33 - 7.65)	2.82 (0.28 - 6.67)
Chlorophyll-*a* (µg/L)		5.4 (1.4 - 18.0)	6.5 (1.0 - 35.0)	3.1 (0.8 - 15.0)	2.1 (0.9 - 7.7)	2.2 (0.7 - 7.2)
E.coli (count/100mL)		200 (14 - 1200)	93 (7 - 3200)	21 (2 - 270)	19 (1 - 220)	22 (2 - 180)
Faecal Coliforms (count/100mL)		480 (30 - 2900)	210 (26 - 7300)	56 (6 - 900)	50 (2 - 1000)	50 (8 - 450)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.
4. N/A (Not Applicable) indicates the measurement was not made due to shallow water.

Summary of water quality data for the North Western WCZ in 2024

Parameter		Lantau Island (North) NM1	Pearl Island NM2	Pillar Point NM3	Urmston Road NM5	Chek Lap Kok (North) NM6	Chek Lap Kok (West) NM8
Number of samples		12	12	12	12	12	12
Temperature (°C)		24.8 (19.6 - 29.0)	25.1 (19.5 - 29.0)	25.0 (19.5 - 28.8)	25.1 (19.6 - 29.0)	25.2 (19.5 - 29.2)	25.3 (19.7 - 29.9)
Salinity		28.7 (19.6 - 33.3)	26.3 (13.9 - 33.0)	25.5 (8.8 - 32.8)	25.3 (12.5 - 32.1)	23.9 (6.2 - 31.6)	24.7 (3.0 - 34.0)
Dissolved Oxygen (mg/L)		5.6 (4.3 - 6.8)	6.0 (4.4 - 7.5)	5.9 (4.8 - 7.6)	5.8 (4.6 - 7.4)	6.3 (5.1 - 7.7)	6.1 (5.0 - 7.1)
	Bottom	5.1 (2.9 - 7.4)	5.4 (3.7 - 7.5)	5.4 (4.1 - 7.5)	5.4 (3.9 - 7.6)	5.9 (3.9 - 7.4)	5.5 (4.1 - 7.4)
Dissolved Oxygen (% Saturation)		79 (66 - 95)	84 (60 - 102)	82 (64 - 101)	80 (63 - 98)	87 (74 - 110)	86 (69 - 100)
	Bottom	73 (45 - 98)	76 (55 - 99)	76 (59 - 100)	75 (55 - 100)	82 (58 - 103)	77 (58 - 98)
pH		7.9 (7.6 - 8.3)	8.0 (7.6 - 8.4)	8.0 (7.6 - 8.3)	7.9 (7.7 - 8.3)	8.0 (7.7 - 8.3)	8.0 (7.7 - 8.3)
Secchi Disc Depth (m)		2.1 (1.6 - 3.3)	2.0 (1.6 - 3.0)	2.0 (1.3 - 3.0)	1.8 (1.2 - 2.5)	1.8 (1.4 - 2.1)	1.7 (1.1 - 2.5)
Turbidity (NTU)		5.4 (0.5 - 10.3)	4.7 (0.5 - 9.1)	7.4 (0.4 - 19.5)	8.7 (0.6 - 17.9)	6.0 (1.2 - 14.9)	11.9 (1.8 - 34.4)
Suspended Solids (mg/L)		6.1 (2.3 - 11.1)	5.9 (2.5 - 9.9)	5.4 (2.4 - 11.6)	6.3 (1.9 - 11.7)	6.8 (3.4 - 14.0)	9.9 (2.6 - 21.8)
5-day Biochemical Oxygen Demand (mg/L)		0.9 (0.1 - 2.6)	0.8 (0.1 - 3.5)	0.8 (<0.1 - 3.6)	0.8 (0.2 - 2.6)	0.8 (<0.1 - 1.9)	0.8 (<0.1 - 1.7)
Ammonia Nitrogen (mg/L)		0.084 (0.017 - 0.167)	0.071 (0.014 - 0.150)	0.071 (0.016 - 0.143)	0.071 (0.023 - 0.133)	0.059 (0.017 - 0.123)	0.038 (0.006 - 0.108)
Unionised Ammonia (mg/L)		0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.005)	0.002 (<0.001 - 0.006)
Nitrite Nitrogen (mg/L)		0.047 (0.003 - 0.120)	0.051 (0.005 - 0.123)	0.052 (0.006 - 0.130)	0.055 (0.007 - 0.147)	0.061 (0.008 - 0.173)	0.042 (<0.002 - 0.127)
Nitrate Nitrogen (mg/L)		0.334 (0.133 - 1.000)	0.446 (0.147 - 0.980)	0.468 (0.180 - 0.933)	0.480 (0.217 - 0.963)	0.591 (0.200 - 1.400)	0.503 (0.054 - 1.370)
Total Inorganic Nitrogen (mg/L)		0.47 (0.27 - 1.14)	0.57 (0.29 - 1.12)	0.59 (0.29 - 1.07)	0.61 (0.33 - 1.11)	0.71 (0.30 - 1.54)	0.58 (0.09 - 1.46)
Total Kjeldahl Nitrogen (mg/L)		0.26 (0.19 - 0.33)	0.24 (0.15 - 0.49)	0.22 (0.16 - 0.30)	0.23 (0.16 - 0.31)	0.24 (0.16 - 0.30)	0.21 (0.14 - 0.31)
Total Nitrogen (mg/L)		0.64 (0.40 - 1.33)	0.74 (0.40 - 1.29)	0.75 (0.40 - 1.21)	0.77 (0.45 - 1.27)	0.89 (0.46 - 1.78)	0.75 (0.24 - 1.70)
Orthophosphate Phosphorus (mg/L)		0.017 (0.005 - 0.030)	0.018 (<0.002 - 0.035)	0.018 (0.003 - 0.033)	0.021 (0.004 - 0.035)	0.020 (0.003 - 0.040)	0.013 (<0.002 - 0.030)
Total Phosphorus (mg/L)		0.08 (0.03 - 0.17)	0.07 (0.04 - 0.16)	0.07 (0.04 - 0.15)	0.08 (0.04 - 0.15)	0.07 (0.04 - 0.14)	0.07 (0.03 - 0.15)
Silica (as SiO_2) (mg/L)		2.04 (0.30 - 6.60)	2.48 (0.36 - 6.43)	2.59 (0.37 - 6.33)	2.73 (0.52 - 6.93)	3.24 (0.56 - 9.10)	2.92 (0.33 - 8.40)
Chlorophyll-*a* (μg/L)		4.8 (0.6 - 19.9)	4.5 (0.4 - 20.3)	4.5 (0.4 - 19.7)	3.2 (0.4 - 14.8)	5.0 (0.6 - 23.0)	3.6 (1.3 - 10.7)
E.coli (count/100mL)		230 (10 - 5300)	68 (8 - 1100)	48 (8 - 360)	54 (5 - 830)	14 (1 - 460)	3 (<1 - 150)
Faecal Coliforms (count/100mL)		410 (17 - 12000)	140 (26 - 1500)	110 (18 - 1100)	130 (14 - 2000)	40 (7 - 1300)	7 (1 - 580)

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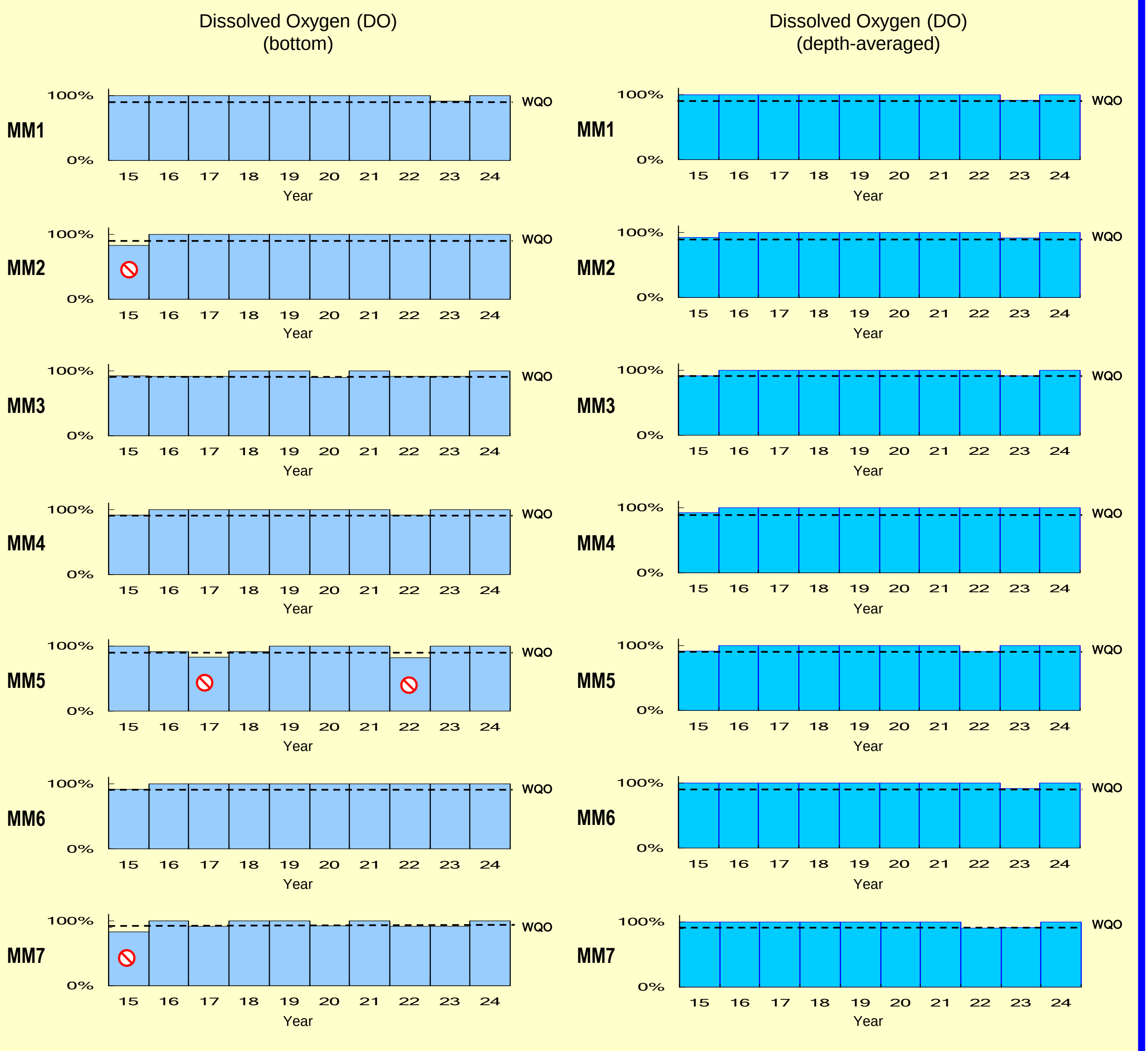
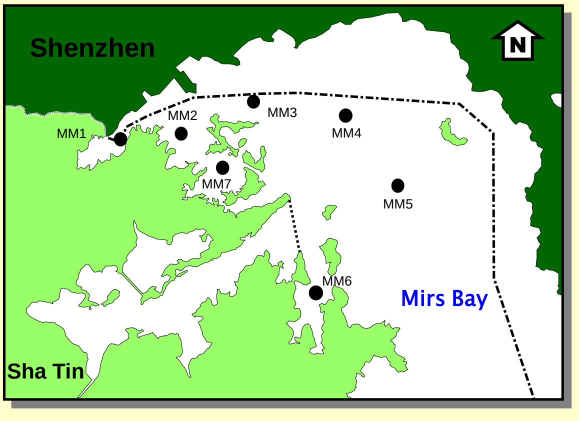
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

WQO compliance rates for the Mirs Bay WCZ





Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

% sample with bottom DO ≥ 2 mg/L

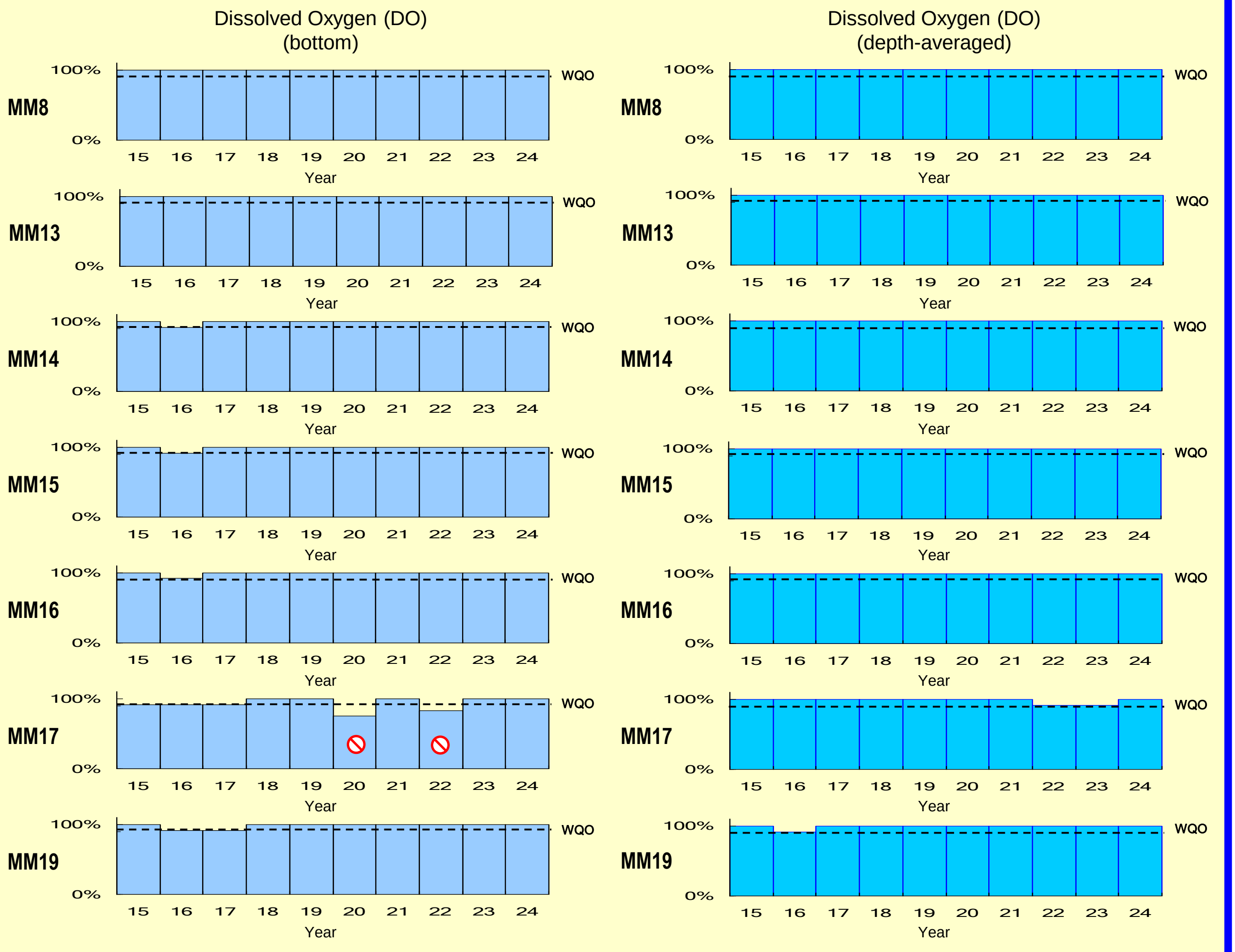
2. Depth-averaged

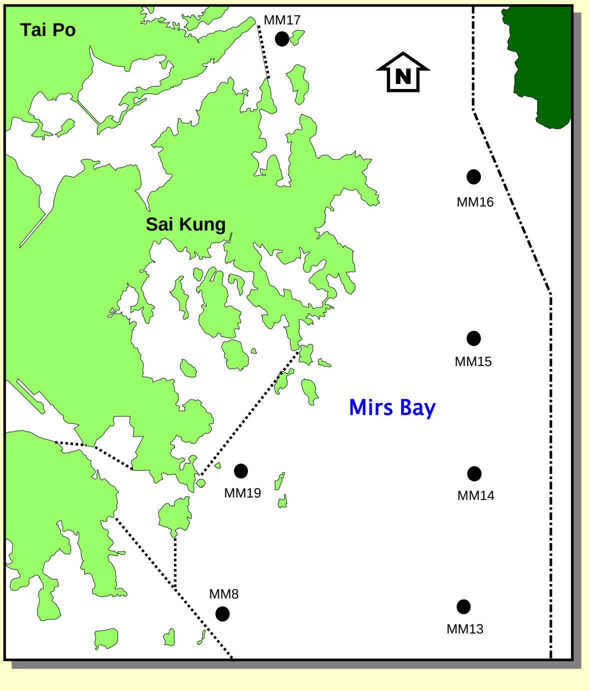
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

% sample with depth-averaged DO ≥ 4 mg/L

Non-compliance

WQO compliance rates for the Mirs Bay WCZ (continued)





Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

% sample with bottom DO ≥ 2 mg/L

2. Depth-averaged

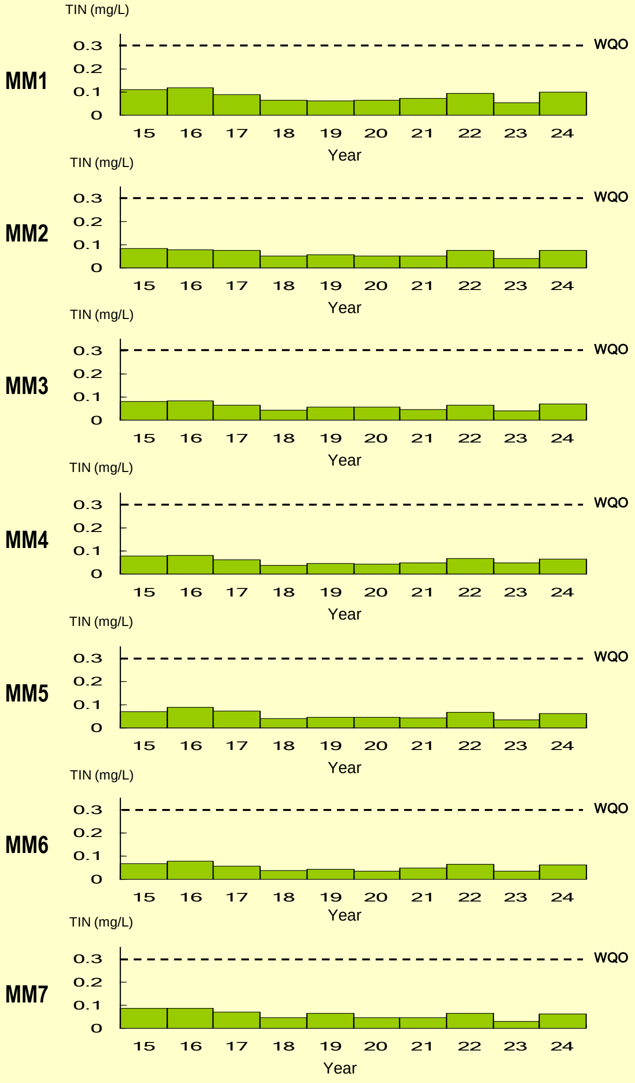
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

% sample with depth-averaged DO ≥ 4 mg/L

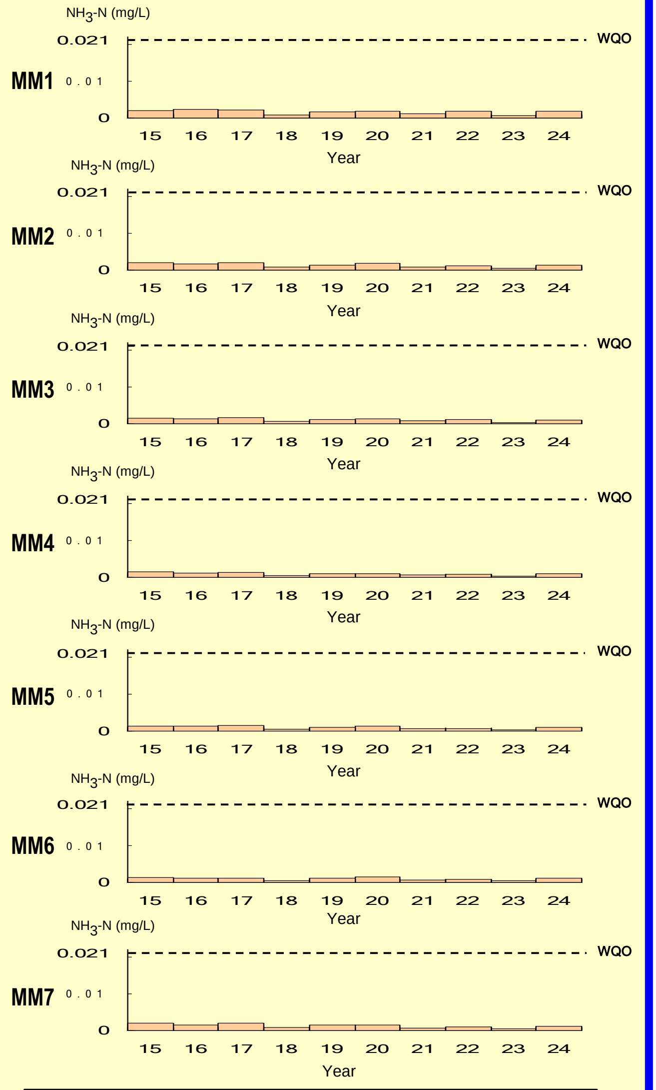
Non-compliance

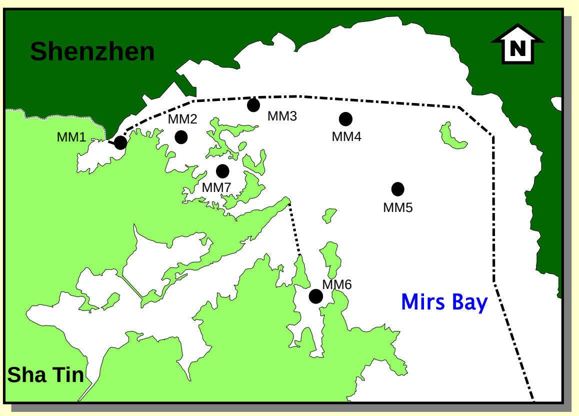
WQO compliance rates for the Mirs Bay WCZ (continued)

Total Inorganic Nitrogen (TIN) (depth-averaged)



Unionised Ammonia (NH_3-N) (depth-averaged)

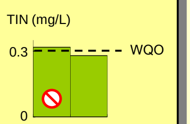




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.3 mg/L

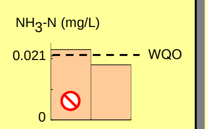
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

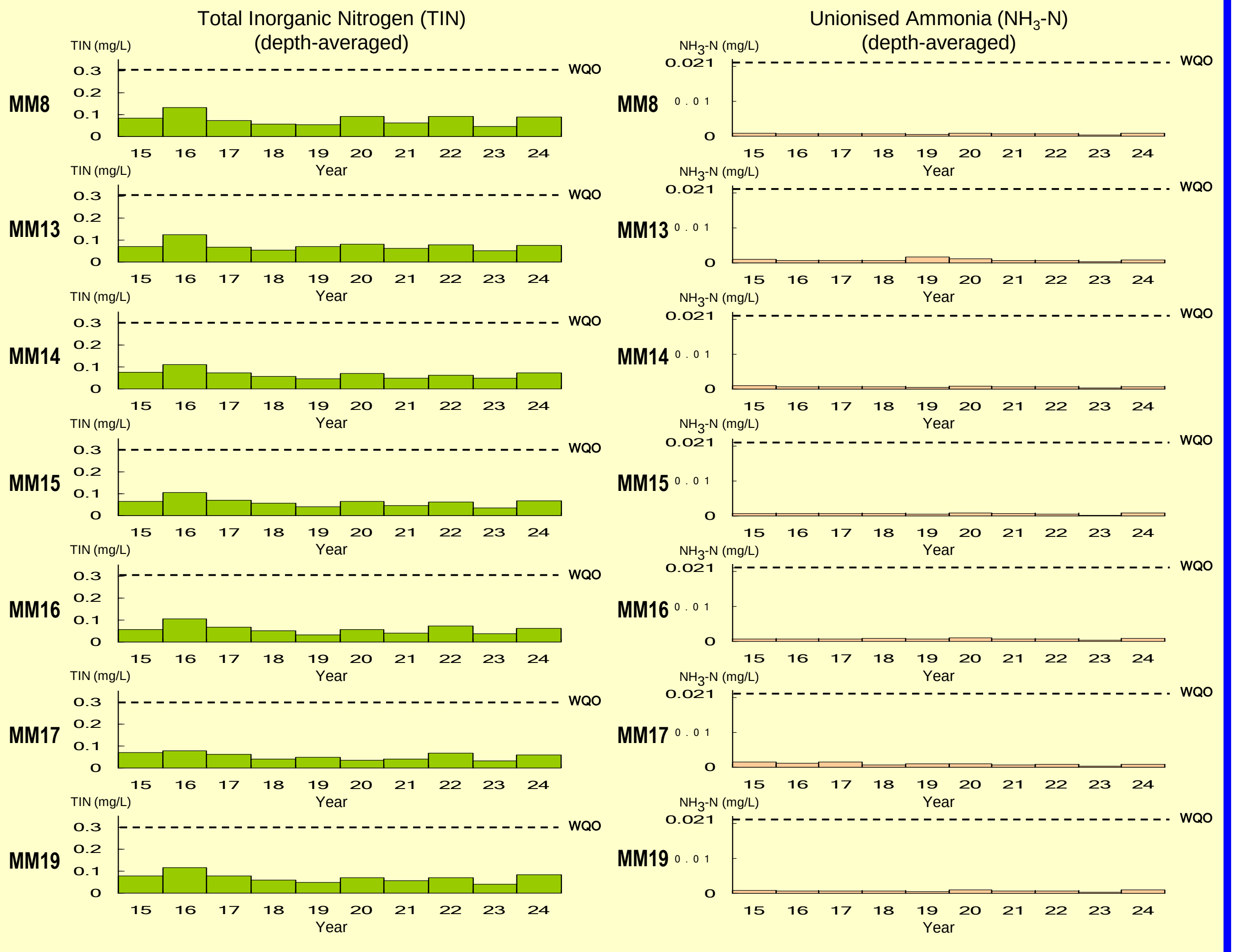
WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

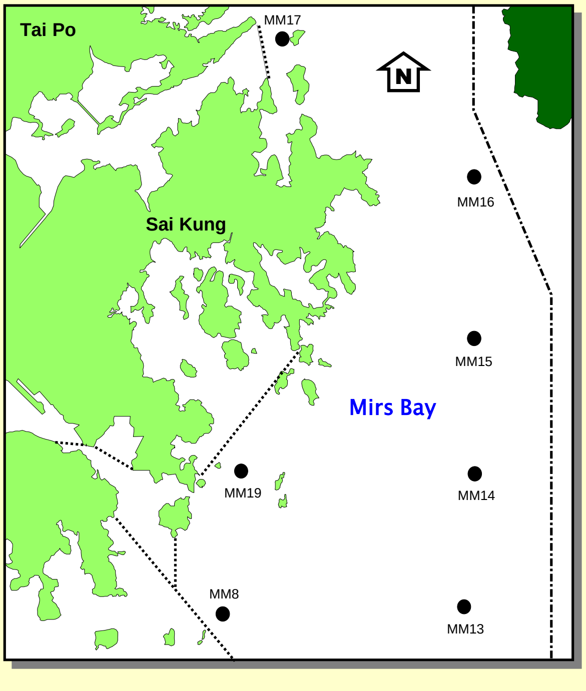
annual mean for depth-averaged NH_3-N

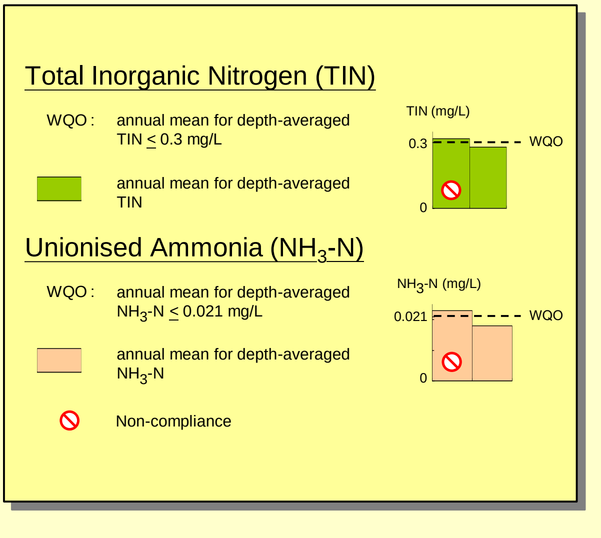


Non-compliance

WQO compliance rates for the Mirs Bay WCZ (continued)



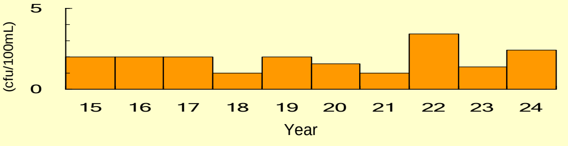




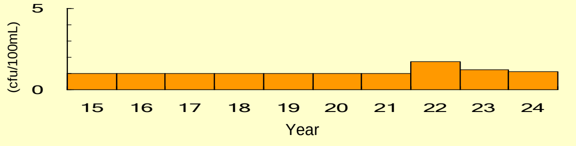
WQO compliance rates for the Mirs Bay WCZ (continued)

E. coli
(annual geometric mean)

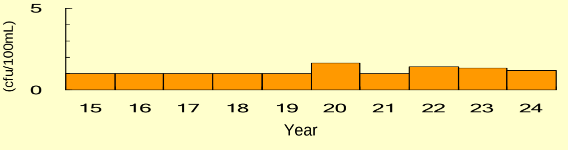
MM2

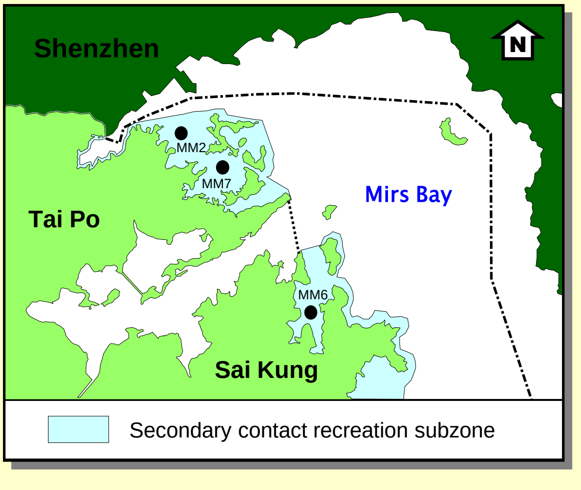


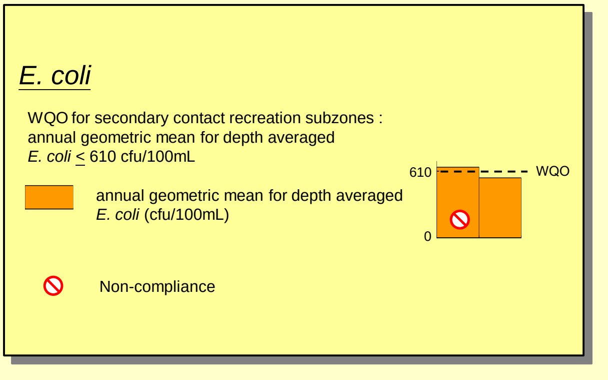
MM7



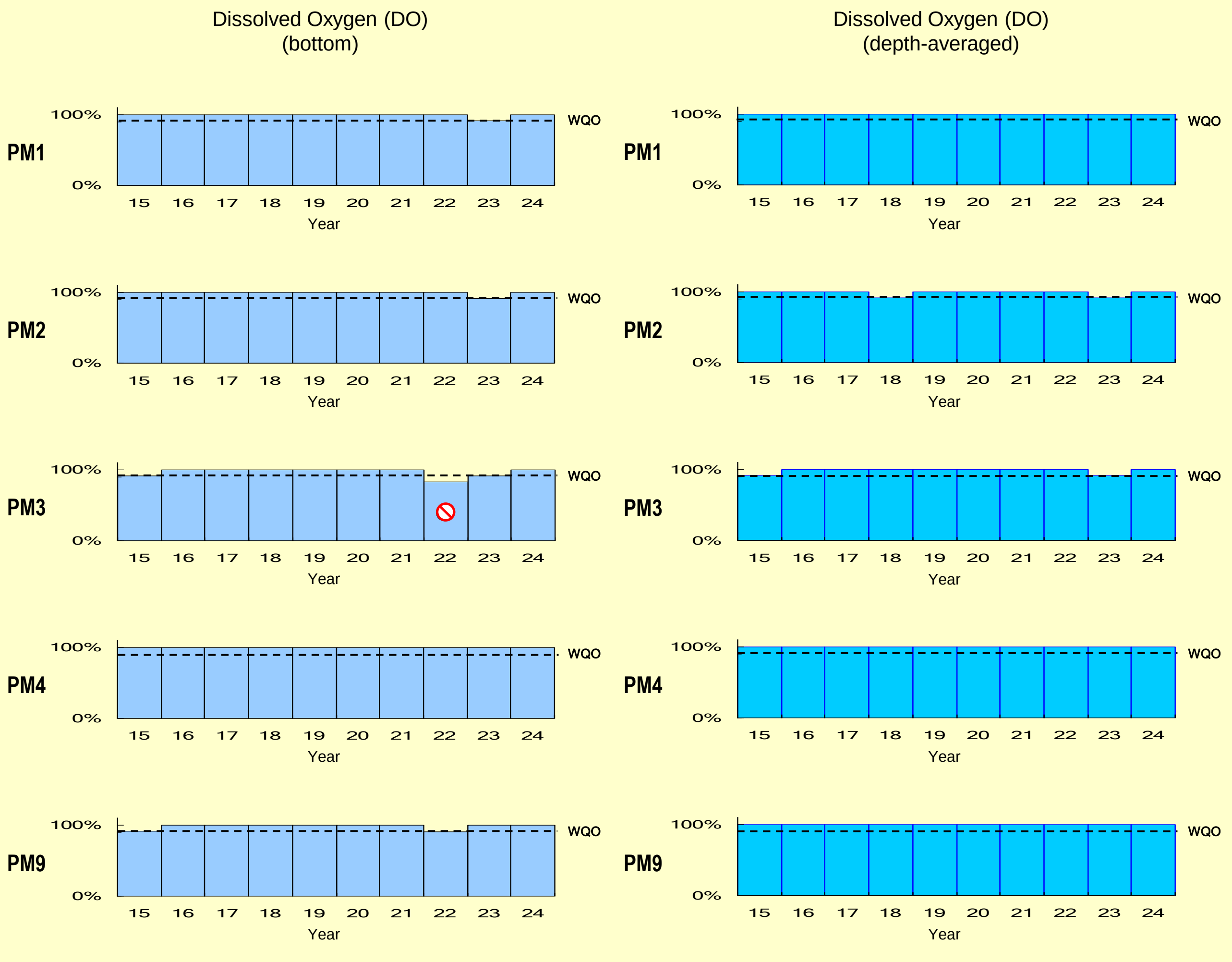
MM6

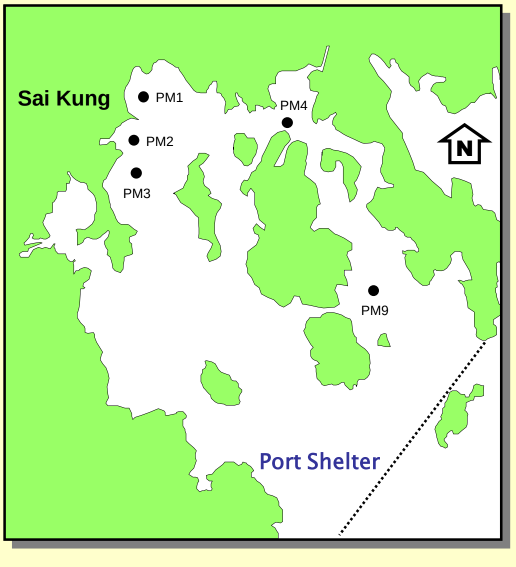


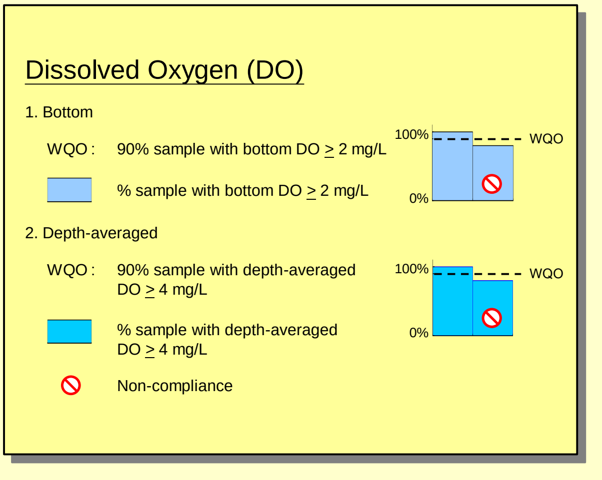




WQO compliance rates for the Port Shelter WCZ



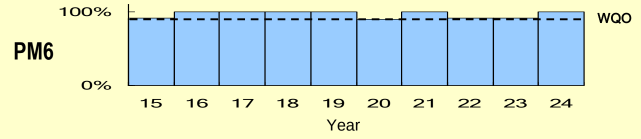


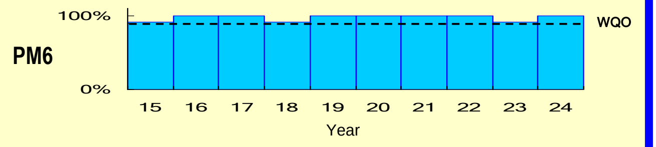


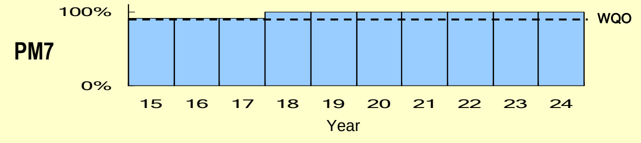
WQO compliance rates for the Port Shelter WCZ (continued)

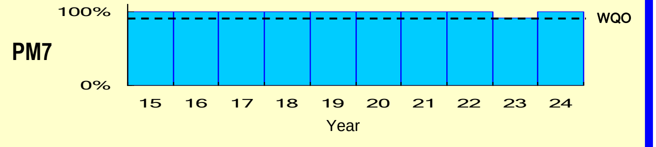
Dissolved Oxygen (DO)
(bottom)

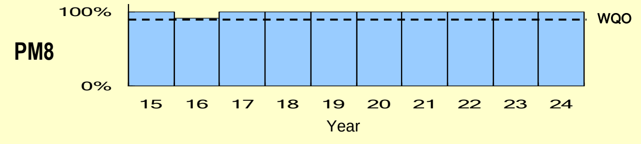
Dissolved Oxygen (DO)
(depth-averaged)

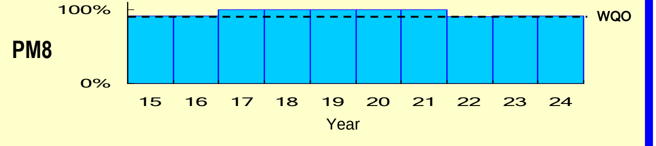


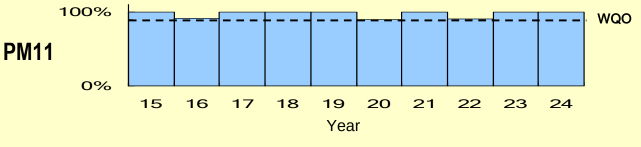


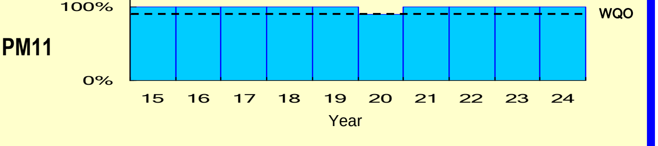


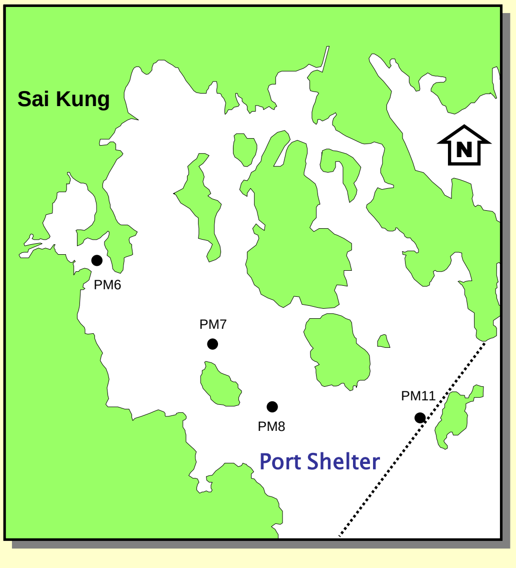












Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

% sample with bottom DO ≥ 2 mg/L

100% WQO 0%

2. Depth-averaged

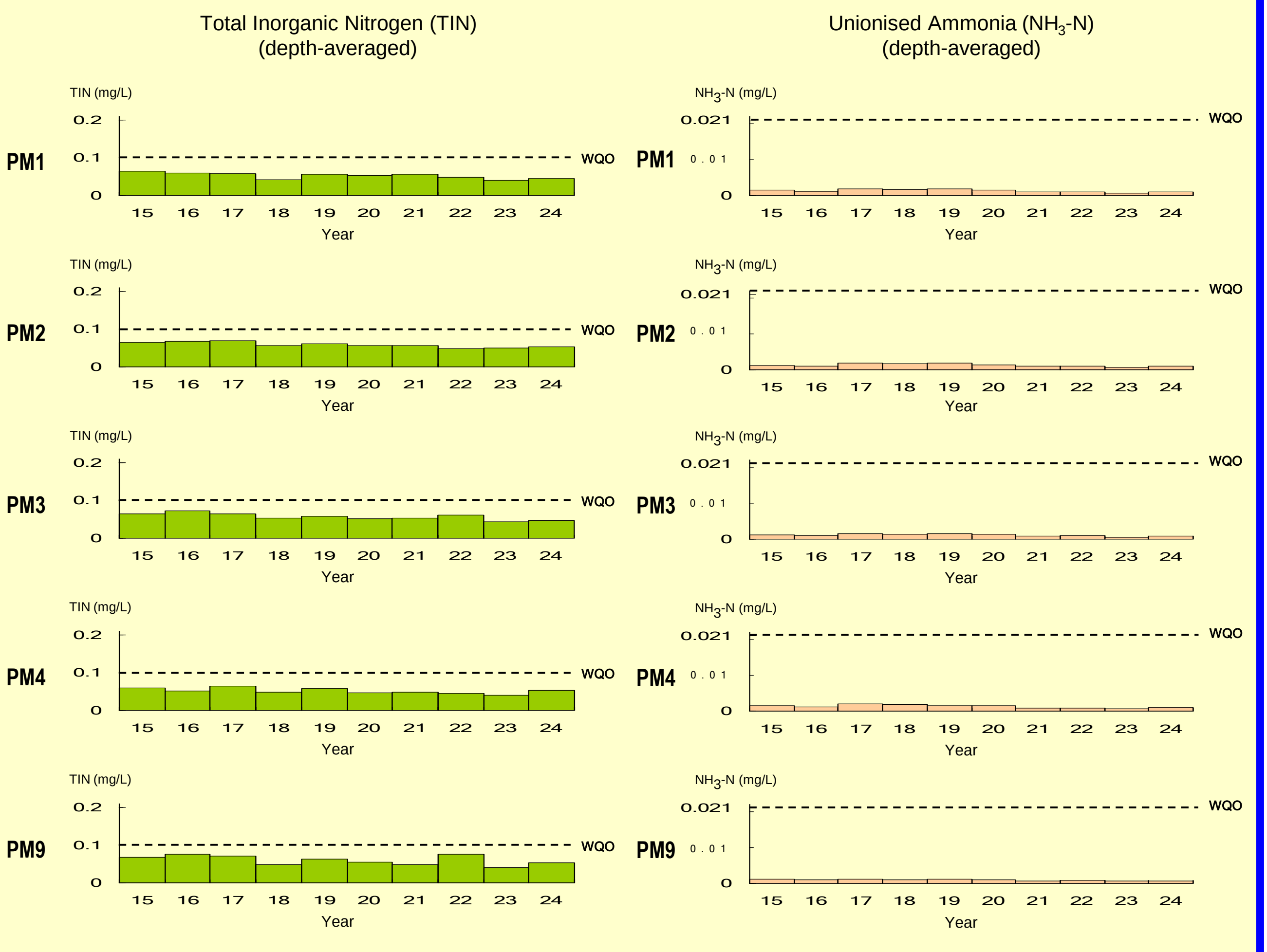
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

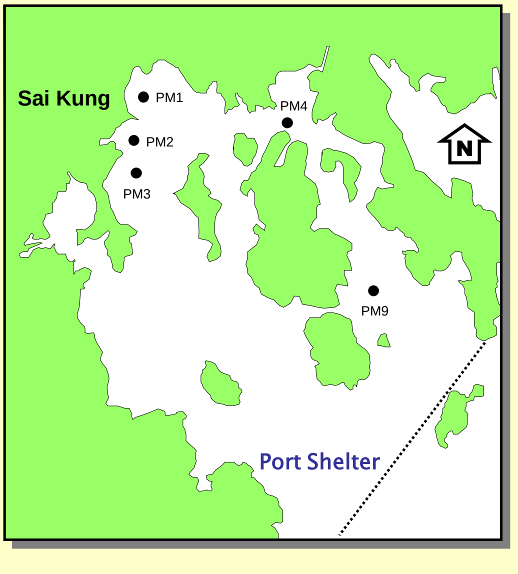
% sample with depth-averaged DO ≥ 4 mg/L

100% WQO 0%

Non-compliance

WQO compliance rates for the Port Shelter WCZ (continued)

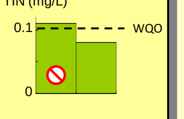




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.1 mg/L

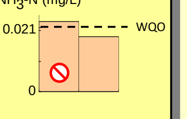
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

annual mean for depth-averaged NH_3-N

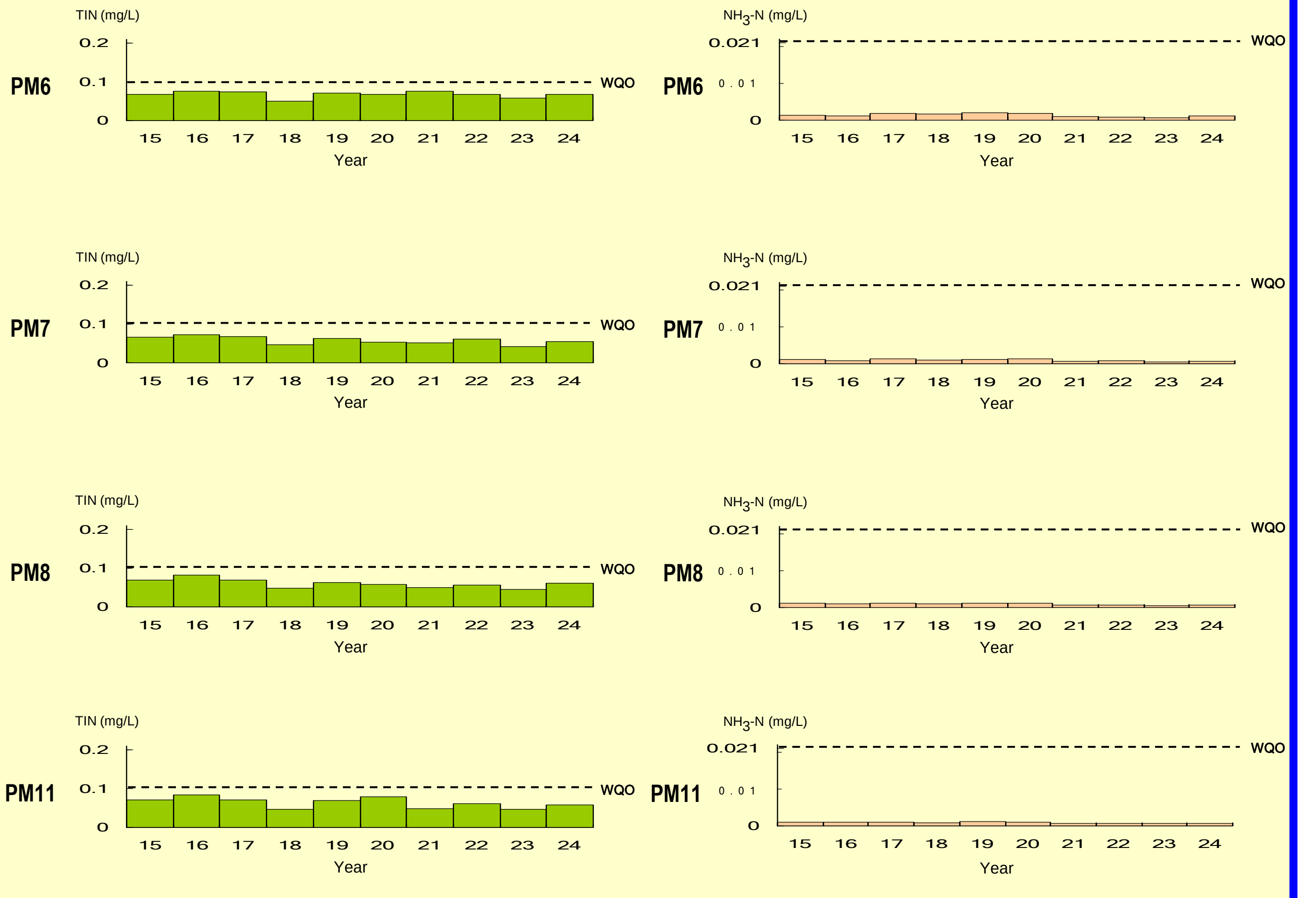


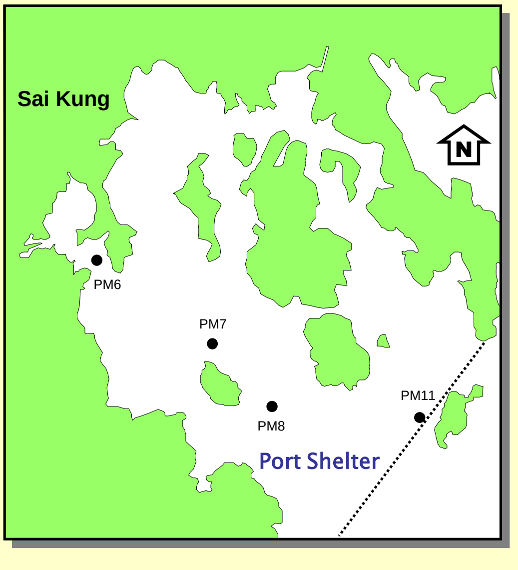
Non-compliance

WQO compliance rates for the Port Shelter WCZ (continued)

Total Inorganic Nitrogen (TIN)
(depth-averaged)

Unionised Ammonia (NH_3-N)
(depth-averaged)

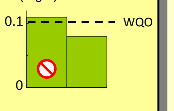




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.1 mg/L

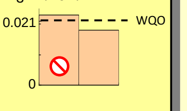
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

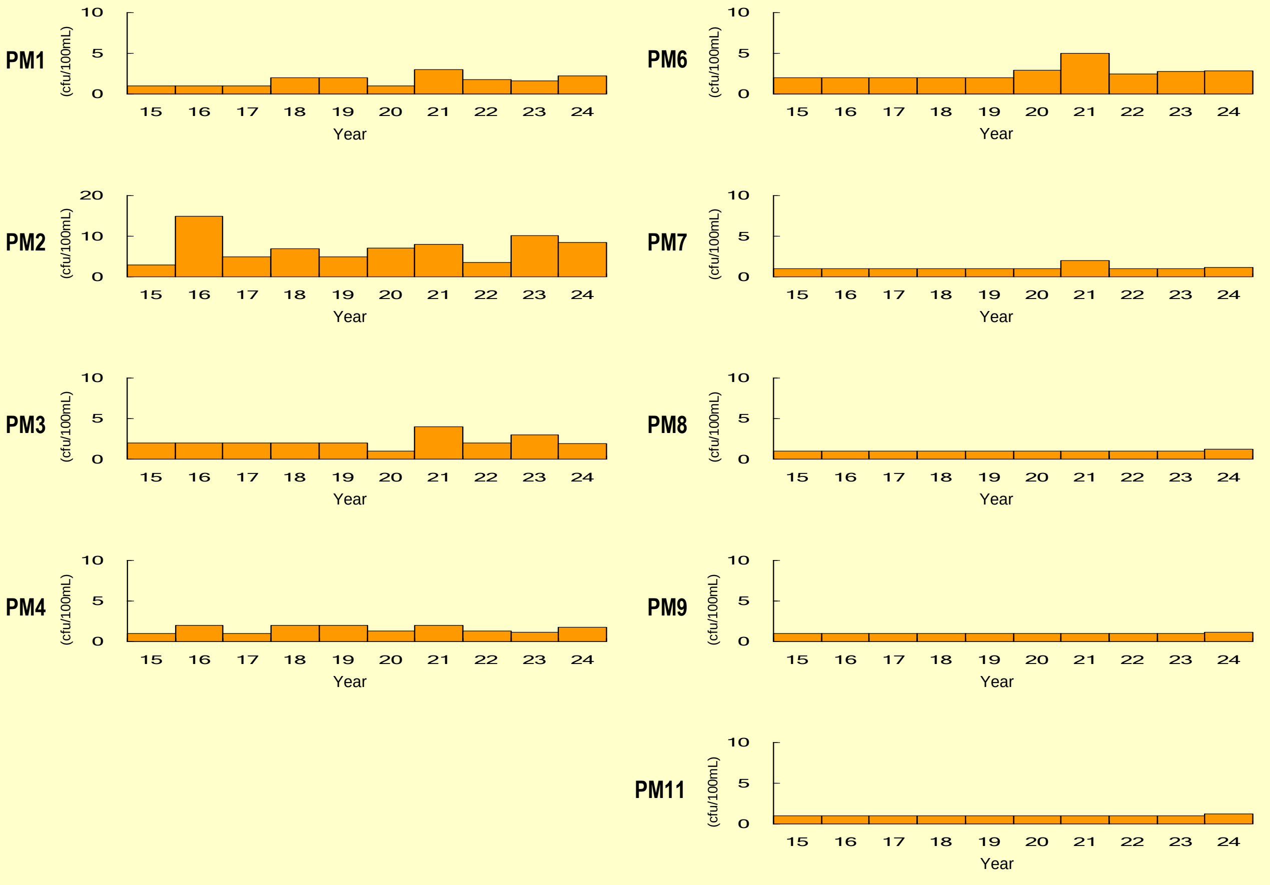
annual mean for depth-averaged NH_3-N

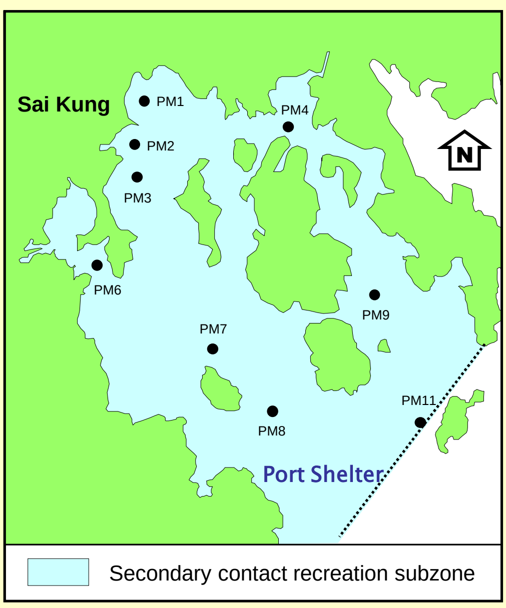


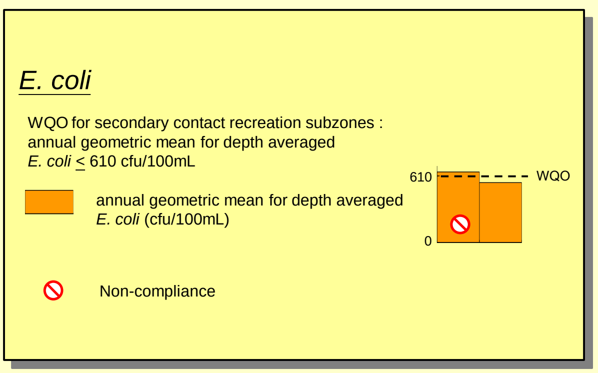
Non-compliance

WQO compliance rates for the Port Shelter WCZ (continued)

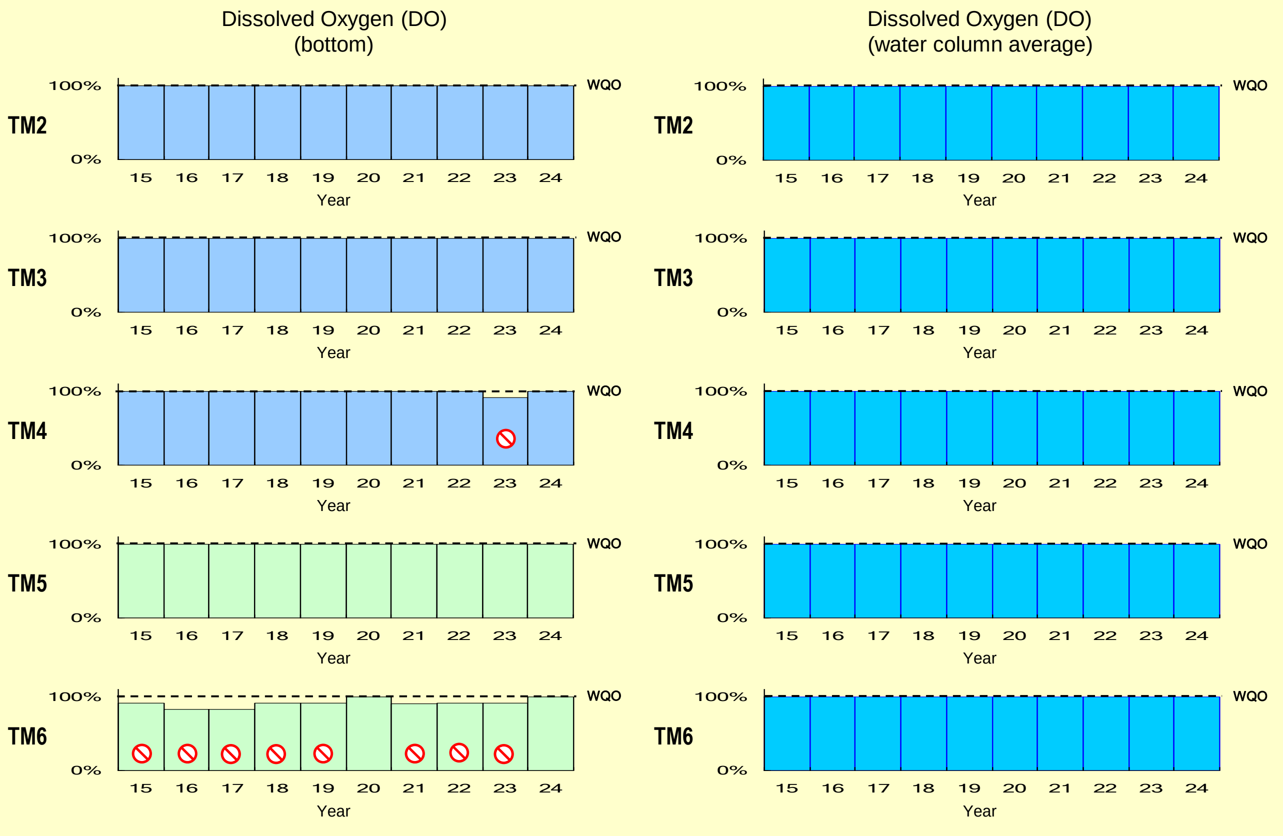
E. coli
(annual geometric mean)

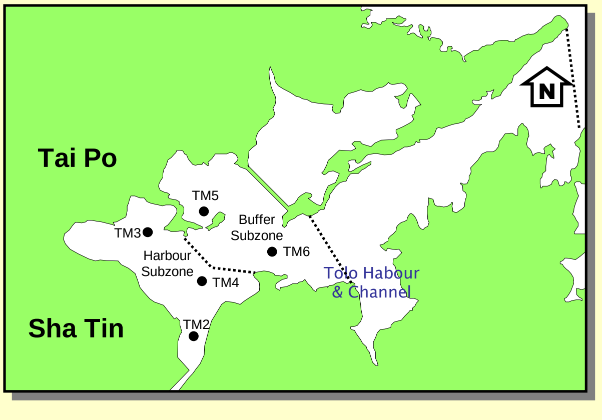






WQO compliance rates for the Tolo Harbour and Channel WCZ





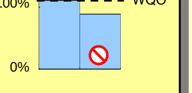
Dissolved Oxygen (DO)

Harbour Subzone (TM2 - TM4)

1. Bottom

WQO : 100% sample with bottom DO $\geq$ 2 mg/L

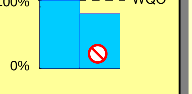
% sample with bottom DO $\geq$ 2 mg/L



2. Water column average (surface to 2m above bottom)

WQO : 100% sample with water column average DO $\geq$ 4 mg/L

% sample with water column average DO $\geq$ 4 mg/L

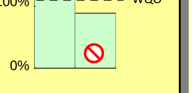


Buffer Subzone (TM5 - TM6)

1. Bottom

WQO : 100% sample with bottom DO $\geq$ 3 mg/L

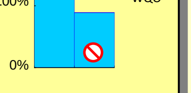
% sample with bottom DO $\geq$ 3 mg/L



2. Water column average (surface to 2m above bottom)

WQO : 100% sample with water column average DO $\geq$ 4 mg/L

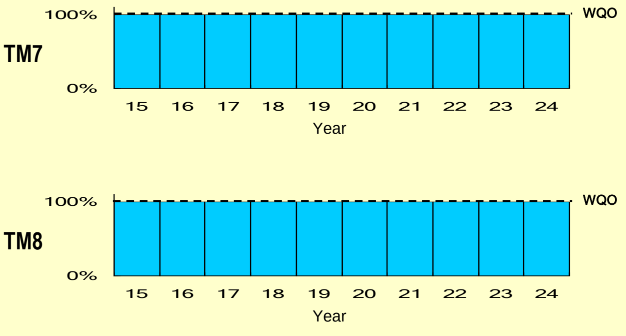
% sample with water column average DO $\geq$ 4 mg/L



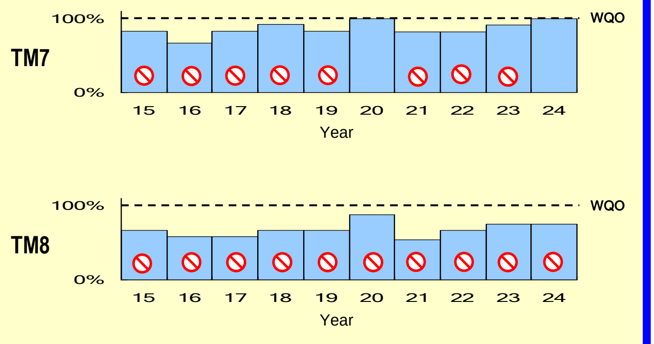
Non-compliance

WQO compliance rates for the Tolo Harbour and Channel WCZ (continued)

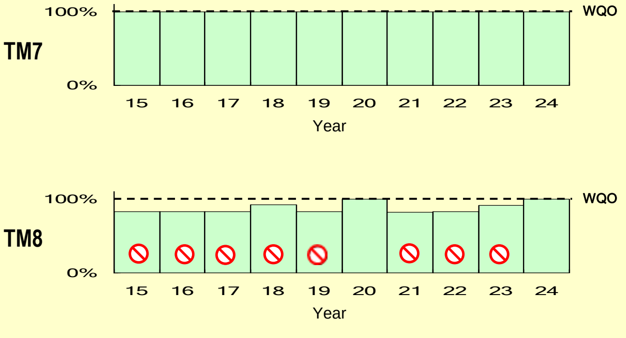
Dissolved Oxygen (DO)
(surface)



Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(middle)



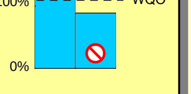
Dissolved Oxygen (DO)

Channel Subzone (TM7 - TM8)

1. Surface

WQO : 100% sample with surface DO $\geq$ 4 mg/L

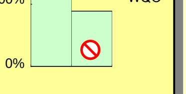
% sample with surface DO $\geq$ 4 mg/L



2. Middle

WQO : 100% sample with middle DO $\geq$ 4 mg/L

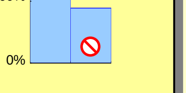
% sample with middle DO $\geq$ 4 mg/L



3. Bottom

WQO : 100% sample with bottom DO $\geq$ 4 mg/L

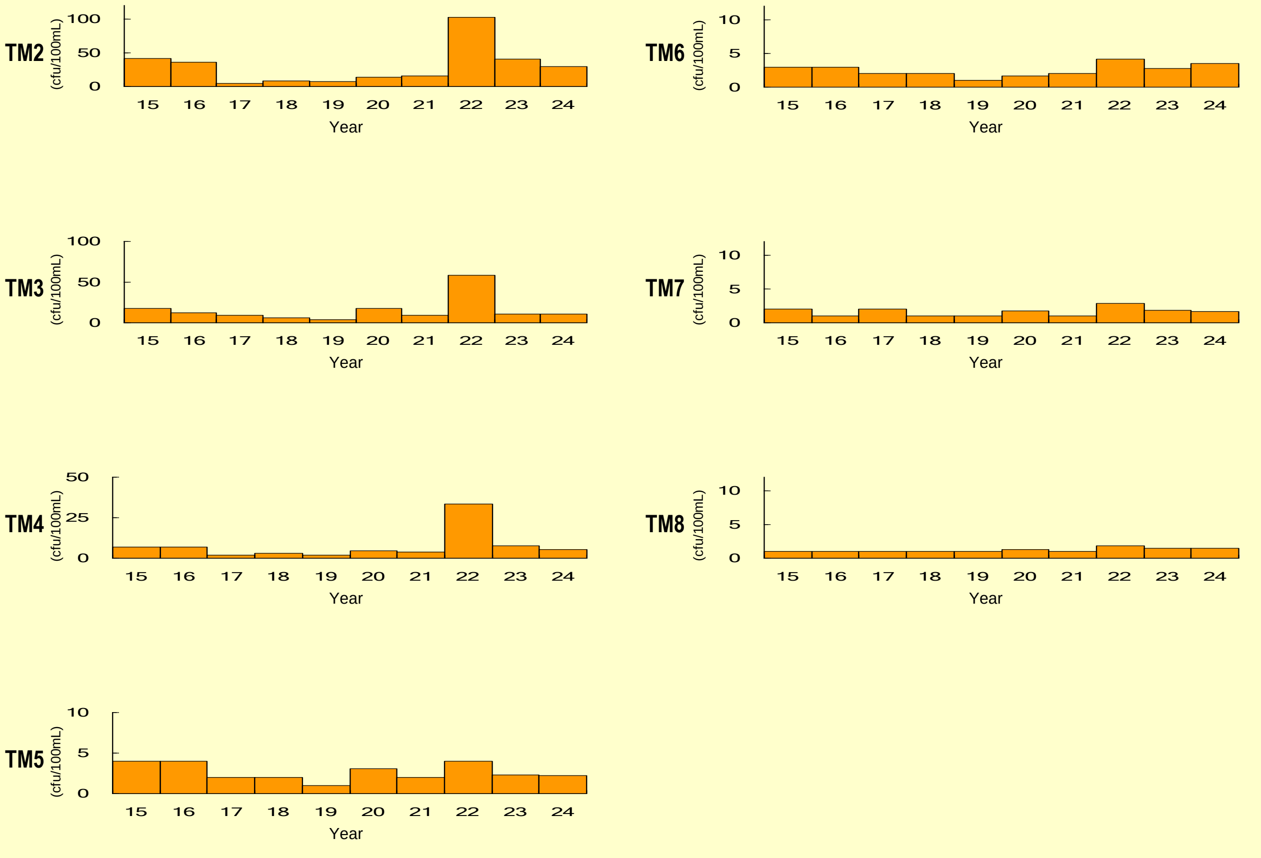
% sample with bottom DO $\geq$ 4 mg/L

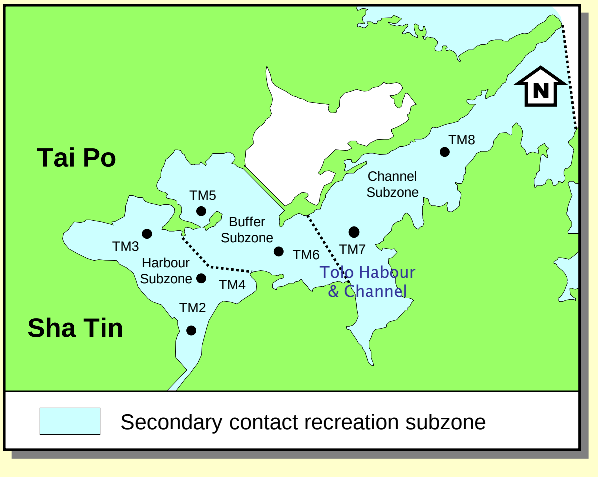


Non-compliance

WQO compliance rates for the Tolo Harbour and Channel WCZ (continued)

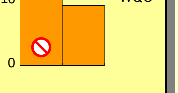
E. coli
(annual geometric mean)





E. coli

WQO for secondary contact recreation subzones :
annual geometric mean for depth averaged
E. coli ≤ 610 cfu/100mL

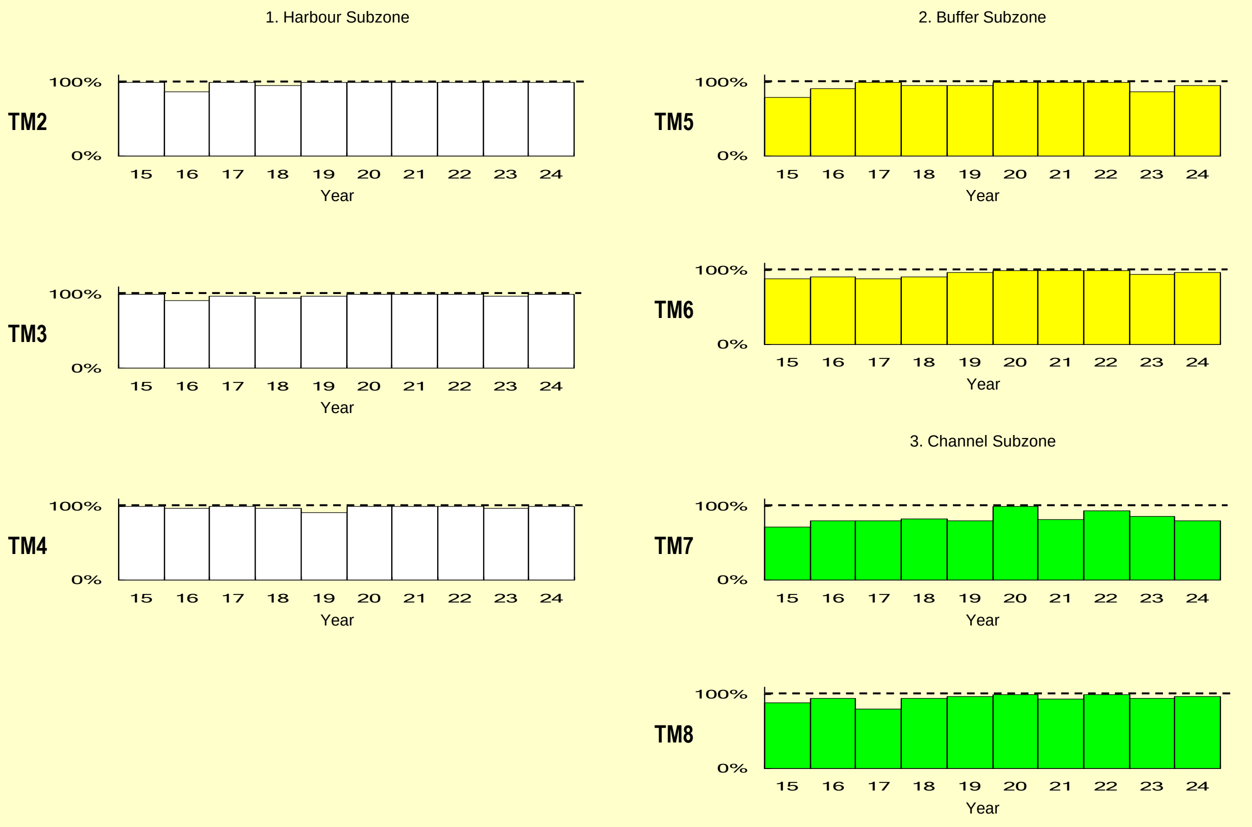


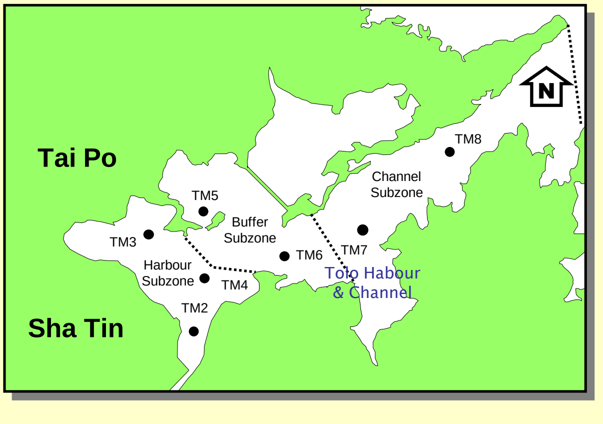
annual geometric mean for depth averaged *E. coli* (cfu/100mL)

Non-compliance

WQO compliance rates of chlorophyll-*a* for the Tolo Harbour and Channel WCZ

Chlorophyll-*a*





Chlorophyll-*a*

1. Harbour Subzone

% sample (S, M, B) with Chlorophyll-*a* ≤ 20 μg/L

WQO : Chlorophyll-*a* ≤ 20 μg/L

2. Buffer Subzone

% sample (S, M, B) with Chlorophyll-*a* ≤ 10 μg/L

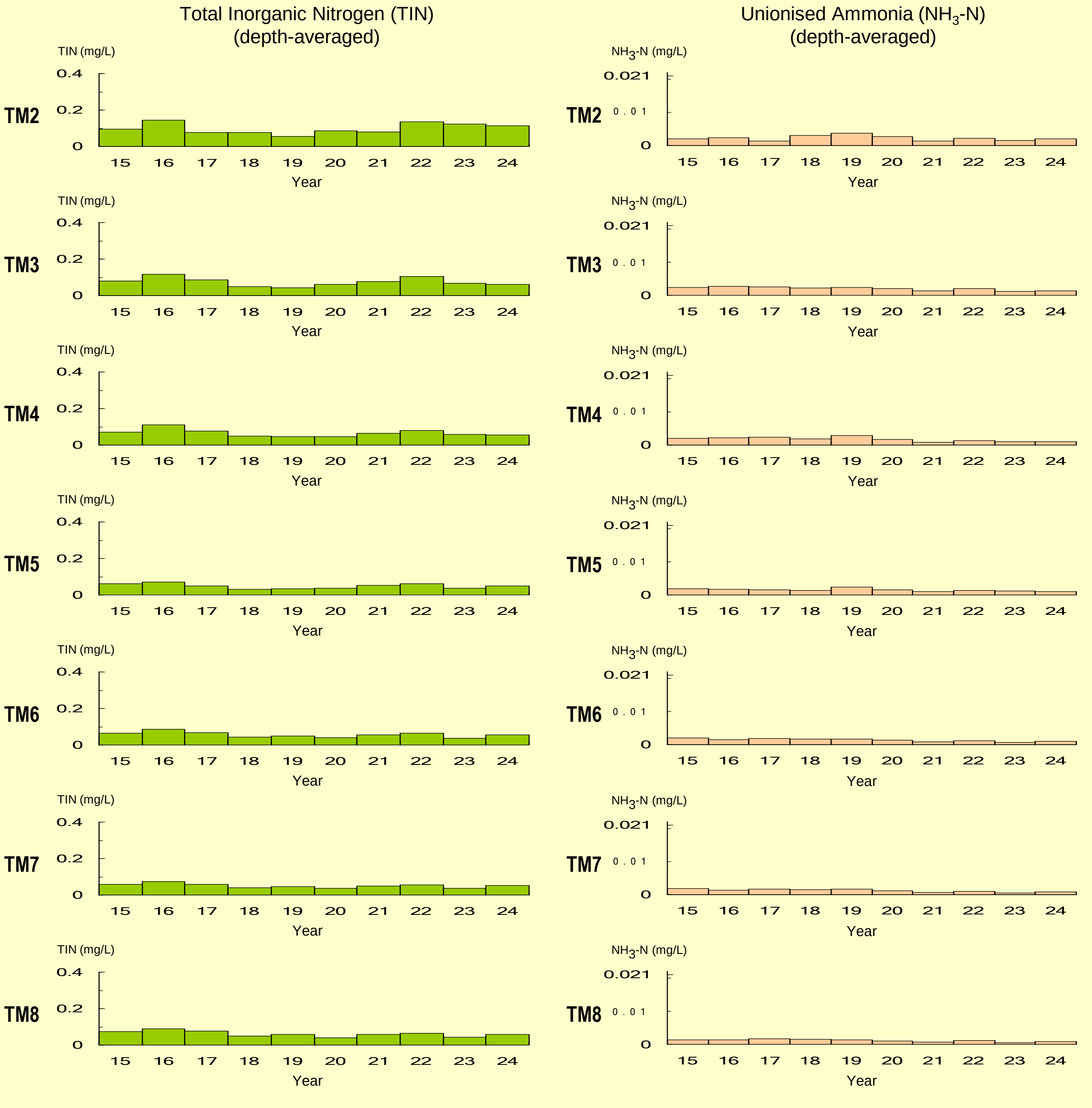
WQO : Chlorophyll-*a* ≤ 10 μg/L

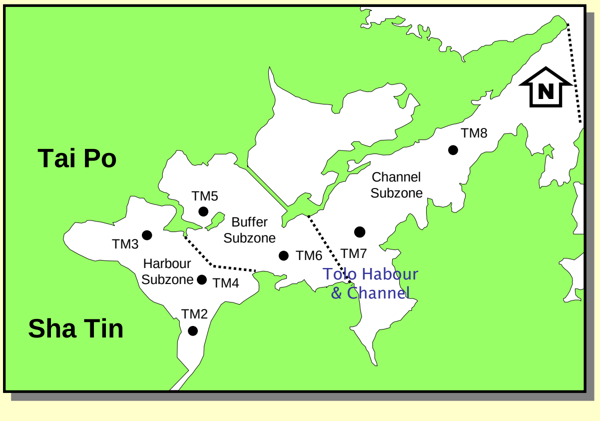
3. Channel Subzone

% sample (S, M, B) with Chlorophyll-*a* ≤ 6 μg/L

WQO : Chlorophyll-*a* ≤ 6 μg/L

Total inorganic nitrogen and unionised ammonia levels in the Tolo Harbour and Channel WCZ





Total Inorganic Nitrogen (TIN)

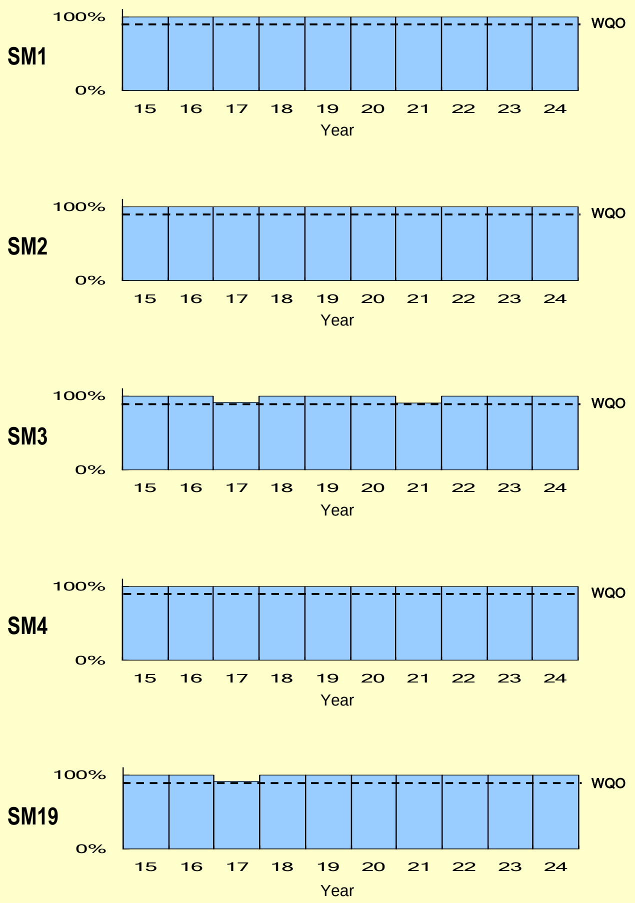
annual mean for depth-averaged TIN (mg/L)

Unionised Ammonia (NH_3-N)

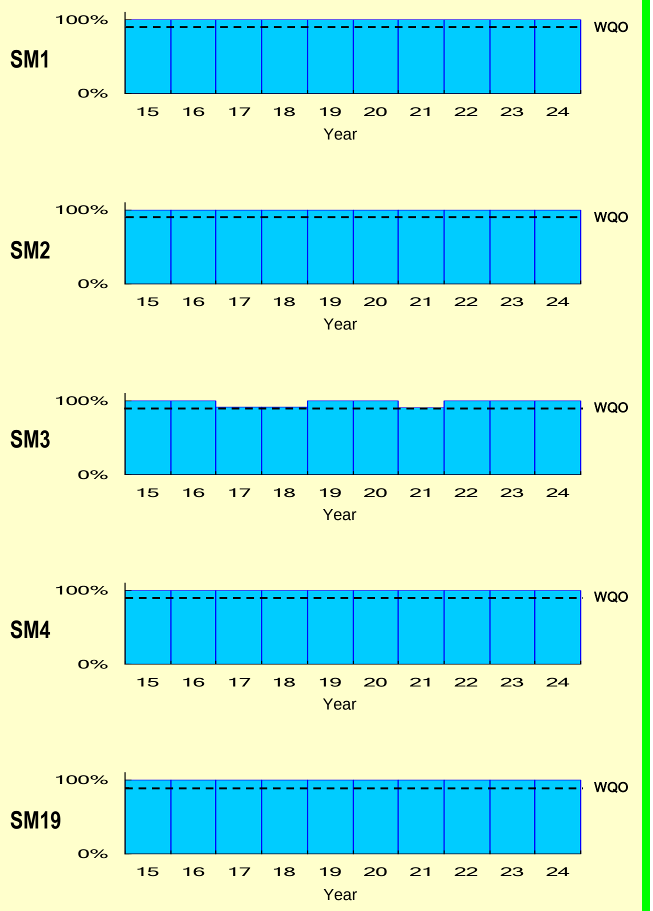
annual mean for depth-averaged NH_3-N (mg/L)

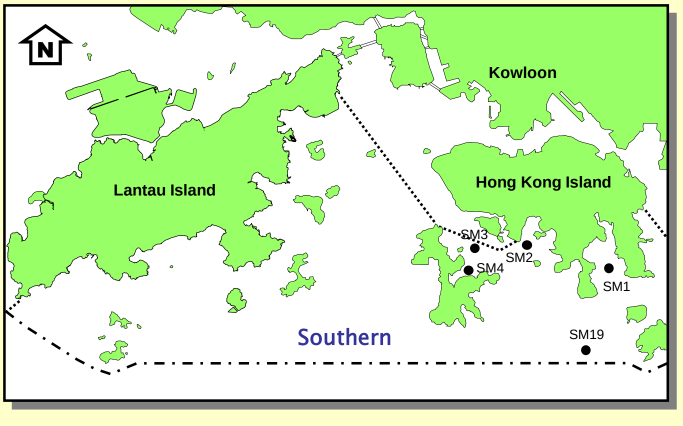
WQO compliance rates for the Southern WCZ

Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(depth-averaged)





Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

% sample with bottom DO ≥ 2 mg/L

2. Depth-averaged

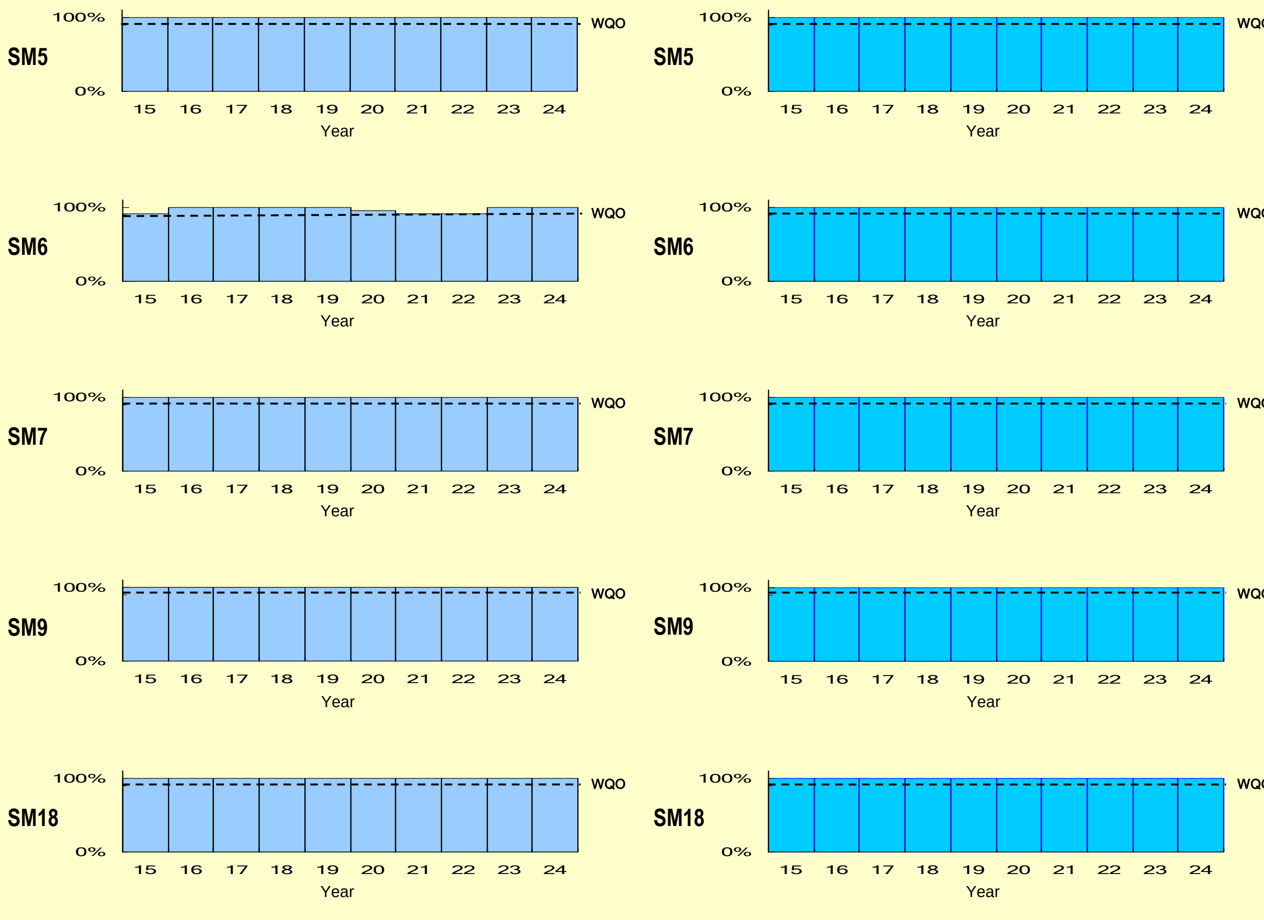
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

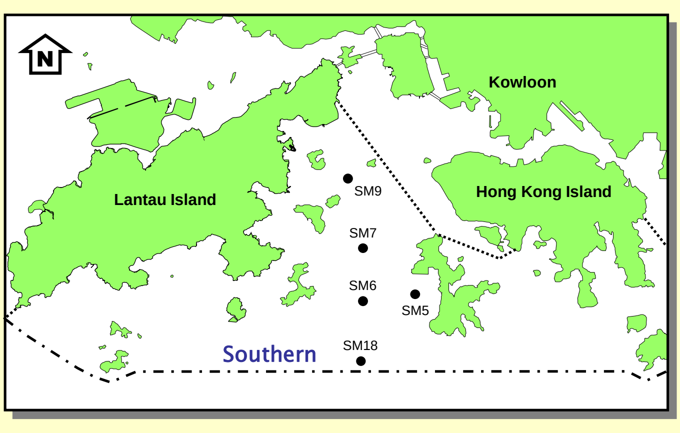
% sample with depth-averaged DO ≥ 4 mg/L

Non-compliance

WQO compliance rates for the Southern WCZ (continued)

Dissolved Oxygen (DO) (bottom) | Dissolved Oxygen (DO) (depth-averaged)



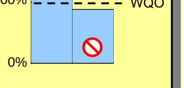


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

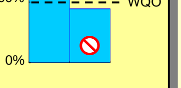
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

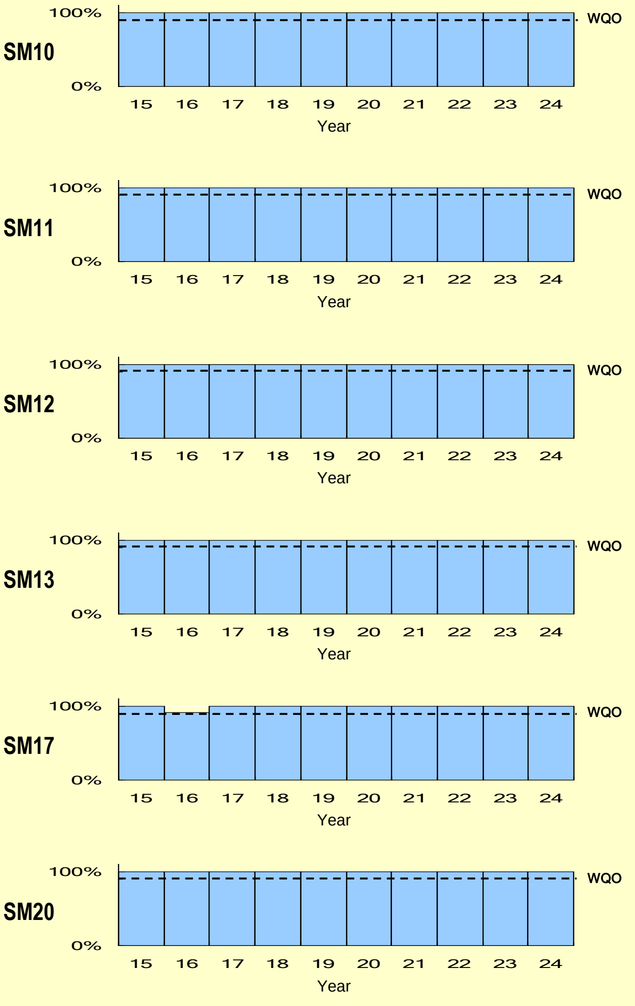
% sample with depth-averaged DO ≥ 4 mg/L



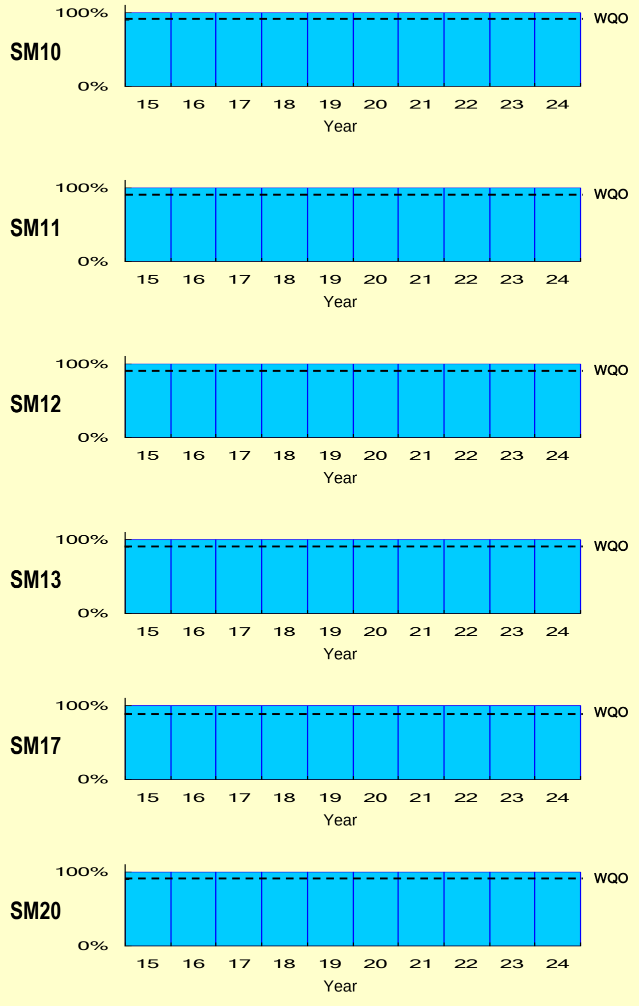
Non-compliance

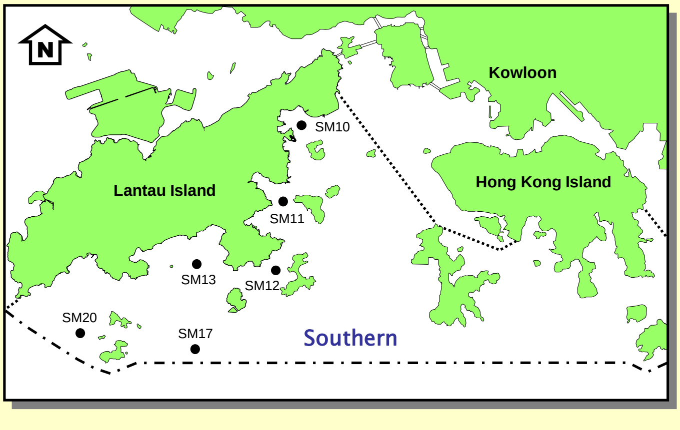
WQO compliance rates for the Southern WCZ (continued)

Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(depth-averaged)



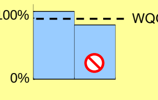


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

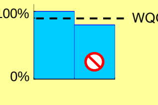
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

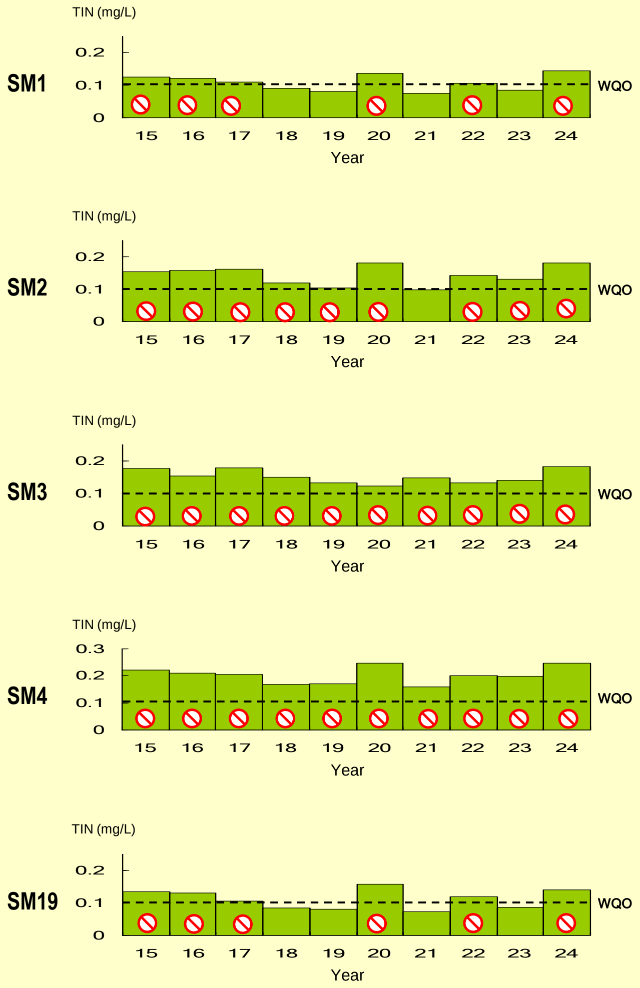
% sample with depth-averaged DO ≥ 4 mg/L



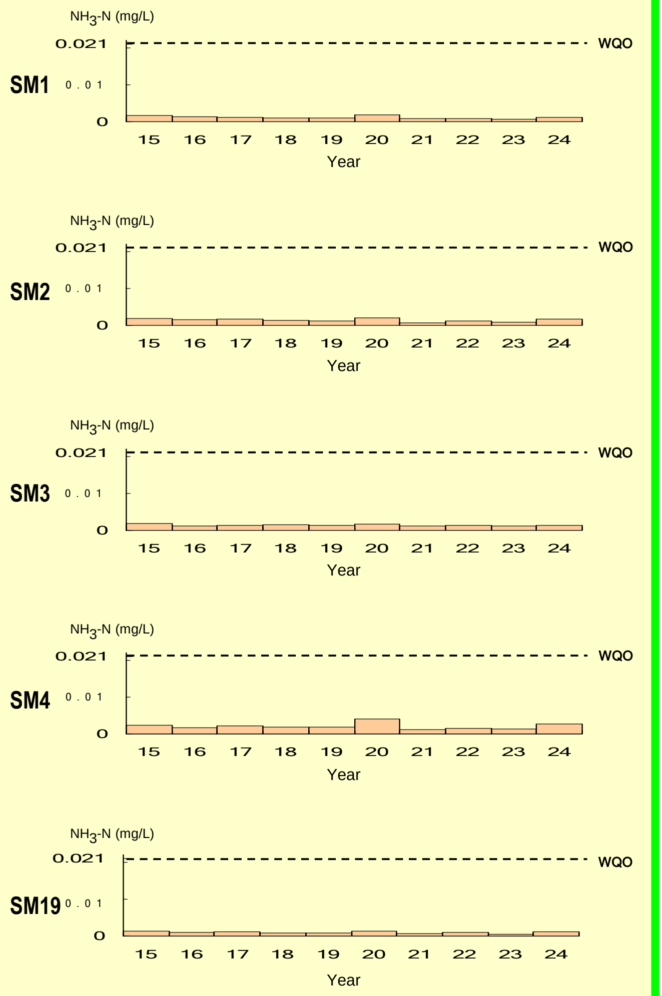
Non-compliance

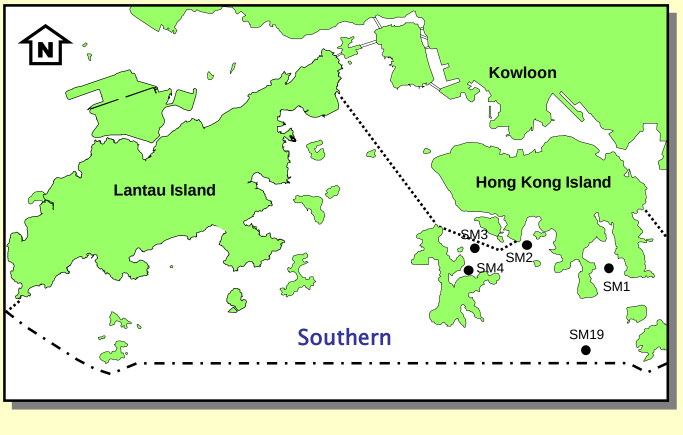
WQO compliance rates for the Southern WCZ (continued)

Total Inorganic Nitrogen (TIN)
(depth-averaged)



Unionised Ammonia (NH_3-N)
(depth-averaged)

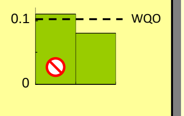




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.1 mg/L

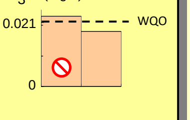
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

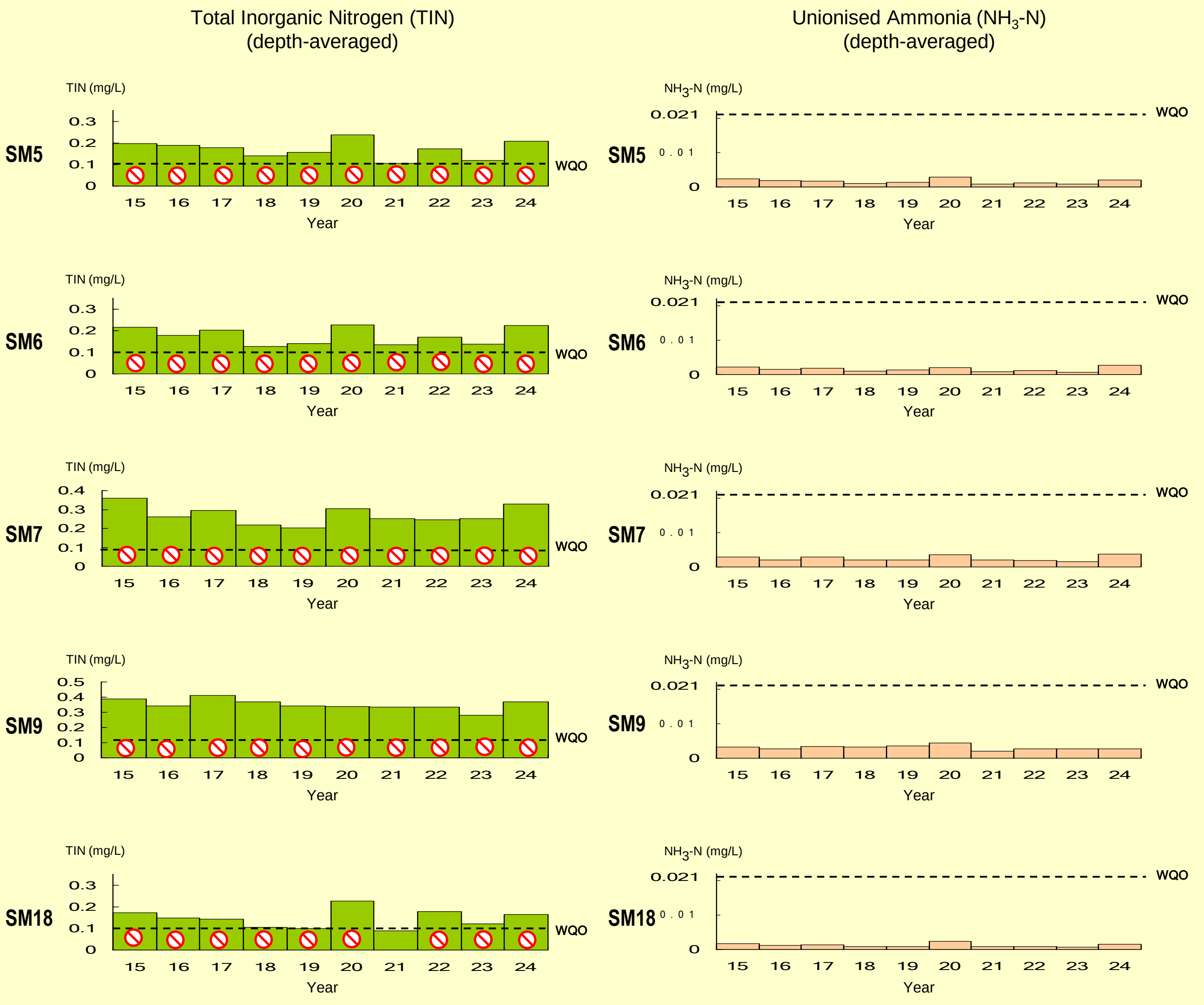
WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

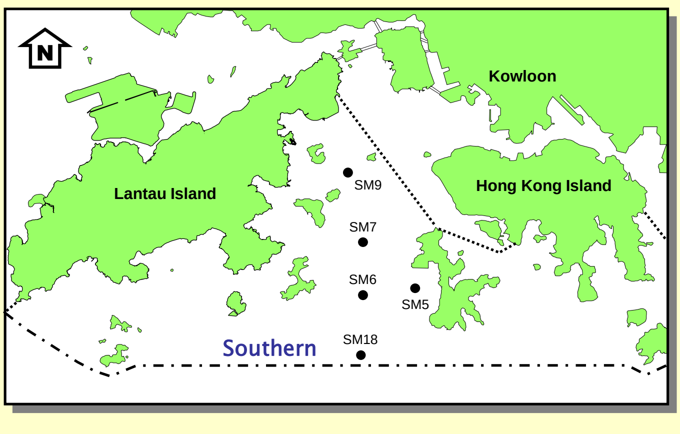
annual mean for depth-averaged NH_3-N

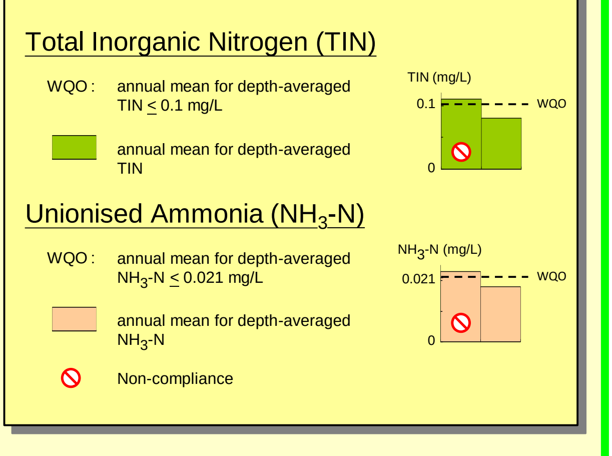


Non-compliance

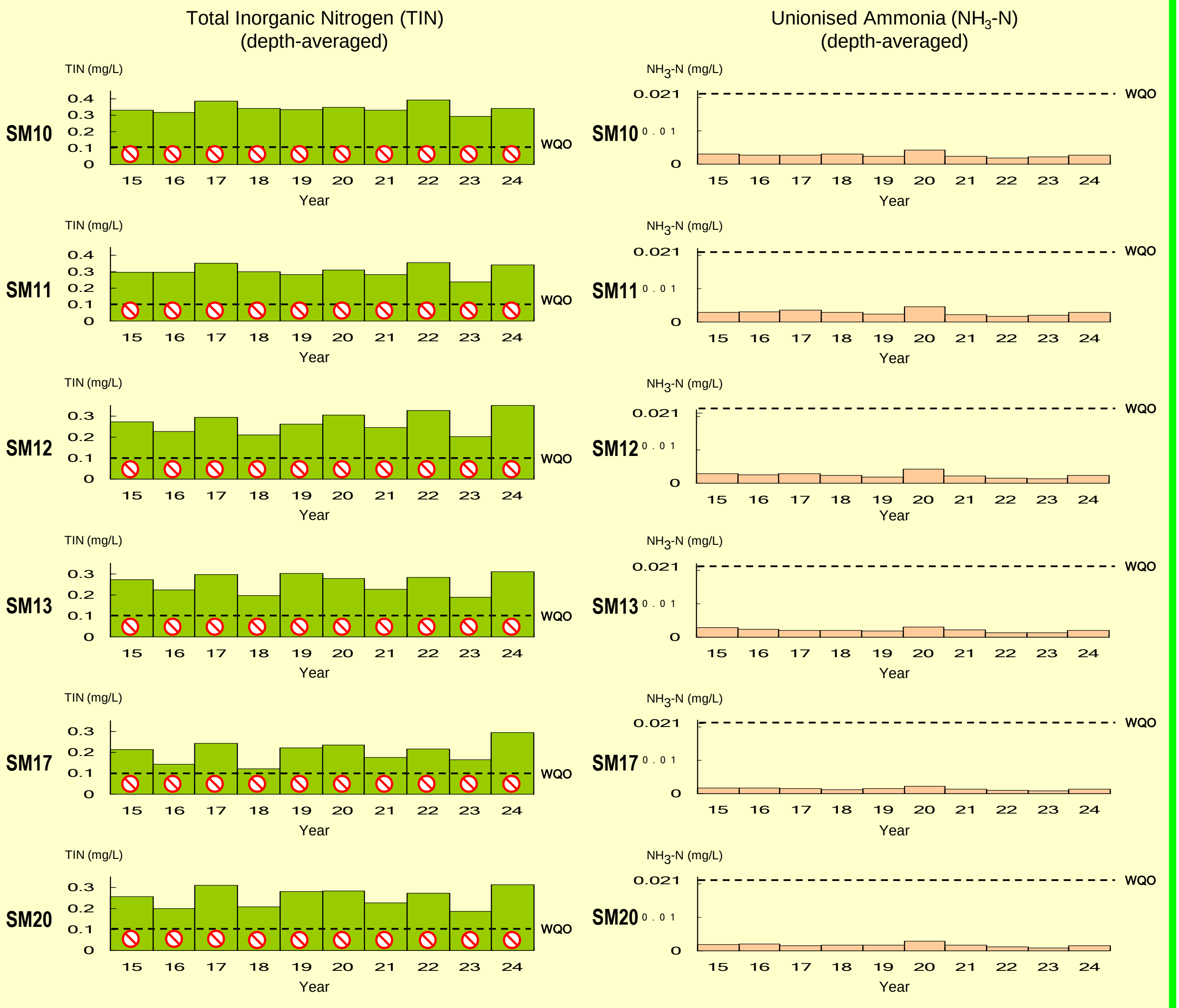
WQO compliance rates for the Southern WCZ (continued)

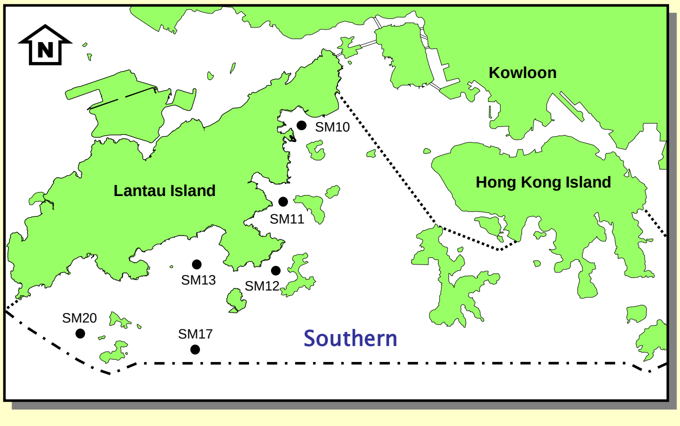






WQO compliance rates for the Southern WCZ (continued)

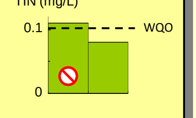




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.1 mg/L

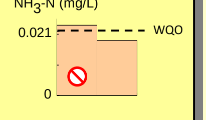
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

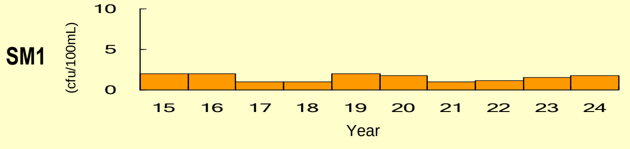
annual mean for depth-averaged NH_3-N

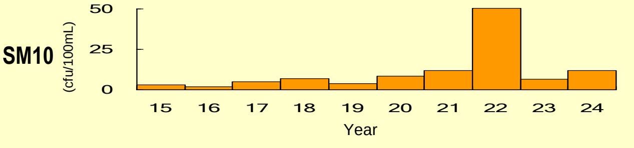


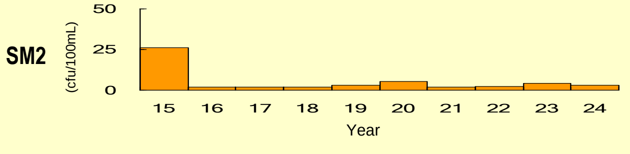
Non-compliance

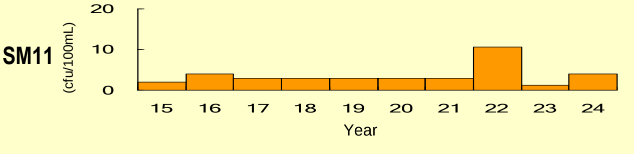
WQO compliance rates for the Southern WCZ (continued)

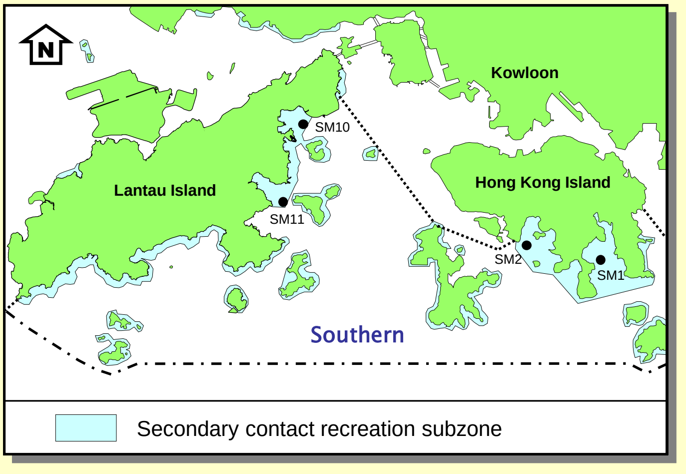
E. coli
(annual geometric mean)

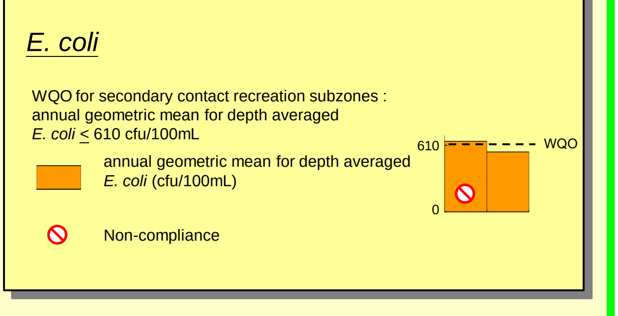






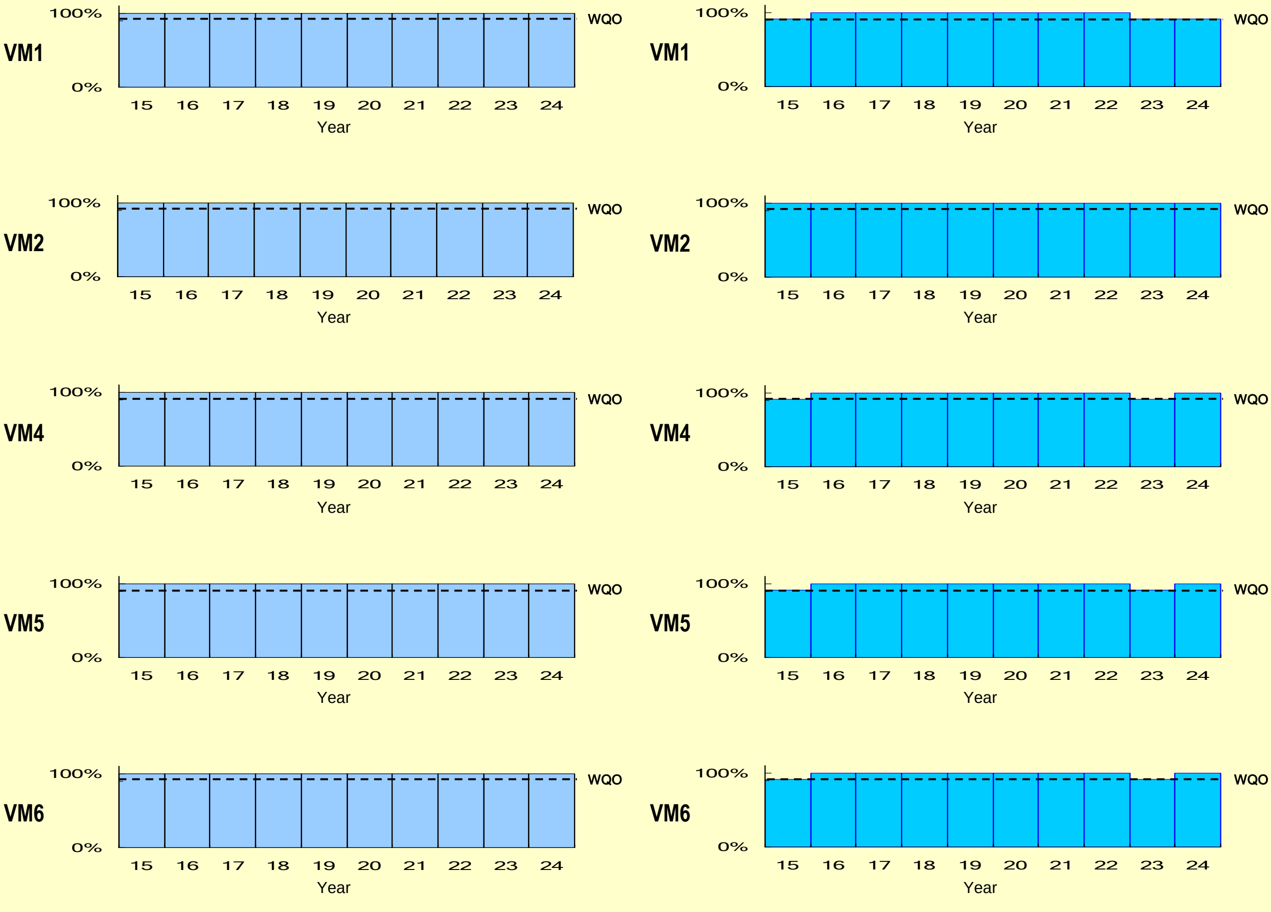


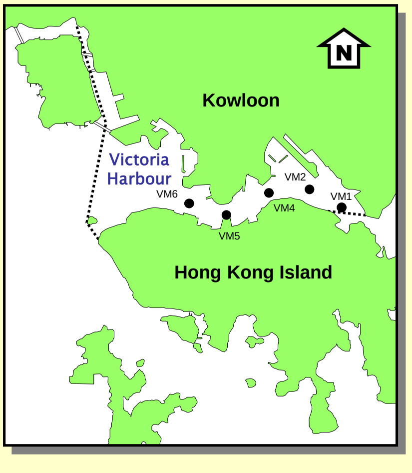




WQO compliance rates for the Victoria Harbour WCZ

Dissolved Oxygen (DO) (bottom) | Dissolved Oxygen (DO) (depth-averaged)



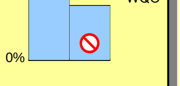


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO $\geq$ 2 mg/L

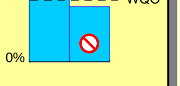
% sample with bottom DO $\geq$ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO $\geq$ 4 mg/L

% sample with depth-averaged DO $\geq$ 4 mg/L



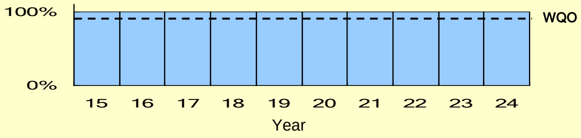
Non-compliance

WQO compliance rates for the Victoria Harbour WCZ (continued)

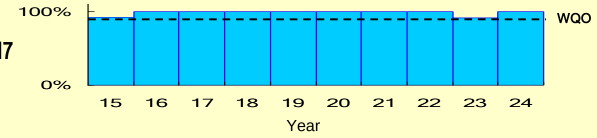
Dissolved Oxygen (DO)
(bottom)

Dissolved Oxygen (DO)
(depth-averaged)

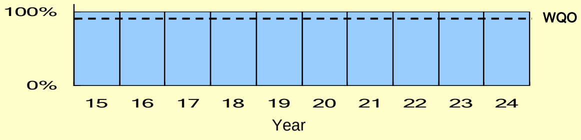
VM7



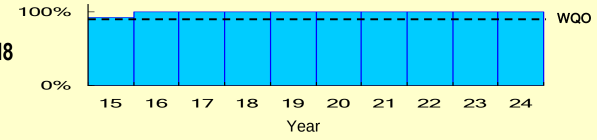
VM7



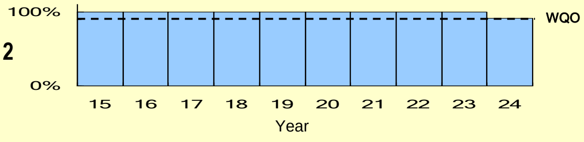
VM8



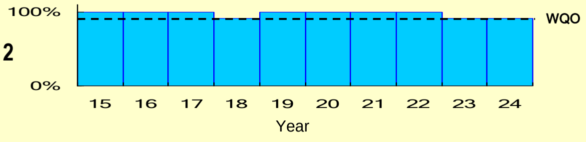
VM8



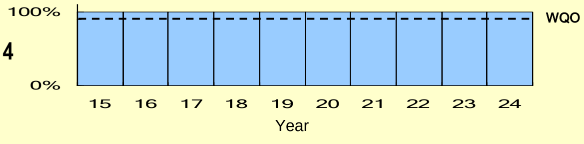
VM12



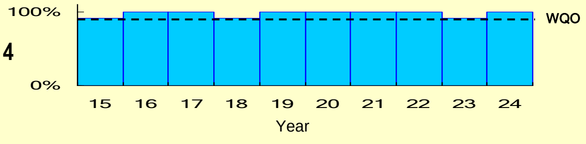
VM12



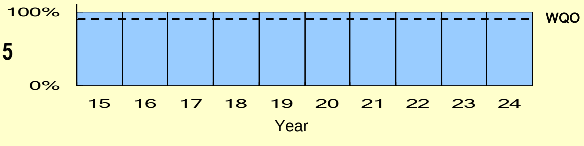
VM14



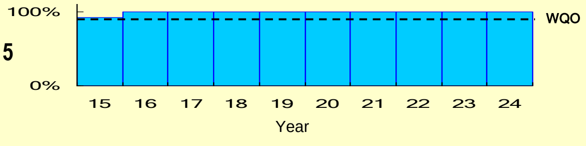
VM14



VM15



VM15

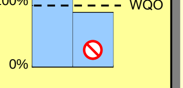


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

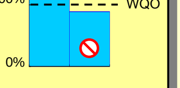
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

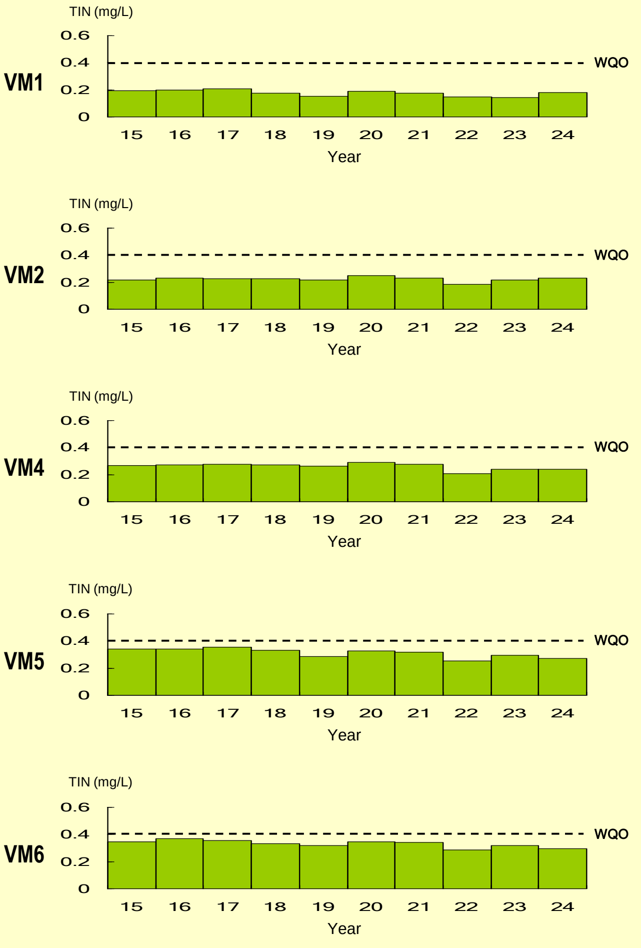
% sample with depth-averaged DO ≥ 4 mg/L



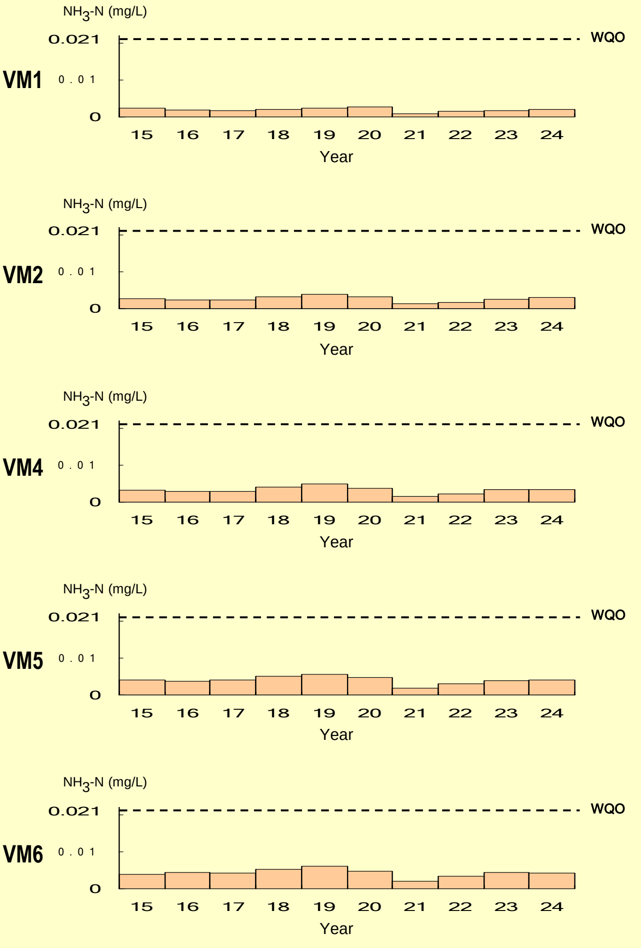
Non-compliance

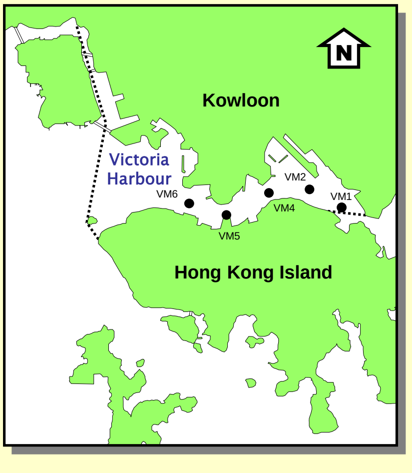
WQO compliance rates for the Victoria Harbour WCZ (continued)

Total Inorganic Nitrogen (TIN) (depth-averaged)



Unionised Ammonia (NH_3-N) (depth-averaged)

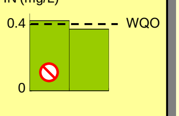




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.4 mg/L

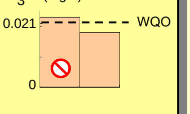
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

annual mean for depth-averaged NH_3-N

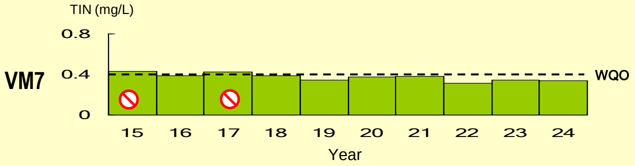


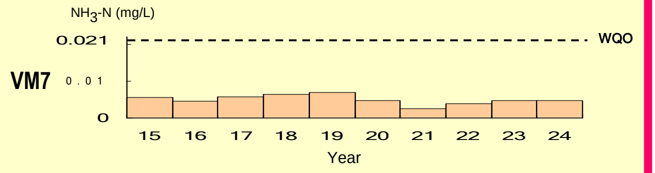
Non-compliance

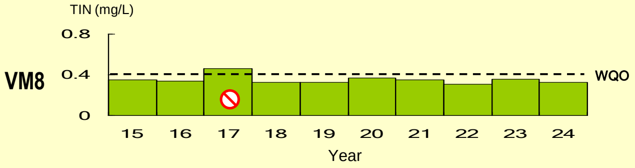
WQO compliance rates for the Victoria Harbour WCZ (continued)

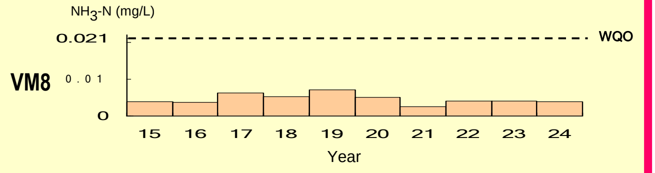
Total Inorganic Nitrogen (TIN) (depth-averaged)

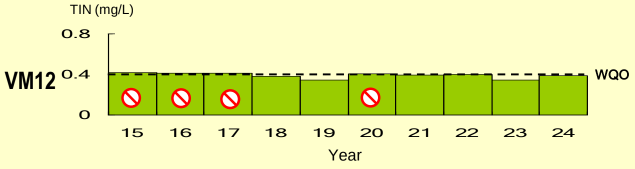
Unionised Ammonia (NH_3-N) (depth-averaged)

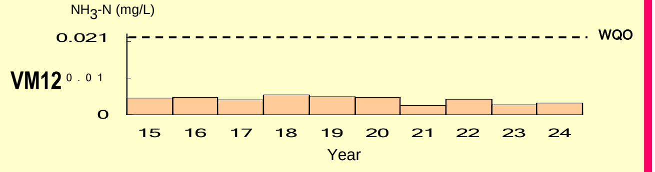


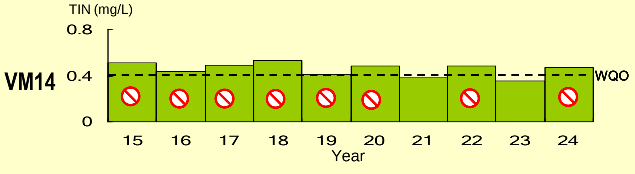


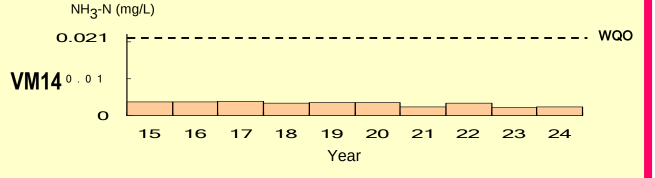


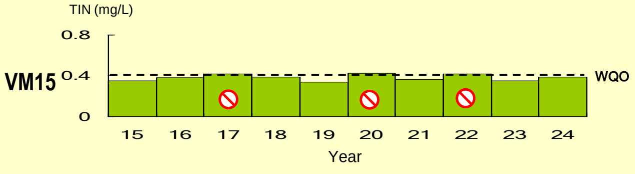


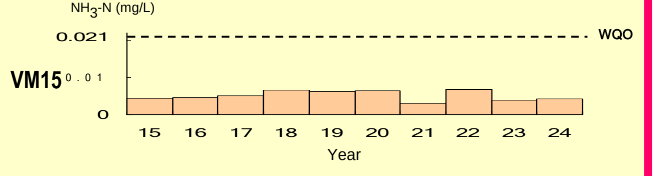








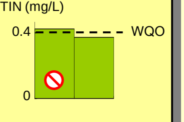




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN ≤ 0.4 mg/L

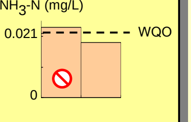
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

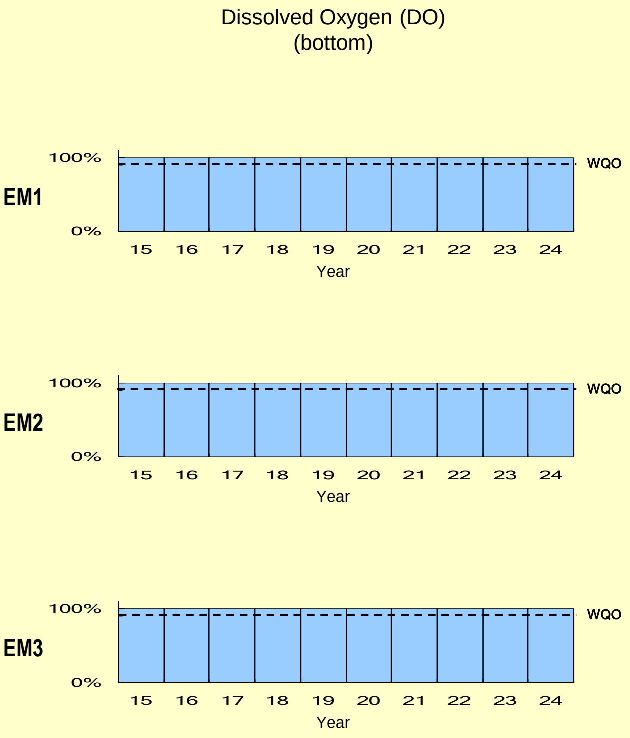
WQO : annual mean for depth-averaged NH_3-N ≤ 0.021 mg/L

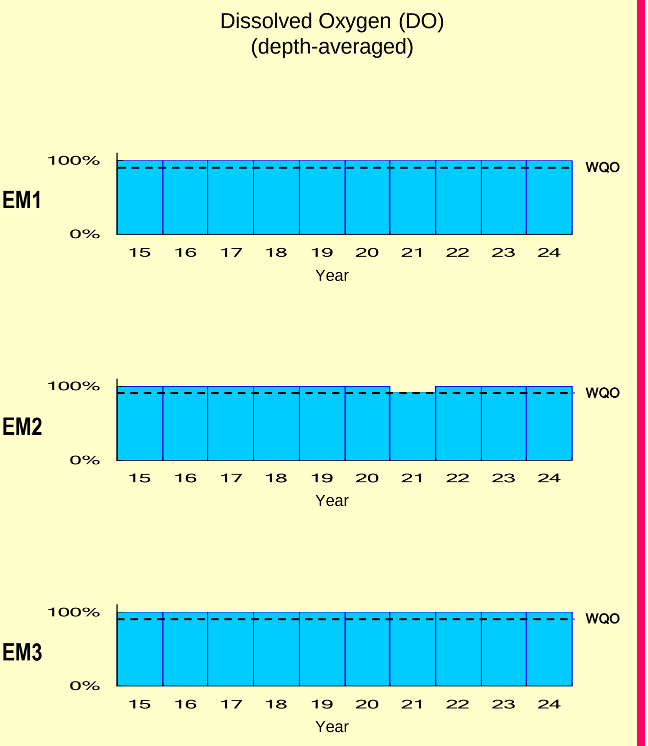
annual mean for depth-averaged NH_3-N

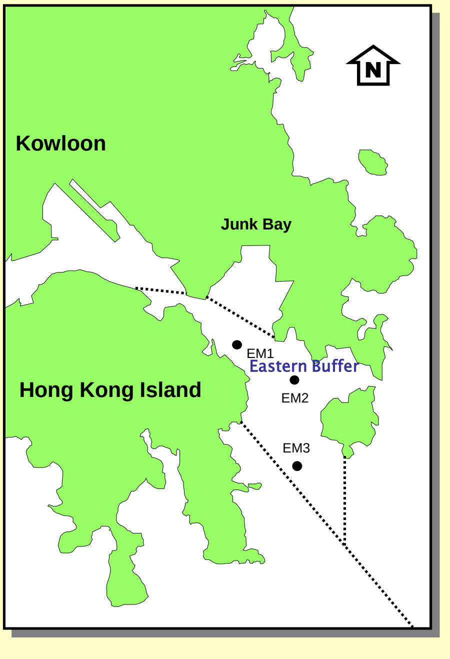


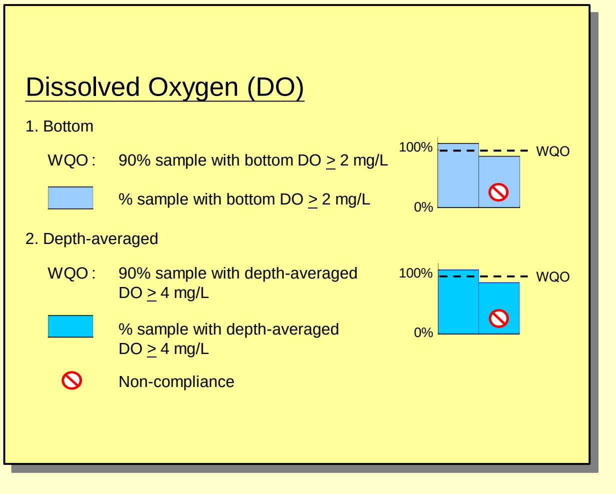
Non-compliance

WQO compliance rates for the Eastern Buffer WCZ







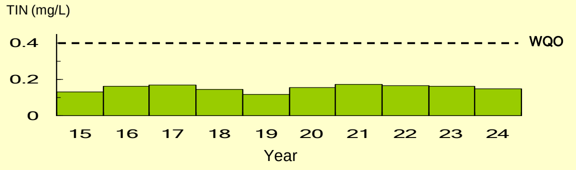


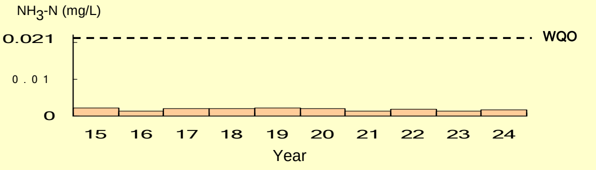
WQO compliance rates for the Eastern Buffer WCZ (continued)

Total Inorganic Nitrogen (TIN)
(depth-averaged)

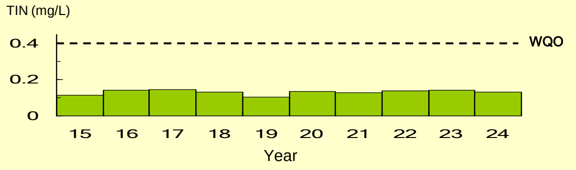
Unionised Ammonia (NH_3-N)
(depth-averaged)

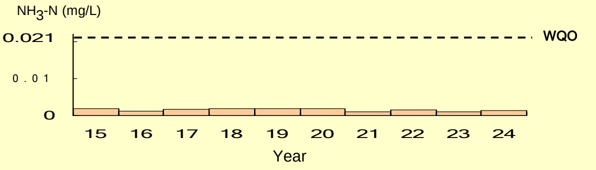
EM1



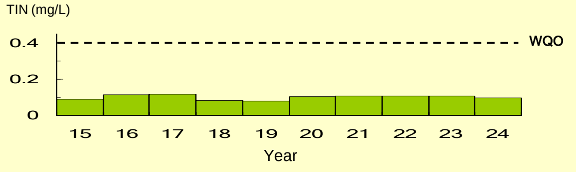


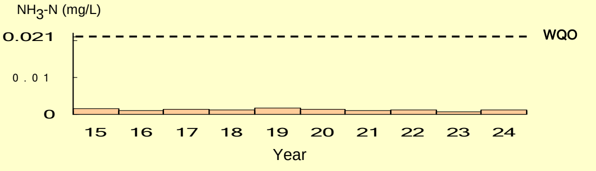
EM2

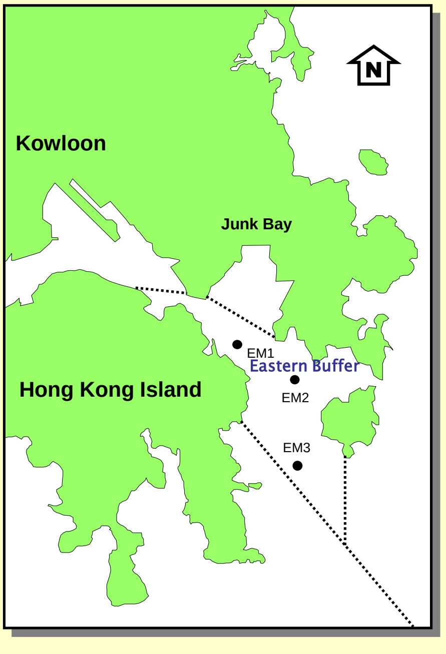




EM3







Total Inorganic Nitrogen (TIN)

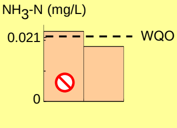
WQO : annual mean for depth-averaged TIN ≤ 0.4 mg/L

annual mean for depth-averaged TIN

Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N ≤ 0.021 mg/L

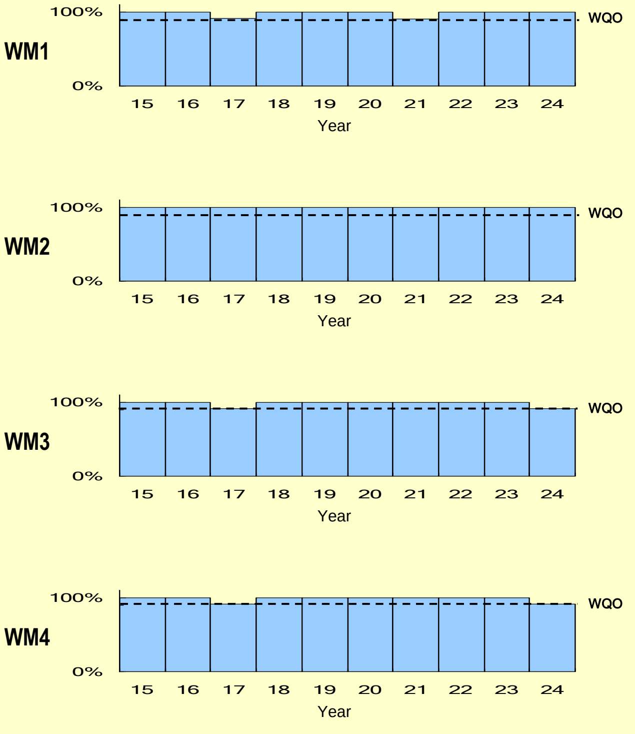
annual mean for depth-averaged NH_3-N



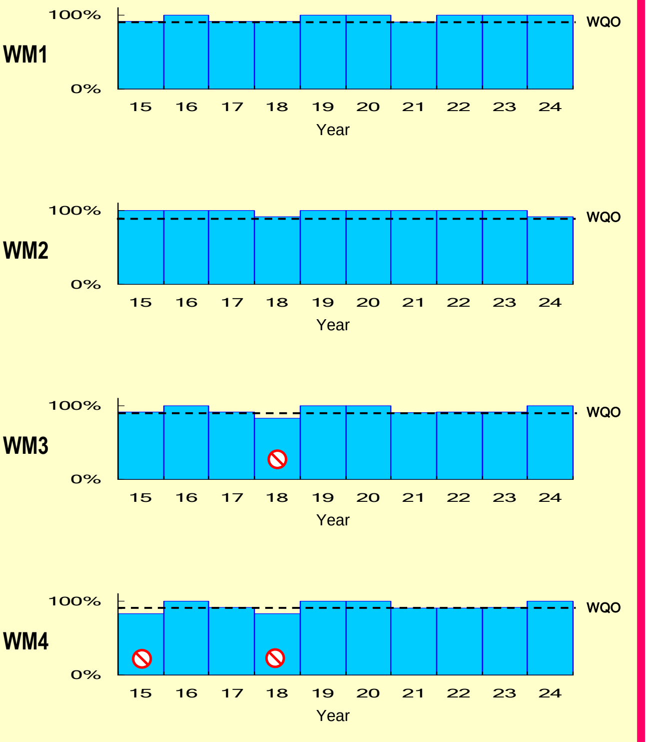
Non-compliance

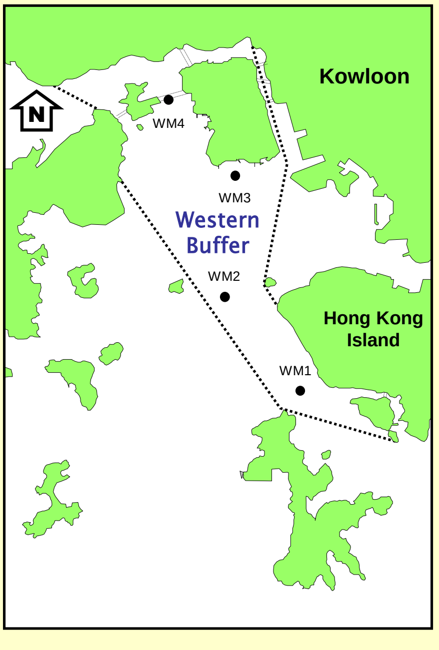
WQO compliance rates for the Western Buffer WCZ

Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(depth-averaged)



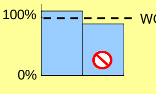


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

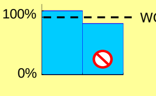
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

% sample with depth-averaged DO ≥ 4 mg/L

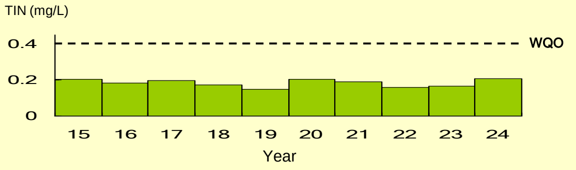


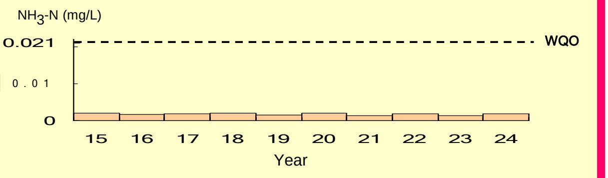
Non-compliance

WQO compliance rates for the Western Buffer WCZ (continued)

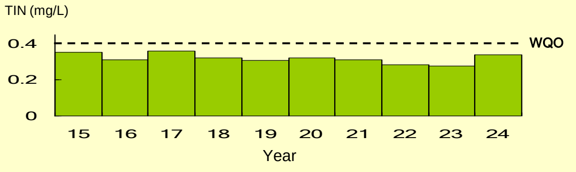
Total Inorganic Nitrogen (TIN) (depth-averaged)	Unionised Ammonia (NH_3-N) (depth-averaged)

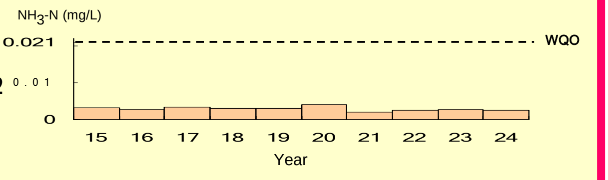
WM1



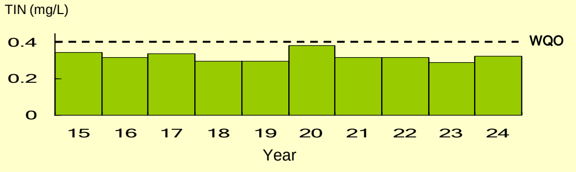


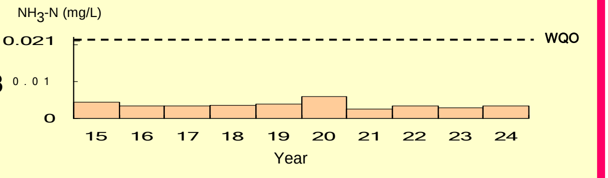
WM2



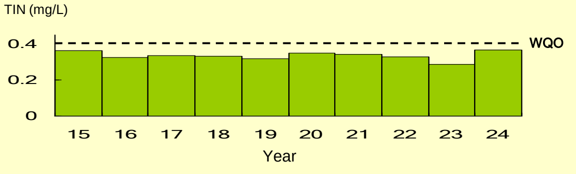


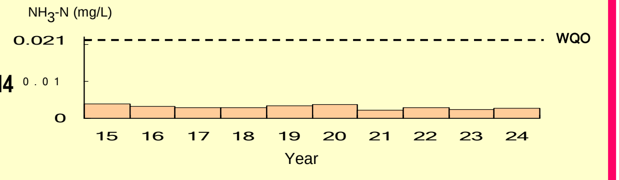
WM3

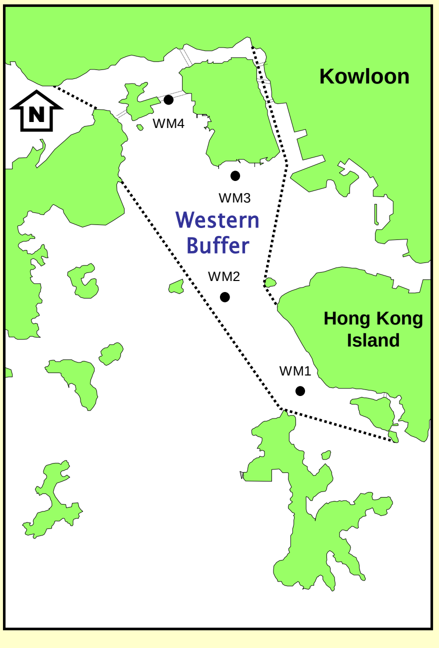




WM4



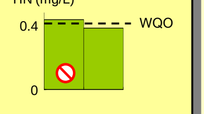




Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.4 mg/L

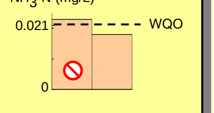
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

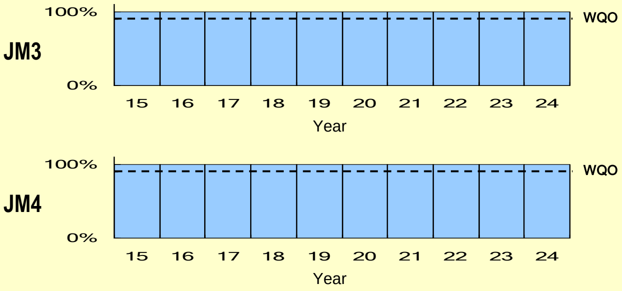
annual mean for depth-averaged NH_3-N



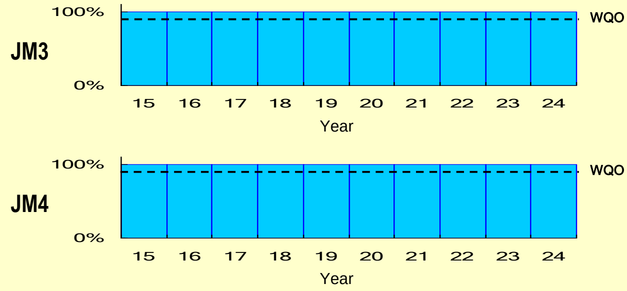
Non-compliance

WQO compliance rates for the Junk Bay WCZ

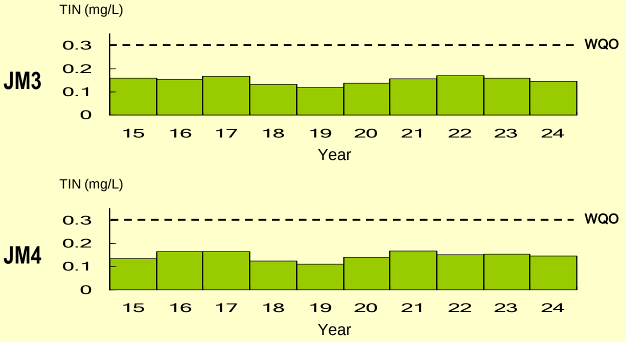
Dissolved Oxygen (DO)
(bottom)



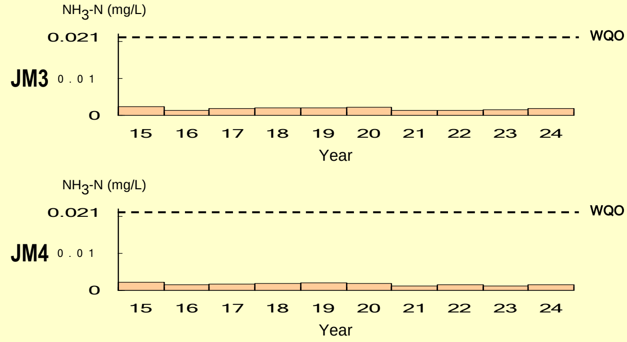
Dissolved Oxygen (DO)
(depth-averaged)

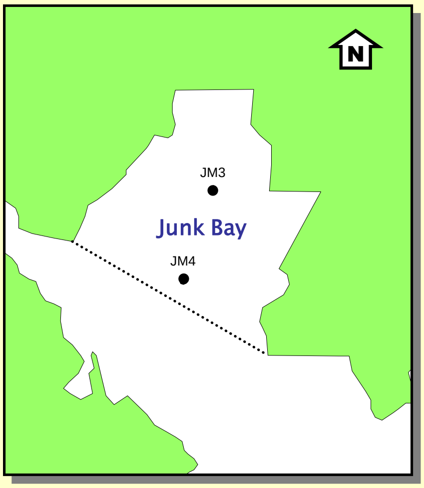


Total Inorganic Nitrogen (TIN)
(depth-averaged)



Unionised Ammonia (NH_3-N)
(depth-averaged)



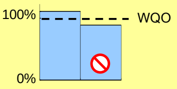


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO $\geq$ 2 mg/L

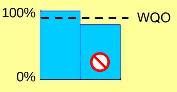
% sample with bottom DO $\geq$ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO $\geq$ 4 mg/L

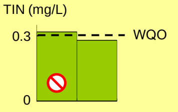
% sample with depth-averaged DO $\geq$ 4 mg/L



Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged TIN $\leq$ 0.3 mg/L

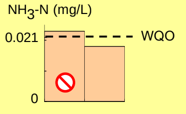
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

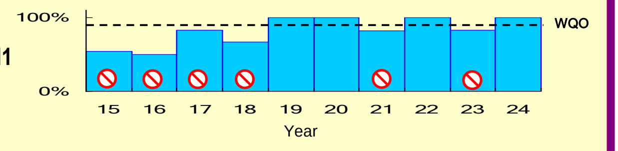
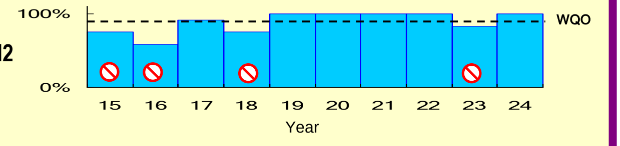
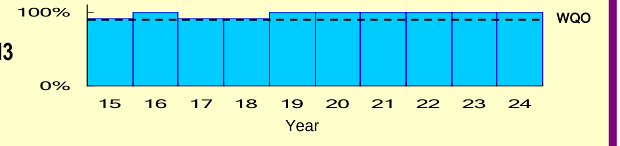
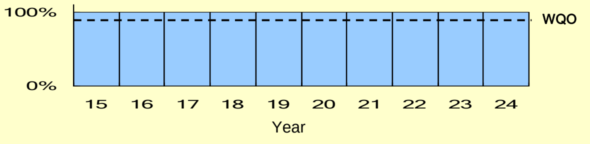
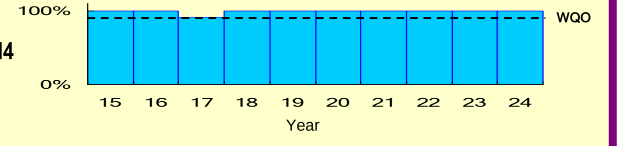
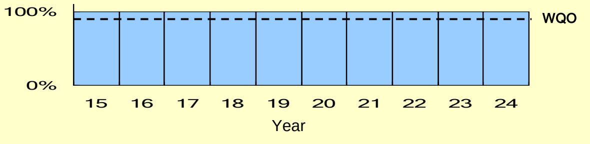
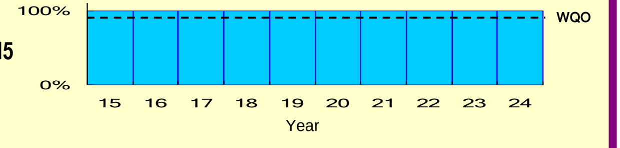
WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

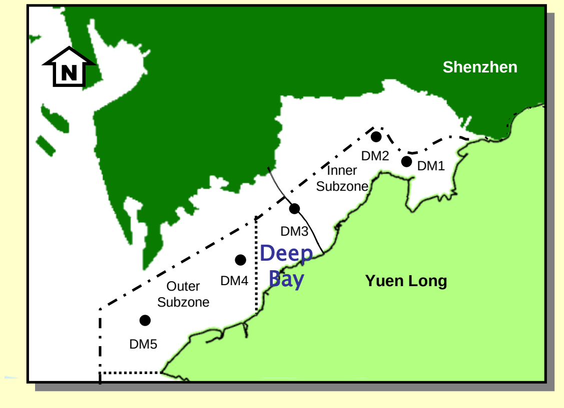
annual mean for depth-averaged NH_3-N



Non-compliance

WQO compliance rates for the Deep Bay WCZ

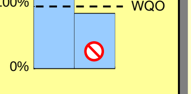
	Dissolved Oxygen (DO) (bottom)		Dissolved Oxygen (DO) (depth-averaged)
DM1	DO (bottom) not measured, water depth < 3m	**DM1**	
DM2	DO (bottom) not measured, water depth < 3m	**DM2**	
DM3	DO (bottom) not measured, water depth < 3m	**DM3**	
DM4		**DM4**	
DM5		**DM5**	



Dissolved Oxygen (DO)

1. Bottom

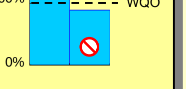
WQO : 90% sample with bottom DO ≥ 2 mg/L



% sample with bottom DO ≥ 2 mg/L

2. Depth-averaged

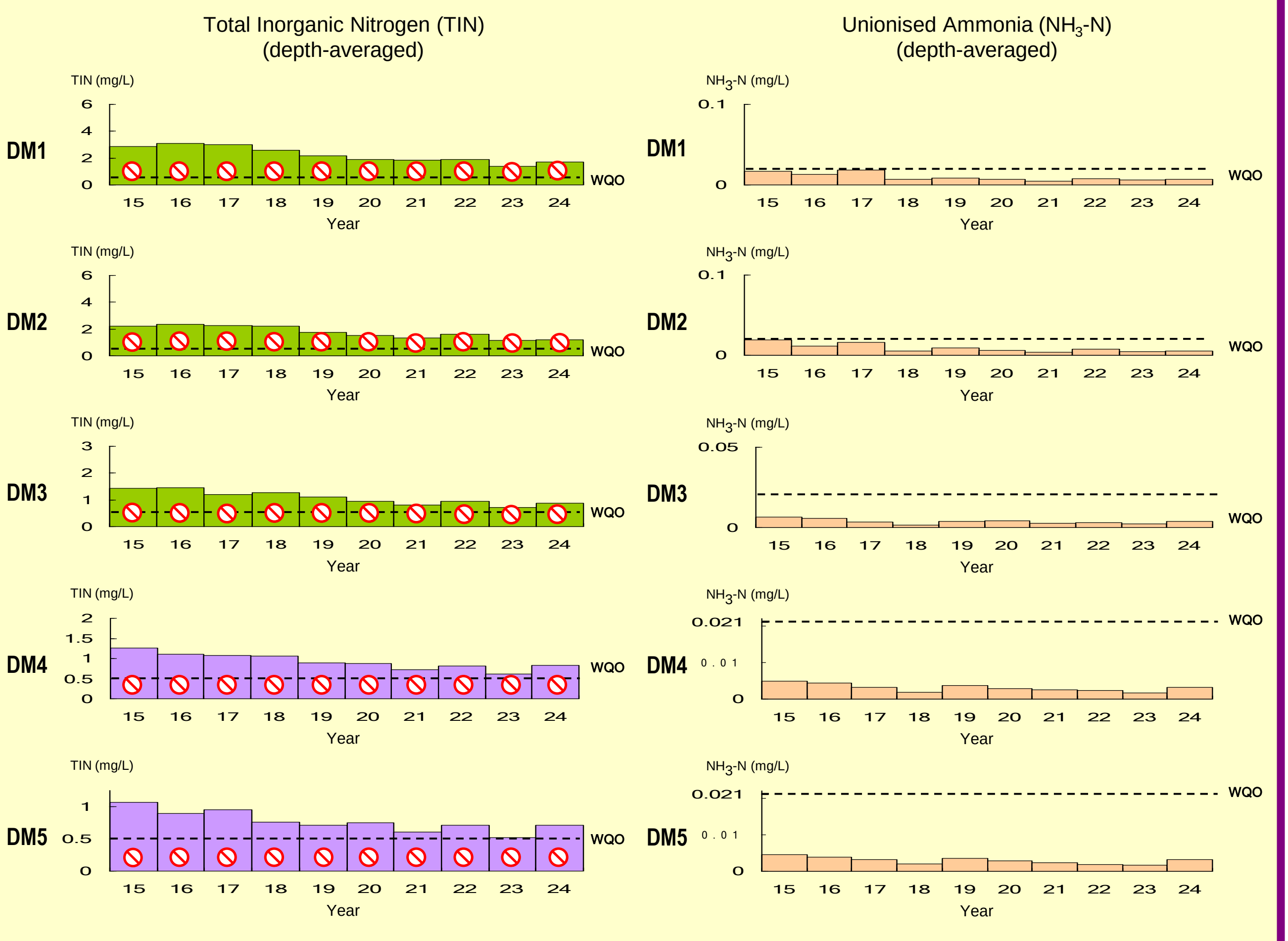
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

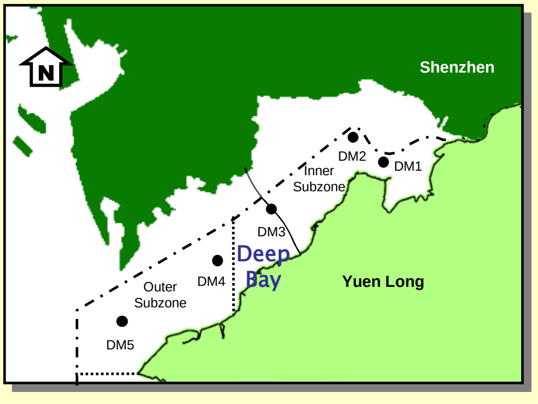


% sample with depth-averaged DO ≥ 4 mg/L

Non-compliance

WQO compliance rates for the Deep Bay WCZ (continued)



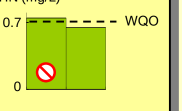


Total Inorganic Nitrogen (TIN)

Inner Subzone (DM1 - DM3)

WQO : annual mean for depth-averaged TIN $\leq$ 0.7 mg/L

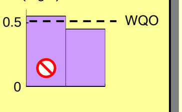
annual mean for depth-averaged TIN



Outer Subzone (DM4 - DM5)

WQO : annual mean for depth-averaged TIN $\leq$ 0.5 mg/L

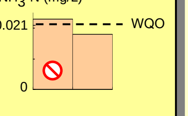
annual mean for depth-averaged TIN



Unionised Ammonia (NH_3-N)

WQO : annual mean for depth-averaged NH_3-N $\leq$ 0.021 mg/L

annual mean for depth-averaged NH_3-N

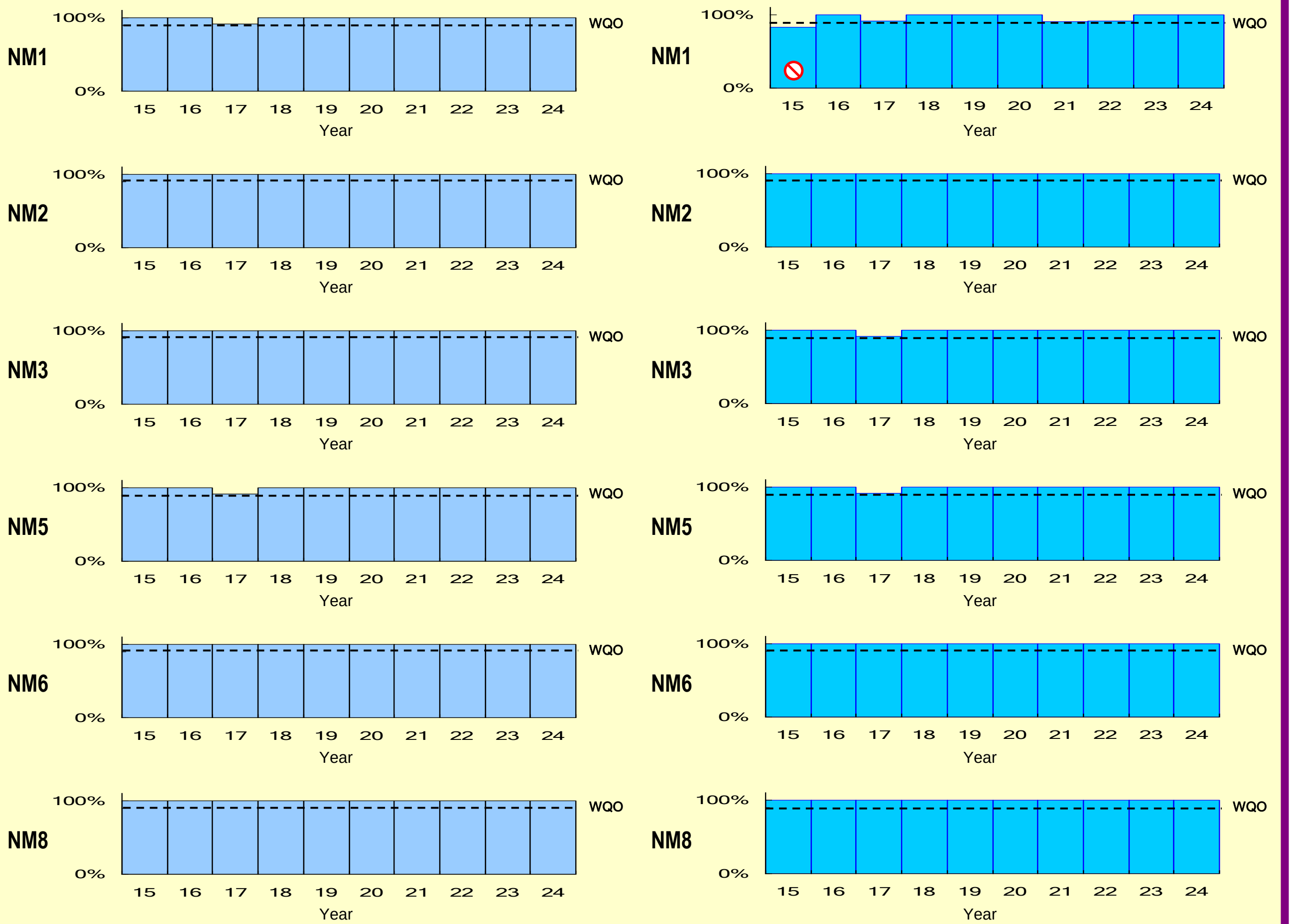


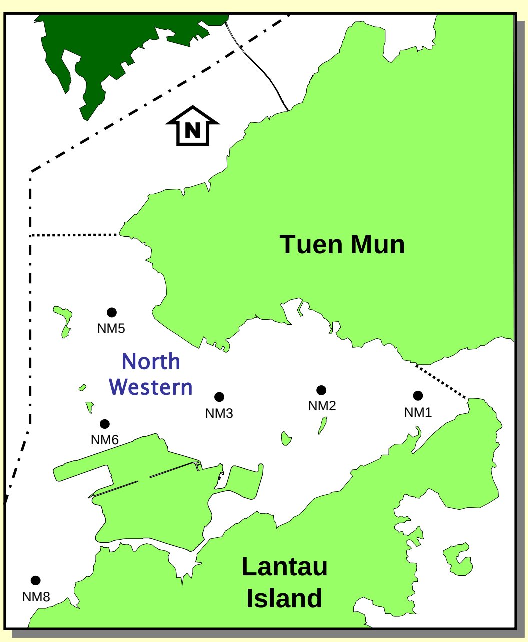
Non-compliance

WQO compliance rates for the North Western WCZ

Dissolved Oxygen (DO)
(bottom)

Dissolved Oxygen (DO)
(depth-averaged)



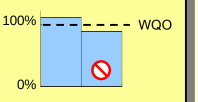


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

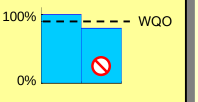
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

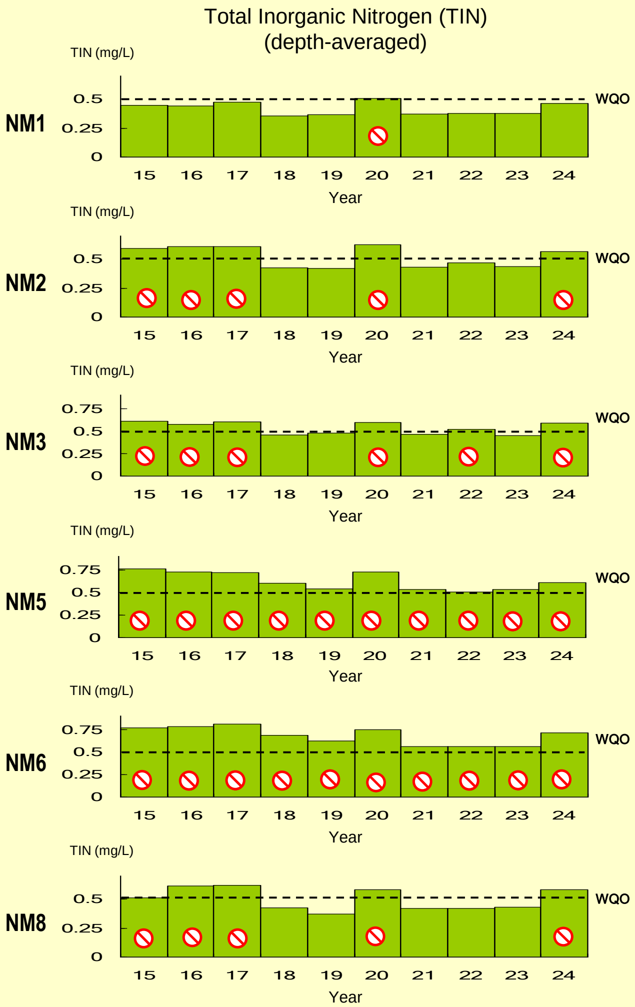
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

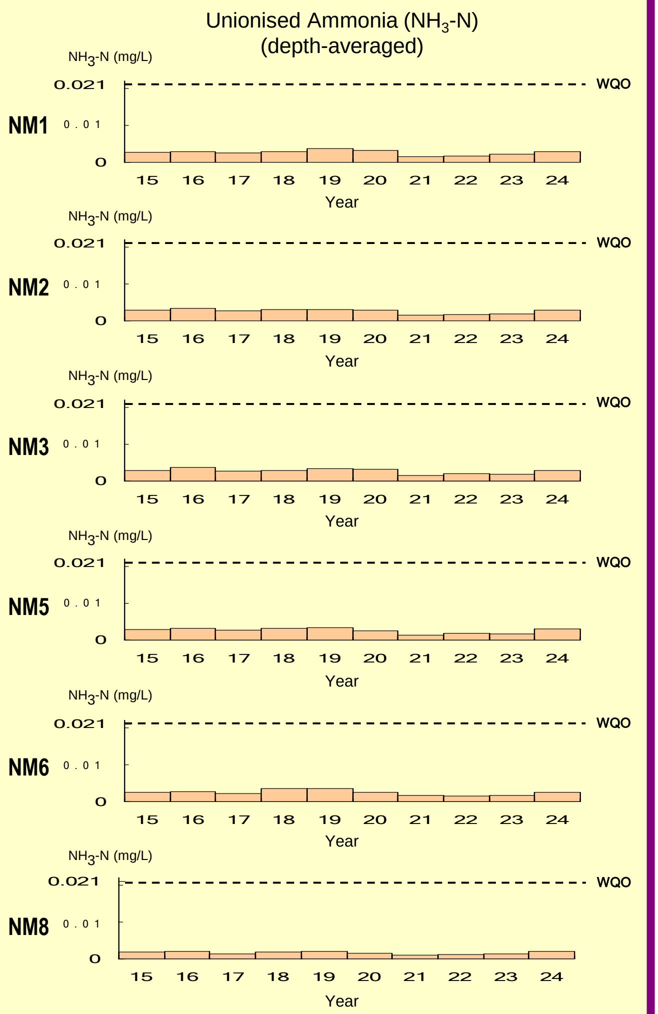
% sample with depth-averaged DO ≥ 4 mg/L

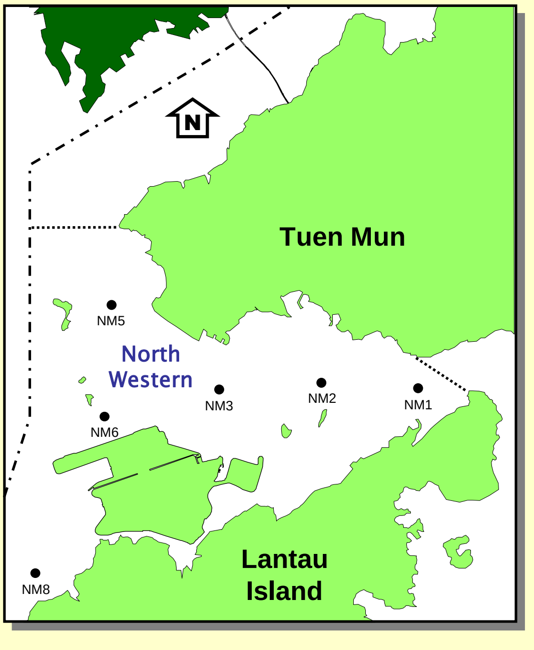


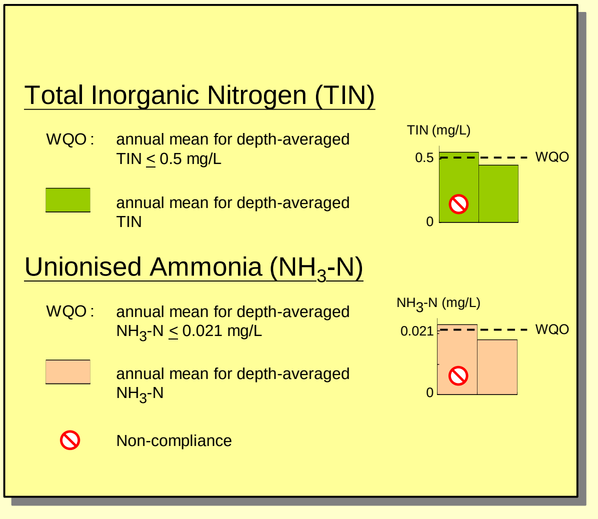
Non-compliance

WQO compliance rates for the North Western WCZ (continued)









Long-term water quality trends in the Mirs Bay WCZ, 1991 - 2024

Monitoring Station		MM1	MM2	MM3	MM4	MM5	MM6	MM7
Monitoring Period		1991 – 2024	1991 – 2024	1991 – 2024	1991 – 2024	1991 – 2024	1991 – 2024	1991 – 2024
Parameter	Water Depth							
Temperature (℃)	Surface	-	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	-	-	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Dissolved Oxygen (%)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	-	-	↘	↘
	Bottom	-	-	-	↘	↘	-	↘
	Average	↘	↘	↘	↘	↘	↘	↘
pH	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	-	↘	↘	↘	-	↘
Turbidity (NTU)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	↘	-
	Average	↘	-	-	-	-	-	-
Suspended Solids (mg/L)	Surface	-	↗	↗	↗	↗	↗	↗
	Middle	-	↗	↗	↗	↗	↗	↗
	Bottom	-	↗	↗	↗	↗	-	↗
	Average	-	↗	↗	↗	↗	↗	↗
Total volatile solids (mg/L)	Surface	-	↗	↗	↗	↗	↗	↗
	Middle	-	↗	↗	↗	↗	↗	↗
	Bottom	-	↗	↗	↗	↗	↗	↗
	Average	-	↗	↗	↗	↗	↗	↗
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	-	↘	-	-	↘	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	↘	-
	Average	↘	-	↘	-	-	↘	-
Ammonia nitrogen (mg/L)	Surface	↘	↘	-	-	-	-	-
	Middle	↘	-	-	-	-	↘	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	↘	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	-	-	-	-
	Middle	↗	↗	-	-	-	-	↗
	Bottom	↗	↗	-	-	-	-	-
	Average	↗	↗	-	-	-	-	-
Total inorganic nitrogen (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	-	-	↗	-	-	↗
	Middle	-	-	-	↗	↗	-	-
	Bottom	-	-	-	↗	-	↗	-
	Average	-	-	-	↗	-	-	↗
Total nitrogen (mg/L)	Surface	↘	-	-	↗	-	-	↗
	Middle	-	-	-	↗	↗	-	-
	Bottom	-	-	-	-	↗	↗	-
	Average	-	-	-	↗	-	-	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Silica (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	↘	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Chlorophyll-*a* (µg/L)	Surface	↘	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	↗
	Bottom	-	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
E. coli (cfu/100mL)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trends in the Mirs Bay WCZ, 1986 - 2024

Monitoring Station		MM8	MM13	MM14	MM15	MM16	MM17	MM19
Monitoring Period		1991 - 2024	1991 - 2024	1994 - 2024	1994 - 2024	1994 - 2024	1986 - 2024	2001 - 2024
Parameter	Water Depth							
Temperature (℃)	Surface	↗	↗	-	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗	-
	Bottom	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	-	-	↘	-	-
	Bottom	↘	↘	↘	↘	↘	-	-
	Average	↘	↘	↘	↘	↘	↘	↘
Dissolved Oxygen (%)	Surface	-	-	↘	↘	↘	↘	↘
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	↘	↘	-	-
	Average	-	-	↘	↘	↘	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		-	↘	↘	↘	↘	↘	-
Turbidity (NTU)	Surface	-	-	-	-	-	-	↘
	Middle	-	-	↘	↘	↘	-	↘
	Bottom	-	↘	↘	↘	↘	-	↘
	Average	-	-	↘	↘	↘	-	↘
Suspended Solids (mg/L)	Surface	↗	↗	↗	↗	↗	-	↗
	Middle	↗	↗	↗	↗	↗	-	↗
	Bottom	↗	-	-	-	-	-	-
	Average	↗	↗	↗	↗	↗	-	-
Total volatile solids (mg/L)	Surface	↗	↗	↗	↗	-	-	↗
	Middle	-	-	↗	-	-	-	↗
	Bottom	↗	-	-	-	-	-	-
	Average	↗	↗	↗	↗	↗	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	↘	-	-	↘	↘
	Middle	↘	↘	-	-	-	↘	↘
	Bottom	↘	↘	↘	-	-	↘	↘
	Average	↘	↘	↘	-	-	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	↘	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	↘	-
Nitrate nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Total inorganic nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	↗	↗	↗	-	↗
	Middle	-	-	↗	↗	↗	-	↗
	Bottom	-	-	↗	↗	↗	-	↗
	Average	-	-	↗	↗	↗	-	↗
Total nitrogen (mg/L)	Surface	-	-	↗	↗	↗	-	↗
	Middle	-	-	↗	↗	↗	-	↗
	Bottom	-	-	↗	↗	↗	-	↗
	Average	-	-	↗	↗	↗	-	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	-
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	↗
Silica (mg/L)	Surface	-	↘	-	-	-	↘	-
	Middle	-	-	-	-	-	↘	-
	Bottom	-	-	-	-	-	↘	-
	Average	-	-	-	-	-	↘	-
Chlorophyll-*a* (µg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	↘	↘	-	↗	-
	Bottom	-	-	↘	↘	↘	-	↘
	Average	-	-	↘	↘	-	-	↘
E. coli (cfu/100mL)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trends in the Port Shelter WCZ, 1986 - 2024

Monitoring Station		PM1	PM2	PM3	PM4	PM6	PM7	PM8	PM9	PM11
Monitoring Period		1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1993 I 2024
Parameter	Water Depth									
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗	↗	↗	-
	Middle	↗	↗	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	↘
	Average	-	-	↘	↘	↘	↘	↘	↘	↘
Dissolved Oxygen (%)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	↗	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		-	↗	-	-	↗	-	↘	↘	-
Turbidity (NTU)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	-	-
Suspended Solids (mg/L)	Surface	↗	↗	↗	↗	-	-	↗	↗	↗
	Middle	-	-	↗	-	-	↗	-	↗	↗
	Bottom	-	-	↗	-	↗	↗	↗	↗	↗
	Average	-	-	↗	-	-	↗	↗	↗	↗
Total volatile solids (mg/L)	Surface	↗	↗	↗	↗	↗	-	-	-	↗
	Middle	↗	-	-	-	-	-	-	-	-
	Bottom	↗	-	-	-	-	-	-	-	-
	Average	↗	-	-	-	-	-	-	-	↗
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	↘	↘	-	↘	↘	↘	-	-
	Middle	-	↘	-	-	↘	-	↘	-	↘
	Bottom	-	↘	-	-	↘	↘	↘	↘	↘
	Average	-	↘	-	-	↘	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	↗
	Middle	-	-	-	-	-	-	↗	↗	↗
	Bottom	-	-	-	-	-	↗	↗	↗	↗
	Average	-	-	-	-	-	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Total nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Silica (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	↘
	Middle	↘	↘	↘	-	-	↘	↘	↘	-
	Bottom	↘	↘	-	-	-	-	-	-	-
	Average	↘	↘	↘	-	-	↘	-	↘	-
Chlorophyll-*a* (μg/L)	Surface	↗	-	↗	↗	-	↗	↗	-	-
	Middle	↗	↗	↗	↗	-	-	-	-	-
	Bottom	↗	↗	↗	↗	-	-	-	-	↘
	Average	↗	-	↗	↗	-	-	-	-	-
E. coli (cfu/100mL)	Surface	-	↘	-	-	↘	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	↘	-	-	-	-	-	-	-
	Average	-	↘	-	-	↘	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	-	-	-	↘	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	↘	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trends in the Tolo Habour and Channel WCZ, 1986 - 2024

Monitoring Station		TM2	TM3	TM4	TM5	TM6	TM7	TM8
Monitoring Period		1986 I 2024	1986 I 2024	1986 I 2024	1988 I 2024	1986 I 2024	1988 I 2024	1986 I 2024
Parameter	Water Depth							
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗	↗
	Middle	NA	↗	↗	NA	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	↘	↘	↘	↘	↘	↘
	Middle	NA	-	-	NA	-	-	-
	Bottom	↗	↗	↗	-	-	-	-
	Average	-	-	-	↘	-	↘	↘
Dissolved Oxygen (%)	Surface	-	↘	↘	↘	↘	↘	↘
	Middle	NA	-	-	NA	-	-	-
	Bottom	↗	↗	↗	-	↗	-	-
	Average	-	-	-	-	-	-	↘
pH	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	-
Turbidity (NTU)	Surface	↘	-	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	↘	-	-	↘	-	-	-
	Average	↘	-	-	-	-	-	-
Suspended Solids (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	↘	↗	-	-	-	-	↗
	Average	↘	-	-	-	-	-	-
Total volatile solids (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	-	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	-
	Bottom	↘	↘	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↘	↘	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	↘	↘	↘	-	↘	↘	-
	Average	↘	↘	↘	-	↘	↘	↘
Nitrate nitrogen (mg/L)	Surface	↘	↘	↘	-	↘	-	-
	Middle	NA	↘	↘	NA	↘	↘	-
	Bottom	↘	↘	↘	-	↘	↘	-
	Average	↘	↘	↘	-	↘	↘	-
Total inorganic nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	-
	Bottom	↘	↘	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	↘	↘	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	NA	-	-	-
	Bottom	↘	↘	↘	↘	-	-	-
	Average	↘	↘	↘	↘	↘	-	-
Total nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	NA	↘	-	-
	Bottom	↘	↘	↘	↘	↘	-	-
	Average	↘	↘	↘	↘	↘	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	-
	Bottom	↘	↘	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	-	↗	↗	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Chlorophyll-*a* (µg/L)	Surface	↘	↘	↘	↘	↘	↘	-
	Middle	NA	↘	↘	NA	-	-	-
	Bottom	-	-	-	↘	↗	↗	↗
	Average	↘	↘	↘	↘	↘	-	-
E. coli (cfu/100mL)	Surface	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	NA	-	-	-
	Bottom	↘	↘	↘	↘	-	-	-
	Average	↘	↘	↘	↘	↘	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	NA	↘	-	-
	Bottom	↘	-	↘	↘	↘	-	-
	Average	↘	↘	↘	↘	↘	↘	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. – indicates no significant trend

3. NA (Not Applicable) indicates the measurement was not made due to shallow water

4. ↗ significant increase

5. ↘ significant decrease

Long-term water quality trends in the Southern WCZ, 1986 - 2024

Monitoring Station		SM1	SM2	SM3	SM4	SM5	SM6	SM7	SM9
Monitoring Period		1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1988 I 2024
Parameter	Water Depth								
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	↘	-	↘	↘	↘	↘	-
	Middle	-	-	↘	↘	-	-	-	-
	Bottom	↘	-	↘	-	-	↘	-	-
	Average	↘	↘	↘	↘	-	↘	-	-
Dissolved Oxygen (%)	Surface	↘	↘	-	↘	↘	-	↘	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-	-	-	-	↘
	Middle	-	-	-	-	-	-	↘	↘
	Bottom	↘	↘	↘	↘	↘	-	↘	↘
	Average	-	-	-	-	-	-	↘	↘
Suspended Solids (mg/L)	Surface	↗	-	-	-	-	↗	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	↘
Total volatile solids (mg/L)	Surface	-	-	-	-	-	↗	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	↘	-	-	↘	-	-
	Average	-	-	-	-	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	-	-	↗	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	-	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	↗	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	-	-	↗	↗
	Middle	↗	↗	-	↗	-	-	↗	↗
	Bottom	-	-	-	↗	-	-	↗	↗
	Average	↗	↗	↗	↗	-	-	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	↗	↗	↗	-	-	↗	↗
	Middle	-	-	-	↗	-	-	↗	↗
	Bottom	-	-	-	↗	-	-	↗	↗
	Average	-	↗	↗	↗	-	-	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Total nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	↗
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Silica (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Chlorophyll-*a* (μg/L)	Surface	-	-	↗	-	-	-	-	↗
	Middle	-	-	-	-	-	-	-	↗
	Bottom	-	-	-	↗	↗	-	↗	↗
	Average	-	-	-	-	↗	-	↗	↗
E. coli (cfu/100mL)	Surface	-	↘	↘	-	-	-	-	-
	Middle	-	↘	↘	↘	-	-	-	-
	Bottom	-	↘	↘	↘	-	-	-	-
	Average	↘	↘	↘	↘	-	-	↘	-
Faecal coliforms (cfu/100mL)	Surface	-	↘	-	-	-	-	-	-
	Middle	-	↘	↘	-	-	-	-	-
	Bottom	-	↘	↘	-	-	-	-	-
	Average	↘	↘	↘	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trend analyses in the Southern WCZ, 1986 - 2024 (continued)

Monitoring Station		SM10	SM11	SM12	SM13	SM17	SM18	SM19	SM20
Monitoring Period		1986 – 2024	1986 – 2024	1986 – 2024	1986 – 2024	1989 – 2024	1989 – 2024	1989 – 2024	1999 – 2024
Parameter	Water Depth								
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	NA	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	↘	-	-	-
	Bottom	-	-	-	↘	-	-	-	↘
	Average	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	-
	Middle	NA	-	↘	↘	-	↘	↘	-
	Bottom	-	↘	↘	-	-	↘	↘	-
	Average	↘	↘	↘	↘	↘	↘	↘	-
Dissolved Oxygen (%)	Surface	↘	↘	↘	↘	↘	↘	↘	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	↘	-	-
	Average	-	-	↘	-	-	-	↘	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	↘	↘	-	-	-	-	-	↘
	Middle	NA	↘	-	-	↘	↘	-	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	-	-	↘	↘	-	↘
Suspended Solids (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	↘
	Bottom	↘	-	-	-	-	-	-	↘
	Average	↘	-	-	-	-	-	-	↘
Total volatile solids (mg/L)	Surface	-	-	-	-	-	↗	↗	-
	Middle	NA	-	-	-	-	↗	↗	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	↗	↗	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	↘	-	-	-	-	-
	Middle	NA	-	↘	-	-	-	-	-
	Bottom	-	↘	-	-	-	-	↘	-
	Average	↘	-	↘	-	-	-	↘	-
Ammonia nitrogen (mg/L)	Surface	↗	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	↗	↗	-	-	-	-	-	-
	Average	↗	↗	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	↗	↗	↗	↗	-	-	-
	Middle	NA	-	-	↗	↗	-	-	-
	Bottom	-	-	-	↗	↗	-	-	-
	Average	-	-	↗	↗	↗	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	-	-	-
	Middle	NA	↗	↗	↗	↗	-	-	↗
	Bottom	↗	↗	↗	↗	↗	-	-	↗
	Average	↗	↗	↗	↗	↗	-	-	-
Total inorganic nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	-	-	-
	Middle	NA	↗	↗	↗	↗	-	-	↗
	Bottom	↗	↗	↗	↗	↗	-	-	↗
	Average	↗	↗	↗	↗	↗	-	-	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	↗
	Middle	NA	-	-	-	-	-	-	↗
	Bottom	-	-	-	-	-	-	-	↗
	Average	-	-	-	-	-	-	-	↗
Total nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	-	-	↗
	Middle	NA	↗	↗	↗	↗	-	-	↗
	Bottom	↗	↗	↗	↗	↗	-	-	↗
	Average	↗	↗	↗	↗	↗	-	-	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	↗
Silica (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Chlorophyll-*a* (µg/L)	Surface	-	↗	↗	↗	↗	-	-	-
	Middle	NA	-	↗	↗	↗	-	-	-
	Bottom	-	-	-	↗	↗	-	-	-
	Average	-	-	↗	↗	↗	-	-	-
E. coli (cfu/100mL)	Surface	↘	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	↘	↘	-	-	-	-	-	-
	Average	↘	↘	-	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	-	-	↗	-	-	-	-
	Middle	NA	↘	-	↗	-	-	-	-
	Bottom	↘	↘	↗	↗	-	-	-	-
	Average	↘	↘	-	↗	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. NA (Not Applicable) indicates the measurement was not made due to shallow water
4. ↗ significant increase
5. ↘ significant decrease

Long-term water quality trend analyses in the Victoria Harbour WCZ, 1986 - 2024

Monitoring Station		VM1	VM2	VM4	VM5	VM6
Monitoring Period		1988 – 2024	1988 – 2024	1988 – 2024	1986 – 2024	1988 – 2024
Parameter	Water Depth					
Temperature (℃)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-
	Middle	↗	↗	-	-	-
	Bottom	↗	↗	↗	-	-
	Average	↗	↗	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Dissolved Oxygen (%)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
pH	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	↘	↘	↘	↘
	Average	-	-	-	-	-
Suspended Solids (mg/L)	Surface	-	↘	-	-	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	-	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total volatile solids (mg/L)	Surface	-	-	↘	-	↘
	Middle	↘	-	↘	↘	↘
	Bottom	-	↘	↘	↘	↘
	Average	-	↘	↘	↘	↘
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-
	Middle	↘	-	-	-	-
	Bottom	↘	↘	↘	-	-
	Average	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	-	↗	↗	↗	↗
	Bottom	-	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-
	Middle	↘	↘	↘	-	-
	Bottom	↘	↘	-	-	-
	Average	↘	↘	-	-	-
Chlorophyll-*a* (μg/L)	Surface	-	-	-	-	-
	Middle	-	-	↗	↗	↗
	Bottom	-	↗	-	-	↗
	Average	-	-	-	-	-
E. coli (cfu/100mL)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trend analyses in the Victoria Harbour WCZ, 1986 - 2024 (continued)

Monitoring Station		VM7	VM8	VM12	VM14	VM15
Monitoring Period		1986 I 2024	1986 I 2024	1986 I 2024	1986 I 2024	1993 I 2024
Parameter	Water Depth					
Temperature (℃)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	↗	↗	↗
	Middle	↗	-	↗	↗	↗
	Bottom	↗	-	↗	↗	↗
	Average	↗	-	↗	↗	↗
Dissolved Oxygen (%)	Surface	↗	-	↗	↗	↗
	Middle	↗	-	↗	↗	↗
	Bottom	↗	-	↗	↗	↗
	Average	↗	-	↗	↗	↗
pH	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-	-
	Middle	↘	↘	-	-	↘
	Bottom	↘	↘	↘	-	↘
	Average	-	↘	↘	-	↘
Suspended Solids (mg/L)	Surface	↘	↘	-	↘	↘
	Middle	↘	↘	-	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total volatile solids (mg/L)	Surface	↘	↘	↘	↘	-
	Middle	↘	↘	↘	↘	-
	Bottom	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	-	↘	↘	-
	Middle	↘	-	↘	↘	-
	Bottom	↘	-	↘	↘	↘
	Average	↘	-	↘	↘	-
Ammonia nitrogen (mg/L)	Surface	↘	↗	↘	↘	↘
	Middle	↘	↗	↘	↘	↘
	Bottom	↘	-	↘	↘	↘
	Average	↘	-	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-
	Middle	-	-	-	↗	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	↗	-	-	-
	Middle	-	↗	-	↗	↘
	Bottom	-	↗	-	↗	↘
	Average	-	↗	-	-	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	-	↘	↘	↘
	Middle	↘	-	↘	↘	↘
	Bottom	↘	-	↘	↘	-
	Average	↘	-	↘	↘	-
Total nitrogen (mg/L)	Surface	-	-	↘	↘	-
	Middle	↘	-	-	-	-
	Bottom	↘	-	-	-	-
	Average	-	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Chlorophyll-*a* (μg/L)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
E. coli (cfu/100mL)	Surface	-	-	↘	↘	↘
	Middle	↘	-	↘	↘	↘
	Bottom	↘	↗	-	↘	↘
	Average	-	-	↘	↘	↘
Faecal coliforms (cfu/100mL)	Surface	-	↗	-	↘	↘
	Middle	-	↗	-	↘	↘
	Bottom	-	↗	-	↘	↘
	Average	-	↗	-	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trend analyses in the Eastern Buffer WCZ, 1986 - 2024

Monitoring Station		EM1	EM2	EM3
Monitoring Period		1986 – 2024	1986 – 2024	1988 – 2024
Parameter	Water Depth			
Temperature (℃)	Surface	↗	↗	↗
	Middle	↗	↗	↗
	Bottom	↗	↗	↗
	Average	↗	↗	↗
Salinity	Surface	-	-	-
	Middle	↗	-	-
	Bottom	↗	-	-
	Average	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	-
	Middle	↗	-	-
	Bottom	-	-	-
	Average	↗	-	-
Dissolved Oxygen (%)	Surface	↗	-	-
	Middle	↗	↗	-
	Bottom	↗	-	-
	Average	↗	-	-
pH	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Secchi disc depth (m)		↗	↗	-
Turbidity (NTU)	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
Suspended Solids (mg/L)	Surface	-	-	↗
	Middle	-	-	-
	Bottom	↘	↗	-
	Average	-	-	-
Total volatile solids (mg/L)	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	-
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	↘	-
	Average	↘	↘	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
Total inorganic nitrogen (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	↘	-
	Average	↘	↘	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	-	-
	Average	↘	↘	-
Total nitrogen (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	-	-
	Average	↘	↘	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	-	-
	Average	↘	↘	-
Silica (mg/L)	Surface	↘	↘	↘
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	↘
Chlorophyll-*a* (μg/L)	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
E. coli (cfu/100mL)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trend analyses in the Western Buffer WCZ, 1986 - 2024

Monitoring Station		WM1	WM2	WM3	WM4
Monitoring Period		1988 I 2024	1988 I 2024	1986 I 2024	1986 I 2024
Parameter	Water Depth				
Temperature (℃)	Surface	↗	↗	↗	↗
	Middle	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗
	Average	↗	↗	↗	↗
Salinity	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	↗	-	-	-
	Average	↗	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	↘	-	-
	Middle	-	-	-	↘
	Bottom	-	-	-	-
	Average	↘	-	-	↘
Dissolved Oxygen (%)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
pH	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-
	Middle	-	↘	↘	↘
	Bottom	-	↘	↘	↘
	Average	-	↘	↘	↘
Suspended Solids (mg/L)	Surface	-	-	↘	-
	Middle	-	-	↘	↘
	Bottom	↗	-	↘	-
	Average	-	-	↘	-
Total volatile solids (mg/L)	Surface	-	-	-	-
	Middle	-	-	-	↘
	Bottom	-	-	-	-
	Average	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	-	-
	Middle	↘	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	-	-	-	-
	Middle	-	↗	-	↗
	Bottom	↘	-	-	-
	Average	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↗	↗	↗	↗
	Middle	-	↗	-	-
	Bottom	-	-	-	-
	Average	-	↗	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗
	Middle	-	↗	↗	↗
	Bottom	-	↗	↗	↗
	Average	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↗	↗	↗	↗
	Middle	-	↗	↗	↗
	Bottom	-	↗	-	↗
	Average	-	↗	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Total nitrogen (mg/L)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	↘	-
	Middle	-	-	↘	-
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	-
Silica (mg/L)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Chlorophyll-*a* (μg/L)	Surface	-	-	-	-
	Middle	↗	↗	-	↗
	Bottom	↗	↗	-	-
	Average	↗	↗	-	-
E. coli (cfu/100mL)	Surface	↘	-	-	-
	Middle	↘	-	-	-
	Bottom	↘	-	-	-
	Average	↘	-	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	-	-	-
	Middle	↘	-	-	-
	Bottom	↘	↗	↗	-
	Average	↘	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trend analyses in the Junk Bay WCZ, 1986 - 2024

Monitoring Station		JM3	JM4
Monitoring Period		1986 – 2024	1986 – 2024
Parameter	Water Depth		
Temperature (℃)	Surface	↗	↗
	Middle	↗	↗
	Bottom	↗	↗
	Average	↗	↗
Salinity	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Dissolved Oxygen (mg/L)	Surface	-	-
	Middle	↗	↗
	Bottom	↗	-
	Average	↗	-
Dissolved Oxygen (%)	Surface	↗	↗
	Middle	↗	↗
	Bottom	↗	↗
	Average	↗	↗
pH	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Secchi disc depth (m)		↗	↗
Turbidity (NTU)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Suspended Solids (mg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Total volatile solids (mg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-
	Middle	-	↘
	Bottom	↘	↘
	Average	↘	↘
Nitrate nitrogen (mg/L)	Surface	↗	↗
	Middle	↗	↗
	Bottom	-	-
	Average	-	↗
Total inorganic nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	-
	Average	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Silica (mg/L)	Surface	↘	-
	Middle	-	-
	Bottom	↘	-
	Average	↘	-
Chlorophyll-*a* (μg/L)	Surface	-	-
	Middle	-	↗
	Bottom	-	-
	Average	-	-
E. coli (cfu/100mL)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Long-term water quality trend analyses in the Deep Bay WCZ, 1986 - 2024

Monitoring Station		DM1	DM2	DM3	DM4	DM5
Monitoring Period		1986 - 2024	1986 - 2024	1986 - 2024	1986 - 2024	1991 - 2024
Parameter	Water Depth					
Temperature (℃)	Surface	↗	↗	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	↗	↗	↗	↗	↗
Salinity	Surface	↘	↘	-	-	-
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↘	↘	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	↘	↘	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	-
	Average	↗	-	↘	↘	↘
Dissolved Oxygen (%)	Surface	↗	-	↘	↘	↘
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	↘	-
	Average	↗	-	↘	↘	-
pH	Surface	↘	↘	↘	↘	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	↘	-	-	-	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↘	-	↘	↘	↘
Suspended Solids (mg/L)	Surface	-	-	-	-	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	-	-	-	↘	↘
Total volatile solids (mg/L)	Surface	-	-	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	↘	↘
	Average	-	-	-	-	↘
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	↘	↘	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	-	-	↘	↘	-
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	↘	↘	↘	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↘	↘	-	↗	-
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	↘	↘	-	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	↘	-
	Average	↘	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	↘	-	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	↗
	Average	↘	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	-	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	-	↘
	Average	↘	↘	↘	-	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	-	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	↗	↗	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	↗
	Average	↗	↗	-	-	-
Chlorophyll-*a* (μg/L)	Surface	↗	↗	-	-	-
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	-	↗
	Average	↗	↗	-	-	↗
E. coli (cfu/100mL)	Surface	↘	-	↘	-	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	-	↘
	Average	↘	-	↘	-	↘
Faecal coliforms (cfu/100mL)	Surface	↘	-	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	↘
	Average	↘	-	-	-	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. NA (Not Applicable) indicates the measurement was not made due to shallow water
4. ↗ significant increase
5. ↘ significant decrease

Long-term water quality trend analyses in the North Western WCZ, 1986 - 2024

Monitoring Station		NM1	NM2	NM3	NM5	NM6	NM8
Monitoring Period		1988 – 2024	1986 – 2024	1986 – 2024	1988 – 2024	1991 – 2024	1999 – 2024
Parameter	Water Depth						
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	↘	-	-	-
	Middle	-	↘	↘	-	↘	↘
	Bottom	-	-	↘	-	↘	↘
	Average	-	-	↘	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	↘	↘	↘	-
	Middle	↘	↘	-	↘	↘	-
	Bottom	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	-
Dissolved Oxygen (%)	Surface	-	-	↘	↘	-	-
	Middle	-	↘	-	-	↘	-
	Bottom	-	↘	-	-	-	↘
	Average	-	↘	↘	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘
Suspended Solids (mg/L)	Surface	-	-	-	↘	-	-
	Middle	-	-	↘	-	-	-
	Bottom	-	↘	↘	-	↘	-
	Average	-	↘	↘	↘	↘	-
Total volatile solids (mg/L)	Surface	-	-	-	-	-	-
	Middle	-	-	-	-	-	-
	Bottom	-	-	↘	-	-	-
	Average	-	-	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	↘	-	-	-	-
	Middle	-	-	-	-	-	-
	Bottom	-	↘	-	↘	↘	-
	Average	-	↘	-	↘	-	-
Ammonia nitrogen (mg/L)	Surface	-	-	-	-	-	-
	Middle	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-
	Average	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	↗	↗	-	-	-
	Middle	↗	-	↗	↗	-	-
	Bottom	-	↗	↗	-	↗	-
	Average	-	↗	↗	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	↘	↘	-	-
	Middle	-	-	-	-	-	↗
	Bottom	-	-	-	-	-	↗
	Average	-	-	-	↘	-	↗
Total nitrogen (mg/L)	Surface	↗	↗	↗	-	↗	↗
	Middle	↗	↗	↗	↗	↗	↗
	Bottom	-	-	↗	-	↗	↗
	Average	↗	-	↗	↗	↗	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-
	Middle	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-
	Average	-	-	-	-	-	-
Silica (mg/L)	Surface	-	-	-	-	-	↗
	Middle	-	↗	-	-	↗	↗
	Bottom	-	-	-	-	-	↗
	Average	-	-	-	-	-	↗
Chlorophyll-*a* (μg/L)	Surface	-	-	-	-	-	-
	Middle	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-
	Average	-	-	-	-	-	-
E. coli (cfu/100mL)	Surface	-	↘	↘	↘	-	-
	Middle	-	↘	↘	↘	-	-
	Bottom	-	↘	↘	-	-	-
	Average	-	↘	↘	↘	-	-
Faecal coliforms (cfu/100mL)	Surface	-	↘	↘	↘	-	-
	Middle	-	↘	↘	-	-	-
	Bottom	-	↘	↘	-	-	-
	Average	-	↘	↘	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. ↗ significant increase
4. ↘ significant decrease

Summary of marine sediment quality data for the Tolo Harbour and Channel WCZ and Southern WCZ, 2020 - 2024

	Tolo Harbour and Channel				Hong Kong Island (South)		West Lamma Channel	
	Harbour Subzone		Buffer Subzone	Channel Subzone				
Parameter	TS2	TS3	TS4	TS5	SS1	SS2	SS3	SS4
Number of samples	**10**	**10**	**10**	**10**	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	66 (2 - 88)	76 (20 - 98)	59 (9 - 93)	77 (6 - 97)	60 (32 - 80)	72 (14 - 90)	76 (21 - 87)	72 (6 - 94)
Electrochemical Potential (mV)	-249 (-353 - -85)	-263 (-341 - -102)	-295 (-379 - -204)	-246 (-322 - -137)	-112 (-192 - -57)	-106 (-183 - -50)	-112 (-205 - -62)	-106 (-169 - -46)
Total Solids (%w/w)	38 (33 - 43)	35 (31 - 41)	37 (33 - 45)	32 (28 - 39)	54 (48 - 60)	49 (44 - 59)	51 (47 - 54)	45 (40 - 51)
Total Volatile Soilds (%TS)	8.6 (7.5 - 9.9)	8.8 (7.4 - 10.0)	9.5 (6.3 - 11.0)	10.6 (9.1 - 15.0)	6.0 (5.0 - 7.4)	7.1 (5.4 - 7.8)	6.2 (5.2 - 7.2)	6.7 (5.2 - 7.4)
Chemical Oxygen Demand (mg/kg)	20200 (16000 - 25000)	19100 (16000 - 22000)	17960 (9600 - 24000)	16400 (11000 - 20000)	11760 (9600 - 15000)	11600 (7200 - 16000)	11070 (8000 - 14000)	12800 (11000 - 14000)
Total Carbon (%w/w)	0.8 (0.7 - 0.9)	0.7 (0.5 - 0.8)	0.9 (0.7 - 1.0)	0.8 (0.7 - 1.1)	0.9 (0.8 - 1.1)	0.9 (0.7 - 1.4)	0.8 (0.7 - 0.9)	0.7 (0.6 - 1.1)
Ammonical Nitrogen (mg/kg)	5.88 (2.80 - 8.50)	4.37 (0.05 - 7.70)	8.03 (2.50 - 16.00)	8.78 (5.10 - 13.00)	6.46 (3.50 - 11.00)	10.04 (2.70 - 51.00)	5.19 (3.30 - 8.00)	7.09 (3.80 - 12.00)
Total Kjeldahl Nitrogen (mg/kg)	560 (500 - 600)	500 (330 - 590)	640 (580 - 680)	700 (640 - 830)	550 (390 - 1200)	490 (400 - 590)	470 (390 - 550)	570 (460 - 920)
Total Phosphorus (mg/kg)	180 (170 - 200)	170 (140 - 200)	210 (180 - 240)	210 (180 - 260)	280 (240 - 430)	240 (220 - 270)	240 (230 - 260)	240 (220 - 270)
Total Sulphide (mg/kg)	56.5 (7.4 - 130.0)	53.4 (3.1 - 140.0)	49.6 (7.1 - 120.0)	53.6 (20.0 - 120.0)	27.4 (2.7 - 99.0)	55.3 (4.5 - 260.0)	23.0 (5.2 - 41.0)	36.4 (12.0 - 94.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)
Arsenic (mg/kg)	13.0 (4.0 - 18.0)	13.6 (7.9 - 18.0)	11.1 (5.0 - 16.0)	9.7 (7.5 - 13.0)	8.3 (6.2 - 9.8)	9.5 (6.1 - 11.0)	8.8 (4.9 - 11.0)	10.0 (6.6 - 12.0)
Cadmium (mg/kg)	0.5 (0.1 - 0.7)	0.5 (0.3 - 0.6)	0.4 (0.2 - 0.5)	0.3 (0.2 - 0.4)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)
Chromium (mg/kg)	27 (20 - 35)	28 (21 - 37)	31 (15 - 57)	39 (31 - 55)	28 (19 - 36)	39 (27 - 57)	33 (24 - 45)	40 (28 - 55)
Copper (mg/kg)	41 (21 - 68)	45 (30 - 87)	39 (12 - 77)	28 (17 - 56)	23 (9 - 82)	34 (20 - 50)	24 (14 - 70)	29 (20 - 40)
Lead (mg/kg)	95 (34 - 180)	100 (74 - 170)	73 (33 - 96)	56 (39 - 83)	33 (24 - 42)	40 (30 - 56)	37 (28 - 62)	40 (35 - 45)
Mercury (mg/kg)	0.07 (0.05 - 0.09)	0.06 (0.05 - 0.07)	0.06 (0.05 - 0.09)	0.06 (0.05 - 0.08)	0.07 (0.05 - 0.18)	0.08 (0.06 - 0.13)	0.07 (0.05 - 0.09)	0.11 (0.08 - 0.13)
Nickel (mg/kg)	16 (12 - 22)	16 (10 - 21)	18 (10 - 27)	25 (20 - 35)	18 (14 - 21)	23 (19 - 31)	21 (17 - 27)	24 (17 - 31)
Silver (mg/kg)	0.3 (0.2 - 0.4)	0.3 (0.2 - 0.5)	0.2 (0.2 - 0.3)	0.2 (0.2 - 0.3)	0.2 (0.2 - 0.3)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)
Zinc (mg/kg)	220 (95 - 310)	270 (170 - 410)	190 (66 - 340)	160 (90 - 240)	95 (63 - 140)	120 (86 - 170)	110 (78 - 250)	120 (75 - 180)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) (3)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 21)	18 (18 - 18)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) (4) (6)	98 (90 - 150)	100 (90 - 150)	99 (90 - 170)	110 (90 - 170)	93 (90 - 110)	90 (90 - 94)	90 (90 - 94)	92 (90 - 97)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) (5) (6)	62 (27 - 110)	60 (22 - 96)	56 (34 - 79)	66 (40 - 100)	52 (25 - 160)	80 (32 - 110)	56 (27 - 92)	98 (51 - 150)

Note:
1. Data presented are arithmetic means ; data in brackets indicate ranges.
2. All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.
3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4. Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5. High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary of marine sediment quality data for the Southern, Junk Bay WCZ and Deep Bay WCZ, 2020 - 2024

	Lantau Island (East)	Lantau Island (South)	Junk Bay	Inner Deep Bay		Outer Deep Bay	
Parameter	SS5	SS6	JS2	DS1	DS2	DS3	DS4
Number of samples	10	10	10	10	10	10	10
Particle Size Fractionation <63μm (%w/w)	77 (7 - 98)	67 (48 - 81)	74 (2 - 98)	78 (17 - 98)	71 (15 - 88)	75 (10 - 93)	68 (20 - 91)
Electrochemical Potential (mV)	-137 (-225 - -33)	-123 (-190 - -82)	-135 (-265 - -93)	-130 (-203 - -74)	-113 (-217 - -55)	-118 (-202 - -76)	-128 (-198 - -89)
Total Solids (%w/w)	38 (35 - 44)	55 (41 - 64)	41 (39 - 43)	49 (43 - 62)	47 (36 - 51)	48 (43 - 66)	44 (40 - 49)
Total Volatile Soilds (%TS)	7.5 (5.9 - 8.8)	4.6 (3.6 - 6.1)	7.1 (4.7 - 8.2)	5.6 (4.1 - 7.0)	6.2 (4.7 - 8.4)	6.3 (2.8 - 8.3)	6.6 (5.6 - 7.4)
Chemical Oxygen Demand (mg/kg)	13100 (11000 - 17000)	9940 (5200 - 13000)	14800 (10000 - 18000)	21000 (14000 - 30000)	16400 (13000 - 21000)	14800 (11000 - 19000)	13900 (11000 - 18000)
Total Carbon (%w/w)	0.5 (0.5 - 0.7)	0.5 (0.4 - 0.6)	0.6 (0.6 - 0.7)	0.6 (0.5 - 0.8)	0.6 (0.4 - 0.7)	0.5 (0.2 - 0.7)	0.5 (0.5 - 0.6)
Ammonical Nitrogen (mg/kg)	8.76 (4.80 - 13.00)	9.01 (2.80 - 20.00)	5.81 (1.10 - 14.00)	8.36 (0.63 - 23.00)	4.03 (0.30 - 7.80)	8.94 (0.29 - 36.00)	10.37 (0.25 - 64.00)
Total Kjeldahl Nitrogen (mg/kg)	550 (480 - 620)	400 (330 - 510)	490 (420 - 570)	520 (400 - 650)	450 (350 - 510)	450 (290 - 530)	460 (390 - 530)
Total Phosphorus (mg/kg)	200 (180 - 240)	220 (190 - 320)	200 (180 - 210)	320 (270 - 440)	290 (210 - 330)	260 (210 - 320)	240 (200 - 320)
Total Sulphide (mg/kg)	21.5 (0.4 - 45.0)	27.8 (0.2 - 79.0)	28.7 (6.2 - 98.0)	130.1 (14.0 - 250.0)	46.1 (0.2 - 270.0)	40.4 (0.2 - 87.0)	56.3 (4.6 - 200.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)
Arsenic (mg/kg)	8.6 (3.2 - 12.0)	7.4 (3.2 - 11.0)	9.1 (5.2 - 12.0)	12.7 (8.9 - 18.0)	15.3 (8.8 - 22.0)	13.8 (8.7 - 18.0)	14.1 (12.0 - 16.0)
Cadmium (mg/kg)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.7 (0.1 - 1.3)	0.4 (0.2 - 0.6)	0.3 (0.1 - 0.6)	0.1 (0.1 - 0.2)
Chromium (mg/kg)	41 (26 - 52)	27 (17 - 45)	43 (24 - 54)	47 (35 - 71)	49 (28 - 77)	49 (29 - 67)	43 (32 - 54)
Copper (mg/kg)	34 (24 - 50)	22 (9 - 54)	75 (47 - 130)	63 (37 - 100)	55 (34 - 84)	53 (35 - 69)	39 (31 - 46)
Lead (mg/kg)	43 (35 - 51)	26 (19 - 37)	49 (33 - 58)	54 (38 - 78)	76 (40 - 250)	53 (40 - 59)	52 (36 - 110)
Mercury (mg/kg)	0.13 (0.09 - 0.16)	0.07 (0.05 - 0.08)	0.19 (0.10 - 0.24)	0.20 (0.10 - 0.46)	0.16 (0.10 - 0.35)	0.12 (0.06 - 0.16)	0.10 (0.06 - 0.14)
Nickel (mg/kg)	26 (17 - 31)	16 (11 - 25)	22 (12 - 27)	27 (20 - 40)	28 (16 - 42)	29 (18 - 38)	26 (19 - 32)
Silver (mg/kg)	0.3 (0.2 - 0.4)	<0.2 (0.2 - <0.2)	0.9 (0.6 - 1.3)	0.6 (0.3 - 1.1)	0.4 (0.3 - 0.6)	0.4 (0.3 - 0.7)	0.2 (0.2 - 0.3)
Zinc (mg/kg)	150 (97 - 180)	82 (55 - 130)	170 (96 - 240)	280 (120 - 430)	230 (140 - 340)	190 (120 - 280)	150 (93 - 200)
Total Polychlorinated Biphenyls (PCBs) (μg/kg) (3)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (μg/kg) (4) (6)	95 (90 - 140)	96 (90 - 150)	95 (90 - 100)	120 (90 - 180)	110 (90 - 230)	96 (90 - 110)	93 (90 - 97)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (μg/kg) (5) (6)	67 (29 - 110)	50 (17 - 110)	230 (65 - 350)	300 (130 - 690)	260 (42 - 1100)	120 (28 - 220)	96 (56 - 130)

Note:
1 Data presented are arithmetic means ; data in brackets indicate ranges.
2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.
3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4 Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5 High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary of marine sediment quality data for the Port Shelter WCZ and Mirs Bay WCZ, 2020 - 2024

	Inner Port Shelter	Outer Port Shelter		Starling Inlet	Crooked Island		Port Island	Mirs Bay (North)
Parameter	PS3	PS5	PS6	MS1	MS2	MS7	MS17	MS3
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63μm (%w/w)	72 (10 - 96)	70 (9 - 96)	65 (9 - 86)	72 (26 - 98)	73 (4 - 99)	57 (<1 - 96)	68 (6 - 93)	75 (7 - 94)
Electrochemical Potential (mV)	-182 (-234 - -103)	-129 (-188 - -71)	-113 (-143 - -71)	-141 (-257 - -54)	-237 (-311 - -182)	-238 (-326 - -137)	-167 (-239 - -58)	-141 (-182 - -65)
Total Solids (%w/w)	37 (31 - 43)	45 (33 - 53)	47 (36 - 54)	42 (37 - 47)	34 (30 - 36)	32 (28 - 36)	34 (31 - 36)	39 (36 - 42)
Total Volatile Soilds (%TS)	11.0 (7.1 - 13.0)	8.8 (6.4 - 12.0)	8.6 (7.3 - 9.7)	7.3 (5.9 - 8.8)	8.5 (6.6 - 10.0)	9.1 (7.3 - 10.0)	9.1 (7.6 - 11.0)	7.6 (6.0 - 10.0)
Chemical Oxygen Demand (mg/kg)	14700 (11000 - 17000)	14700 (13000 - 17000)	12700 (10000 - 16000)	14700 (11000 - 17000)	14500 (12000 - 18000)	15100 (11000 - 18000)	15000 (14000 - 16000)	14200 (10000 - 18000)
Total Carbon (%w/w)	1.1 (0.7 - 1.3)	1.2 (0.7 - 2.3)	1.2 (0.5 - 1.6)	0.6 (0.5 - 0.8)	0.6 (0.5 - 0.7)	0.6 (0.5 - 0.7)	0.7 (0.6 - 1.0)	0.7 (0.6 - 0.9)
Ammonical Nitrogen (mg/kg)	8.52 (4.70 - 14.00)	6.58 (1.20 - 13.00)	5.65 (3.30 - 11.00)	7.58 (2.50 - 17.00)	9.49 (3.70 - 13.00)	8.39 (1.90 - 13.00)	7.60 (0.30 - 13.00)	7.92 (2.60 - 13.00)
Total Kjeldahl Nitrogen (mg/kg)	600 (440 - 720)	540 (400 - 630)	600 (540 - 650)	550 (480 - 740)	560 (500 - 710)	580 (500 - 740)	650 (550 - 770)	530 (460 - 630)
Total Phosphorus (mg/kg)	200 (180 - 210)	190 (150 - 230)	230 (200 - 270)	200 (170 - 230)	180 (150 - 200)	170 (150 - 220)	210 (190 - 250)	190 (140 - 220)
Total Sulphide (mg/kg)	35.4 (3.6 - 150.0)	21.8 (8.1 - 35.0)	25.8 (4.2 - 78.0)	18.8 (4.3 - 32.0)	41.1 (14.0 - 110.0)	35.0 (4.3 - 77.0)	20.8 (3.1 - 50.0)	23.4 (2.9 - 54.0)
Total Cyanide (mg/kg)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)
Arsenic (mg/kg)	7.5 (3.7 - 10.0)	7.6 (4.7 - 10.0)	6.9 (3.8 - 9.4)	9.5 (5.5 - 12.0)	10.4 (7.3 - 12.0)	8.6 (2.8 - 11.0)	9.1 (5.2 - 11.0)	9.2 (4.8 - 11.0)
Cadmium (mg/kg)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	0.2 (0.1 - 0.3)	0.2 (0.1 - 0.4)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)
Chromium (mg/kg)	29 (18 - 39)	27 (13 - 40)	25 (19 - 34)	36 (25 - 46)	44 (33 - 53)	39 (20 - 56)	40 (27 - 51)	36 (26 - 50)
Copper (mg/kg)	22 (13 - 28)	46 (14 - 230)	15 (10 - 23)	22 (18 - 31)	26 (20 - 38)	22 (16 - 28)	19 (12 - 22)	16 (12 - 21)
Lead (mg/kg)	37 (21 - 46)	32 (21 - 40)	32 (26 - 39)	43 (33 - 53)	48 (41 - 58)	46 (27 - 59)	45 (29 - 58)	37 (28 - 54)
Mercury (mg/kg)	0.07 (0.05 - 0.08)	0.07 (0.05 - 0.13)	0.05 (0.05 - 0.06)	0.05 (0.05 - 0.09)	0.05 (0.05 - 0.07)	0.06 (0.05 - 0.08)	0.06 (0.05 - 0.07)	<0.05 (0.05 - <0.05)
Nickel (mg/kg)	17 (9 - 21)	15 (5 - 22)	17 (14 - 20)	22 (15 - 29)	27 (20 - 32)	24 (14 - 29)	26 (17 - 31)	23 (16 - 28)
Silver (mg/kg)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	0.3 (0.2 - 0.5)	0.2 (0.2 - 0.3)	0.2 (0.2 - 0.3)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)
Zinc (mg/kg)	110 (58 - 130)	120 (69 - 230)	88 (62 - 120)	120 (84 - 160)	130 (98 - 150)	110 (78 - 150)	110 (64 - 130)	97 (73 - 130)
Total Polychlorinated Biphenyls (PCBs) (μg/kg) (3)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (μg/kg) (4) (6)	110 (90 - 200)	100 (90 - 160)	100 (90 - 150)	99 (90 - 180)	91 (90 - 94)	97 (90 - 130)	98 (90 - 150)	100 (90 - 140)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (μg/kg) (5) (6)	53 (34 - 92)	43 (16 - 91)	31 (21 - 44)	310 (17 - 2800)	33 (16 - 52)	50 (19 - 150)	39 (19 - 89)	33 (16 - 75)

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3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4. Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5. High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary of marine sediment quality data for the Mirs Bay WCZ, 2020 - 2024

	Mirs Bay (North)		Long Harbour	Waglan Island	Mirs Bay (South)	Mirs Bay (Central)		
Parameter	MS4	MS5	MS6	MS8	MS13	MS14	MS15	MS16
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63μm (%w/w)	82 (5 - 97)	79 (1 - 99)	73 (3 - 95)	55 (3 - 96)	64 (40 - 96)	82 (4 - 95)	77 (8 - 95)	74 (38 - 89)
Electrochemical Potential (mV)	-152 (-285 - -73)	-141 (-243 - -82)	-213 (-304 - -131)	-106 (-241 - -73)	-100 (-255 - -36)	-108 (-272 - -53)	-102 (-303 - -50)	-86 (-217 - -42)
Total Solids (%w/w)	38 (33 - 43)	39 (35 - 46)	33 (30 - 36)	50 (44 - 62)	50 (35 - 65)	50 (43 - 59)	50 (45 - 59)	56 (46 - 62)
Total Volatile Soilds (%TS)	8.2 (6.5 - 11.0)	7.9 (5.3 - 9.9)	10.6 (8.8 - 12.0)	6.3 (5.5 - 7.6)	6.2 (5.3 - 7.3)	6.3 (5.6 - 7.4)	6.6 (5.7 - 8.9)	5.8 (4.9 - 7.4)
Chemical Oxygen Demand (mg/kg)	13700 (11000 - 16000)	14600 (12000 - 17000)	15100 (12000 - 17000)	10290 (8500 - 13000)	10100 (8200 - 14000)	9370 (6400 - 13000)	9940 (6100 - 12000)	9550 (5200 - 13000)
Total Carbon (%w/w)	0.6 (0.5 - 0.7)	0.7 (0.6 - 0.9)	0.9 (0.8 - 1.0)	0.7 (0.5 - 1.4)	0.8 (0.5 - 2.0)	0.7 (0.5 - 2.2)	0.7 (0.5 - 2.2)	0.7 (0.6 - 0.8)
Ammonical Nitrogen (mg/kg)	6.98 (0.75 - 11.00)	6.18 (0.45 - 10.00)	9.50 (0.79 - 16.00)	3.08 (0.16 - 6.70)	3.15 (0.42 - 7.20)	3.74 (0.30 - 7.40)	4.25 (0.45 - 9.80)	7.57 (0.68 - 25.00)
Total Kjeldahl Nitrogen (mg/kg)	560 (410 - 640)	590 (530 - 670)	690 (590 - 750)	470 (370 - 550)	440 (370 - 510)	490 (430 - 530)	490 (410 - 560)	500 (460 - 550)
Total Phosphorus (mg/kg)	200 (170 - 240)	210 (170 - 240)	210 (200 - 230)	220 (180 - 250)	230 (220 - 260)	240 (200 - 270)	240 (180 - 280)	260 (230 - 280)
Total Sulphide (mg/kg)	19.7 (10.0 - 34.0)	18.9 (3.4 - 44.0)	22.5 (9.4 - 37.0)	24.2 (3.2 - 140.0)	15.4 (0.2 - 38.0)	20.8 (2.6 - 110.0)	13.0 (3.3 - 25.0)	15.1 (2.3 - 33.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)
Arsenic (mg/kg)	8.9 (5.3 - 12.0)	8.8 (6.1 - 12.0)	8.3 (5.5 - 11.0)	8.3 (3.7 - 11.0)	8.6 (4.1 - 12.0)	8.6 (5.8 - 10.0)	8.2 (5.4 - 11.0)	7.9 (4.2 - 10.0)
Cadmium (mg/kg)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)	<0.1 (0.1 - <0.1)
Chromium (mg/kg)	38 (23 - 55)	36 (28 - 47)	38 (26 - 75)	33 (17 - 49)	33 (17 - 51)	34 (20 - 42)	31 (20 - 41)	29 (17 - 47)
Copper (mg/kg)	15 (8 - 19)	16 (13 - 23)	18 (13 - 22)	15 (10 - 21)	15 (10 - 24)	13 (7 - 20)	12 (8 - 14)	10 (6 - 16)
Lead (mg/kg)	37 (22 - 46)	41 (33 - 59)	39 (28 - 47)	35 (19 - 41)	32 (19 - 43)	33 (24 - 37)	32 (23 - 40)	29 (17 - 41)
Mercury (mg/kg)	0.05 (0.05 - 0.06)	0.05 (0.05 - 0.06)	0.06 (0.05 - 0.09)	0.06 (0.05 - 0.13)	0.06 (0.05 - 0.10)	0.06 (0.05 - 0.10)	<0.05 (0.05 - <0.05)	<0.05 (0.05 - <0.05)
Nickel (mg/kg)	25 (14 - 32)	24 (19 - 33)	24 (17 - 33)	22 (11 - 30)	22 (11 - 31)	22 (13 - 27)	21 (13 - 27)	19 (12 - 29)
Silver (mg/kg)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)
Zinc (mg/kg)	110 (50 - 170)	110 (78 - 160)	110 (78 - 130)	89 (46 - 120)	91 (48 - 130)	88 (53 - 120)	85 (51 - 130)	73 (43 - 110)
Total Polychlorinated Biphenyls (PCBs) (μg/kg) (3)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (μg/kg) (4) (6)	94 (90 - 130)	110 (90 - 330)	92 (90 - 110)	91 (90 - 97)	93 (90 - 120)	91 (90 - 99)	90 (90 - 90)	92 (90 - 110)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (μg/kg) (5) (6)	35 (17 - 110)	78 (16 - 520)	45 (17 - 73)	43 (19 - 98)	38 (22 - 65)	30 (20 - 69)	25 (18 - 39)	28 (18 - 75)

Note:
1. Data presented are arithmetic means ; data in brackets indicate ranges.
2. All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.
3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4. Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5. High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary of marine sediment quality data for the North Western WCZ and Western Buffer WCZ, 2020 - 2024

Parameter	Pearl Island NS2	Pillar Point NS3	Urmston Road NS4	Chek Lap Kok (North) NS6	Tsing Yi (South) WS1	Hong Kong Island (West) WS2
Number of samples	10	10	10	10	10	10
Particle Size Fractionation <63μm (%w/w)	71 (20 - 97)	68 (25 - 96)	64 (28 - 89)	65 (13 - 89)	66 (17 - 80)	56 (26 - 87)
Electrochemical Potential (mV)	-122 (-158 - -85)	-123 (-180 - -76)	-138 (-197 - -107)	-122 (-222 - -87)	-216 (-353 - -97)	-224 (-293 - -108)
Total Solids (%w/w)	46 (37 - 57)	47 (36 - 59)	50 (34 - 58)	45 (41 - 49)	41 (38 - 44)	42 (38 - 49)
Total Volatile Soilds (%TS)	6.7 (4.9 - 7.8)	7.2 (5.3 - 12.0)	6.2 (4.2 - 10.0)	7.0 (4.6 - 9.5)	7.2 (6.1 - 7.9)	7.0 (5.9 - 8.0)
Chemical Oxygen Demand (mg/kg)	12600 (10000 - 19000)	15170 (6700 - 19000)	12680 (8800 - 22000)	13080 (9800 - 21000)	17270 (8700 - 22000)	14980 (9800 - 18000)
Total Carbon (%w/w)	0.7 (0.5 - 1.6)	0.7 (0.6 - 1.1)	0.6 (0.5 - 0.8)	0.5 (0.4 - 1.3)	0.6 (0.6 - 0.7)	0.6 (0.5 - 0.8)
Ammonical Nitrogen (mg/kg)	7.78 (0.05 - 17.00)	8.41 (0.32 - 16.00)	6.65 (0.48 - 21.00)	6.44 (0.61 - 18.00)	18.08 (8.30 - 35.00)	8.08 (0.58 - 22.00)
Total Kjeldahl Nitrogen (mg/kg)	490 (370 - 760)	490 (380 - 770)	450 (300 - 770)	480 (370 - 750)	580 (500 - 660)	530 (440 - 650)
Total Phosphorus (mg/kg)	220 (190 - 250)	220 (180 - 240)	210 (180 - 250)	230 (190 - 270)	220 (200 - 240)	210 (200 - 230)
Total Sulphide (mg/kg)	35.0 (2.5 - 77.0)	58.8 (3.7 - 120.0)	58.1 (5.4 - 230.0)	52.1 (2.8 - 190.0)	116.3 (9.4 - 290.0)	60.7 (5.6 - 180.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)
Arsenic (mg/kg)	11.0 (6.0 - 15.0)	12.4 (8.0 - 18.0)	11.6 (8.3 - 16.0)	13.8 (4.6 - 18.0)	10.7 (6.2 - 15.0)	10.7 (7.4 - 14.0)
Cadmium (mg/kg)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.3)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.2 (0.1 - 0.2)	0.2 (0.1 - 0.3)
Chromium (mg/kg)	39 (27 - 55)	39 (28 - 51)	35 (17 - 48)	40 (23 - 50)	43 (28 - 59)	44 (34 - 57)
Copper (mg/kg)	34 (27 - 44)	32 (25 - 43)	25 (16 - 36)	30 (21 - 39)	54 (33 - 78)	48 (25 - 72)
Lead (mg/kg)	42 (28 - 52)	39 (32 - 48)	37 (25 - 47)	41 (27 - 52)	43 (29 - 53)	43 (35 - 48)
Mercury (mg/kg)	0.11 (0.06 - 0.14)	0.09 (0.06 - 0.11)	0.11 (0.05 - 0.54)	0.09 (0.05 - 0.17)	0.17 (0.12 - 0.21)	0.18 (0.11 - 0.32)
Nickel (mg/kg)	22 (16 - 31)	23 (16 - 28)	20 (10 - 29)	24 (16 - 30)	25 (17 - 33)	25 (19 - 30)
Silver (mg/kg)	0.2 (0.2 - 0.4)	0.2 (0.2 - 0.3)	<0.2 (0.2 - <0.2)	<0.2 (0.2 - <0.2)	0.6 (0.5 - 0.8)	0.5 (0.3 - 1.0)
Zinc (mg/kg)	140 (85 - 200)	120 (89 - 160)	130 (78 - 200)	120 (74 - 170)	160 (84 - 200)	160 (94 - 180)
Total Polychlorinated Biphenyls (PCBs) (μg/kg) (3)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (μg/kg) (4) (6)	97 (90 - 130)	99 (90 - 150)	98 (90 - 150)	93 (90 - 110)	97 (90 - 100)	97 (90 - 150)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (μg/kg) (5) (6)	180 (51 - 400)	190 (52 - 600)	95 (51 - 170)	110 (31 - 210)	240 (120 - 390)	130 (31 - 300)

Note:
1. Data presented are arithmetic means ; data in brackets indicate ranges.
2. All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.
3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4. Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5. High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary of marine sediment quality data for the Eastern Buffer WCZ and Victoria Harbour WCZ, 2020 - 2024

	Eastern Buffer			Victoria Harbour				
	Chai Wan	Tathong Channel		(East)	(Central)	(West)	Rambler Channel	
Parameter	ES1	ES2	ES4	VS3	VS5	VS6	VS9	VS10
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	71 (24 - 95)	76 (10 - 98)	81 (52 - 97)	70 (19 - 96)	73 (20 - 93)	61 (30 - 98)	76 (10 - 98)	65 (5 - 90)
Electrochemical Potential (mV)	-136 (-220 - -85)	-158 (-243 - -103)	-115 (-187 - -54)	-146 (-313 - -59)	-228 (-340 - -101)	-238 (-356 - -124)	-248 (-333 - -157)	-190 (-333 - -112)
Total Solids (%w/w)	50 (45 - 61)	49 (42 - 67)	49 (44 - 63)	43 (35 - 50)	40 (35 - 48)	40 (35 - 59)	40 (38 - 46)	42 (38 - 58)
Total Volatile Soilds (%TS)	6.7 (4.9 - 7.8)	6.4 (3.8 - 8.0)	6.8 (4.7 - 7.7)	7.8 (6.7 - 9.2)	7.9 (6.0 - 9.3)	7.5 (5.3 - 8.4)	6.9 (5.0 - 8.1)	7.3 (5.4 - 8.2)
Chemical Oxygen Demand (mg/kg)	12490 (8900 - 17000)	11700 (8900 - 15000)	12460 (9600 - 19000)	17400 (13000 - 22000)	17380 (9800 - 21000)	17400 (11000 - 22000)	15720 (7200 - 20000)	17800 (10000 - 23000)
Total Carbon (%w/w)	0.9 (0.7 - 1.2)	0.7 (0.5 - 0.9)	0.8 (0.7 - 1.0)	0.8 (0.7 - 1.0)	0.7 (0.6 - 0.9)	0.7 (0.6 - 0.8)	0.6 (0.5 - 0.9)	0.7 (0.6 - 0.9)
Ammonical Nitrogen (mg/kg)	5.25 (0.25 - 9.30)	8.72 (1.80 - 29.00)	5.78 (2.20 - 13.00)	4.91 (0.10 - 11.00)	9.47 (2.40 - 21.00)	13.57 (1.60 - 25.00)	15.28 (8.80 - 23.00)	7.68 (2.50 - 16.00)
Total Kjeldahl Nitrogen (mg/kg)	460 (360 - 560)	450 (340 - 500)	480 (380 - 570)	500 (450 - 580)	560 (490 - 620)	580 (430 - 640)	520 (440 - 600)	490 (420 - 540)
Total Phosphorus (mg/kg)	210 (180 - 250)	200 (170 - 240)	210 (190 - 240)	210 (190 - 220)	210 (200 - 250)	220 (200 - 270)	220 (190 - 300)	210 (190 - 230)
Total Sulphide (mg/kg)	23.6 (8.2 - 43.0)	29.8 (5.2 - 83.0)	26.8 (9.7 - 46.0)	68.2 (6.1 - 180.0)	121.9 (39.0 - 340.0)	125.8 (32.0 - 240.0)	158.9 (38.0 - 440.0)	70.4 (7.5 - 160.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.3)	0.2 (0.1 - 0.4)	0.2 (0.1 - 0.2)
Arsenic (mg/kg)	7.2 (3.4 - 9.0)	7.4 (4.9 - 11.0)	7.4 (4.2 - 11.0)	9.5 (6.4 - 13.0)	10.7 (8.5 - 14.0)	11.3 (7.3 - 14.0)	10.9 (7.8 - 14.0)	11.0 (6.8 - 16.0)
Cadmium (mg/kg)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.2 (0.1 - 0.4)	0.3 (0.2 - 0.6)	0.3 (0.2 - 0.4)	0.8 (0.2 - 6.1)	0.4 (0.2 - 0.8)
Chromium (mg/kg)	33 (16 - 42)	35 (16 - 50)	35 (17 - 51)	46 (25 - 60)	51 (38 - 62)	48 (34 - 69)	54 (32 - 120)	59 (28 - 93)
Copper (mg/kg)	33 (17 - 50)	39 (15 - 89)	46 (23 - 83)	90 (49 - 120)	110 (73 - 140)	88 (61 - 130)	95 (43 - 400)	130 (48 - 250)
Lead (mg/kg)	34 (19 - 41)	36 (21 - 52)	39 (22 - 51)	50 (37 - 59)	59 (48 - 80)	62 (45 - 130)	47 (37 - 92)	52 (37 - 73)
Mercury (mg/kg)	0.14 (0.07 - 0.41)	0.10 (0.05 - 0.14)	0.14 (0.07 - 0.31)	0.33 (0.19 - 0.47)	0.33 (0.26 - 0.45)	0.34 (0.26 - 0.46)	0.22 (0.12 - 0.50)	0.21 (0.10 - 0.37)
Nickel (mg/kg)	18 (10 - 25)	20 (11 - 27)	18 (9 - 26)	22 (13 - 28)	24 (20 - 29)	25 (16 - 33)	29 (19 - 62)	27 (13 - 37)
Silver (mg/kg)	0.3 (0.2 - 0.6)	0.4 (0.2 - 0.6)	0.6 (0.2 - 1.6)	1.7 (0.8 - 2.8)	1.5 (1.0 - 2.3)	1.3 (0.9 - 2.2)	1.2 (0.6 - 5.4)	1.5 (0.5 - 3.2)
Zinc (mg/kg)	110 (58 - 160)	120 (58 - 220)	130 (63 - 210)	180 (120 - 240)	240 (160 - 340)	220 (170 - 340)	180 (130 - 400)	200 (99 - 290)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) [(3)]	18 (18 - 18)	18 (18 - 18)	18 (18 - 21)	19 (18 - 30)	19 (18 - 22)	18 (18 - 19)	18 (18 - 22)	20 (18 - 29)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) [(4) (6)]	96 (90 - 120)	130 (90 - 290)	100 (90 - 150)	230 (90 - 740)	130 (90 - 220)	110 (90 - 140)	110 (90 - 210)	120 (90 - 200)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) [(5) (6)]	160 (37 - 280)	210 (17 - 700)	270 (43 - 560)	890 (120 - 4700)	590 (160 - 1300)	410 (69 - 550)	200 (44 - 380)	310 (130 - 650)

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3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
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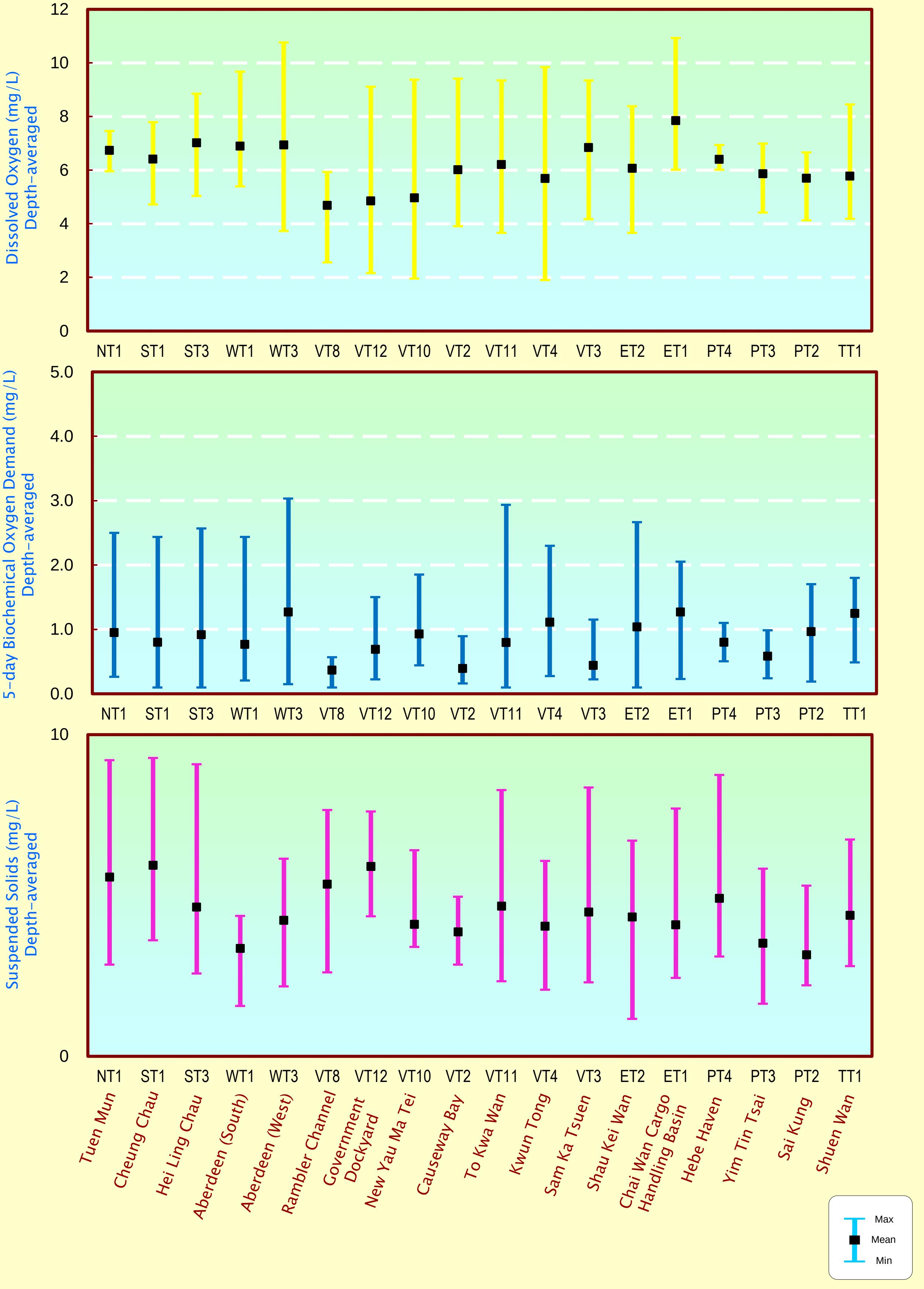
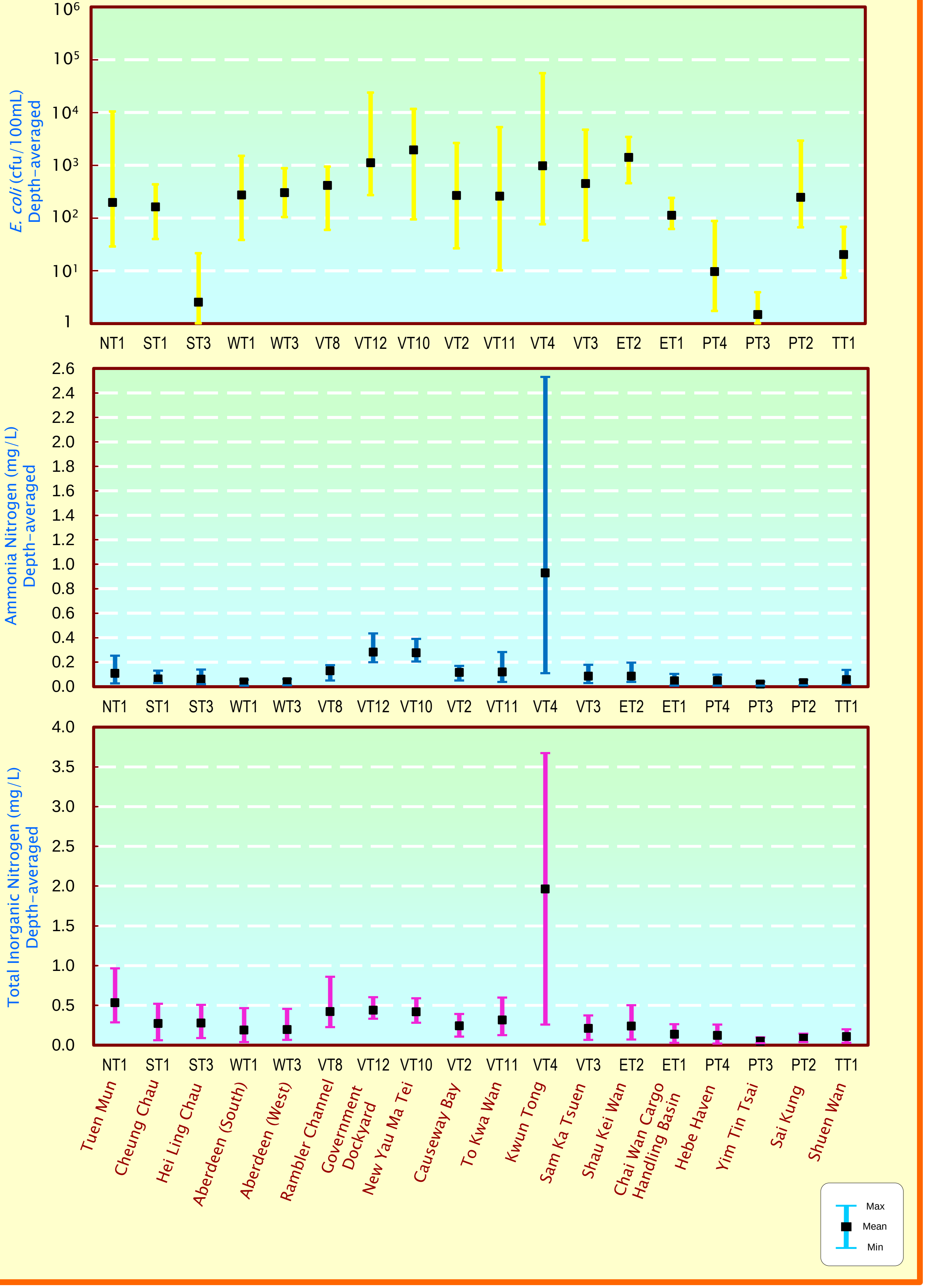
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Water quality of typhoon shelters, sheltered anchorages and Government Dockyard in 2024



Water quality of typhoon shelters, sheltered anchorages and Government Dockyard in 2024 (continued)



Long-term water quality trends in typhoon shelters, sheltered anchorages and Government Dockyard, 1986-2024

Monitoring Station		NT1	ST1	ST3	WT3	WT1	VT8	VT10	VT2	VT11
Monitoring Period		1986 - 2024	1986 - 2024	2000 - 2024	1986 - 2024	1986 - 2024	1986 - 2024	1993 - 2024	1986 - 2024	1994 - 2024
Parameter	Water Depth									
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	NA	↗	↗	↗	↗	NA	NA	NA	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	-	-	-	↗	↗	↗	↗
	Middle	NA	-	-	-	-	NA	NA	NA	↗
	Bottom	↗	-	-	-	↗	↗	↗	↗	↗
	Average	↗	-	-	-	-	↗	↗	↗	↗
Dissolved Oxygen (%)	Surface	↗	-	-	-	-	↗	↗	↗	↗
	Middle	NA	-	-	-	-	NA	NA	NA	↗
	Bottom	↗	-	-	-	↗	↗	↗	↗	↗
	Average	↗	-	-	-	-	↗	↗	↗	↗
pH	Surface	↘	↘	↘	↘	↘	↘	-	-	↘
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	↘
	Bottom	-	↘	↘	↘	↘	↘	-	-	↘
	Average	↘	↘	↘	↘	↘	↘	-	-	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	↗	↗	-
Turbidity (NTU)	Surface	↘	↘	↘	↘	↘	-	↘	-	-
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	-
	Bottom	-	↘	↘	↘	↘	-	↘	-	-
	Average	↘	↘	↘	↘	↘	-	↘	-	-
Suspended Solids (mg/L)	Surface	↘	-	-	-	-	↘	↘	-	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	-	-	-	-	-	↘	↘	-	-
	Average	↘	-	-	-	-	↘	↘	-	-
Total volatile solids (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	-	-	-	-	-	↘	↘	↘	-
	Average	↘	-	-	-	-	↘	↘	↘	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	-	↘	-	↘	-	↘	↘
	Middle	NA	-	-	↘	↘	NA	NA	NA	↘
	Bottom	↘	-	-	↘	↘	↘	↘	↘	↘
	Average	↘	↘	-	↘	↘	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	-	-	↘	↘	↘	↘	↘	↘
	Middle	NA	-	-	↘	↘	NA	NA	NA	↘
	Bottom	↘	-	-	↘	↘	↘	↘	↘	↘
	Average	↘	-	-	↘	↘	↘	↘	↘	↘
Unionised Ammonia(mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↗	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	↗	-	-	-	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	-	↗	-	-	↗	↗	↗	↗
	Middle	NA	-	↗	-	-	NA	NA	NA	↗
	Bottom	↗	-	↗	-	-	↗	↗	↗	↗
	Average	↗	-	↗	-	-	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	-	-	-	-	-	↘	↘	↘
	Middle	NA	-	-	-	-	NA	NA	NA	↘
	Bottom	-	-	-	-	-	-	↘	↘	↘
	Average	-	-	-	-	-	-	↘	↘	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	-	↗	-	↗	↘	-	↘	↘
	Middle	NA	-	↗	-	↗	NA	NA	NA	↘
	Bottom	↘	-	↗	↗	↗	↘	↘	↘	↘
	Average	↘	-	↗	↗	↗	↘	-	↘	↘
Total nitrogen (mg/L)	Surface	-	↗	↗	↗	↗	↘	↘	↘	-
	Middle	NA	↗	↗	↗	↗	NA	NA	NA	-
	Bottom	-	-	↗	↗	↗	↘	↘	↘	-
	Average	-	↗	↗	↗	↗	↘	↘	↘	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	↘
	Middle	NA	-	-	-	-	NA	NA	NA	↘
	Bottom	↘	-	-	-	-	↘	↘	↘	↘
	Average	↘	-	↗	-	-	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-	-	↘	-	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	-	-	-	-	-	-	↘	-	-
	Average	-	-	-	-	-	-	-	-	-
Chlorophyll-*a* (μg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
E. coli (cfu/100mL)	Surface	↘	↘	-	↘	↘	↘	↘	↘	↘
	Middle	NA	-	-	↘	↘	NA	NA	NA	↘
	Bottom	↘	-	-	↘	↘	↘	↘	↘	↘
	Average	↘	↘	-	↘	↘	↘	↘	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	-	-	↘	↘	↘	↘	↘	↘
	Middle	NA	-	-	↘	↘	NA	NA	NA	↘
	Bottom	-	-	-	↘	-	-	↘	↘	↘
	Average	↘	-	-	↘	↘	↘	↘	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. – indicates no significant trend
3. NA (Not Applicable) indicates the measurement was not made due to shallow water
4. ↗ significant increase
5. ↘ significant decrease

Long-term water quality trends in typhoon shelters, sheltered anchorages and Government Dockyard, 1986-2024

Monitoring Station		VT12	VT4	VT3	ET2	ET1	PT4	PT3	PT2	TT1
Monitoring Period		2000 - 2024	1987 - 2024	1986 - 2024	1993 - 2024	1986 - 2024	1986 - 2024	1986 - 2024	1986 - 2024	1986 - 2024
Parameter	Water Depth									
Temperature (℃)	Surface	↗	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	NA	↗	NA	↗	↗	NA	↗	NA	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	NA	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	↘	-	-	-	-	-	-	-
	Middle	NA	-	NA	-	-	NA	-	NA	-
	Bottom	-	-	-	-	-	-	-	NA	-
	Average	-	↘	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	↗	↗	↗	↗	↘	-	-	↘
	Middle	NA	↗	NA	↗	↗	NA	-	NA	↘
	Bottom	-	↗	↗	↗	↗	-	-	NA	-
	Average	-	↗	↗	↗	↗	-	-	-	↘
Dissolved Oxygen (%)	Surface	-	↗	↗	↗	↗	↘	-	-	↘
	Middle	NA	↗	NA	↗	↗	NA	-	NA	↘
	Bottom	-	↗	↗	↗	↗	-	-	NA	-
	Average	-	↗	↗	↗	↗	-	-	-	↘
pH	Surface	↘	-	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	-	NA	↘	↘	NA	↘	NA	↘
	Bottom	↘	-	↘	↘	↘	↘	↘	NA	↘
	Average	↘	-	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	-	↗	-	-	↗
Turbidity (NTU)	Surface	↘	↘	-	-	-	-	↘	↘	-
	Middle	NA	↘	NA	-	↘	NA	↘	NA	-
	Bottom	↘	↘	-	-	↘	-	↘	NA	-
	Average	↘	↘	-	-	-	-	↘	↘	-
Suspended Solids (mg/L)	Surface	-	↘	↘	-	-	-	↗	-	-
	Middle	NA	-	NA	-	-	NA	↗	NA	-
	Bottom	↘	-	-	-	-	-	-	NA	-
	Average	-	-	↘	-	-	-	↗	-	-
Total volatile solids (mg/L)	Surface	-	↘	-	-	-	-	-	-	-
	Middle	NA	-	NA	-	-	NA	↗	NA	-
	Bottom	-	-	-	-	-	-	-	NA	-
	Average	-	↘	-	-	-	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	↘	↘	↘	-	↘	-	↘	↘
	Middle	NA	↘	NA	↘	-	NA	-	NA	↘
	Bottom	-	↘	↘	↘	↘	↘	-	NA	↘
	Average	-	↘	↘	↘	-	↘	-	↘	↘
Ammonia nitrogen (mg/L)	Surface	-	↘	↘	↘	↘	-	-	-	↘
	Middle	NA	↘	NA	↘	↘	NA	↗	NA	↘
	Bottom	↘	↘	↘	↘	↘	-	-	NA	↘
	Average	↘	↘	↘	↘	↘	-	-	-	↘
Unionised Ammonia(mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	NA	↘	↘	NA	-	NA	↘
	Bottom	↘	↘	↘	↘	↘	↘	-	NA	↘
	Average	↘	↘	↘	↘	↘	↘	-	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	↗	-	-	-	-	-	-	-
	Middle	NA	↗	NA	-	-	NA	-	NA	-
	Bottom	↘	↗	-	-	-	-	-	NA	-
	Average	-	↗	-	-	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	-	↗	↗	↗	-	-	-	-	-
	Middle	NA	↗	NA	-	-	NA	-	NA	-
	Bottom	-	↗	↗	-	-	-	-	NA	-
	Average	-	↗	↗	-	-	-	-	-	-
Total inorganic nitrogen (mg/L)	Surface	-	-	↘	↘	-	-	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	↗	NA	↘
	Bottom	↘	↘	↘	↘	↘	-	-	NA	↘
	Average	-	-	↘	↘	-	-	-	-	↘
Total Kjeldahl nitrogen (mg/L)	Surface	-	↘	↘	↘	-	-	↗	↗	↘
	Middle	NA	↘	NA	↘	-	NA	↗	NA	-
	Bottom	-	↘	↘	↘	-	-	↗	NA	-
	Average	↗	↘	↘	↘	-	-	↗	↗	-
Total nitrogen (mg/L)	Surface	↗	-	↘	↘	-	-	↗	↗	-
	Middle	NA	↘	NA	↘	-	NA	↗	NA	-
	Bottom	-	↘	↘	↘	-	-	↗	NA	-
	Average	↗	↘	↘	↘	-	-	↗	↗	-
Orthophosphate phosphorus (mg/L)	Surface	↘	-	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	NA	↘	↘	NA	↘	NA	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	NA	↘
	Average	↘	-	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	↘	↘	-	↘	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	-	NA	↘
	Bottom	-	↘	↘	↘	-	↘	-	NA	↘
	Average	-	↘	↘	↘	-	↘	-	-	↘
Silica (mg/L)	Surface	-	-	↘	↘	-	↗	-	-	↗
	Middle	NA	-	NA	↘	-	NA	-	NA	-
	Bottom	-	-	-	↘	-	-	-	NA	-
	Average	-	-	-	↘	-	↗	-	-	-
Chlorophyll-*a* (μg/L)	Surface	-	↗	-	-	-	↘	-	↘	↘
	Middle	NA	↗	NA	-	-	NA	-	NA	↘
	Bottom	-	↗	-	-	-	-	-	NA	-
	Average	-	↗	-	-	-	↘	-	↘	↘
E. coli (cfu/100mL)	Surface	-	↘	↘	↘	↘	↘	-	-	↘
	Middle	NA	↘	NA	↘	↘	NA	-	NA	↘
	Bottom	-	↘	↘	↘	↘	↘	-	NA	↘
	Average	-	↘	↘	↘	↘	↘	-	-	↘
Faecal coliforms (cfu/100mL)	Surface	-	↘	↘	↘	-	↘	-	-	↘
	Middle	NA	↘	NA	↘	↘	NA	-	NA	↘
	Bottom	-	↘	↘	-	-	-	-	NA	↘
	Average	-	↘	↘	↘	↘	-	-	-	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$
2. - indicates no significant trend
3. NA (Not Applicable) indicates the measurement was not made due to shallow water
4. ↗ significant increase
5. ↘ significant decrease

Summary of water quality data for typhoon shelters, sheltered anchorages and Government Dockyard in 2024

Parameter		Tuen Mun NT1	Cheung Chau ST1	Hei Ling Chau ST3	Aberdeen (South) WT1	Aberdeen (West) WT3	Rambler Channel VT8
Number of samples		6	6	6	6	6	6
Temperature (°C)		24.9 (19.8 - 29.7)	24.5 (18.1 - 29.9)	24.3 (17.8 - 29.9)	24.5 (19.8 - 28.4)	24.3 (19.9 - 27.4)	25.0 (20.0 - 29.4)
Salinity		25.6 (16.5 - 32.3)	30.0 (20.0 - 33.5)	29.6 (20.7 - 33.2)	30.2 (20.2 - 34.2)	29.8 (22.5 - 33.4)	30.5 (27.3 - 33.1)
Dissolved Oxygen (mg/L)		6.7 (6.0 - 7.5)	6.4 (4.7 - 7.8)	7.0 (5.0 - 8.9)	6.9 (5.4 - 9.7)	7.0 (3.7 - 10.8)	4.7 (2.6 - 5.9)
	Bottom	7.2 (5.6 - 8.6)	6.3 (4.1 - 8.4)	6.8 (4.5 - 9.5)	7.0 (4.7 - 11.6)	6.7 (3.5 - 13.6)	4.8 (2.9 - 6.6)
Dissolved Oxygen (% Saturation)		94 (80 - 108)	91 (69 - 113)	99 (73 - 127)	98 (79 - 140)	98 (56 - 154)	67 (38 - 85)
	Bottom	100 (85 - 121)	89 (60 - 122)	97 (65 - 128)	99 (70 - 168)	95 (52 - 195)	68 (43 - 88)
pH		8.0 (7.7 - 8.3)	8.0 (7.7 - 8.3)	8.0 (7.6 - 8.3)	7.8 (7.6 - 8.1)	7.7 (7.4 - 8.1)	7.7 (7.3 - 8.1)
Secchi Disc Depth (m)		1.7 (1.5 - 1.9)	2.1 (1.6 - 2.6)	2.1 (1.3 - 2.7)	2.7 (1.9 - 4.0)	2.5 (1.8 - 3.6)	2.2 (1.8 - 2.9)
Turbidity (NTU)		4.4 (0.6 - 9.8)	3.0 (0.7 - 7.3)	2.9 (0.6 - 9.4)	3.1 (2.5 - 4.8)	3.7 (2.8 - 4.4)	3.7 (2.6 - 5.2)
Suspended Solids (mg/L)		5.6 (2.9 - 9.2)	5.9 (3.6 - 9.3)	4.6 (2.6 - 9.1)	3.4 (1.6 - 4.4)	4.2 (2.2 - 6.1)	5.4 (2.6 - 7.7)
5-day Biochemical Oxygen Demand (mg/L)		1.0 (0.3 - 2.5)	0.8 (<0.1 - 2.4)	0.9 (<0.1 - 2.6)	0.8 (0.2 - 2.4)	1.3 (0.2 - 3.0)	0.4 (0.1 - 0.6)
Ammonia Nitrogen (mg/L)		0.108 (0.028 - 0.255)	0.063 (0.032 - 0.130)	0.061 (0.021 - 0.140)	0.039 (0.010 - 0.058)	0.040 (0.011 - 0.063)	0.129 (0.050 - 0.175)
Unionised Ammonia (mg/L)		0.004 (0.002 - 0.010)	0.003 (<0.001 - 0.007)	0.003 (<0.001 - 0.008)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	0.003 (0.002 - 0.006)
Nitrite Nitrogen (mg/L)		0.039 (0.007 - 0.079)	0.034 (0.004 - 0.100)	0.035 (0.006 - 0.120)	0.014 (<0.002 - 0.035)	0.015 (0.002 - 0.034)	0.059 (0.003 - 0.200)
Nitrate Nitrogen (mg/L)		0.387 (0.180 - 0.875)	0.174 (0.023 - 0.460)	0.183 (0.049 - 0.443)	0.136 (0.017 - 0.407)	0.141 (0.029 - 0.403)	0.233 (0.093 - 0.610)
Total Inorganic Nitrogen (mg/L)		0.53 (0.28 - 0.97)	0.27 (0.06 - 0.52)	0.28 (0.09 - 0.51)	0.19 (0.04 - 0.46)	0.20 (0.06 - 0.45)	0.42 (0.23 - 0.86)
Total Kjeldahl Nitrogen (mg/L)		0.27 (0.19 - 0.34)	0.23 (0.14 - 0.34)	0.24 (0.10 - 0.43)	0.18 (0.11 - 0.22)	0.17 (0.13 - 0.21)	0.27 (0.23 - 0.37)
Total Nitrogen (mg/L)		0.70 (0.43 - 1.22)	0.44 (0.17 - 0.78)	0.46 (0.19 - 0.76)	0.33 (0.16 - 0.65)	0.33 (0.16 - 0.62)	0.57 (0.33 - 1.05)
Orthophosphate Phosphorus (mg/L)		0.021 (0.008 - 0.033)	0.015 (0.005 - 0.028)	0.013 (<0.002 - 0.027)	0.006 (<0.002 - 0.010)	0.007 (0.002 - 0.010)	0.056 (0.008 - 0.249)
Total Phosphorus (mg/L)		0.08 (0.04 - 0.13)	0.08 (0.04 - 0.15)	0.08 (0.03 - 0.17)	0.06 (0.04 - 0.08)	0.06 (0.03 - 0.09)	0.12 (0.04 - 0.31)
Silica (as SiO_2) (mg/L)		2.62 (0.42 - 7.00)	1.00 (0.25 - 1.90)	0.92 (0.07 - 1.77)	0.67 (0.17 - 1.30)	0.73 (0.24 - 1.30)	1.35 (0.29 - 3.65)
Chlorophyll-*a* (µg/L)		7.0 (0.5 - 19.0)	8.9 (0.7 - 28.7)	9.4 (0.6 - 28.3)	4.6 (0.6 - 12.7)	5.0 (0.5 - 16.7)	1.7 (1.2 - 2.3)
E.coli (count/100mL)		200 (29 - 10000)	160 (40 - 430)	3 (1 - 21)	270 (38 - 1500)	300 (100 - 870)	410 (59 - 940)
Faecal Coliforms (count/100mL)		1200 (120 - 41000)	390 (86 - 790)	7 (1 - 230)	790 (86 - 4800)	1000 (210 - 2400)	1000 (88 - 2700)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for typhoon shelters, sheltered anchorages and Government Dockyard in 2024 (continued)

Parameter		Government Dockyard VT12	New Yau Ma Tei VT10	Causeway Bay VT2	To Kwa Wan VT11	Kwun Tong VT4	Sam Ka Tsuen VT3
Number of samples		6	6	6	12	12	6
Temperature (°C)		25.0 (19.5 - 29.7)	24.9 (19.8 - 29.6)	25.0 (19.3 - 30.5)	24.8 (19.6 - 29.4)	25.0 (19.7 - 29.4)	24.8 (19.6 - 29.2)
Salinity		30.9 (27.7 - 33.0)	31.0 (27.9 - 33.3)	30.9 (26.5 - 33.2)	31.3 (25.3 - 33.9)	29.0 (24.0 - 32.2)	31.5 (28.4 - 33.6)
Dissolved Oxygen (mg/L)		4.9 (2.2 - 9.1)	5.0 (2.0 - 9.4)	6.0 (3.9 - 9.4)	6.2 (3.7 - 9.4)	5.7 (1.9 - 9.8)	6.9 (4.2 - 9.4)
	Bottom	4.5 (2.1 - 10.0)	4.5 (1.9 - 10.4)	5.5 (3.7 - 10.3)	6.0 (3.3 - 10.8)	5.5 (1.8 - 11.2)	6.9 (3.5 - 10.2)
Dissolved Oxygen (% Saturation)		68 (32 - 122)	71 (30 - 126)	85 (58 - 126)	89 (57 - 133)	80 (29 - 131)	98 (62 - 130)
	Bottom	64 (33 - 134)	64 (29 - 140)	78 (55 - 139)	87 (51 - 153)	78 (28 - 150)	98 (53 - 146)
pH		7.6 (7.2 - 8.0)	7.6 (7.2 - 8.0)	7.7 (7.4 - 8.1)	7.8 (7.4 - 8.2)	7.7 (7.3 - 8.1)	7.7 (7.3 - 8.0)
Secchi Disc Depth (m)		2.0 (1.5 - 2.3)	2.2 (1.8 - 2.5)	2.3 (1.9 - 3.0)	2.4 (1.6 - 3.6)	2.3 (1.6 - 3.0)	2.5 (2.0 - 3.0)
Turbidity (NTU)		4.2 (3.3 - 7.2)	3.0 (2.3 - 3.4)	3.3 (2.4 - 4.8)	3.4 (0.9 - 5.3)	2.5 (0.7 - 4.3)	2.9 (2.0 - 3.6)
Suspended Solids (mg/L)		5.9 (4.4 - 7.6)	4.1 (3.4 - 6.4)	3.9 (2.9 - 5.0)	4.7 (2.3 - 8.3)	4.0 (2.1 - 6.1)	4.5 (2.3 - 8.4)
5-day Biochemical Oxygen Demand (mg/L)		0.7 (0.2 - 1.5)	0.9 (0.4 - 1.9)	0.4 (0.2 - 0.9)	0.8 (<0.1 - 2.9)	1.1 (0.3 - 2.3)	0.4 (0.2 - 1.2)
Ammonia Nitrogen (mg/L)		0.283 (0.200 - 0.435)	0.278 (0.205 - 0.390)	0.117 (0.052 - 0.170)	0.120 (0.039 - 0.283)	0.930 (0.111 - 2.530)	0.085 (0.030 - 0.180)
Unionised Ammonia (mg/L)		0.006 (0.003 - 0.011)	0.006 (0.002 - 0.015)	0.004 (<0.001 - 0.008)	0.003 (<0.001 - 0.010)	0.030 (0.001 - 0.123)	0.002 (<0.001 - 0.007)
Nitrite Nitrogen (mg/L)		0.027 (0.012 - 0.051)	0.027 (0.010 - 0.055)	0.021 (0.004 - 0.052)	0.028 (0.004 - 0.074)	0.257 (0.007 - 0.670)	0.021 (0.003 - 0.052)
Nitrate Nitrogen (mg/L)		0.129 (0.070 - 0.200)	0.115 (0.065 - 0.180)	0.104 (0.041 - 0.190)	0.168 (0.042 - 0.313)	0.776 (0.130 - 1.690)	0.103 (0.034 - 0.180)
Total Inorganic Nitrogen (mg/L)		0.44 (0.33 - 0.60)	0.42 (0.28 - 0.59)	0.24 (0.11 - 0.39)	0.32 (0.12 - 0.60)	1.96 (0.26 - 3.67)	0.21 (0.07 - 0.37)
Total Kjeldahl Nitrogen (mg/L)		0.50 (0.39 - 0.65)	0.51 (0.32 - 0.90)	0.28 (0.16 - 0.41)	0.29 (0.16 - 0.43)	1.29 (0.23 - 2.87)	0.25 (0.12 - 0.51)
Total Nitrogen (mg/L)		0.66 (0.53 - 0.81)	0.65 (0.42 - 1.04)	0.40 (0.21 - 0.57)	0.49 (0.23 - 0.72)	2.32 (0.37 - 4.08)	0.38 (0.16 - 0.70)
Orthophosphate Phosphorus (mg/L)		0.035 (0.023 - 0.052)	0.027 (0.015 - 0.045)	0.012 (0.004 - 0.019)	0.025 (0.006 - 0.086)	0.209 (0.015 - 0.493)	0.014 (0.006 - 0.033)
Total Phosphorus (mg/L)		0.13 (0.07 - 0.18)	0.12 (0.06 - 0.19)	0.10 (0.04 - 0.14)	0.12 (0.04 - 0.21)	0.36 (0.08 - 0.66)	0.09 (0.03 - 0.15)
Silica (as SiO_2) (mg/L)		1.09 (0.39 - 1.90)	0.94 (0.23 - 2.10)	0.93 (0.30 - 1.75)	0.78 (0.23 - 2.23)	2.42 (0.35 - 4.40)	0.88 (0.25 - 2.00)
Chlorophyll-*a* (µg/L)		3.1 (1.2 - 7.4)	2.7 (0.8 - 6.5)	1.8 (0.5 - 4.4)	5.1 (0.6 - 35.3)	5.8 (1.0 - 20.3)	1.0 (0.4 - 1.5)
E.coli (count/100mL)		1100 (270 - 24000)	2000 (94 - 12000)	270 (26 - 2600)	260 (10 - 5200)	980 (75 - 55000)	450 (37 - 4700)
Faecal Coliforms (count/100mL)		2500 (580 - 41000)	4400 (360 - 28000)	910 (110 - 6300)	660 (24 - 7600)	2800 (250 - 120000)	860 (63 - 6400)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.

Summary of water quality data for typhoon shelters, sheltered anchorages and Government Dockyard in 2024 (continued)

Parameter		Shau Kei Wan ET2	Chai Wan Cargo Handling Basin ET1	Hebe Haven PT4	Yim Tin Tsai PT3	Sai Kung PT2	Shuen Wan TT1
Number of samples		6	6	6	6	6	6
Temperature (°C)		24.8 (19.9 - 29.0)	24.9 (20.1 - 29.7)	24.8 (17.2 - 32.4)	24.9 (17.2 - 32.7)	25.0 (17.0 - 32.6)	26.1 (20.3 - 31.7)
Salinity		30.6 (22.7 - 34.0)	31.5 (27.0 - 33.9)	30.4 (25.9 - 33.7)	30.8 (26.2 - 34.0)	30.8 (26.5 - 33.9)	29.9 (27.5 - 32.6)
Dissolved Oxygen (mg/L)		6.1 (3.7 - 8.4)	7.9 (6.0 - 10.9)	6.4 (6.0 - 6.9)	5.9 (4.4 - 7.0)	5.7 (4.1 - 6.7)	5.8 (4.2 - 8.5)
	Bottom	4.9 (3.3 - 7.6)	7.4 (5.7 - 11.1)	5.7 (4.4 - 6.7)	5.6 (4.1 - 7.3)	#N/A #N/A	5.5 (3.8 - 8.4)
Dissolved Oxygen (% Saturation)		86 (57 - 113)	113 (88 - 160)	92 (82 - 106)	84 (68 - 104)	82 (64 - 100)	84 (61 - 113)
	Bottom	70 (50 - 101)	106 (84 - 149)	82 (68 - 105)	79 (63 - 92)	#N/A #N/A	79 (56 - 112)
pH		7.7 (7.5 - 8.1)	7.9 (7.6 - 8.2)	7.9 (7.6 - 8.1)	8.0 (7.8 - 8.2)	8.1 (7.9 - 8.3)	8.0 (7.6 - 8.3)
Secchi Disc Depth (m)		2.7 (1.6 - 3.8)	2.4 (1.9 - 3.3)	1.9 (1.7 - 2.2)	2.6 (1.6 - 3.2)	2.1 (1.6 - 2.6)	2.5 (1.8 - 3.8)
Turbidity (NTU)		2.4 (1.5 - 3.7)	2.6 (2.2 - 3.2)	1.9 (0.6 - 3.0)	1.5 (0.3 - 3.0)	1.7 (0.4 - 4.0)	1.3 (0.7 - 1.9)
Suspended Solids (mg/L)		4.3 (1.2 - 6.7)	4.1 (2.4 - 7.7)	4.9 (3.1 - 8.8)	3.5 (1.6 - 5.8)	3.2 (2.2 - 5.3)	4.4 (2.8 - 6.7)
5-day Biochemical Oxygen Demand (mg/L)		1.0 (<0.1 - 2.7)	1.3 (0.2 - 2.1)	0.8 (0.5 - 1.1)	0.6 (0.2 - 1.0)	1.0 (0.2 - 1.7)	1.2 (0.5 - 1.8)
Ammonia Nitrogen (mg/L)		0.086 (0.038 - 0.197)	0.049 (0.011 - 0.105)	0.050 (0.010 - 0.097)	0.022 (0.008 - 0.043)	0.034 (0.012 - 0.057)	0.057 (0.013 - 0.135)
Unionised Ammonia (mg/L)		0.002 (<0.001 - 0.007)	0.002 (<0.001 - 0.008)	0.002 (<0.001 - 0.004)	0.001 (<0.001 - 0.003)	0.002 (<0.001 - 0.004)	0.004 (<0.001 - 0.010)
Nitrite Nitrogen (mg/L)		0.021 (0.002 - 0.055)	0.011 (<0.002 - 0.028)	0.003 (<0.002 - 0.004)	0.002 (<0.002 - 0.002)	0.002 (<0.002 - 0.003)	0.006 (<0.002 - 0.016)
Nitrate Nitrogen (mg/L)		0.132 (0.031 - 0.423)	0.078 (0.016 - 0.227)	0.069 (0.009 - 0.161)	0.026 (0.011 - 0.053)	0.052 (0.023 - 0.085)	0.046 (0.013 - 0.102)
Total Inorganic Nitrogen (mg/L)		0.24 (0.07 - 0.50)	0.14 (0.03 - 0.26)	0.12 (0.02 - 0.26)	0.05 (0.02 - 0.10)	0.09 (0.04 - 0.14)	0.11 (0.03 - 0.20)
Total Kjeldahl Nitrogen (mg/L)		0.26 (0.17 - 0.37)	0.23 (0.20 - 0.26)	0.19 (0.14 - 0.23)	0.15 (0.14 - 0.18)	0.18 (0.13 - 0.24)	0.23 (0.12 - 0.30)
Total Nitrogen (mg/L)		0.41 (0.23 - 0.81)	0.32 (0.25 - 0.51)	0.26 (0.15 - 0.39)	0.18 (0.16 - 0.20)	0.24 (0.17 - 0.31)	0.28 (0.14 - 0.34)
Orthophosphate Phosphorus (mg/L)		0.011 (<0.002 - 0.026)	0.006 (<0.002 - 0.017)	0.004 (<0.002 - 0.007)	0.004 (<0.002 - 0.008)	0.004 (<0.002 - 0.006)	0.004 (<0.002 - 0.009)
Total Phosphorus (mg/L)		0.08 (0.04 - 0.15)	0.08 (0.04 - 0.14)	0.06 (0.04 - 0.12)	0.06 (0.03 - 0.14)	0.06 (0.03 - 0.12)	0.07 (0.03 - 0.13)
Silica (as SiO_2) (mg/L)		0.58 (0.17 - 1.30)	0.39 (0.14 - 0.62)	1.09 (0.40 - 2.80)	0.61 (0.05 - 1.37)	0.58 (0.13 - 0.98)	0.68 (0.18 - 1.47)
Chlorophyll-*a* (μg/L)		8.1 (0.4 - 38.0)	6.9 (0.5 - 15.3)	3.1 (1.3 - 4.5)	2.0 (1.2 - 4.2)	2.5 (1.4 - 4.8)	3.6 (1.2 - 6.5)
E.coli (count/100mL)		1400 (460 - 3400)	110 (62 - 240)	10 (2 - 87)	2 (<1 - 4)	250 (66 - 2900)	20 (7 - 67)
Faecal Coliforms (count/100mL)		2700 (900 - 6300)	330 (140 - 1400)	35 (5 - 420)	3 (1 - 11)	1000 (180 - 4600)	57 (18 - 240)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).
2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.
3. Data in brackets indicate the ranges.
4. N/A (Not Applicable) indicates the measurement was not made due to shallow water.

Summary of marine sediment quality data for typhoon shelters, sheltered anchorages and Government Dockyard, 2020 - 2024

	Tuen Mun	Cheung Chau	Hei Ling Chau	Rambler Channel	Government Dockyard	New Yau Ma Tei	Causeway Bay
Parameter	NS5	SS7	SS8	VS17	VS21	VS19	VS12
Number of samples	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	78 (4 - 97)	58 (12 - 95)	78 (3 - 97)	70 (8 - 86)	72 (2 - 99)	76 (30 - 96)	63 (23 - 98)
Electrochemical Potential (mV)	-153 (-211 - -79)	-254 (-366 - -149)	-185 (-251 - -132)	-190 (-306 - -64)	-276 (-349 - -145)	-328 (-367 - -259)	-222 (-307 - -127)
Total Solids (%w/w)	48 (38 - 58)	37 (30 - 47)	36 (33 - 40)	41 (36 - 54)	38 (34 - 44)	46 (39 - 52)	49 (41 - 58)
Total Volatile Soilds (%TS)	6.8 (5.0 - 8.9)	7.2 (5.0 - 9.3)	7.1 (5.7 - 8.2)	8.2 (5.0 - 9.7)	7.4 (6.0 - 8.9)	7.3 (6.3 - 7.9)	7.2 (4.8 - 8.9)
Chemical Oxygen Demand (mg/kg)	17900 (12000 - 25000)	15100 (11000 - 25000)	13680 (9800 - 20000)	22700 (17000 - 29000)	17700 (13000 - 26000)	19500 (14000 - 26000)	23100 (18000 - 29000)
Total Carbon (%w/w)	0.8 (0.5 - 1.8)	0.6 (0.5 - 0.8)	0.5 (0.5 - 0.6)	1.1 (0.8 - 1.6)	0.7 (0.5 - 1.1)	0.9 (0.7 - 1.2)	1.2 (0.8 - 1.7)
Ammonical Nitrogen (mg/kg)	10.88 (0.29 - 23.00)	15.77 (6.60 - 28.00)	10.27 (4.10 - 18.00)	7.32 (1.60 - 15.00)	9.33 (2.60 - 16.00)	19.76 (6.00 - 29.00)	4.27 (0.26 - 8.70)
Total Kjeldahl Nitrogen (mg/kg)	460 (370 - 750)	490 (420 - 570)	500 (430 - 580)	510 (440 - 590)	420 (340 - 510)	540 (430 - 730)	570 (430 - 790)
Total Phosphorus (mg/kg)	220 (170 - 260)	260 (190 - 310)	190 (160 - 210)	220 (210 - 250)	200 (170 - 300)	240 (220 - 260)	250 (190 - 340)
Total Sulphide (mg/kg)	48.3 (7.5 - 120.0)	45.4 (4.3 - 96.0)	29.1 (8.4 - 86.0)	90.5 (9.7 - 290.0)	50.7 (0.5 - 180.0)	71.8 (29.0 - 130.0)	128.3 (18.0 - 410.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.2 (0.1 - 0.4)	<0.1 (0.1 - <0.1)	0.1 (0.1 - 0.2)	0.2 (0.1 - 0.3)
Arsenic (mg/kg)	9.6 (6.2 - 13.0)	9.0 (4.4 - 14.0)	9.0 (3.5 - 12.0)	12.0 (5.5 - 17.0)	10.9 (6.7 - 13.0)	8.7 (5.4 - 11.0)	9.1 (6.5 - 12.0)
Cadmium (mg/kg)	0.2 (0.1 - 0.3)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.6 (0.4 - 0.8)	0.4 (0.4 - 0.5)	0.4 (0.2 - 0.8)	0.6 (0.2 - 1.1)
Chromium (mg/kg)	37 (28 - 52)	46 (29 - 70)	44 (25 - 65)	110 (56 - 160)	50 (39 - 65)	45 (29 - 58)	51 (31 - 100)
Copper (mg/kg)	39 (23 - 61)	87 (39 - 140)	33 (19 - 45)	190 (110 - 260)	170 (130 - 240)	120 (50 - 210)	200 (74 - 580)
Lead (mg/kg)	45 (31 - 89)	52 (29 - 99)	43 (27 - 51)	76 (51 - 100)	56 (45 - 63)	55 (32 - 96)	89 (38 - 120)
Mercury (mg/kg)	0.07 (0.05 - 0.11)	0.14 (0.07 - 0.24)	0.11 (0.07 - 0.14)	0.24 (0.17 - 0.29)	0.25 (0.18 - 0.34)	0.31 (0.15 - 0.81)	0.73 (0.26 - 1.70)
Nickel (mg/kg)	20 (12 - 28)	23 (16 - 33)	32 (15 - 80)	39 (26 - 51)	25 (20 - 30)	23 (16 - 27)	21 (16 - 36)
Silver (mg/kg)	0.3 (0.2 - 0.5)	0.3 (0.2 - 0.4)	0.2 (0.2 - 0.3)	2.1 (1.3 - 3.6)	1.3 (1.2 - 1.7)	1.3 (0.7 - 3.2)	1.8 (1.0 - 2.7)
Zinc (mg/kg)	160 (120 - 200)	170 (81 - 250)	150 (80 - 210)	350 (230 - 460)	300 (170 - 430)	260 (120 - 420)	320 (110 - 430)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) (3)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	57 (25 - 260)	22 (18 - 27)	23 (18 - 38)	42 (18 - 89)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) (4) (6)	93 (90 - 110)	100 (90 - 170)	100 (90 - 170)	300 (110 - 820)	210 (90 - 720)	260 (90 - 930)	1800 (100 - 3800)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) (5) (6)	220 (55 - 650)	160 (94 - 280)	65 (31 - 130)	3000 (530 - 14000)	1900 (150 - 11000)	2900 (130 - 12000)	21000 (540 - 46000)

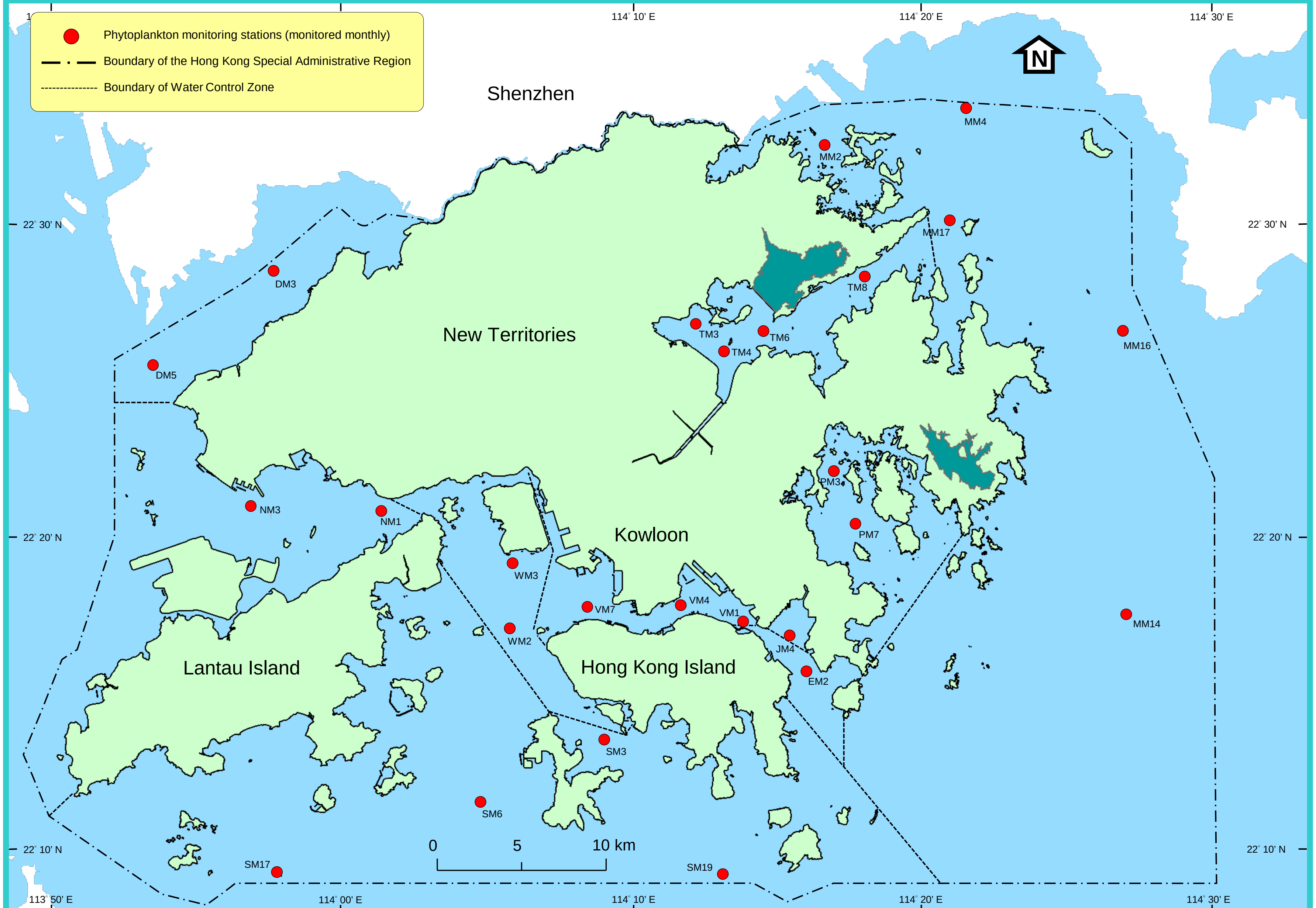
Note:
1. Data presented are arithmetic means ; data in brackets indicate ranges.
2. All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.
3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4. Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5. High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary of marine sediment quality data for typhoon shelters, sheltered anchorages and Government Dockyard, 2020 - 2024 (continued)

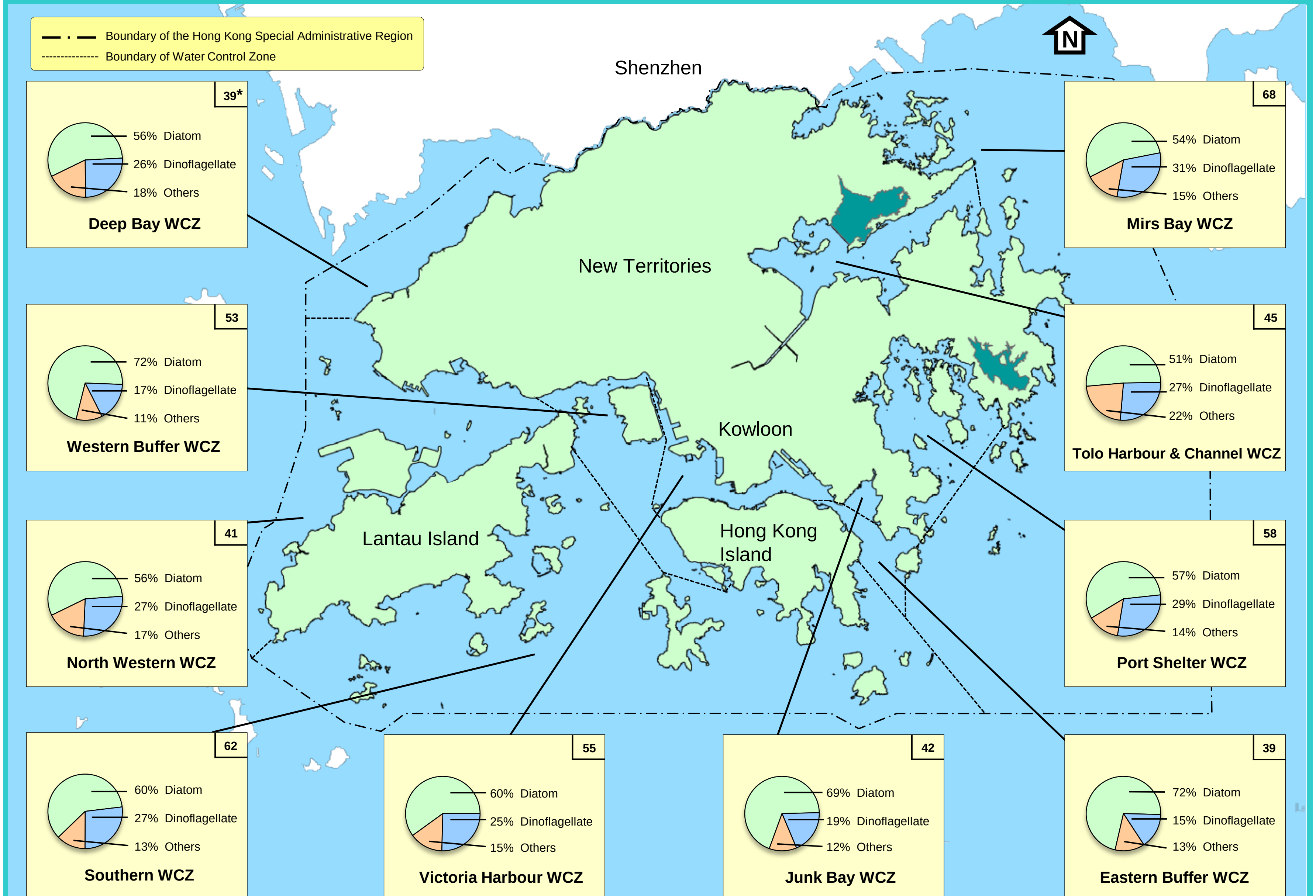
Parameter	To Kwa Wan VS20	Kwun Tong VS14	Sam Ka Tsuen VS13	Shau Kei Wan ES5	Chai Wan Cargo Handling Basin ES3	Hebe Haven PS4	Yim Tin Tsai PS2	Shuen Wan TS7
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	71 (47 - 97)	80 (13 - 91)	68 (3 - 93)	68 (1 - 96)	81 (9 - 95)	75 (18 - 95)	73 (10 - 91)	66 (5 - 93)
Electrochemical Potential (mV)	-224 (-338 - -47)	-275 (-363 - -130)	-207 (-344 - -82)	-321 (-360 - -208)	-146 (-227 - -74)	-171 (-245 - -101)	-149 (-232 - -101)	-209 (-346 - 318)
Total Solids (%w/w)	48 (43 - 58)	37 (25 - 58)	40 (34 - 46)	30 (27 - 37)	49 (41 - 53)	40 (35 - 50)	45 (38 - 53)	38 (32 - 44)
Total Volatile Soilds (%TS)	6.8 (5.0 - 9.0)	9.3 (6.2 - 12.0)	8.3 (6.9 - 9.1)	9.4 (7.3 - 11.0)	7.3 (6.3 - 8.1)	9.0 (6.4 - 11.0)	9.6 (6.6 - 12.0)	8.3 (5.8 - 10.0)
Chemical Oxygen Demand (mg/kg)	20490 (8000 - 29000)	24900 (19000 - 32000)	22400 (18000 - 34000)	16800 (10000 - 20000)	17850 (9500 - 22000)	18300 (11000 - 25000)	15100 (11000 - 19000)	17300 (13000 - 28000)
Total Carbon (%w/w)	1.0 (0.8 - 1.3)	1.1 (0.7 - 1.6)	1.0 (0.8 - 1.1)	0.8 (0.6 - 1.5)	0.9 (0.9 - 1.2)	0.8 (0.6 - 1.0)	1.5 (0.5 - 2.1)	0.9 (0.7 - 1.3)
Ammonical Nitrogen (mg/kg)	12.58 (3.80 - 23.00)	25.34 (4.80 - 46.00)	11.09 (5.80 - 25.00)	26.06 (5.60 - 35.00)	5.58 (2.30 - 12.00)	5.67 (0.93 - 12.00)	7.68 (4.30 - 11.00)	6.92 (2.50 - 12.00)
Total Kjeldahl Nitrogen (mg/kg)	480 (330 - 670)	500 (350 - 770)	530 (380 - 720)	480 (380 - 670)	450 (410 - 500)	570 (350 - 730)	580 (340 - 690)	600 (410 - 940)
Total Phosphorus (mg/kg)	240 (200 - 250)	240 (170 - 430)	300 (260 - 350)	160 (140 - 180)	240 (200 - 270)	190 (170 - 210)	190 (180 - 220)	200 (170 - 240)
Total Sulphide (mg/kg)	58.8 (24.0 - 110.0)	149.7 (5.0 - 390.0)	152.8 (13.0 - 480.0)	279.9 (49.0 - 450.0)	29.0 (0.2 - 86.0)	32.2 (6.1 - 69.0)	27.7 (4.4 - 82.0)	66.4 (10.0 - 120.0)
Total Cyanide (mg/kg)	0.1 (0.1 - 0.2)	0.2 (0.1 - 0.5)	0.2 (0.1 - 0.4)	0.1 (0.1 - 0.2)	0.2 (0.1 - 0.3)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)	0.1 (0.1 - 0.2)
Arsenic (mg/kg)	9.1 (5.7 - 11.0)	10.8 (6.9 - 15.0)	10.5 (6.6 - 13.0)	9.7 (5.2 - 13.0)	9.8 (4.1 - 13.0)	11.4 (7.1 - 13.0)	6.3 (3.8 - 9.4)	12.3 (8.2 - 16.0)
Cadmium (mg/kg)	0.7 (0.2 - 0.9)	1.9 (1.2 - 2.9)	0.9 (0.3 - 1.1)	0.4 (0.1 - 0.5)	0.3 (0.1 - 0.4)	0.2 (0.1 - 0.2)	<0.1 (0.1 - <0.1)	0.4 (0.2 - 0.5)
Chromium (mg/kg)	77 (46 - 100)	240 (130 - 440)	85 (42 - 130)	56 (37 - 74)	64 (34 - 82)	29 (18 - 48)	22 (13 - 31)	27 (17 - 38)
Copper (mg/kg)	540 (240 - 1200)	1400 (600 - 2400)	310 (71 - 520)	150 (85 - 180)	170 (86 - 220)	69 (45 - 79)	17 (10 - 25)	99 (33 - 150)
Lead (mg/kg)	110 (50 - 290)	120 (89 - 200)	91 (53 - 110)	62 (42 - 78)	72 (45 - 140)	41 (30 - 47)	34 (25 - 43)	94 (56 - 110)
Mercury (mg/kg)	1.07 (0.32 - 1.70)	0.64 (0.36 - 1.00)	0.85 (0.30 - 1.30)	0.20 (0.09 - 0.27)	0.40 (0.14 - 0.90)	0.12 (0.06 - 0.15)	0.06 (0.05 - 0.08)	0.11 (0.05 - 0.21)
Nickel (mg/kg)	28 (19 - 36)	63 (37 - 100)	26 (17 - 35)	26 (18 - 32)	22 (13 - 27)	12 (7 - 32)	13 (8 - 17)	14 (9 - 19)
Silver (mg/kg)	2.8 (0.7 - 4.5)	5.8 (3.4 - 9.2)	1.9 (0.7 - 2.4)	1.2 (0.8 - 1.5)	3.0 (1.4 - 6.6)	0.2 (0.2 - 0.3)	<0.2 (0.2 - <0.2)	0.3 (0.2 - 0.4)
Zinc (mg/kg)	280 (220 - 380)	460 (230 - 740)	380 (260 - 480)	310 (170 - 480)	240 (130 - 360)	200 (110 - 330)	90 (56 - 130)	300 (200 - 500)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) [(3)]	77 (32 - 110)	130 (40 - 240)	41 (18 - 96)	18 (18 - 19)	28 (18 - 34)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) [(4) (6)]	8100 (560 - 17000)	150 (96 - 340)	130 (90 - 190)	99 (90 - 140)	140 (90 - 240)	110 (90 - 180)	110 (90 - 190)	130 (90 - 320)
High Molecular Weight Polycylic Aromatic Hydrocarbons (PAHs) (µg/kg) [(5) (6)]	88000 (9900 - 150000)	890 (580 - 1500)	670 (460 - 1000)	370 (210 - 510)	580 (320 - 1100)	98 (18 - 380)	61 (23 - 220)	220 (45 - 720)

Note:
1. Data presented are arithmetic means ; data in brackets indicate ranges.
2. All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.
3. Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.
4. Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.
5. High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Ben
6. Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

The 26 phytoplankton monitoring stations in Hong Kong marine waters

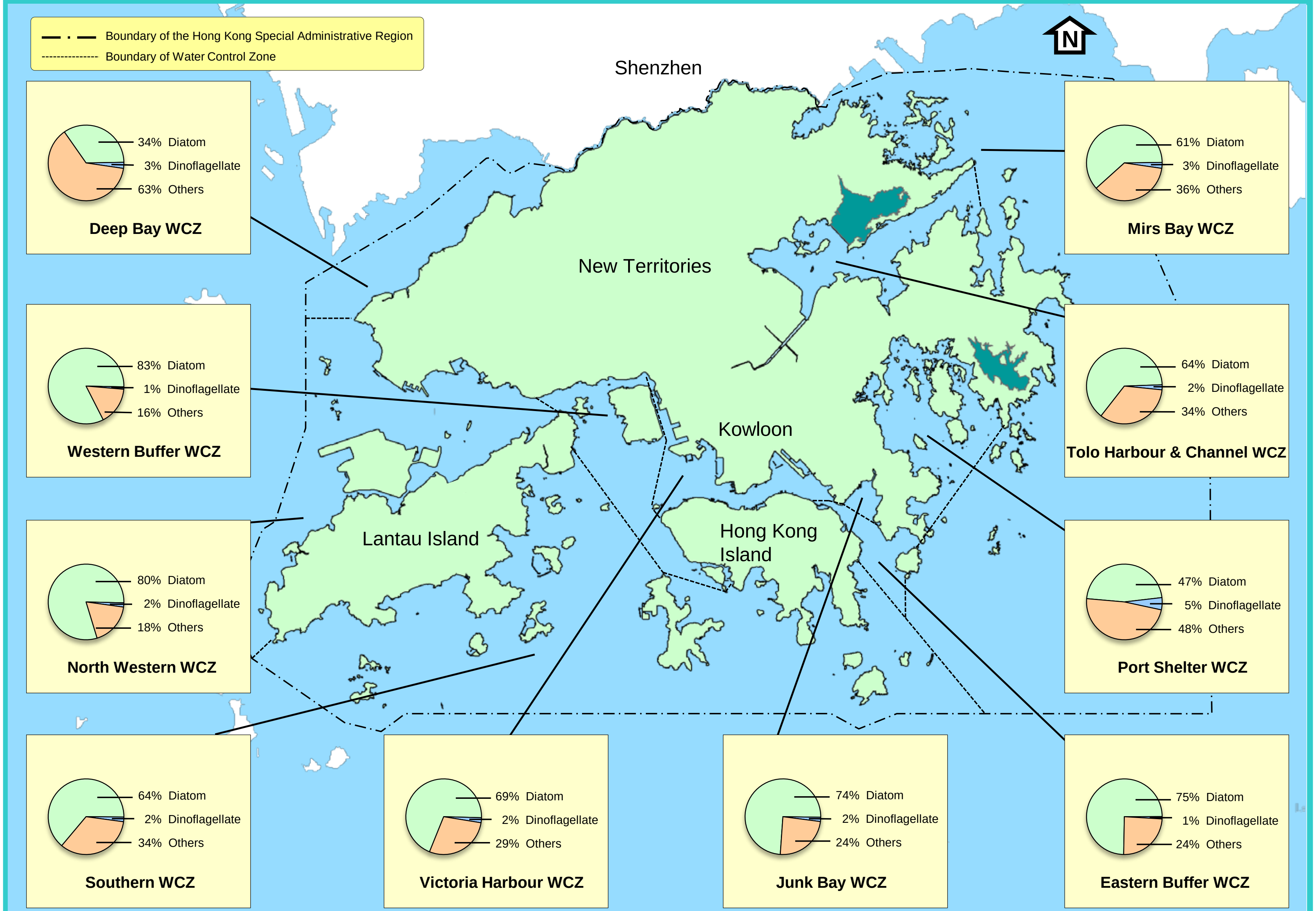


Composition (%) of phytoplankton groups in terms of total number of species in Hong Kong marine waters in 2024

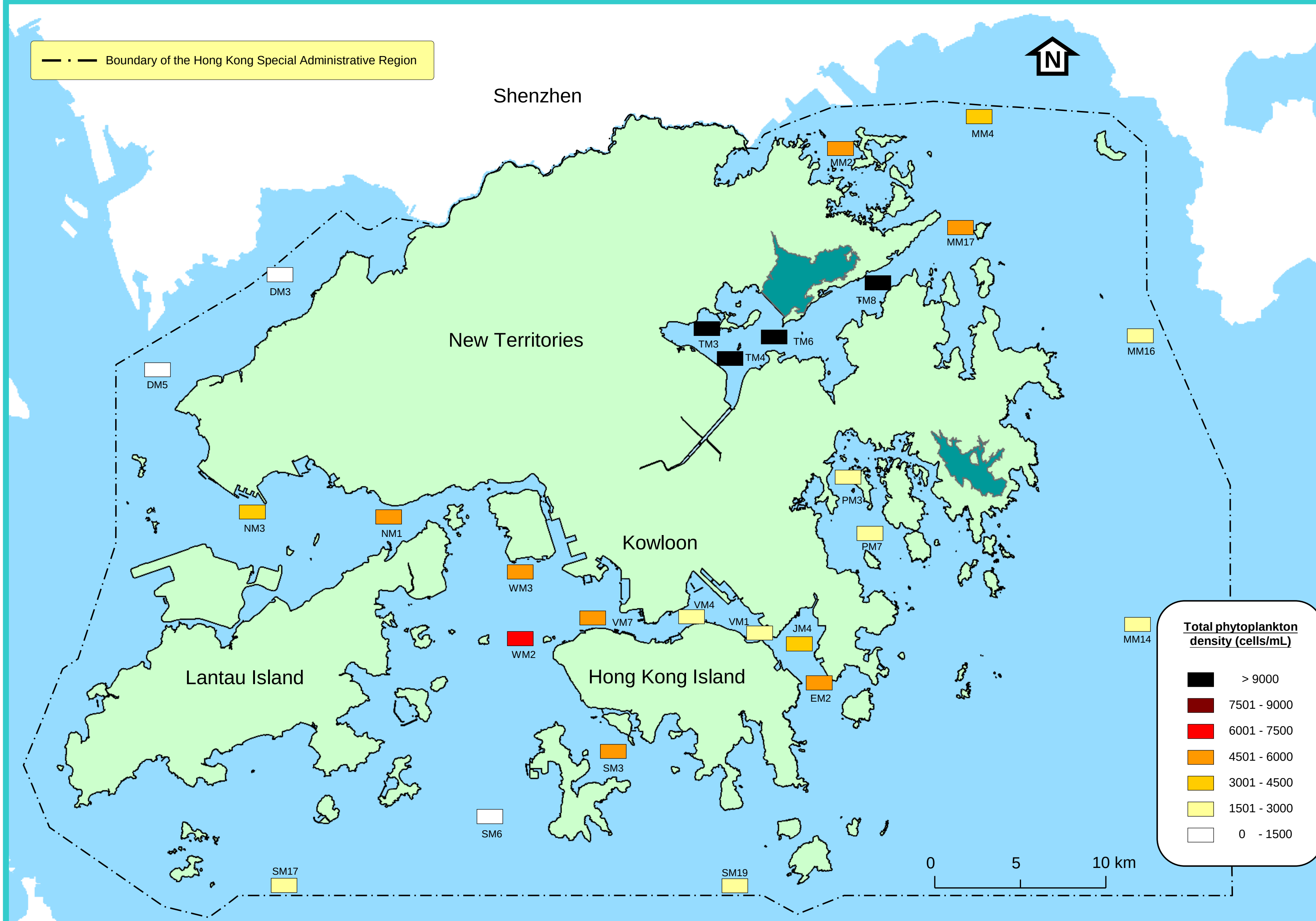


* Total number of species in the corresponding WCZ

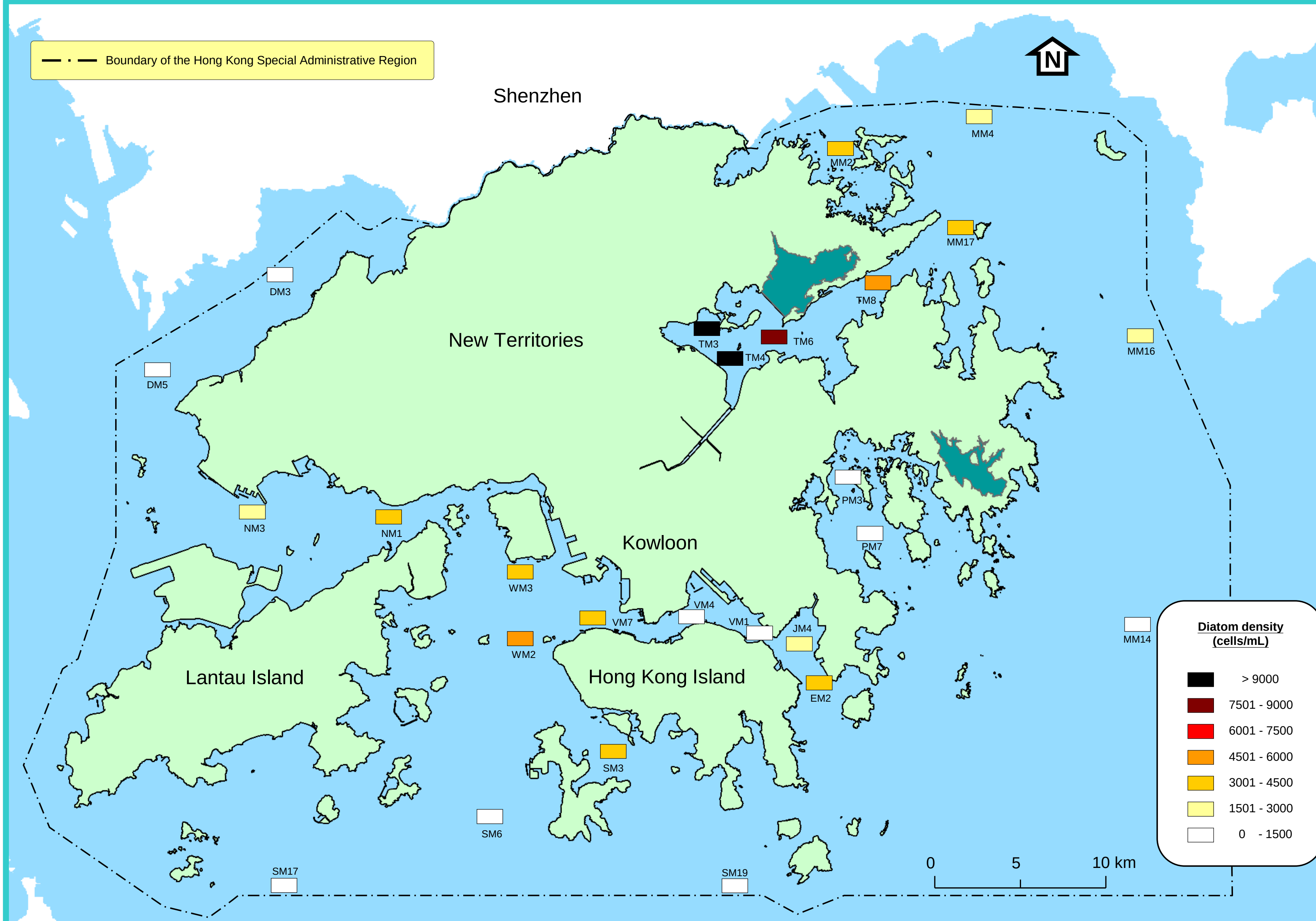
Composition (%) of phytoplankton groups in terms of total density in Hong Kong marine waters in 2024



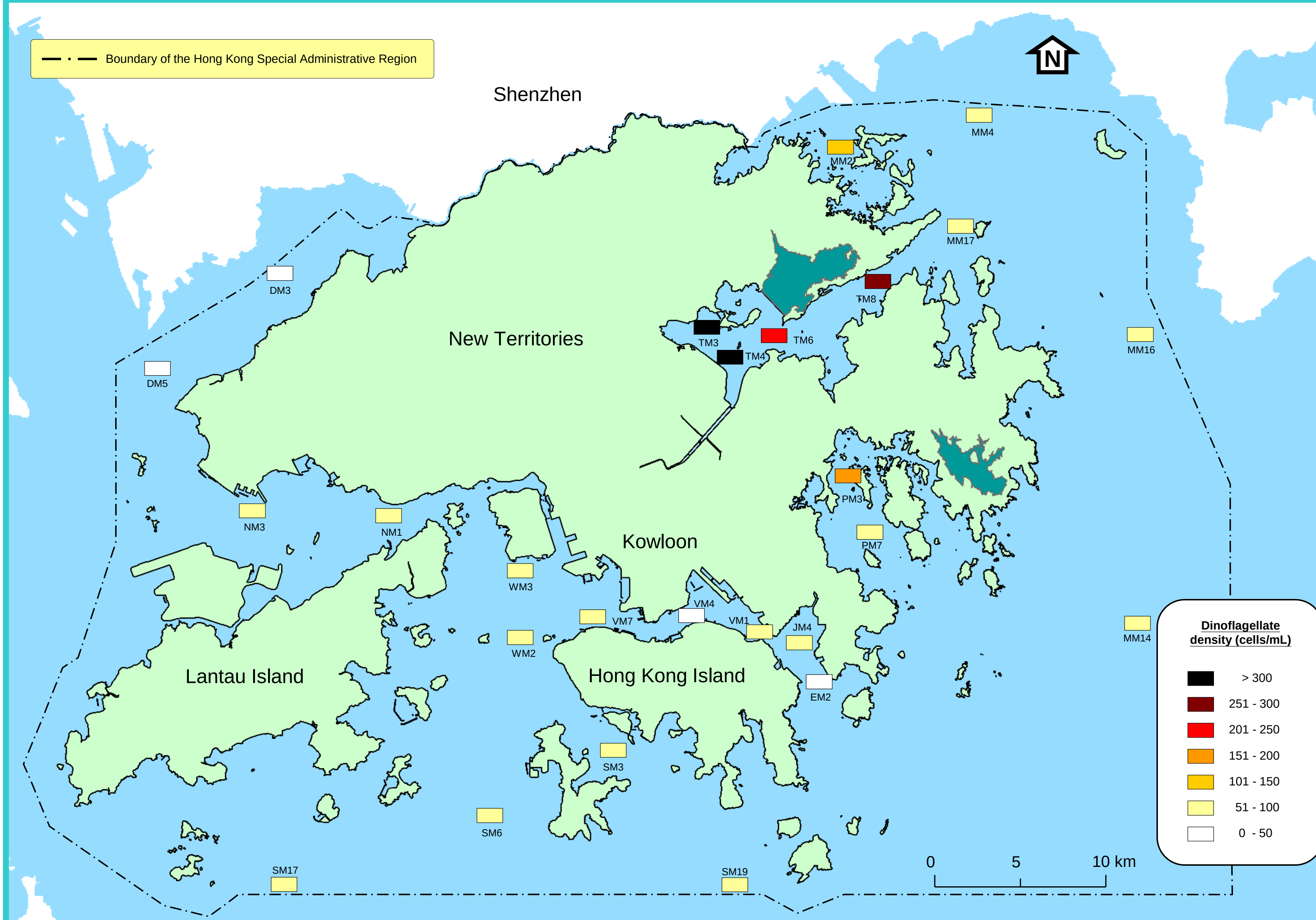
Annual mean total phytoplankton densities at 26 monitoring stations in Hong Kong marine waters in 2024

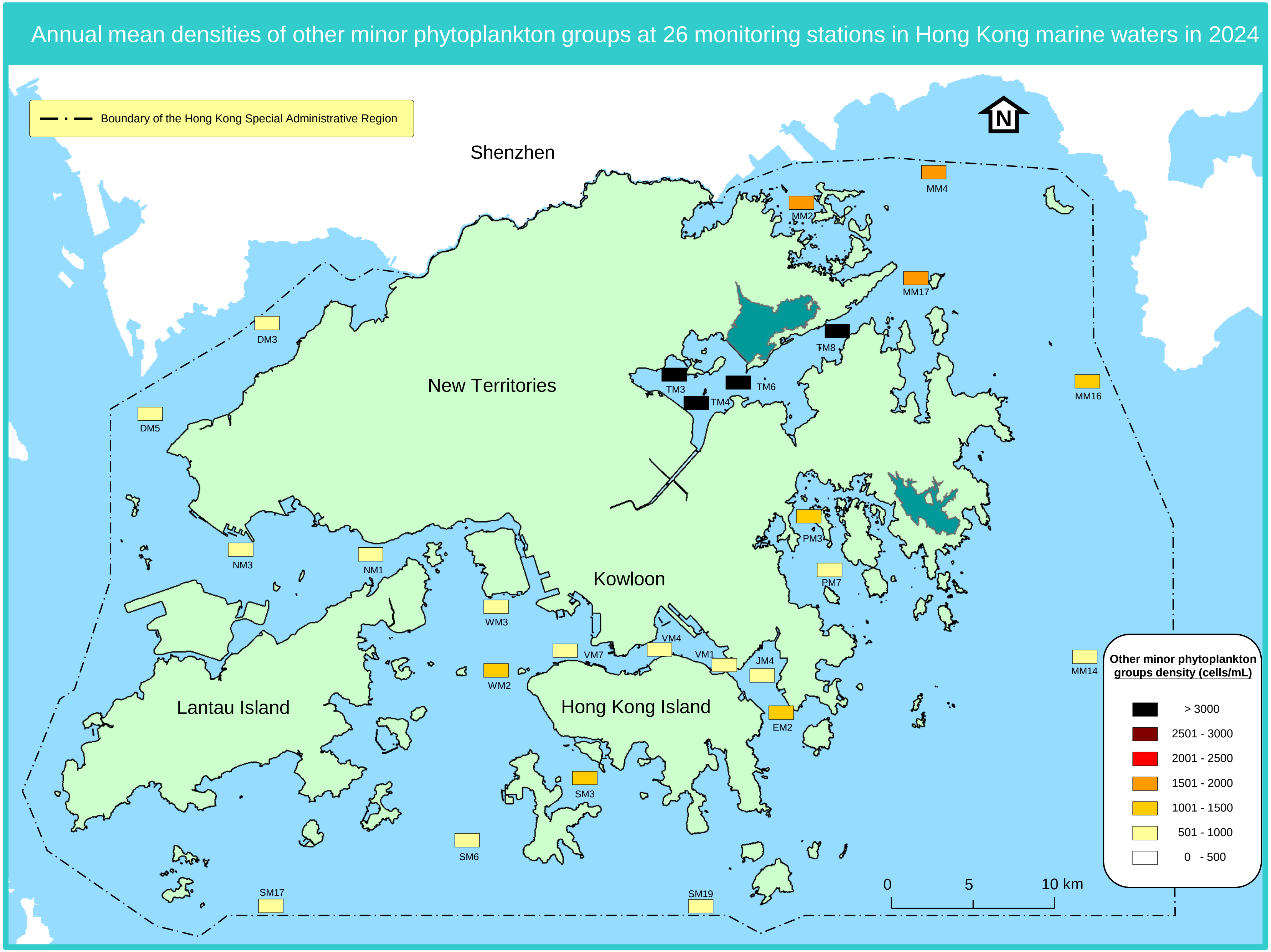


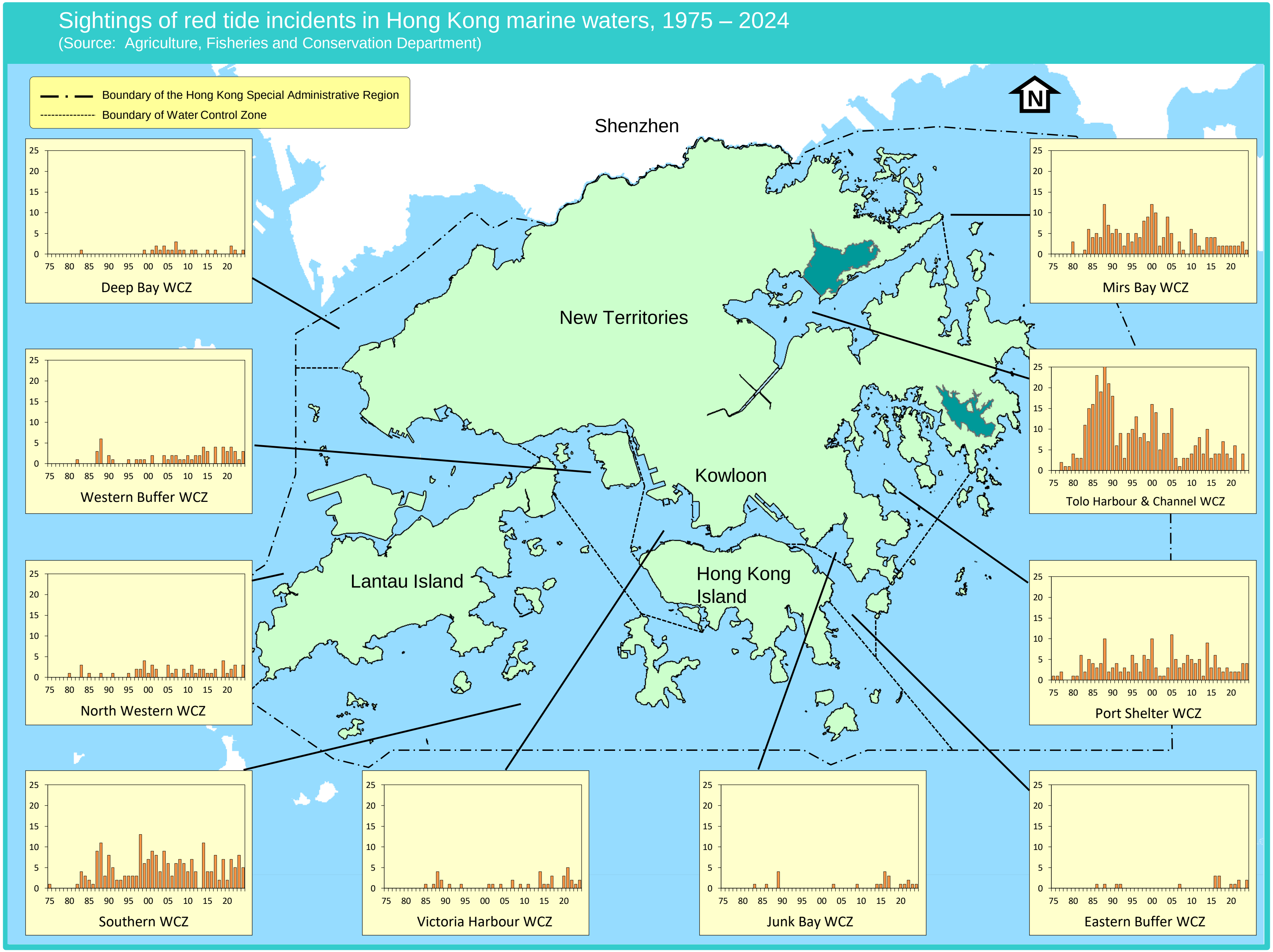
Annual mean diatoms densities at 26 monitoring stations in Hong Kong marine waters in 2024



Annual mean dinoflagellates densities at 26 monitoring stations in Hong Kong marine waters in 2024



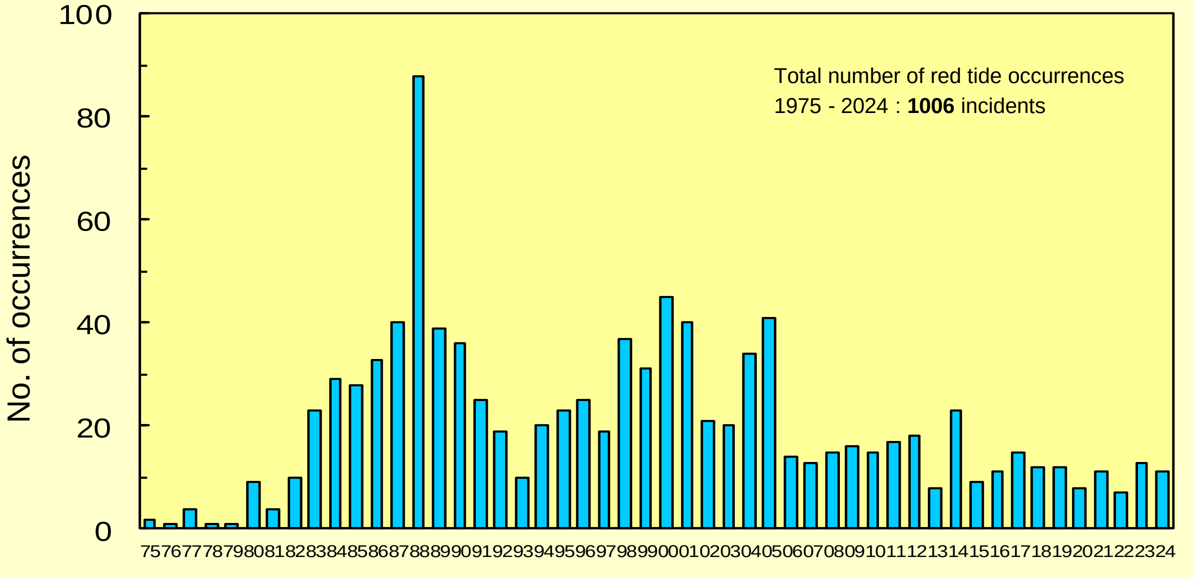

Annual mean densities of other minor phytoplankton groups at 26 monitoring stations in Hong Kong marine waters in 2024
Boundary of the Hong Kong Special Administrative Region
Shenzhen
New Territories
Kowloon
Lantau Island
Hong Kong Island
N
MM4
MM2
MM17
TM8
TM6
TM3
TM4
MM16
DM3
DM5
NM3
NM1
PM3
PM7
WM3
VM4
VM7
VM1
JM4
WM2
MM14
EM2
SM3
SM6
SM17
SM19
0
5
10 km
Other minor phytoplankton groups density (cells/mL)
> 3000
2501 - 3000
2001 - 2500
1501 - 2000
1001 - 1500
501 - 1000
0 - 500


Sightings of red tide incidents in Hong Kong marine waters, 1975 – 2024
(Source: Agriculture, Fisheries and Conservation Department)
Boundary of the Hong Kong Special Administrative Region
Boundary of Water Control Zone
N
Shenzhen
New Territories
Kowloon
Lantau Island
Hong Kong Island
Deep Bay WCZ
Western Buffer WCZ
North Western WCZ
Southern WCZ
Victoria Harbour WCZ
Junk Bay WCZ
Eastern Buffer WCZ
Port Shelter WCZ
Tolo Harbour & Channel WCZ
Mirs Bay WCZ

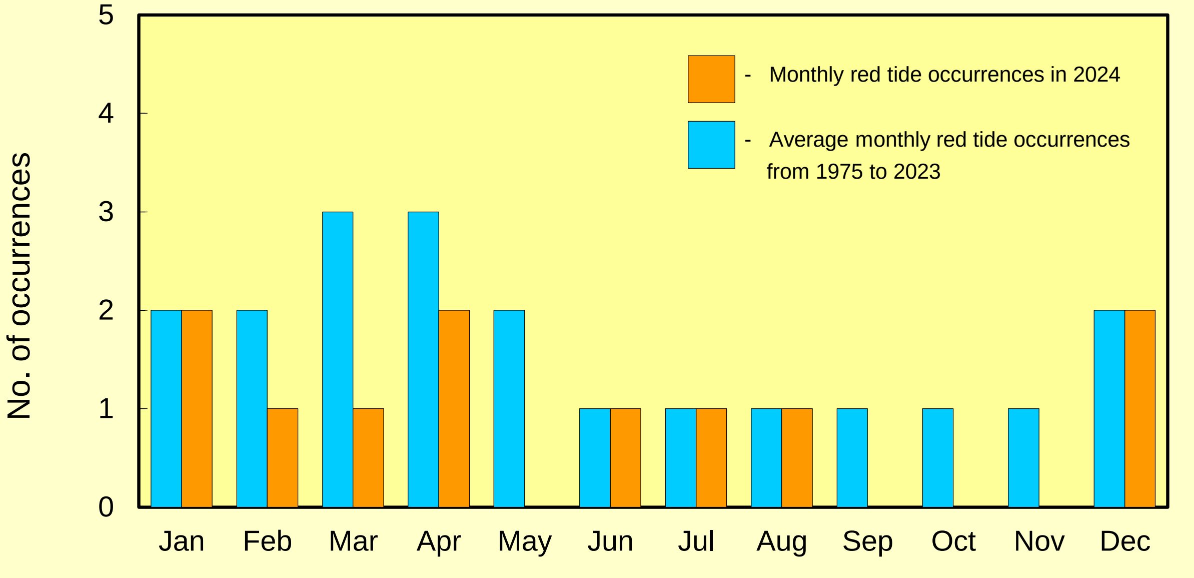
Occurrence of red tides in Hong Kong marine waters, 1975-2024

(Source: Agriculture, Fisheries and Conservation Department)

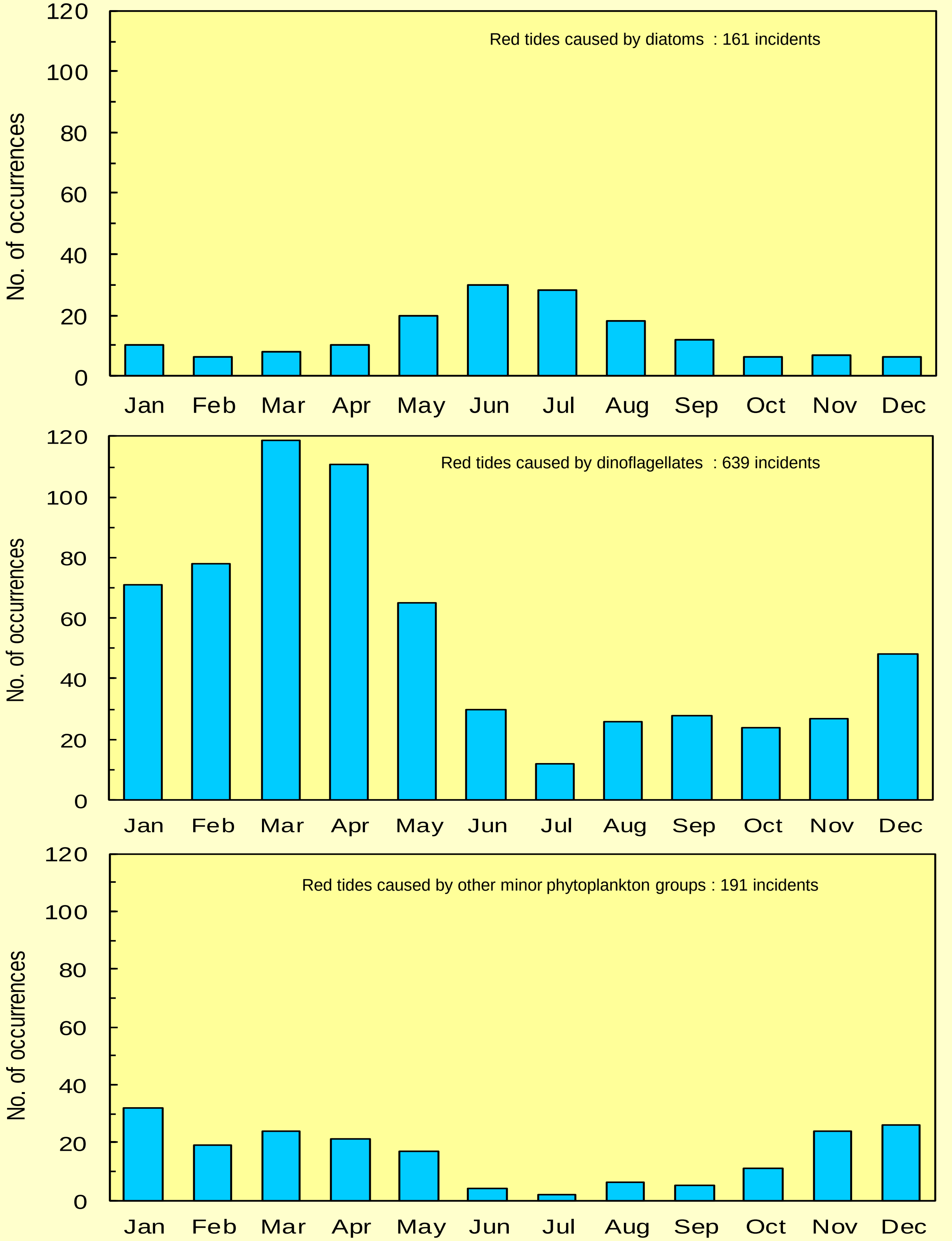
Yearly Distribution



Monthly Distribution



Seasonal distribution of red tides caused by different phytoplankton groups in Hong Kong marine waters, 1975-2024 (Source: Agriculture, Fisheries and Conservation Department)



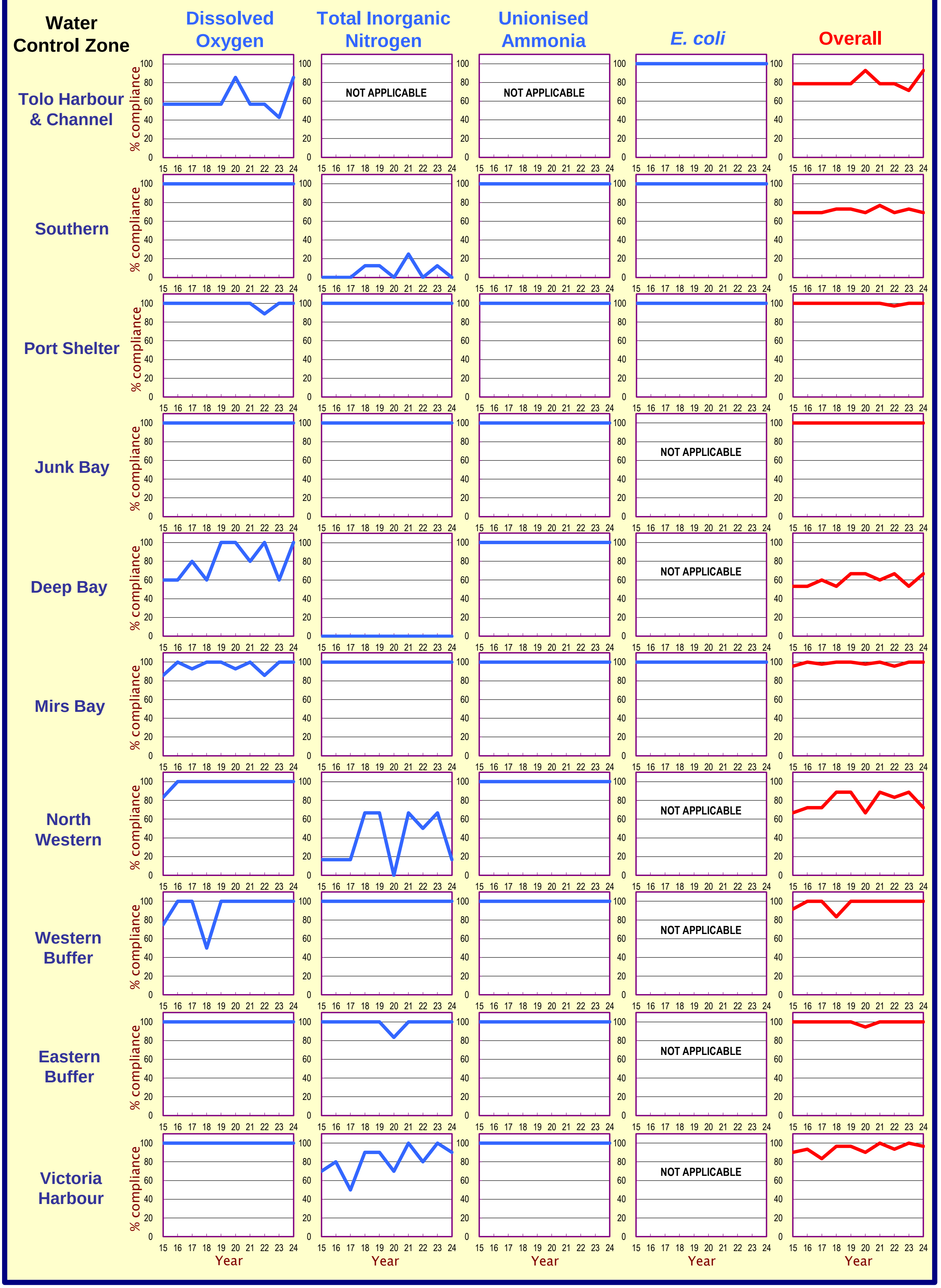
Percentage abundance of the three most dominant phytoplankton species in Hong Kong marine waters in 2024

Species	% Abundance[1]	Species	% Abundance
Deep Bay WCZ		**Mirs Bay WCZ**	
Diatoms		Diatoms	
Thalassiosira spp.	42.85	*Pseudo-nitzschia* spp.	33.81
Skeletonema costatum	28.76	*Dactyliosolen fragilissimus*	21.44
Chaetoceros spp.	13.95	*Chaetoceros* spp.	17.22
Dinoflagellates		Dinoflagellates	
Gymnodinium spp.	56.57	*Gymnodinium* spp.	74.08
Amphidinium spp.	14.48	*Scrippsiella* spp.	6.42
Scrippsiella spp.	8.18	*Gyrodinium* spp.	5.14
Others[2]		Others	
small flagellates	62.74	small flagellates	78.70
Plagioselmis prolonga	20.52	*Plagioselmis prolonga*	16.65
Teleaulax acuta	14.68	*Teleaulax acuta*	4.20
Western Buffer WCZ		**Tolo Harbour & Channel WCZ**	
Diatoms		Diatoms	
Chaetoceros spp.	55.83	*Pseudo-nitzschia* spp.	27.79
Cerataulina dentata	13.82	*Leptocylindrus danicus*	22.48
Thalassiosira spp.	13.35	*Chaetoceros* spp.	13.19
Dinoflagellates		Dinoflagellates	
Gymnodinium spp.	62.39	*Gymnodinium* spp.	55.52
Scrippsiella spp.	14.77	*Gyrodinium* spp.	10.64
Amphidinium spp.	10.30	*Prorocentrum cordatum*	7.41
Others		Others	
small flagellates	75.89	small flagellates	79.97
Plagioselmis prolonga	11.88	*Plagioselmis prolonga*	14.95
Teleaulax acuta	11.58	*Teleaulax acuta*	3.82
North Western WCZ		**Port Shelter WCZ**	
Diatoms		Diatoms	
Skeletonema costatum	41.85	*Chaetoceros* spp.	47.02
Chaetoceros spp.	31.95	*Pseudo-nitzschia* spp.	12.82
Thalassiosira spp.	16.43	*Thalassiosira* spp.	11.30
Dinoflagellates		Dinoflagellates	
Gymnodinium spp.	69.02	*Gymnodinium* spp.	54.52
Gyrodinium spp.	16.82	*Scrippsiella* spp.	14.33
Amphidinium spp.	7.75	*Amphidinium* spp.	7.93
Others		Others	
small flagellates	74.08	small flagellates	74.40
Plagioselmis prolonga	14.09	*Plagioselmis prolonga*	20.79
Teleaulax acuta	11.40	*Teleaulax acuta*	4.37
Southern WCZ		**Eastern Buffer WCZ**	
Diatoms		Diatoms	
Chaetoceros spp.	40.35	*Chaetoceros* spp.	53.78
Thalassiosira spp.	19.76	*Thalassiosira* spp.	17.94
Pseudo-nitzschia spp.	17.11	*Leptocylindrus danicus*	7.43
Dinoflagellates		Dinoflagellates	
Gymnodinium spp.	64.37	*Gymnodinium* spp.	74.23
Amphidinium spp.	13.53	*Gyrodinium* spp.	12.43
Scrippsiella spp.	6.65	*Scrippsiella* spp.	6.31
Others		Others	
small flagellates	76.99	small flagellates	78.64
Plagioselmis prolonga	15.16	*Plagioselmis prolonga*	16.59
Teleaulax acuta	7.36	*Teleaulax acuta*	4.58
Victoria Harbour WCZ		**Junk Bay WCZ**	
Diatoms		Diatoms	
Thalassiosira spp.	42.53	*Chaetoceros* spp.	38.43
Pseudo-nitzschia spp.	17.83	*Thalassiosira* spp.	17.13
Chaetoceros spp.	16.63	*Skeletonema costatum*	13.56
Dinoflagellates		Dinoflagellates	
Gymnodinium spp.	62.46	*Gymnodinium* spp.	57.38
Amphidinium spp.	15.36	*Gyrodinium* spp.	22.37
Gyrodinium spp.	5.99	*Amphidinium* spp.	7.96
Others		Others	
small flagellates	70.37	small flagellates	75.35
Plagioselmis prolonga	22.02	*Plagioselmis prolonga*	18.21
Teleaulax acuta	6.44	*Teleaulax acuta*	5.75

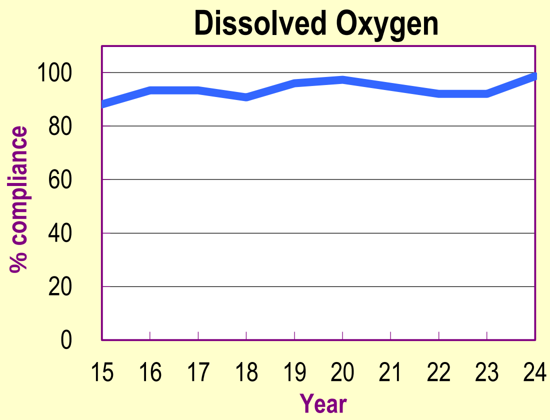
Note: 1 The % abundance of dominant species of diatoms, dinoflagellates and other minor phytoplankton groups in different WCZs.

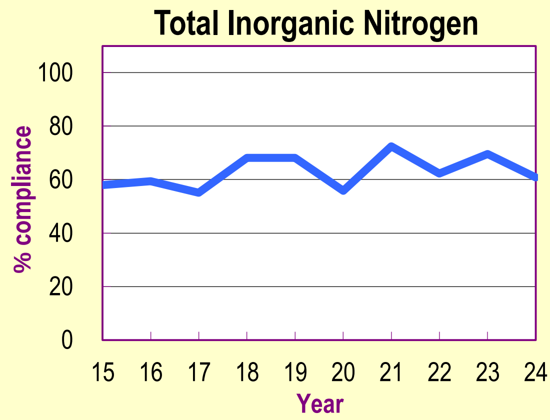
2 Other minor phytoplankton groups.

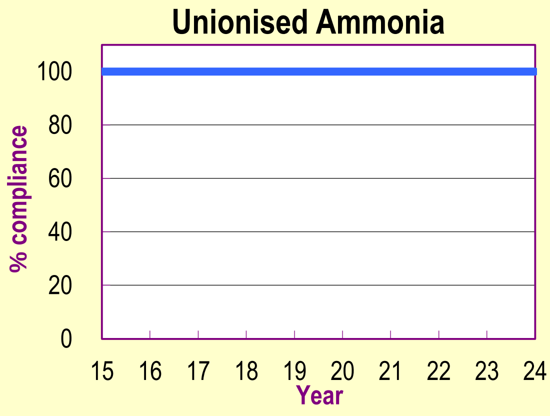
Marine WQO compliance rates for the 10 WCZs, 2015 - 2024

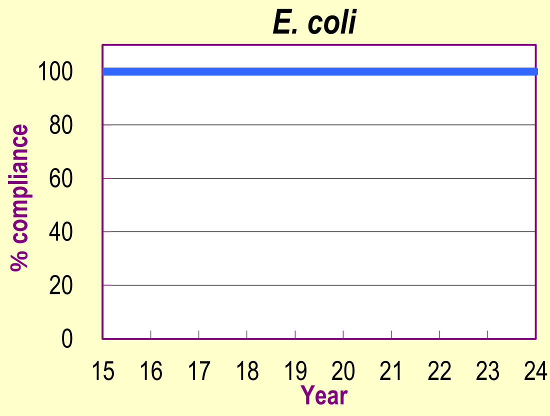


Compliance rates for key marine WQOs, 2015 - 2024

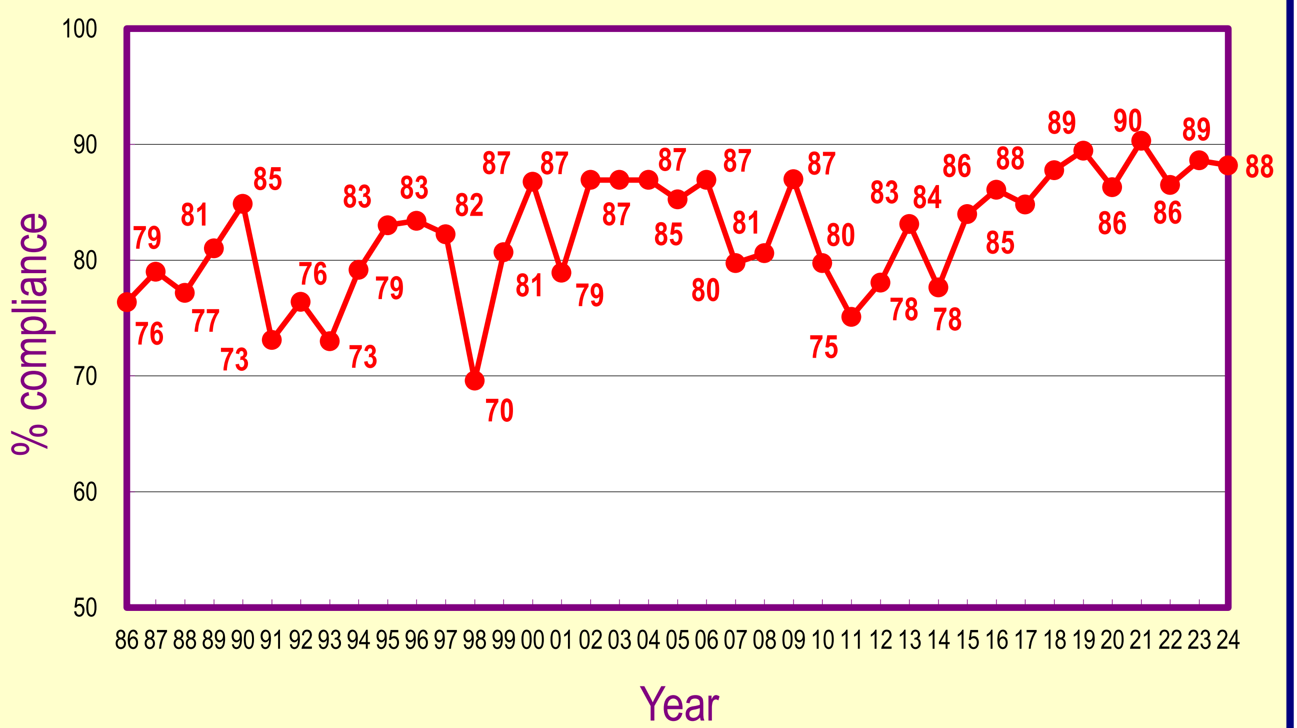




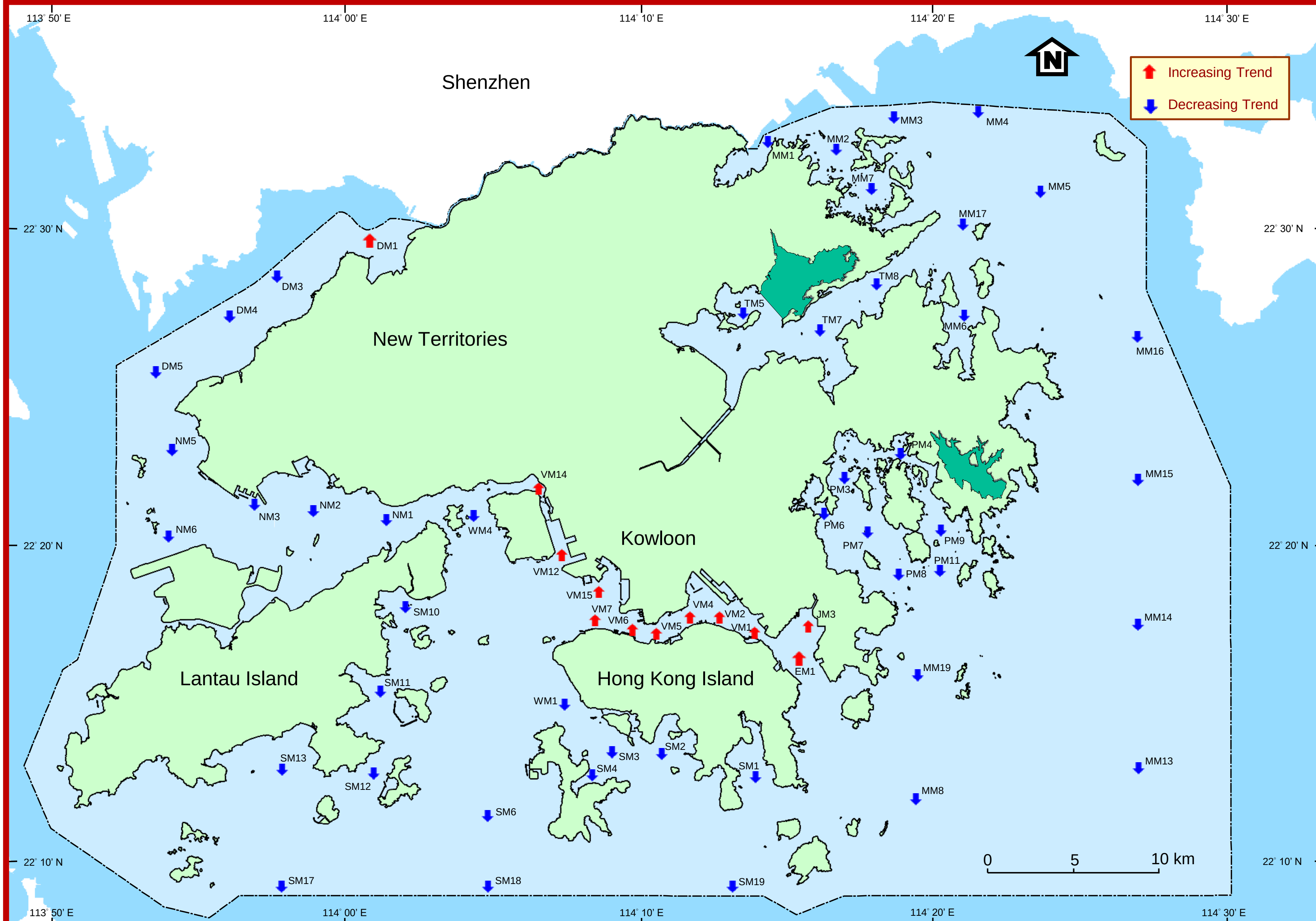




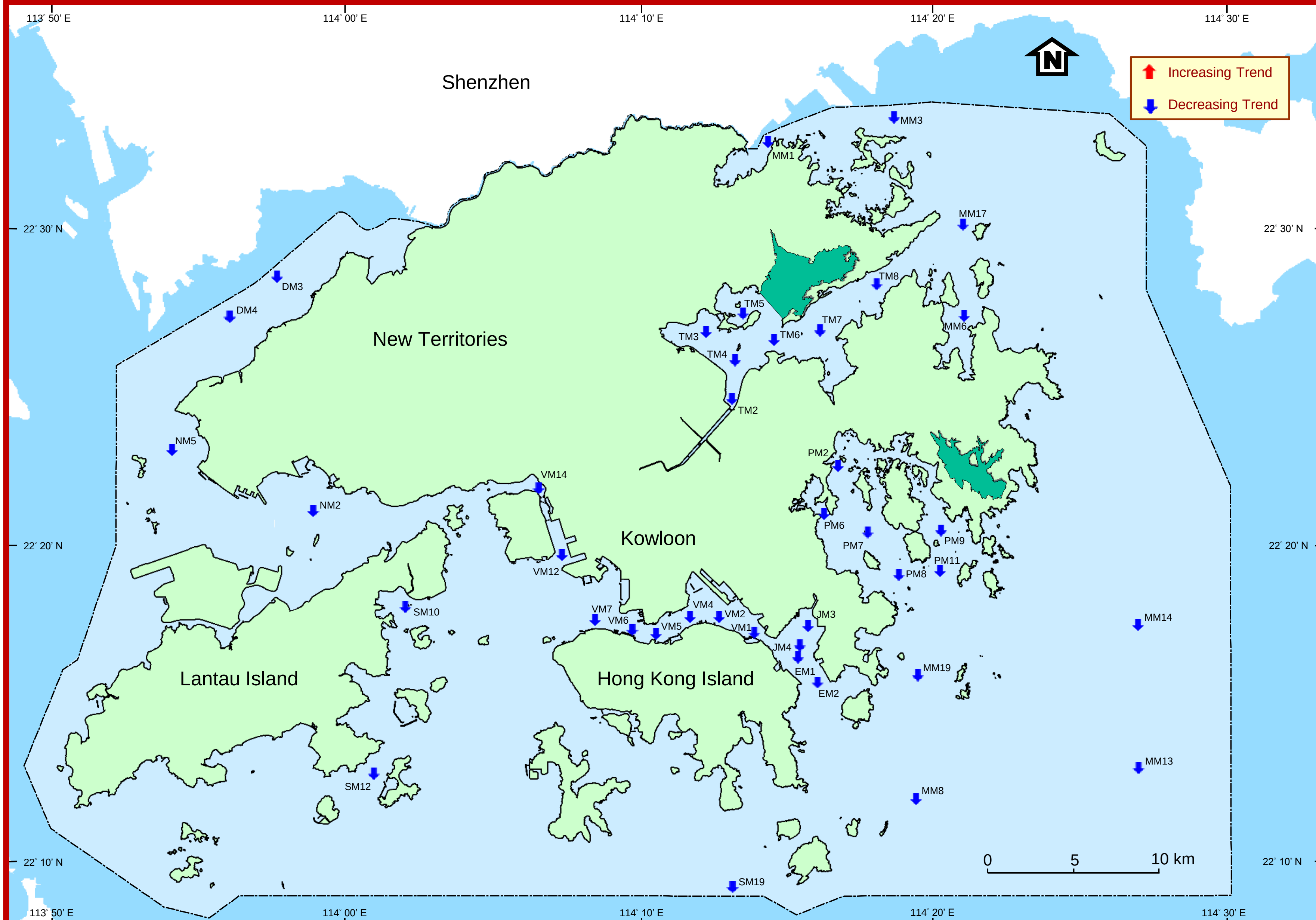
Overall compliance rates for key marine WQOs, 1986 - 2024



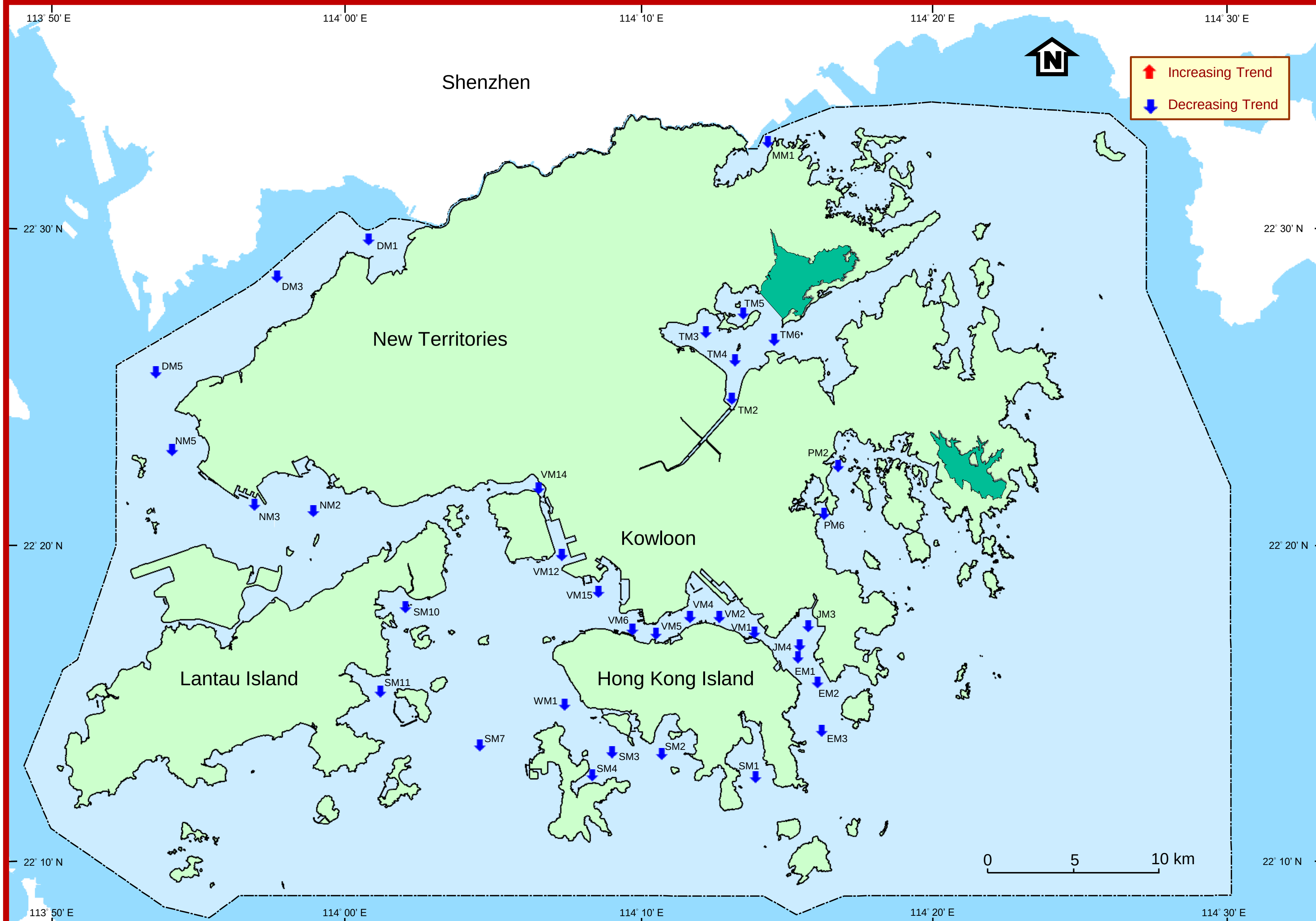
Long-term changes in dissolved oxygen levels in Hong Kong marine waters, 1986-2024



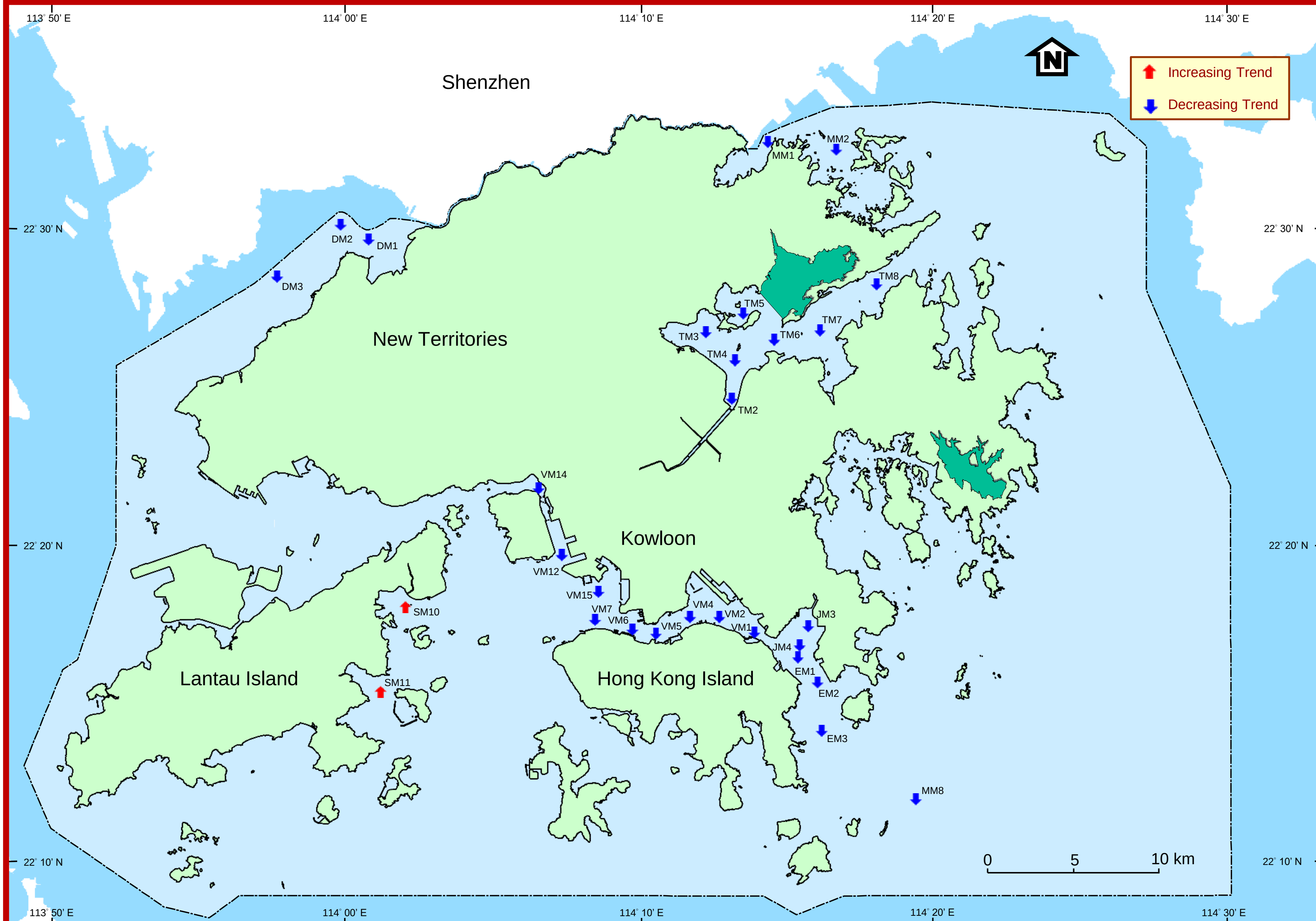
Long-term changes in 5-day biochemical oxygen demand levels in Hong Kong marine waters, 1986-2024



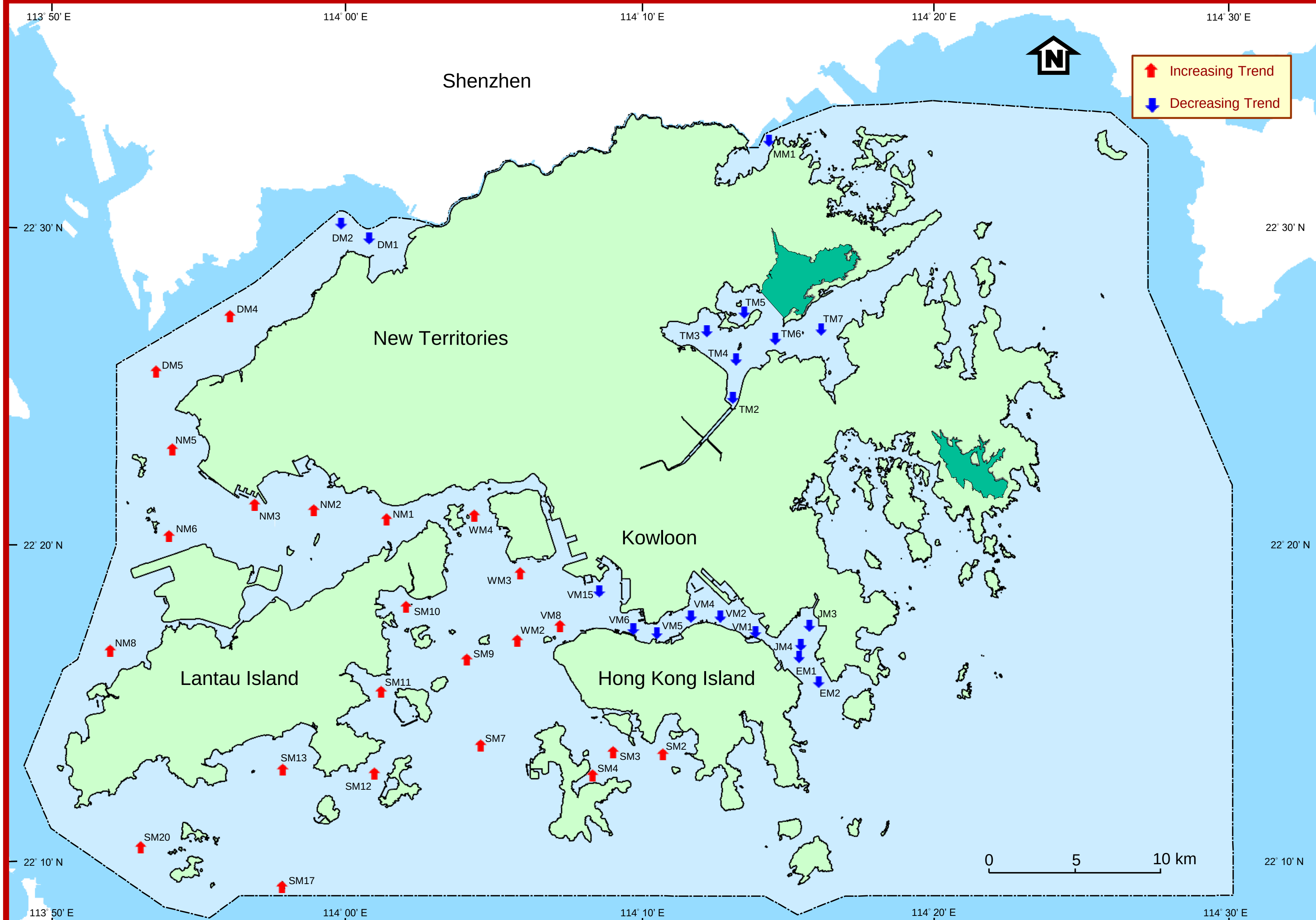
Long-term changes in *E.coli* levels in Hong Kong marine waters, 1986-2024



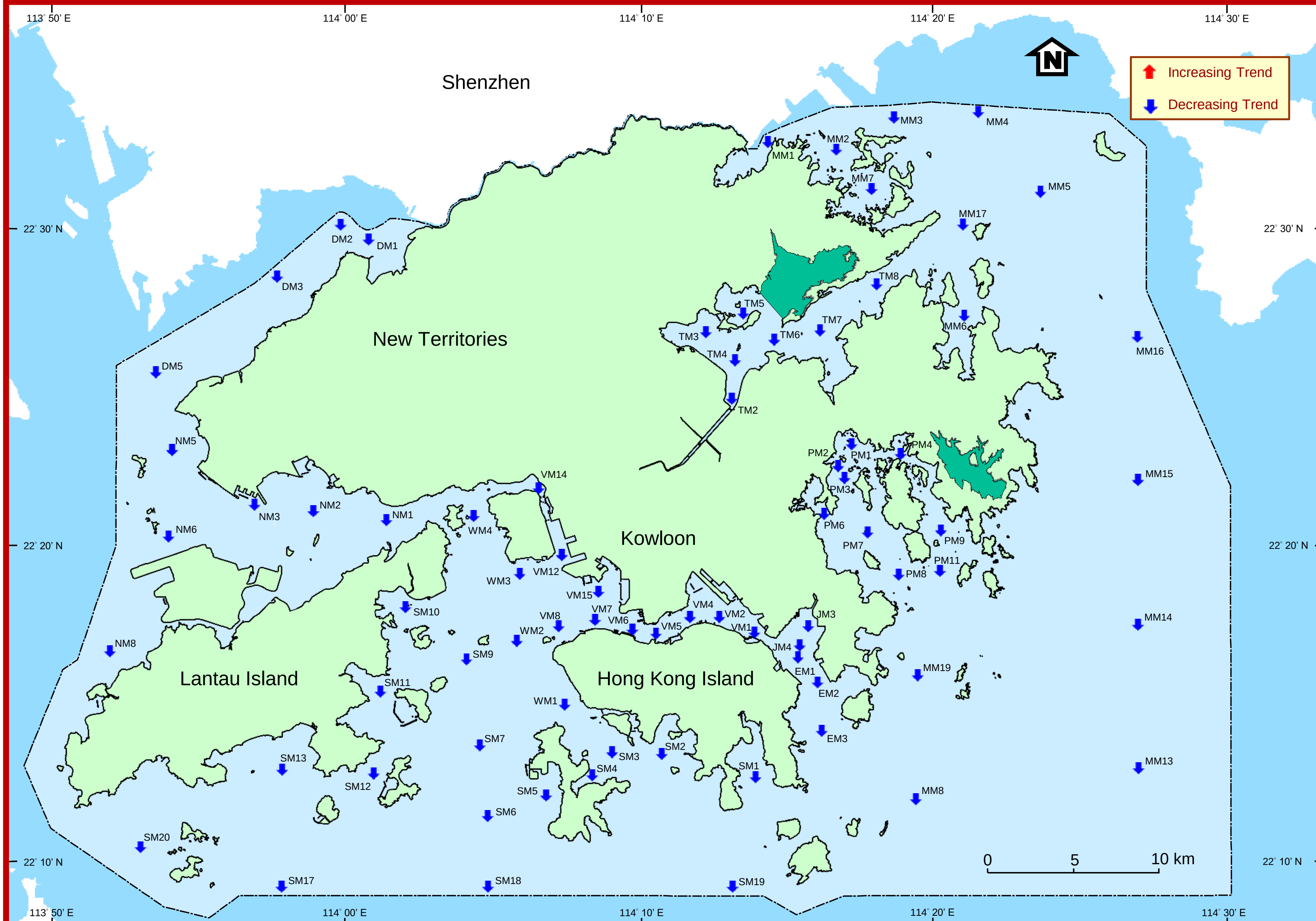
Long-term changes in ammonia nitrogen levels in Hong Kong marine waters, 1986-2024



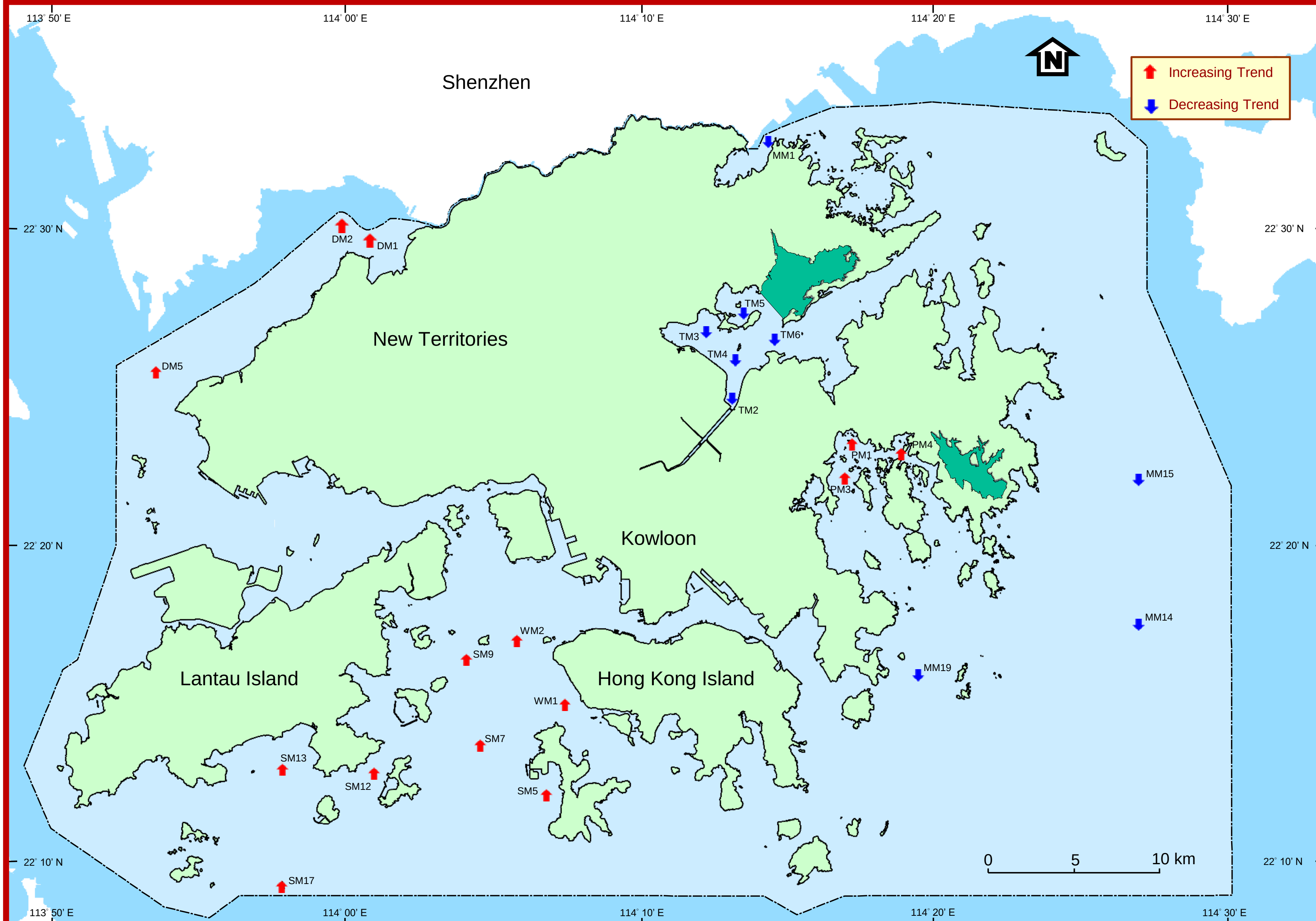
Long-term changes in total inorganic nitrogen levels in Hong Kong marine waters, 1986-2024



Long-term changes in orthophosphate phosphorus levels in Hong Kong marine waters, 1986-2024



Long-term changes in chlorophyll-*a* levels in Hong Kong marine waters, 1986-2024



Long-term changes in temperature in Hong Kong marine waters, 1986-2024